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**TECHNIQUES FOR TACTICAL RADIOS AND  
RETRANSMISSION**

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**JULY 2025**

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# TECHNIQUES FOR TACTICAL RADIOS AND RETRANSMISSION

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## Preface

ATP 6-02.53 is the primary doctrine for tactical radios and tactical radio networks. This publication describes the non-prescriptive methods to perform missions, functions, and tasks for the employment of tactical radios and tactical radio networks to support every warfighting function and enable command and control of forces. This publication includes doctrine on new communications and networking capabilities. It addresses the employment of interdependent and interoperable tactical radios.

The principal audience for ATP 6-02.53 is leaders, staffs, supervisors, planners, and radio operators responsible for operating tactical radios or employing tactical radio networks at echelons corps and below. This ATP is a standard reference for system planners. It provides network planning guidance, interoperability considerations, and equipment capabilities. Commanders and staffs of Army headquarters serving as joint task force or multinational headquarters should also refer to applicable joint or multinational doctrine concerning the range of military operations and joint or multinational forces. Trainers and educators throughout the Army will also use this publication.

Commanders, staffs, and subordinates ensure that their decisions and actions comply with applicable United States, international, and in some cases host-nation laws and regulations. Commanders at all levels ensure that their Soldiers operate in accordance with the law of armed conflict and the rules of engagement (see FM 6-27). They also adhere to Army Ethic as described in ADP 6-22.

ATP 6-02.53 implements the following interoperability agreements:

STANAG 4204, *Technical Standards for Single Channel VHF Radio Equipment* (Edition 3).

STANAG 4372, SATURN – *A Fast Frequency Hopping EPM Mode for UHF Radio* (Edition 4).

STANAG 4538, *Technical Standards for an Automated Radio Control System (ARCS) for HF communication Links* (Edition 1).

This publication uses joint terms where applicable. Selected joint and Army terms and definitions appear in both the glossary and the text. Terms for which ATP 6-02.53 is the proponent publication (the authority) are italicized in the text and are marked with an asterisk (\*) in the glossary. Terms and definitions for which ATP 6-02.53 is the proponent publication are boldfaced in the text. For other definitions shown in the text, the term is italicized, and the number of the proponent publication follows the definition.

This publication applies to the Active Army, Army National Guard/Army National Guard of the United States, and United States Army Reserve unless otherwise stated.

The proponent of ATP 6-02.53 is the United States Army Cyber Center of Excellence. The preparing agency is Doctrine Division, United States Army Cyber Center of Excellence. Send comments and recommendations on DA Form 2028 (*Recommended Changes to Publications and Blank Forms*) to Commander, United States Army Cyber Center of Excellence, ATZH-OPD (ATP 6-02.53), 419 B Street, Fort Gordon, GA 30905-5735, or e-mail to [usarmy.gordon.cyber-coe.mbx.gord-fg-doctrine@army.mil](mailto:usarmy.gordon.cyber-coe.mbx.gord-fg-doctrine@army.mil).

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# Introduction

ATP 6-02.53 expands on the foundational information about tactical radios and tactical radio networks in FM 6-02. Tactical radios are critical command and control enablers. Tactical radio networks empower subordinate decision making and allow integration and synchronization across all domains. Tactical radios and networks support the Army's mission across the competition continuum.

Fluid tactical situations in sustained operations require versatile and adaptable tactical radio networks. Tactical radios allow forces to move and communicate from higher echelons down to the squad level. Squad-level radios enable communication and situational awareness at the lowest level possible.

This publication contains important developments regarding tactical radio operations with greater emphasis on the employment of tactical radio retransmission at echelons corps and below.

ATP 6-02.53 has three chapters and nine appendixes:

- **Chapter 1** discusses the operational environment and examines challenges in a contested and congested environment. It includes an explanation of how tactical networks and radios enable command and control and situational awareness during Army operations.
- **Chapter 2** addresses the employment of tactical radios and signal support by Army echelon. It also addresses the importance of integrating tactical radios into command posts.
- **Chapter 3** introduces network planning and explains planning requirements and considerations for secure tactical radio operations. This chapter provides an overview of retransmission planning and employment and includes retransmission pre-combat checks and pre-combat inspections. This chapter also describes communications staff elements—planners, spectrum managers, and communications security account managers—and the importance of coordinated activities among staff elements during the planning process.
- **Appendix A** includes information on how to identify and respond to electromagnetic attack. It addresses techniques to overcome threat capabilities including information on how to report and prevent the effects of suspected electromagnetic attack. This appendix also describes types of electromagnetic interference and appropriate actions to minimize the impact to operations or friendly communications.
- **Appendix B** presents training and certification standards for retransmission teams. This appendix also contains signal assessment tables, battle drills, and team drills to include assigned tasks supporting the unit's mission.
- **Appendix C** addresses the combatant command communications team and their responsibility in providing the combatant commander and theater army commander secure, reliable communications support.
- **Appendix D** discusses aspects of waveform and waveform application functional components within the tactical networking environment. It also describes the properties of radio waves and wave modulation.
- **Appendix E** describes high frequency radio networks and how they support command and control. It also discusses establishing high frequency radio networks and how propagation, antenna choice, and site selection affect signal transmission. Included in this appendix is information on how each echelon uses high frequency radio networks to enable communications and fulfill mission objectives during operations.
- **Appendix F** addresses the Julian day, synchronization time, and ZULU time. It also includes a sample time zone conversion chart and world time zone map.
- **Appendix G** addresses radio operations in adverse weather conditions. Included in this appendix is information on techniques for better communication including how radio system maintenance based on environmental factors enhances radio capabilities and ensures successful communication.
- **Appendix H** provides information on antenna selection and techniques with examples of field repair of antennas and antenna support equipment. This appendix also contains important concepts and

terms beneficial to understanding tactical radio operations and implementing effective tactical radio communications.

- **Appendix I** discusses tactical radios and radio platforms. Also addressed are enhanced capabilities and interoperability features for handheld, manpack, vehicle-mounted, and airborne radios. This appendix includes information on the integrated tactical network, a key enabler for robust communication for units facing disconnected, intermittent, or limited communication.

# Chapter 1

## Overview

*A piece of paper makes you an officer, a radio makes you a commander.*

General of the Army Omar N. Bradley

This chapter discusses the operational environment and challenges encountered in a congested and contested environment. It includes information on tactical radio fundamentals covering the topic of radios in Army operations, common tactical radio frequency bands, and capabilities. The chapter concludes with a brief overview of Army and joint networks.

### SECTION I – OPERATIONAL ENVIRONMENT AND THREAT

1-1. Commanders and their staffs seek to develop and maintain an understanding of their operational environment and its effects across the competition continuum. A thorough understanding of the operational environment helps identify opportunities to coordinate with mission partners and converge effects when operating throughout a multidomain battlespace.

1-2. Threats are a fundamental part of an overall operational environment. For mission success, commanders and staffs must identify potential threats—both state and non-state actors—within the context of an operational environment that might disrupt, destroy, or intercept the flow of information to command and control systems and Army communications networks.

### THE OPERATIONAL ENVIRONMENT

1-3. Army forces conduct operations in specific operational environments. An *operational environment* is the aggregate of the conditions, circumstances, and influences that affect the employment of capabilities and bear on the decisions of the commander (JP 3-0). The operational environment encompasses the physical areas of the air, land, maritime, and space domains, as well as cyberspace and the electromagnetic spectrum (EMS). Signal leaders must understand the available communications capabilities and interoperability challenges of communications support in each domain.

1-4. Signal formations' delivery of communications services varies based on the mission. As commanders operate across the competition continuum, signal capabilities must also be tailorable to meet the commander's requirements. Commanders rely on signal leaders to clearly articulate available communications capabilities to support operations and accomplish the mission.

### CONTESTED ENVIRONMENT

1-5. Communications and information networks are continually targeted by threats. Threats understand the extent of U.S. forces' reliance on communications and automated information systems. Consequently, threats will contest the operational environment and seek to diminish the effectiveness of U.S. and allied forces' communications networks. Threats will deliberately attempt to deny, degrade, compromise, neutralize, and destroy friendly use of the EMS, space, and cyberspace capabilities, including terrestrial communications systems.

1-6. Successfully integrating signal support with electromagnetic warfare and intelligence capabilities is essential to obtaining and maintaining freedom of action in the operating environment while denying the same to adversaries. Synchronizing capabilities across multiple domains and warfighting functions enables command and control and mitigates the threat's ability to disrupt tactical radio networks.

### CONGESTED ENVIRONMENT

1-7. Competition for available bandwidth results in a congested EMS. Army elements and supporting agencies depend on the EMS for communications and information collection, and electromagnetic warfare capabilities to support operations in the air, land, maritime, space, and cyberspace domains.

1-8. Signal systems rely on the EMS for network transport. Gaining and maintaining access to the EMS is critical for signal support to operations. Signal elements plan communications and network capabilities to support all anticipated requirements in their assigned areas. In a congested operational environment, however, there may not be adequate spectrum availability to support all missions. Signal leaders must clearly articulate the limitations and expected level of degradation, so commanders can make appropriate risk decisions and align the available capabilities with their priorities. In addition, signal leaders should incorporate a primary, alternate, contingency, and emergency (PACE) communication plan to establish redundant communications means if the primary capability is unavailable. See chapter 3 for additional information on PACE communication plans and planning considerations.

1-9. Units that use the full extent of their communications capabilities during training improve their ability to understand conditions, circumstances, and influences impacting the electromagnetic operational environment. Signal planners should leverage resources within their support channels during unit training and exercises to include capabilities, expertise, and tools to further maximize performance effectiveness during mission execution. Implementing realistic training scenarios fosters proficiency enabling team members to achieve greater accuracy in characterizing congested environments.

## FREEDOM OF MANEUVER

1-10. Maintaining dominance is increasingly dependent upon the Army's ability to establish and maintain freedom of maneuver within cyberspace and the EMS. Advances in technology and a rise in the number of spectrum-dependent devices used by nonmilitary and military entities have resulted in greater competition for usable bandwidth and an exponential increase in electromagnetic emitters within the electromagnetic operational environment. The *electromagnetic operational environment* is a composite of the actual and potential electromagnetic energy radiation, conditions, circumstances, and influences that affect the employment of capabilities and the decisions of the commander (JP 3-85).

1-11. The EMS spans every domain in the operational environment; threats attempt to deny friendly ability to use it successfully. Disruptions present significant challenges that threaten the Army's ability to successfully accomplish objectives and mission success. For forces to effectively operate in an increasingly congested and contested environment, signal soldiers must identify malicious activity that compromises network integrity and implement techniques to secure critical communications networks and data.

1-12. Through surveillance and target acquisition, threats attempt to deny U.S. forces the use of EMS, thus compromising tactical network access, disrupting radio networks, and preventing electromagnetic signature characterization efforts of the U.S. and its mission partners. Threats with intent to defeat U.S. forces target key national assets within the United States and abroad using technical ingenuity and electromagnetic warfare equipment with enhanced detection, interception, and geolocation capabilities.

1-13. Signal soldiers and planners must maintain network trust, reliability, and resilience across the EMS. Techniques offering identity and location protection prevent attacks based on distinct signature, enhances force protection, and enables network planners and spectrum managers to develop actionable mission planning strategies to improve Soldier emission awareness and enable leader decision making.

1-14. Threat capabilities specific to tactical radio operations include systems designed to physically destroy U.S. communications equipment. To increase their understanding of threat capabilities used against spectrum-dependent devices and electrical equipment, U.S. forces align cyberspace electromagnetic activities (CEMA) and spectrum management operations to enhance spectrum efficiency, maximize spectrum compatibility, and ensure EMS superiority against current and potential threats. U.S. forces also focus on integration, interoperability, and collaboration with Federal departments and agencies, allies, and mission partners to further synchronize coalition activities, increase EMS capability and capacity, and deter strategic competition. The three distinct divisions of electromagnetic warfare—electromagnetic attack, electromagnetic protection, and electromagnetic support—aid in creating force dominance while spectrum management operations prevent unacceptable interference and enable freedom of action in and through the EMS. Refer to ATP 3-12.3 for more information about electromagnetic warfare techniques.

1-15. Every proactive measure taken by U.S. forces to prevent signal interception and communications systems discovery results in increased Soldiers situational understanding and awareness of their own radio emissions when operating in an increasingly complex, densely populated electromagnetic operational

environment. To mitigate activities that place the mission at risk and deter threats directly impacting tactical radio communications and user operations, signal personnel apply electromagnetic protection measures to protect, detect, characterize, counter, and mitigate specific effects. By taking measures to prevent signal detection, location, and emanation, command post survivability is enhanced and the efforts by attackers to discover and target key infrastructure resource are minimized or defeated. Implementing offensive and defensive actions, active and passive countermeasures, and decoying techniques and capabilities also offsets EMS operating environment constraints and allows sufficient freedom of action to carry out mission objectives successfully. See appendix A for signal threats and how to recognize and respond to spectrum interference.

## THREAT

1-16. A *threat* is any combination of actors, entities, or forces that have the capability and intent to harm United States forces, United States national interests, or the homeland (ADP 3-0). Threats can be categorized as adversaries, enemies, or insider threats. An *adversary* is a party acknowledged as potentially hostile to a friendly party and against which the use of force may be envisaged (JP 3-0). An *enemy* is a party identified as hostile against which the use of force is authorized (ADP 3-0). Insider threats present unique challenges because they are trusted individuals with access to Army capabilities and sensitive operational information. Threats may include individuals or groups, organized, or not organized.

1-17. Threats are a fundamental part of an overall operational environment. A threat actor can disrupt, destroy, or intercept the flow of information while operating from the other side of the world. Threat actors use an array of capabilities and methods to threaten Army systems and networks.

## THREAT EFFECTS ON SIGNAL SUPPORT

1-18. Based on the system and capabilities and the spectrum management practices implemented, commanders project power through delivery of effects. Threat effects created by spectrum-enabled systems focus on operations for the purposes of destruction, degradation, disruption, and deception. Hostile entities target key communications capabilities, nodes, sensors, and networks including on-orbit, link, and terrestrial segments of U.S. positioning, navigation, and timing (PNT) satellites creating potentially significant impacts across all warfighting functions and weapon platforms.

1-19. Executing successful operations requires U.S. forces to effectively neutralize threat activities in the EMS and distinguish threat and friendly capabilities. To deter and combat threats that directly impact tactical radio communications and user operations, signal personnel coordinate effects and implement risk mitigation activities or techniques. Defensive measures include employing—

- Radar jamming technology.
- Self-propelled and manual decoys.
- Reflective materiel or objects such as metallic strips or aluminum-coated tape used to—
  - Reflect frequencies.
  - Cause signal interference.
  - Clutter the radar with false targets.
  - Re-radiate radar energy toward the originating source.
- Other actions include using—
  - Emission control (EMCON).
  - Masking and network countermeasures.

1-20. These measures prevent the threat's effectiveness or either degrade or neutralize friendly capabilities creating impacting, incapacitating hazards in the form of electromagnetic interference, radiation, system and platform capability issues, and environmental and personnel hazards.

1-21. Tactical radio networks also encounter exposure to a broad range of peer threats with signals intelligence, electromagnetic warfare, electromagnetic attack, and cyberspace capabilities. For successful electromagnetic and cyberspace warfare operations, signal personnel combine electromagnetic warfare jamming techniques with surveillance and eavesdropping techniques including signals intelligence designed to locate threat transmissions. Working in concert with signals intelligence counterparts and Soldiers in the CEMA section, signal planners and support elements extract and assess signals technical data correlating

unidentified signals of interest to threat emitters. Their collaborative efforts ensure appropriate defense mechanisms are implemented to maintain continued access to communications networks and capabilities.

1-22. Preventive measures that align key operations of signals intelligence, electromagnetic attack, and cyberspace operations help to secure the network, build network trust, and guarantee the ability to quickly adjust capabilities based on mission needs and threat actions. Operations and planning efforts are further maximized by implementing electromagnetic warfare concepts and incorporating appropriate tactics, techniques, and procedures at all training levels—individual, collective, institutional, and operational.

## **SECTION II – TACTICAL RADIO FUNDAMENTALS**

1-23. This section covers tactical radio communications including common frequency bands and radio assets available to signal elements for use in Army operations. It also contains a general description of tactical radios and radio capabilities.

### **TACTICAL RADIOS IN ARMY OPERATIONS**

1-24. Tactical radio systems are the primary means for Soldiers to communicate during all phases of Army operations. Squad leaders and higher use tactical radios to communicate and share a common view of the operational environment. Tactical radios provide Army Soldiers the ability to send and receive voice, data, and video communication and may include commercial off-the-shelf radios. The systems use high frequency (HF), ultrahigh frequency (UHF), and very high frequency (VHF) bands.

1-25. Army forces employ a variety of tactical radios and waveforms. Each system has unique capabilities and characteristics. Signal planners determine how to employ each system depending on the mission.

### **COMMON TACTICAL RADIO FREQUENCY BANDS**

1-26. Radio frequency (RF) waves are a subcomponent of the EMS, a continuum of all electromagnetic waves arranged according to frequency and wavelength. The most common RF ranges for tactical radios are—

- HF (3–30 MHz).
- VHF (30–300 MHz).
- UHF (300–3,000 MHz).

1-27. The electromagnetic waves within each range have different characteristics that are significant in determining their practical application. A wide variety of tactical radios support communications through several configurations including airborne, maritime, terrestrial fixed station, vehicular, handheld, and manpack.

### **TACTICAL RADIO CAPABILITIES**

1-28. Army forces employ a variety of portable, lightweight tactical radios and waveforms. The type of tactical radio system chosen for employment depends on mission requirements. Tactical radio capabilities include—

- Automatic link establishment (ALE).
- Anti-jamming.
- Beyond line of sight.
- Data rates.
- Global Positioning System (GPS).
- Programmable (software defined).
- Secure communications.

## **SECTION III – NETWORK OVERVIEW**

1-29. Networks are key enablers of successful operations. They consist of end-user applications, information services and data, and network transport and management devices and systems. Network capabilities enable

leaders to receive, process, and disseminate information supporting coordinated planning, preparation, execution, and synchronization with global mission partners.

1-30. Networks are one of four components of a command and control system established by commanders at every echelon to support decision making, facilitate communication of mission objectives, and enable execution of command and control across all warfighting domains. The command and control system, including people, processes, command posts, and networks, are arranged to provide continuity of command and allow integration and synchronization of combat power during sustained operations.

1-31. Along with networks, commanders have access to a collection of communications network resources and systems supporting each warfighting function to include tactical radios, data networks, command and control information systems, and on-the-move packages. For a detailed description of networks supporting each warfighting function, refer to FM 6-02.

## JOINT NETWORK

1-32. Networks provide the connectivity between Army forces and its mission partners in a joint operations area. Joint forces depend on the Department of Defense information network (DODIN) to connect strategic, operational, and tactical commanders across the globe.

1-33. The *Department of Defense information network* is the information capabilities and associated processes owned or leased by the Department of Defense for collecting, processing, storing, disseminating, and managing information, whether interconnected or stand-alone (JP 6-0). The DODIN includes owned and leased communications and computing systems and services, software (including applications), data, security services, other associated services, and national security systems.

1-34. The DODIN supports military operations by enabling seamless and secure information sharing. It encompasses all DOD cyberspace—Army, Navy, Air Force, Marine Corps, and Space Force Service-specific enclaves as well as joint capabilities provided by the Defense Information System Agency.

## ARMY NETWORK

1-35. The *Department of Defense information network-Army* (DODIN-A) is an Army-operated enclave of the Department of Defense information network that encompasses all Army information capabilities that collect, process, store, display, disseminate, and protect information worldwide (ATP 6-02.71). The DODIN-A and the services it extends enable collaboration and synchronization across all domains and among mission partners.

1-36. The DODIN-A provides strategic voice and data capabilities and prepositioned connection points to extend communications services to geographically dispersed personnel, both fixed station and deployed expeditionary forces.

1-37. As part of the command and control system, the network provides the infrastructure for voice, data, and video. The DODIN-A includes tactical radio networks supporting intelligence, sustainment, special operations forces, and medical. These networks enhance situational understanding and enable resource and information sharing and collaboration among units, staff, and support personnel to support the commander's intent and allow informed decision making.

## TACTICAL RADIOS

1-38. The primary function of tactical radio networks is to provide voice communication and situational awareness at all echelons, across all phases of operations. Tactical radio networks enable command and control, communications, information sharing, and coordination between higher and lower echelons as well as adjacent units. Tactical radio networks support command post operations and command and control of mounted and dismounted forces operating while either static or on the move. They enable air-ground communications with Army and joint aviation platforms and other combat enablers such as fires, casualty evacuation, and sustainment. Tactical radio networks assume a secondary support role for data transmission where other data capabilities do not exist. The tactical radio network includes—

- Combat net radios.
- Vehicular and dismounted radios.

- Joint Battle Command Platform.
- Nett Warrior.
- Blue Force Tracker (I, II, and III).
- Mobile User Objective System (MUOS).
- Integrated Tactical Network (ITN).

### **Integrated Tactical Network**

1-39. The ITN provides an independent mobile network solution supporting command and control, situational awareness, and air-to-ground integration. The ITN is not a single system or program of record, but a combination of developing tactical network capabilities augmented with emerging commercial and non-developmental items, capabilities, and technologies. ITN capabilities provide echelons brigade and below with smaller, lighter, faster, and more flexible communications systems. With the fielding of the ITN, new commercial components, network transport capabilities, applications, and devices are introduced into the Army's tactical network environment through network baselines. The network baselines focus on infantry, Stryker, and armored brigade combat teams. ITN capabilities include the ability to enable multipath network diversity using the tactical scalable mobile ad hoc network (TSM) commercial waveforms. This feature enables highly mobile operations and provides tactical agility for units facing disconnected, intermittent, or limited communications. The ITN offers enhanced communications resiliency and security in a contested or congested environment.

### **Tactical Radio Networks**

1-40. Tactical radios enable every mission, from training the force to executing tactical tasks to influence the operational environment in large-scale combat operations. Radios enable secure transmission of real-time tactical information and provide connectivity to designated communications networks or stations.

1-41. Army forces rely on tactical radio networks to enable communications, situational understanding, and decision making. Tactical radio networks commonly used by maneuver elements include—

- Command.
- Sustainment.
- Operations and intelligence.
- Medical.
- Fires.
- Surveillance.

1-42. A communications network is an organization of stations capable of intercommunications, but not necessarily on the same channel. Table 1-1 on pages 7 through 9 shows typical tactical radio networks employed at echelons corps and below and the network stations established within each network. The networks shown merely serve as a guide for establishing radio networks. The actual networks established depend on the existing situation, command guidance, and equipment available.

Table 1-1. Typical tactical networks

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Movement and Maneuver |            | Intelligence |              | Fires        |              |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------|--------------|--------------|--------------|--------------|-----------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Command and Control   | Operations | Current      | Surveillance | Coordination | Instructions | Emergency |
| <b>Commanders</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |            |              |              |              |              |           |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | X                     |            | X            |              |              |              |           |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | X                     |            | X            |              |              |              |           |
| Brigade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | X                     |            | X            |              |              |              |           |
| <b>Executive Officers</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                       |            |              |              |              |              |           |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | X                     | X          |              |              |              |              |           |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | X                     | X          |              |              |              |              |           |
| Brigade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | X                     | X          |              |              |              |              |           |
| <b>Command Sergeants Major</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                       |            |              |              |              |              |           |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | X                     | X          |              |              |              |              |           |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | X                     | X          |              |              |              |              |           |
| Brigade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | X                     | X          |              |              |              |              |           |
| <b>G-3 or S-3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |            |              |              |              |              |           |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | X                     | X          | X            |              | X            |              | X         |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | X                     | X          | X            |              | X            |              | X         |
| Brigade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | X                     | X          | X            |              | X            |              | X         |
| <b>G-1 or S-1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |            |              |              |              |              |           |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | X                     |            |              |              |              |              |           |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | X                     |            |              |              |              |              |           |
| Brigade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | X                     |            |              |              |              |              |           |
| <b>G-2 or S-2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |            |              |              |              |              |           |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | X                     |            | X            | X            | X            |              | X         |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | X                     |            | X            | X            | X            |              | X         |
| Brigade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | X                     |            | X            | X            | X            |              | X         |
| <b>Legend:</b><br>G-1      assistant chief of staff, personnel<br>G-2      assistant chief of staff, intelligence<br>G-3      assistant chief of staff, operations<br>G-4      assistant chief of staff, logistics<br>G-5      assistant chief of staff, plans<br>G-6      assistant chief of staff, signal<br>S-1      battalion or brigade personnel staff officer<br>S-2      battalion or brigade intelligence staff officer<br>S-3      battalion or brigade operations staff officer<br>S-4      battalion or brigade logistics staff officer<br>S-6      battalion or brigade signal staff officer |                       |            |              |              |              |              |           |

Table 1-1. Typical tactical networks (continued)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <i>Movement<br/>and<br/>Maneuver</i> |                   | <i>Intelligence</i> |                     | <i>Fires</i>        |                     |                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------|---------------------|---------------------|---------------------|---------------------|------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <i>Command<br/>and<br/>Control</i>   | <i>Operations</i> | <i>Current</i>      | <i>Surveillance</i> | <i>Coordination</i> | <i>Instructions</i> | <i>Emergency</i> |
| <b>G-4 or S-4</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                      |                   |                     |                     |                     |                     |                  |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | X                                    |                   |                     |                     |                     |                     |                  |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | X                                    |                   |                     |                     |                     |                     |                  |
| Brigade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | X                                    |                   |                     |                     |                     |                     |                  |
| <b>G-5</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                      |                   |                     |                     |                     |                     |                  |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | X                                    |                   |                     |                     | X                   |                     |                  |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | X                                    |                   |                     |                     | X                   |                     |                  |
| <b>G-6 or S-6</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                      |                   |                     |                     |                     |                     |                  |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | X                                    |                   |                     |                     |                     |                     |                  |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | X                                    |                   |                     |                     |                     |                     |                  |
| Brigade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | X                                    |                   |                     |                     |                     |                     |                  |
| <b>Legend:</b><br>G-1      assistant chief of staff, personnel<br>G-2      assistant chief of staff, intelligence<br>G-3      assistant chief of staff, operations<br>G-4      assistant chief of staff, logistics<br>G-5      assistant chief of staff, plans<br>G-6      assistant chief of staff, signal<br>S-1      battalion or brigade personnel staff officer<br>S-2      battalion or brigade intelligence staff officer<br>S-3      battalion or brigade operations staff officer<br>S-4      battalion or brigade logistics staff officer<br>S-6      battalion or brigade signal staff officer |                                      |                   |                     |                     |                     |                     |                  |

Table 1-1. Typical tactical networks (continued)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <i>Sustainment</i> |                    |               |                  | <i>Protection</i> |                        |                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|---------------|------------------|-------------------|------------------------|--------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <i>Medical</i>     | <i>Maintenance</i> | <i>Supply</i> | <i>Personnel</i> | <i>Engineer</i>   | <i>Military Police</i> | <i>Guard Force</i> |
| <b>Commanders</b>                                                                                                                                                                                                                                                                                                                                                                                                                                   |                    |                    |               |                  |                   |                        |                    |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                               | X                  |                    |               | X                |                   |                        |                    |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                            | X                  |                    |               | X                |                   |                        |                    |
| Brigade                                                                                                                                                                                                                                                                                                                                                                                                                                             | X                  |                    |               | X                |                   |                        |                    |
| <b>Executive Officers</b>                                                                                                                                                                                                                                                                                                                                                                                                                           |                    |                    |               |                  |                   |                        |                    |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                               | X                  | X                  | X             | X                |                   |                        |                    |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                            | X                  | X                  | X             | X                |                   |                        |                    |
| Brigade                                                                                                                                                                                                                                                                                                                                                                                                                                             | X                  | X                  | X             | X                |                   |                        |                    |
| <b>Command Sergeants Major</b>                                                                                                                                                                                                                                                                                                                                                                                                                      |                    |                    |               |                  |                   |                        |                    |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                               | X                  |                    | X             | X                |                   |                        | X                  |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                            | X                  |                    | X             | X                |                   |                        | X                  |
| Brigade                                                                                                                                                                                                                                                                                                                                                                                                                                             | X                  |                    | X             | X                |                   |                        | X                  |
| <b>G-1 or S-1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                   |                    |                    |               |                  |                   |                        |                    |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                               | X                  |                    |               | X                |                   |                        |                    |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                            | X                  |                    |               | X                |                   |                        |                    |
| Brigade                                                                                                                                                                                                                                                                                                                                                                                                                                             | X                  |                    |               | X                |                   |                        |                    |
| <b>G-3 or S-3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                   |                    |                    |               |                  |                   |                        |                    |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                    | X             |                  | X                 | X                      |                    |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                            |                    |                    | X             |                  | X                 | X                      |                    |
| Brigade                                                                                                                                                                                                                                                                                                                                                                                                                                             |                    |                    | X             |                  | X                 | X                      |                    |
| <b>G-4 or S-4</b>                                                                                                                                                                                                                                                                                                                                                                                                                                   |                    |                    |               |                  |                   |                        |                    |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                               | X                  | X                  | X             | X                |                   |                        |                    |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                            | X                  | X                  | X             | X                |                   |                        |                    |
| Brigade                                                                                                                                                                                                                                                                                                                                                                                                                                             | X                  | X                  | X             | X                |                   |                        |                    |
| <b>G-5</b>                                                                                                                                                                                                                                                                                                                                                                                                                                          |                    |                    |               |                  |                   |                        |                    |
| Corps                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                    |               |                  | X                 |                        |                    |
| Division                                                                                                                                                                                                                                                                                                                                                                                                                                            |                    |                    |               |                  | X                 |                        |                    |
| <b>Legend:</b><br>G-1      assistant chief of staff, personnel<br>G-3      assistant chief of staff, operations<br>G-4      assistant chief of staff, logistics<br>G-5      assistant chief of staff, plans<br>S-1      battalion or brigade personnel staff officer<br>S-2      battalion or brigade intelligence staff officer<br>S-3      battalion or brigade operations staff officer<br>S-4      battalion or brigade logistics staff officer |                    |                    |               |                  |                   |                        |                    |

## COMMAND NETWORK

1-43. Command networks are protected communications networks supporting commanders' requirements to send and receive critical information during operations. Command networks link adjacent, higher, and lower echelons of command. The single-channel ground and airborne radio system (SINCGARS) is the

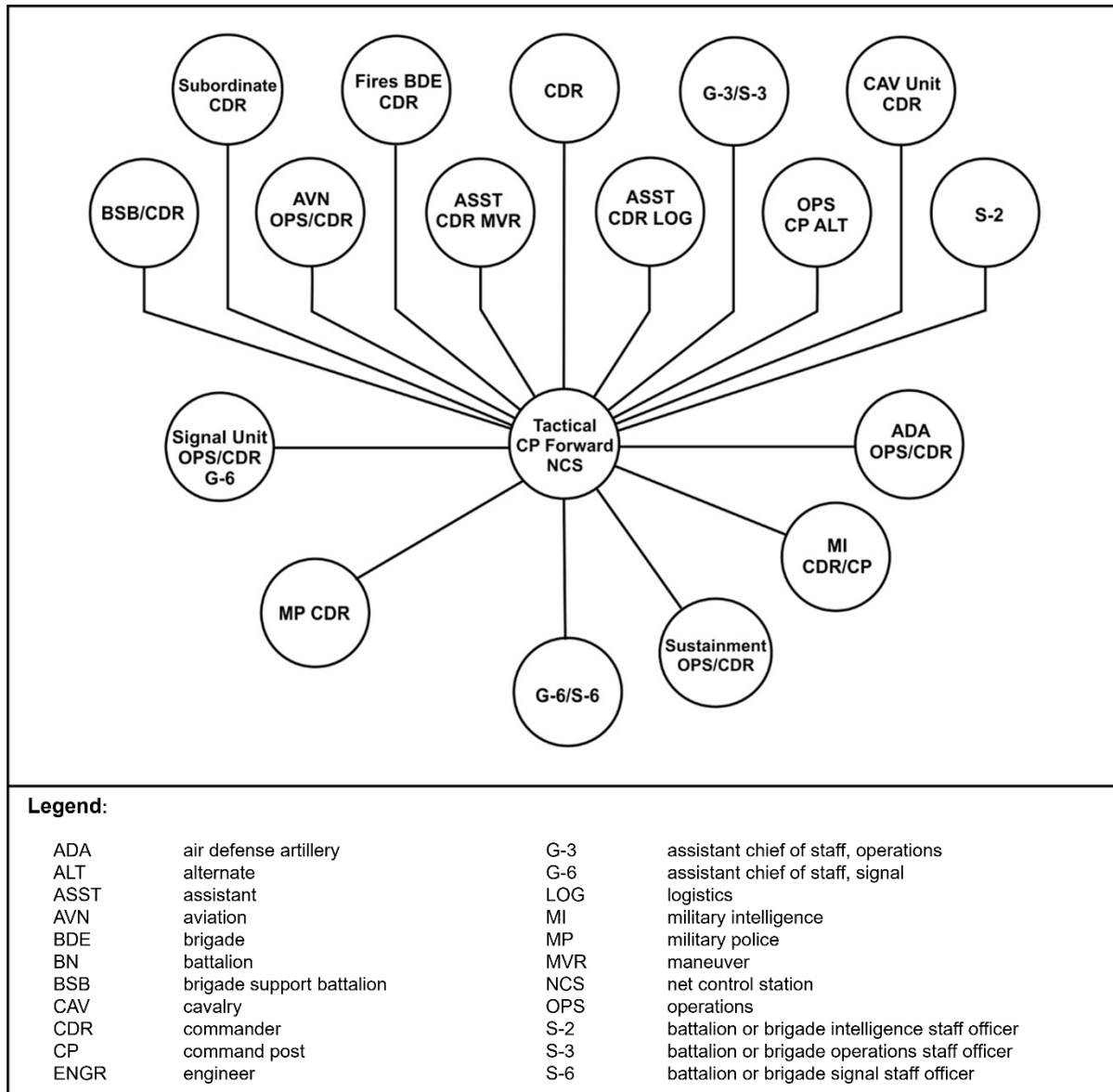
predominant means of operating command networks, and is a prevalent communications means for tactical operations at echelons division and below. Additional means of communication supporting command networks may include—

- HF.
- Narrowband (single-channel) tactical satellite (TACSAT).
- MUOS.
- Networking waveforms.
  - TSM.
  - Warrior Robust Enhanced Network (WREN).

1-44. Planners must ensure that radios use an approved cryptological key or encryption device. The primary radio source is determined based on warfighting function considerations and the mission variables of mission, enemy, terrain and weather, troops and support available, time available, civil considerations, and informational considerations—METT-TC (I) criteria as documented in the PACE plan published in Annex H of the operation order.

1-45. Units establish and rehearse a PACE plan to ensure a viable alternate, contingency, and emergency means of communication is available for communication between team members should the primary means of communication fail. A PACE plan designates the order in which an element will transition to redundant communications means in the event of threat attack or network failure. PACE plan rehearsals should include the commander's priorities for allocation of bandwidth and order of restoration when communications links are denied or degraded.

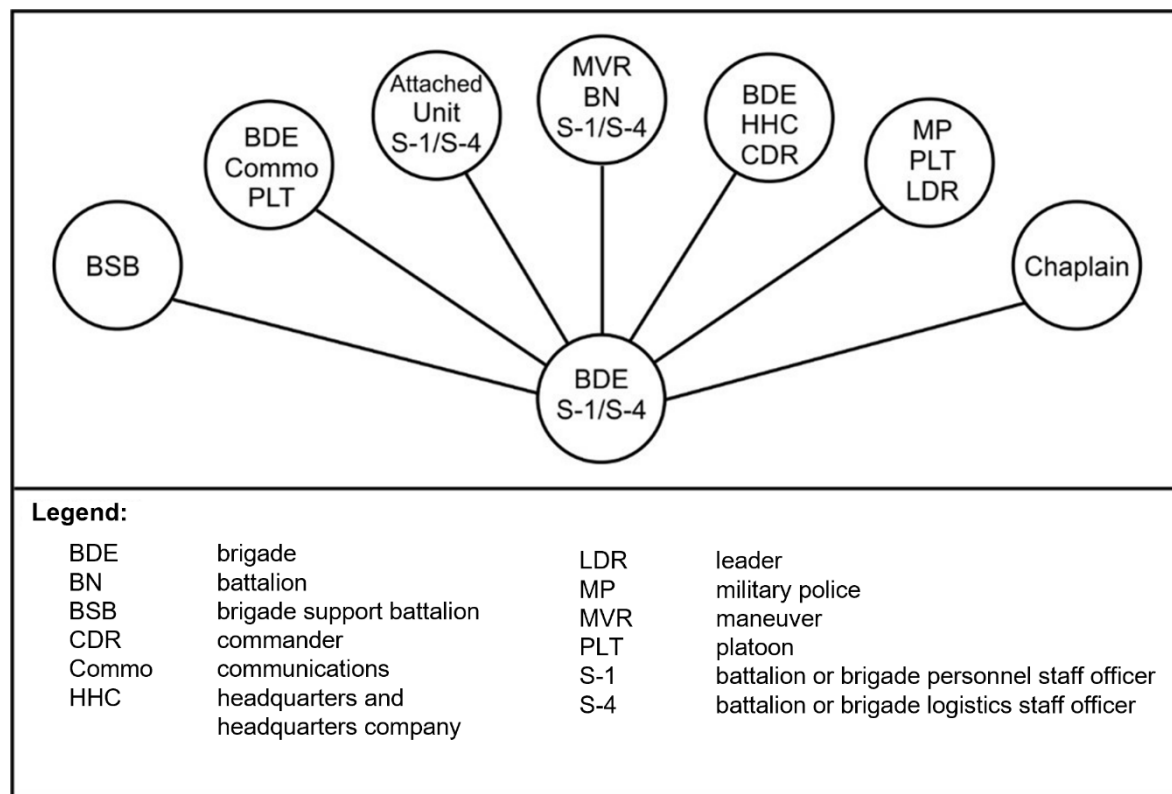
1-46. When necessary, units use retransmission teams to extend the range of the command network and overcome communication obstacles between higher and lower units. Command networks have the highest installation priority. Figure 1-1 on page 11 is an example of the structure of a typical command network.



**Figure 1-1. Typical command network structure**

### Sustainment Network

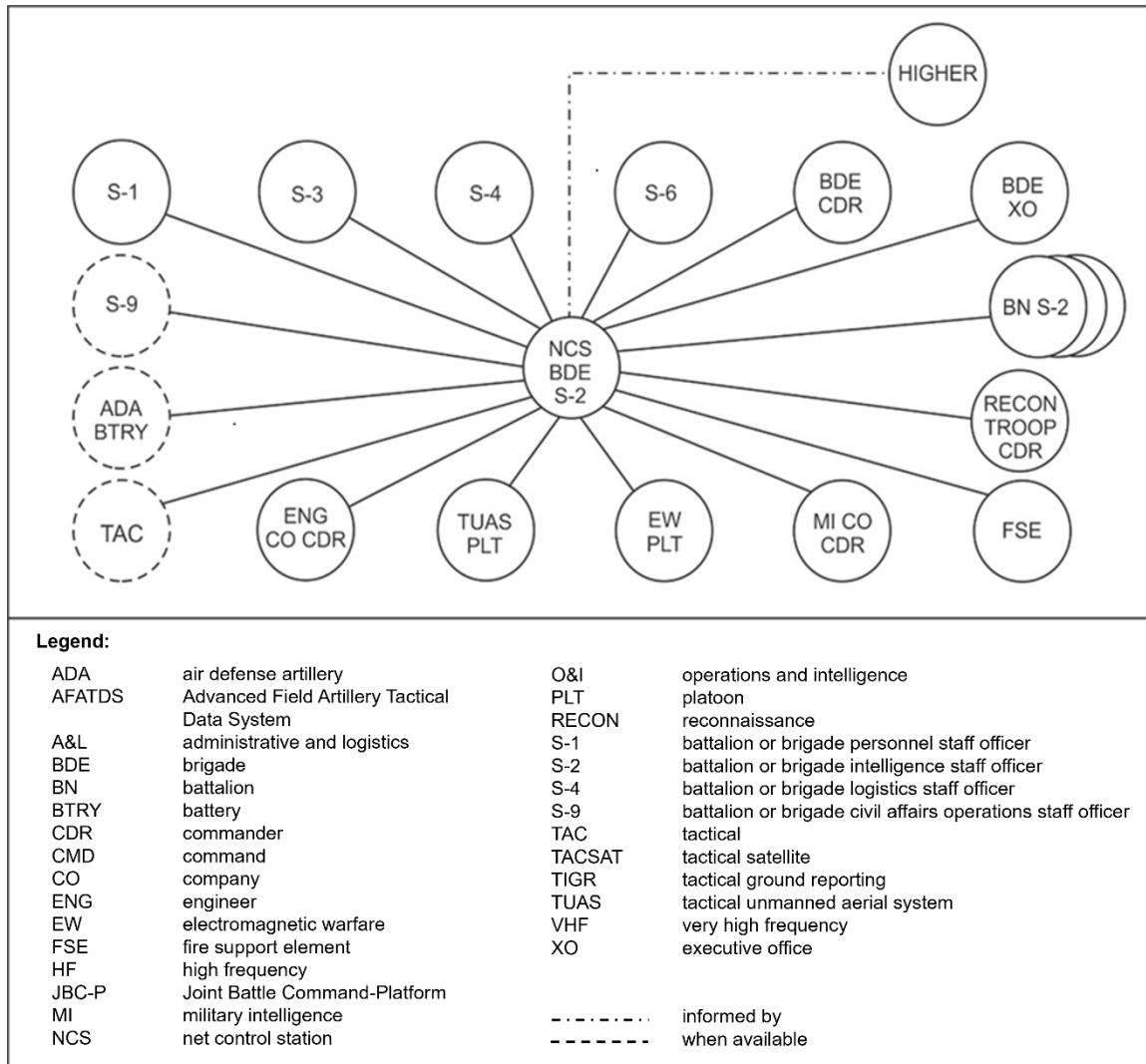
1-47. The sustainment network is a dedicated network used by commanders and staff officers during training exercises, sustainment operations, or military engagements. Its purpose is status reporting of sustainment, logistics, and administrative information to enable unit survivability and to ensure U.S. forces possess the supplies, personnel, medical, and force protection needed during tactical engagements. The sustainment network also facilitates horizontal communication with units or subordinates and provides leaders connectivity to their next higher echelon for situational awareness and coordinated operational objectives. An existing sustainment network may provide infiltration routes and information related to sources of supply or transportation. Figure 1-2 on page 12 is an example of a brigade sustainment network.



**Figure 1-2. Example of a brigade sustainment network**

### Operations and Intelligence Network

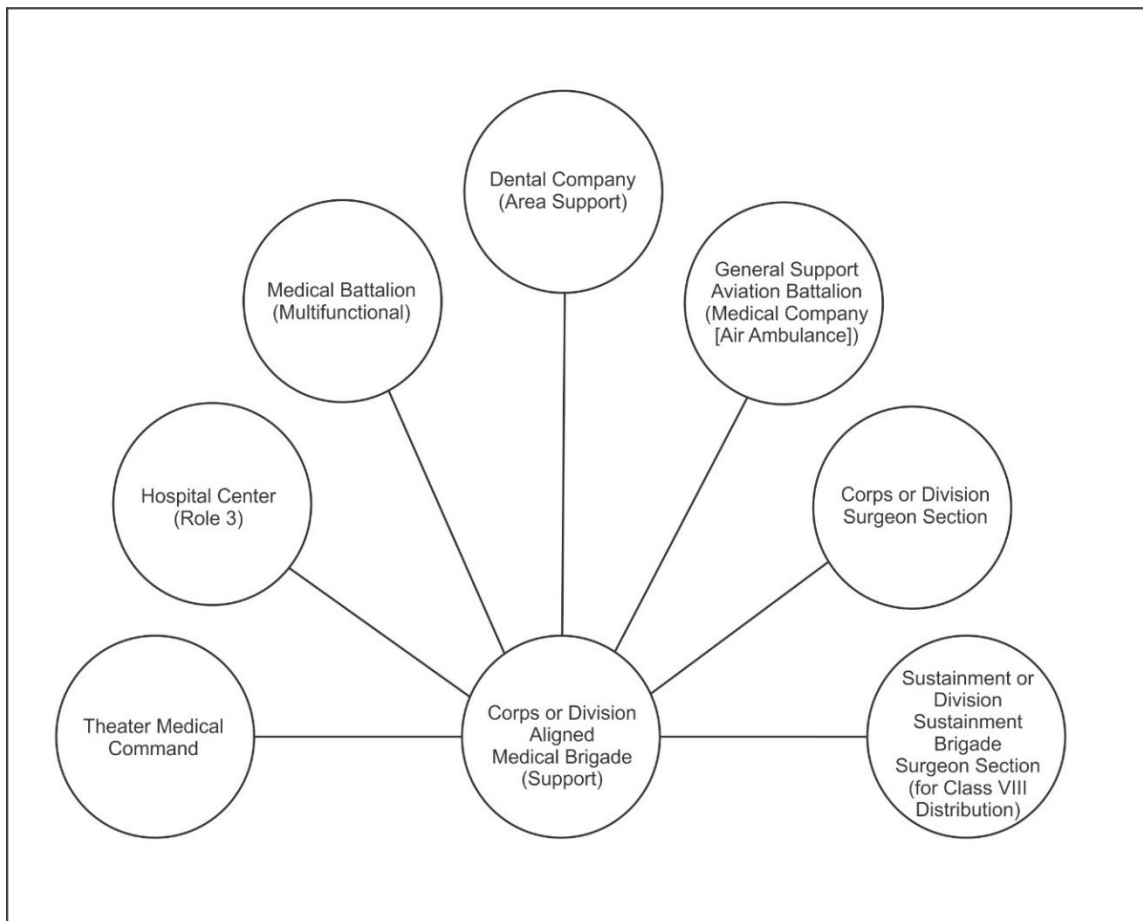
1-48. Operations and intelligence networks enable the transmission of observations, details, and discussions that feed intelligence analysis. The G-2 or S-2 controls operations and intelligence networks. Subordinate elements may monitor the operations and intelligence networks to develop situational awareness. Operations and intelligence networks usually include transmission and reception of surveillance reports related to enemy movement and disposition. A unit information control center may also use this network to coordinate and control surveillance capabilities and sensor teams. Subordinate elements may monitor the operations and intelligence network to develop situational awareness. Each echelon's intelligence section resides on the operations and intelligence networks of their higher echelon allowing reporting and pertinent information exchange. The information transmitted over operations and intelligence networks may require a secondary network to prevent network overloading. Figure 1-3 on page 13 is an example operations and intelligence network.



**Figure 1-3. Example of an operations and intelligence network**

## Medical Network

1-49. The medical network provides internal collaboration and support to tactical units to include coordinated Army Health System support in the form of health service support and force health protection. Health service support includes medical evacuations, medical treatments, hospitalization, and medical logistics medical functions. Force health protection encompasses operational public health, veterinary services, combat and operational stress control, dental services, and laboratory services; and may include the collection of medical intelligence. Figure 1-4 on page 14 is an example of a medical network.



**Figure 1-4. Example medical network**

### **Fires Network**

1-50. The fires network is the highest priority network for fires units. The fires network enables the exchange of technical and firing data. The fires network enables leaders to plan and execute the delivery of lethal and nonlethal fires, provides situational awareness, and enhances collaboration and information exchange among fires personnel. Fire support teams use the battalion fires network to send calls for fires based on assigned priorities. Retransmission teams support the fires network as required.

### **Surveillance Network**

1-51. Surveillance networks enable the transmission and reception of reports related to enemy movement and massing. The battalion battlefield information control center establishes this network to coordinate and control the ground surveillance radar and unattended ground sensor teams. The information from this network is vital to commanders and given high priority for activation.

## Chapter 2

# Tactical Radio Employment by Echelon

This chapter addresses the employment of tactical radios at various Army echelons. Tactical radios support all warfighting functions at all echelons across the competition continuum.

## SIGNAL SUPPORT BY ECHELON

2-1. A thorough understanding of support requirements allows signal elements to plan, implement, and employ networks to enable information exchange. Signal support teams are resourced with sufficient assets to enable reliable functioning of communications systems and capabilities within command posts at all echelons. Signal soldiers ensure command posts have adequate communications connectivity at each echelon by—

- Assessing communications requirements for all warfighting functions.
- Determining capabilities and limitations of available communications systems.
- Determining potential joint, interorganizational, and multinational communications requirements.
- Performing a detailed line of sight analysis.
- Implementing a PACE communication plan.
- Integrating all available signal assets.
- Developing a method of deployment so assets are sequenced to coincide with the arrival of forces.
- Mapping the locations of all communications systems.
- Implementing retransmission, digital network links, and node placement.
- Assessing spectrum requirements for emitters, sensors, radars, or any other spectrum-dependent devices.
- Assessing initial requirements and implementing flexibility for expected changes.
- Implementing the standard operating procedure (SOP) and communications security (COMSEC) procedures.
- Conducting communications rehearsals.
- Assessing the electromagnetic operational environment.
- Identifying critical vulnerabilities to be exploited by threats.
- Developing risk assessment for the commander.

## ROLES AND RESPONSIBILITIES

2-2. Signal elements and capabilities exist to meet the commander's information requirements and enable control of forces. Signal teams at each echelon provide communications connectivity within their assigned area to support continuous operations and collaboration between higher, subordinate, and adjacent elements.

2-3. Tactical radios provide communications and network support at the corps and division levels; echelons brigade, battalions, company, and platoon employ tactical radios to collaborate with adjacent units and higher reporting elements. Maneuver elements have limited organic signal capabilities consisting primarily of single-channel secure radios down to squad level. Their communications systems are mostly general purpose, user-operated equipment. Brigade units rely on tactical radio retransmission networks to extend the effective range of line of sight communication when executing tactical enabling tasks beyond the range of VHF single-channel radios, or when terrain prevents line of sight communications.

## CORPS AND DIVISION G-6

2-4. Signal staffs plan communications and network capabilities to support all anticipated requirements in their assigned area. Signal leaders and staff work closely to align available capabilities with the commander's priorities. They must clearly articulate the limitations and expected level of degradation so commanders can make appropriate risk decisions.

2-5. The G-6 coordinates with the G-3 to direct the action and movement of signal elements to support operations through the orders process. Additional G-6 responsibilities relevant to tactical radios include—

- Developing the scheme of signal support through the military decision-making process, including the PACE communication plan for each phase of the operation.
- Planning COMSEC.
- Planning tactical radio networks and retransmission.
- Conducting spectrum management operations.
- Coordinate with the space operations officer to prioritize satellite communications (SATCOM) links used for monitoring.
- Deconflicting spectrum usage with the CEMA section.
- Overseeing signal maintenance and logistical readiness of tactical radio equipment.
- Providing tactical radio training management and support to units, as required.

## **CORPS AND DIVISION SIGNAL, INTELLIGENCE, AND SUSTAINMENT COMPANY**

2-6. Operating under the technical control of the G-6, the signal portion of the corps and division signal, intelligence, and sustainment companies provides organic signal capabilities including tactical radio retransmission for communication between higher, adjacent, and subordinate elements.

### **Capabilities**

2-7. Organic signal capabilities are similarly employed by both the corps and the division signal, intelligence, and sustainment company to support operations within their specific echelon. These capabilities support intra-network communication between upper and lower echelons and data fusion into a shareable common operational picture.

- The corps employs—
  - HF radios.
  - VHF radios.
  - Narrowband (single-channel) TACSAT.
- The division employs—
  - HF radios.
  - SINCGARS.
  - VHF radios.
  - Narrowband (single-channel) TACSAT.
  - MUOS.

2-8. Tactical radios and signal capabilities enable secure transmission, assured command and control, and multipath diversity. Assured command and control enables commanders to send, receive, and display information in near-real time for rapid decision-making. Multipath diversity supports transport flexibility allowing multiple communications solutions and routing path options to expand the Army's network capability.

2-9. Integrating these enabling capabilities is critical to uninterrupted global tactical communications transport and enhances Force employment, lethality, and effectiveness in accomplishing mission objectives. By leveraging multiple communications pathways commanders take advantage of secure information exchange using both commercial and military transport without self-limiting units to encrypted-only networks and have wider network environment options in which data can be categorized and transmitted according to its classification.

### **Roles and Responsibilities**

- 2-10. Additional responsibilities relevant to tactical radios at echelons corps and division are—
- Coordinating with the division G-6 spectrum manager for frequency assignments prior to operations.
  - Collaborating with G-6 to ensure the availability of communications resources and capabilities.
  - Establishing narrowband (single-channel) TACSAT communication.
  - Establishing retransmission networks and sites to support beyond line of sight communications.

## **BRIGADE COMBAT TEAM S-6**

2-11. The S-6 interacts closely with the deputy commander, S-3, and other staff sections to define communications requirements and command post locations through the military decision-making process. The S-6 directs planning and coordination for tactical radio networks, spectrum management operations, retransmission, and COMSEC. To ensure adequate network support, the S-6 collaborates with higher, subordinate, and adjacent units; and recommends network changes, as needed, to support the commander's intent. The brigade S-6 coordinates with the S-3 to direct the actions and movements of signal company elements to support operations through the orders process.

### **Capabilities**

2-12. The organic signal assets of the brigade combat team support continuous operations to sustain the brigade's communications networks. Signal capabilities not provided by organic signal elements are requested from division. Brigade combat teams employ—

- VHF.
- Narrowband (single-channel) TACSAT and MUOS.
- HF.
- SINCGARS.
- Blue Force Tracker (Joint Battle Command Platform)
- Tactical networking waveforms.
  - TSM.
  - WREN.

### **Roles and Responsibilities**

2-13. The S-6 prepares the PACE plan and exercises technical control over the brigade signal company in executing the plan. Their responsibility includes maintaining the running estimate of communications capabilities and providing signal support plans to build and configure the brigade's communications capabilities. Additional roles and responsibilities of the S-6 for tactical radio networks include—

- Developing the scheme of signal support through the military decision-making process, including PACE plans for each phase of the operation.
- Coordinating spectrum management.
- Deconflicting spectrum usage with the CEMA section.
- COMSEC account management.
- Planning retransmission operations.
- Overseeing maintenance and logistical readiness of tactical radios.

## **BRIGADE SIGNAL COMPANY**

2-14. The brigade signal company provides voice and data retransmission capabilities. Retransmission teams extend the brigade's tactical radio network within the brigade's assigned area.

### **Capabilities**

2-15. Brigade combat teams employ combat net radios to enable communications exchange and situational awareness with the next higher echelon and adjacent units and rely on single-channel communications capabilities to communicate with maneuver elements. Tactical radio systems are employed for mobile and dismounted communications. Tactical radio retransmission extends communications range beyond line of sight.

### **Roles and Responsibilities**

2-16. Operating under the technical control of the brigade S-6, the brigade signal company provides communications and information support at the brigade level. Additional roles and responsibilities of the brigade signal company include—

- Establishing tactical radio retransmission sites.

- Establishing narrowband (single-channel) TACSAT at the brigade.

## **BATTALION S-6**

2-17. The S-6 interacts closely with the executive officer, S-3, and other staff sections to define communications requirements at command post locations through the military decision-making process. They consult with higher, lower, and adjacent headquarters to ensure effective communications throughout the assigned area. The S-6 should maintain a close working relationship with the commander and battalion S-3 to ensure the communication plan supports the commander's intent.

## **Capabilities**

2-18. Handheld and manpack radios are the primary communications capabilities employed by the maneuver battalions and companies. These capabilities allow Soldiers to share information across the squad, platoon, and company up to higher headquarters. The battalion's organic tactical radio capabilities are the main means of communications for maneuver elements away from the command posts.

## **Roles and Responsibilities**

2-19. The primary role of the S-6 is to support the commander's communication requirements by providing advice to ensure effective communications connectivity within their assigned area. Additional roles and responsibilities of the battalion S-6 include—

- Managing tactical radio networks.
- Assisting the S-3 in identifying user requirements.
- Maintaining the signal running estimate.
- Creating the PACE plan.
- Integrating and managing communications resources at battalion and subordinate companies.
- Coordinating spectrum management for the battalion's communications systems.

## **TACTICAL RADIO RETRANSMISSION AND RANGE EXTENSION**

2-20. Retransmission allows bridging between separate radio nets and modes of operation (example frequency hopping to single-channel) and may be used to support mission partner networking or as part of a counter electromagnetic warfare plan. Successful retransmission operations requires retransmission teams to anticipate emerging communications requirements in order to maintain a fully operational network to support a commander's evolving mission.

2-21. Signal teams ensure adequate coverage of the assigned area by performing a physical reconnaissance of the proposed retransmission site when possible and a full rehearsal to build team cohesion and competence in executing retransmission operations. See chapter 3 for retransmission planning and appendix B for retransmission training and certification.

## **RETRANSMISSION TEAMS**

2-22. Corps, division, and brigade elements use retransmission when executing tactical enabling tasks beyond the range of VHF single-channel radios, or when terrain prevents line of sight communications. Retransmission capabilities used for communications between units and command posts include single-channel HF radios, narrowband (single-channel) TACSAT radios, two channel manpack cross banding, and tactical messaging. A typical retransmission team has two crew members and the team chief. Two retransmission teams are assigned for echelons corps and brigade. Division elements are authorized up to three retransmission teams.

## **RETRANSMISSION EMPLOYMENT**

2-23. Retransmission sites support maneuver forces by increasing the effective range between radio stations without increasing transmit power. Signal staffs employ retransmission teams to provide complete coverage of the assigned area. Platoon sergeants and platoon leaders conduct pre-combat checks and inspections prior to each mission to ensure tactical responsiveness and support maneuver operations.

## SITE SELECTION

2-24. Proposed site selections are based on G-6 or S-6 signal site analysis. The G-6 or S-6 recommends signal site locations for each proposed course of action during the military decision-making process. Refer to FM 6-02 for detailed information about signal site selection.

2-25. Selecting the right site location is critical to retransmission team operations and survivability. Poor site placement can result in limited flexibility, limited mobility, threat detection, degraded survivability, and reduced effectiveness. Teams establish and implement contingency plans for alternate site locations considering security and defense requirements. See chapter 3 for additional planning considerations for tactical radio retransmission.

## PRE-COMBAT CHECKS AND PRE-COMBAT INSPECTIONS

2-26. The platoon leader or platoon sergeant must evaluate retransmission teams' preparation for combat. They conduct pre-combat checks to validate the teams' readiness once the team chiefs report their teams and vehicles are prepared. Because of the difficulty of reaching and resupplying retransmission teams operating from remote sites, it is vital that the platoon leader and platoon sergeant conduct detailed pre-combat inspections of retransmission teams before teams leave the staging area. Refer to ATP 6-02.2 and GTA 11-02-001 for more information about retransmission team pre-combat checks and inspection.

## PRE-MISSION BRIEFING

2-27. The platoon leader and platoon sergeant present retransmission teams an updated mission briefing immediately before a mission to ensure team chiefs and team members understand the plan. Leaders should have the team chief back brief portions of the plan to verify their understanding of the mission and procedures. The pre-mission briefing includes—

- Commander's intent.
- Convoy plan and procedures.
- PACE plan.
- Frequencies.
- COMSEC.
- Jamming plan.
- Supported unit's scheme of maneuver.
- Location and role of supporting units.
- Enemy situation—
  - Primary enemy location—update daily.
  - Enemy manned and unmanned aircraft.
  - Enemy intelligence, surveillance, and reconnaissance capabilities.
  - Enemy indirect fire capabilities.
  - Most likely enemy actions.
  - Enemy tactics, techniques, and procedures against remote sites like retransmission sites.
  - Primary and alternate enemy avenues of approach.
  - If the enemy requires an escort, identify escorting unit.
  - Timeline of operations.
  - Weather and terrain.
  - Location of site.
- Terrain masking techniques—
  - Avoiding skylining.
  - Using terrain to mask transmissions from enemy detection.
- Primary and alternate retransmission site locations—explain movement triggers.
- Antenna placement plan—
  - Knowing where the enemy is in relation to the operation.
  - Shielding transmissions from the enemy.
- Site security and defense plan—vital and continuous.
  - Supporting security elements.
  - Internal security plan.

- Site concealment and camouflage.
- Concertina wire.
- Range card and fires plan.
- Compromise plan—
  - Emergency destruction plan—COMSEC, documents, maps, and equipment.
  - Escape and evasion plan.
- Resupply plan.
- Reporting procedures.

### MOVEMENT

2-28. Leaders initiate movements necessary to continue mission preparation or to posture the unit for starting the mission. This step includes execution of priorities of work; movement to an assembly area, battle position, or a new operational environment; or the movement of guides or quartering parties.

### SITE SETUP

2-29. Planning the retransmission network is a critical element in site setup. Retransmission teams, who generally work from isolated or remote positions, use network diagrams to give an overview of the proposed location and equipment. Teams conduct risk assessments to determine the likelihood of mission success at the chosen sites. Analysis of terrain explains the terrain's effect on the mission. Enhancing protection and survivability includes identifying sites that enable reliable signal transmission plus defensible terrain that provides cover concealment and aids the team's ability to provide site security and defense. To facilitate planning and aid in briefing subordinates, teams identify potential terrain interferences with line of sight and retransmission capabilities which may require SATCOM or HF radios for beyond line of sight communication. See chapter 3 for additional site selection considerations.

### RETRANSMISSION OPERATIONS

2-30. Retransmission supports the scheme of maneuver and is dictated by the physical size and terrain of the assigned area. The active engagement of the G-6 or S-6 throughout the military decision-making process helps identify communications requirements early in planning and ensures the retransmission plan supports the commander's intent and scheme of maneuver. The operation order includes specified tasks for retransmission teams.

2-31. Retransmission is particularly critical during offensive and defensive operations when maneuver elements cannot access the communications capabilities present in the command posts. Retransmission normally requires—

- Two tactical radios.
- One retransmission cable.
- Two power amplifiers.
- Two antennas.
- Appropriate COMSEC material.

2-32. The tactical situation determines the type of retransmission to use. Planners may configure retransmission networks as either the same network identifier or different network identifier. Using different network identifiers provides a relay to extend the communications range and allow both radios to transmit and receive. Three options exist for retransmission using SINCGARS:

- Single-channel to single-channel.
- Frequency hopping to single-channel (mixed mode).
- Frequency hopping to frequency hopping.

2-33. Single-channel to single-channel retransmission operation requires at least a 10 MHz frequency separation (phase line separation) to avoid interference. This method requires net control station monitoring for continual signal dispersion. Frequency hopping to single-channel retransmission is not limited by a phase line separation. A **net control station** is a communications station designated to control traffic and enforce circuit discipline within a given net.

2-34. There are two modes for frequency hopping retransmission: F1:F1, and F1:F2. With F1:F1, the radio systems operate on the same network identifier for retransmission. Newer radios implement a time delay. This upgrade allows one radio to receive signals and the other to transmit. This time delay allows a sufficient time lapse for two retransmission sites to operate using the same network identifier for distant stations. This eliminates the need for a phase line separation and allows users to pass the geographic separation without adjusting their radios.

2-35. F1:F2 retransmission operates using two radios with different network identifiers. This method of retransmission considers the fact that legacy SINCGARS radios lack some of the advanced tracking capabilities and functionality. Additional means of retransmission employment include—

- **Single band retransmission (frequency modulation-frequency modulation).** Single band retransmission is a common employment for retransmission. A frequency modulation signal is received and retransmitted as a second frequency modulation signal. The SINCGARS is the most common platform employed for this retransmission choice, but many of the Army's radios are capable. The AN/PRC-148 Multiband Inter/Intra Team Radio (MBITR) or AN/PRC-152, for example, can retransmit even in the higher VHF and UHF ranges.
- **Crossband retransmission.** It is also possible to establish retransmission from one band to another and with completely different means of transmission. This is a powerful option as it gives the commander with only a frequency modulation radio, limited by range and terrain, potentially global reach and the ability to communicate easily in exceptionally challenging terrain.
- **Narrowband (single-channel) TACSAT.** Satellite communications can be global. With the appropriate cabling, a SINCGARS radio can be connected to a AN/PRC-162 manpack with MUOS for retransmission operations. If available, the AN/PRC-117 or AN/PRC-148 MBITR system can be used with the SINCGARS radio for range extension. A pair of AN/PRC-117s or AN/PRC-148 MBITRs can also perform this function with the proper mounts and cables.

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*Note.* The Army plans to divest its inventory of the AN/PSC family of radios (replacing them with the AN/PRC-158 or AN/PRC-162) and the AN/PRC-117F/G radios by fiscal year 28 to support cryptographic modernization.

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- **Aerial retransmission.** One form of aerial retransmission is called battlefield airborne communications node. Battlefield airborne communications node mounts a retransmission radio capability on an unmanned aircraft. Manned aerial platforms can also provide retransmission capability. The challenge in using aerial retransmission is the continual movement of an aircraft means that radio coverage also moves and may make knowing where to change from one net to another to maintain communication quite challenging. As a result, aerial retransmission employment is normally used for data networks rather than voice.
- **Manpack retransmission.** Foregoing the use of a vehicle and a whole team to perform a retransmission mission may be desirable in some cases. If reduced range is not a concern, connecting two dismounted radios with appropriate cabling and an enhanced antenna for maximum range can still achieve the required range extension.

2-36. Throughout the mission, one team member continuously monitors radio traffic and serves as the shift relief for the retransmission operation and site defense. Depending on the duration of the mission, personnel should consider constructing a remote position where they can observe the vehicle and that offers protection from indirect fire.

2-37. The retransmission team specified tasks should include criteria for ending or changing the mission in the event of loss of contact with higher authority. Criteria should cover triggers for displacing a retransmission team to an alternate location.

2-38. Reporting supply issues prevents circumstances that would jeopardize efforts of the retransmission team. Unit SOPs dictate how often to send status reports, what items to track, and the means used to send the reports. Reports on basic supplies, such as food, water, fuel, and ammunition allow close monitoring of the operational environment and increase survivability. Other reports, including a commander's critical information requirement report, may document enemy size, activity, location, unit, time, equipment, and observed activity.

2-39. Retransmission must be practiced in unit training in preparation for large-scale combat operations. Training events provide opportunities for teams to assess speed and proficiency in executing their mission-essential tasks. Unit leaders should factor in time for battle drills, such as gas attack, indirect fire, or emergency destruction drills to develop the team's ability to operate independently, demonstrate their ability to conduct movement to an alternate site, re-establish tactical radio connectivity, and set up camouflage quickly in preparation for deployment. See appendix B for detailed information on retransmission team training and certification.

### **POST MISSION BRIEF—ON SITE**

2-40. At the conclusion of a mission, team chiefs conduct a post-mission briefing and inspection before leaving the retransmission site. If the platoon leader or platoon sergeant is not available to conduct the on-site briefing, the team chief performs post-mission checks, including—

- Ammunition.
- Individual equipment.
  - Weapons.
  - Sensitive items.
  - Eye protection.
  - Water.
  - Hand protection.
  - Identification tags and identification cards.
  - Body armor.
- Chock blocks and drip pans.
- Trailer is connected, brakes are functional, and parking brake is off.
- Vehicular antennas are tied down.
- Convoy frequency plan loaded.
  - Sheriff net (quick-reaction force).
  - Medical evacuation.
- Communications checks with supporting units and security elements.
- Convoy briefing.

### **POST MISSION BRIEF—AFTER RETURN**

2-41. Upon end of mission and return to the staging area or command post, the retransmission team should prepare for immediate redeployment if it becomes necessary. The post-mission briefing and equipment checks after return help the unit collect lessons learned and recover equipment so it is ready for immediate use. The post-mission briefing and after action review include—

- Weapons, sensitive items, and ammunition checked immediately upon return.
- After operation preventive maintenance checks and services performed on vehicles and generators.
  - Petroleum, oils, and lubricants resupplied.
  - Trucks, generators, and fuel cans refilled.
  - Trucks and trailers cleaned as needed.
- Communications equipment.
  - Zeroized as necessary.
  - After operation preventive maintenance checks and services performed on all communications equipment.
  - Inventory-missing or broken equipment identified and replaced.
- All classes of supply (including food and water) replenished and load plan updated as necessary.
- Missing personal equipment identified and replaced.
- Internal after action review conducted.
  - Input from team.
  - Consolidation of notes.
  - After action review comments submitted to higher headquarters for lessons learned and SOP updates.
- Debriefing to higher headquarters conducted.

## Chapter 3

# Planning

This chapter describes tactical radio network planning and the roles of signal staff elements in coordinating requirements for effective signal support. It also provides planning considerations and essential retransmission planning requirements so planners can effectively assess resource capabilities and limitations.

### SECTION I – Tactical Radio Network Planning

3-1. Information from the higher headquarters operation order establishes the basis for tactical radio network planning. Communications planners must understand the commander's intent and concept of operations defined in the base plan or order. The type, size, and complexity of the radio system establishes ideal scenarios of employment and the method of control.

### PLANNING CONSIDERATIONS

3-2. Signal planners are active in the operations process. They define support requirements based on the mission variables of METT-TC (I). Planners must understand their organization's authorized and available communications capabilities and limitations, including personnel and logistical requirements needed to employ and sustain the capabilities. Commanders and leaders develop this understanding during unit training by employing organic equipment, correcting deficiencies, and implementing maintenance and supply programs.

### RUNNING ESTIMATE

3-3. Comprehensive running estimates provide valuable information on communications capabilities and are essential to formulating the scheme of signal support. A running estimate continuously assesses operations and is used to determine if they are proceeding according to the commander's intent and are supportable and feasible. Commanders use recommendations from running estimates to enhance further decision making. Signal staffs use running estimates to advise the commander and make recommendations. Planners consider several additional factors when planning tactical radio networks including—

- Communications requirements for each warfighting function.
- Capabilities and limitations of available communications systems.
- Requirements for mission partner interoperability.
- Available frequencies.
- Waveform applications required.
- Detailed line of sight analysis.
- HF analysis.
- Redundancy in means to communicate.
- Integration of all available signal assets.
- Method of deployment—sequence assets to coincide with the arrival of forces.
- Command post locations.
- The use of retransmission, digital network links, and node placement.
- Priorities of support to manage available communications bandwidth.
- Spectrum requirements for tactical radios.
- COMSEC distribution plans.
- Signal operating instructions.
- Conduct of communications rehearsals.

### SITE CONSIDERATIONS

3-4. The reliability of radio communications depends largely on the selection of a good radio site. A unit must select an acceptable location that enables them to accomplish their assigned mission. Since it is difficult

to select a radio site that satisfies all technical, tactical, and security requirements, signal planners should select the most suitable site available. Site selection considers defensibility and logistical supportability as well as communications suitability.

3-5. Perfect cover and concealment may impair communications. The location selected should provide the best cover and concealment possible, consistent with good transmission and reception. The permissible amount of impairment depends upon the range required, the power of the transmitter, the sensitivity of the receiver, the efficiency of the antenna system, and the nature of the terrain.

3-6. Leaders and planners also consider that communications systems with specific operating characteristics, capabilities, and limitations create unique circumstances that directly impact site selection. To obtain efficiency of transmission and reception, the following factors should also be considered:

- For operation at frequencies above 30 MHz, planners should select a location that allows line of sight communications.
- Dry ground has high resistance and limits radio range. If possible, planners should locate the station near moist ground, which has much less resistance. Water, especially fresh water, greatly increases the distance covered.
- Operators should install antennas away from foliage and dense brush.
- When located near man-made obstructions—
  - Site selection should avoid tunnels, underpasses, and steel bridges due to the high absorption of RF energy.
  - Radio sites should avoid buildings between radio stations, particularly steel and reinforced concrete structures since they interfere with transmission and reception.
  - Radio sites should avoid suspended telephone, telegraph, and high-tension power lines. Wire lines absorb RF energy from radiating antennas in their vicinity. They also introduce electrical interference to receiving antennas.
  - Radio sites should avoid heavily traveled roads and highways. In addition to the noise and confusion caused by tanks and trucks, vehicle ignition systems may cause electrical interference.
  - Operators should locate battery charging units and generators away from radio sets.
  - Operators should not locate radio stations close to one other.
  - Planners should select radio sites in relatively quiet areas to avoid distraction by outside noises. The copying of low-level received messages requires great concentration by the radio operator.

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**Note.** See appendix H for information related to antenna selection and techniques and appendix I for detailed information about specific tactical radio systems and platforms.

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## STAFF COORDINATION

3-7. Effective command and control requires continuous close coordination, synchronization, and information sharing across staff sections. As a coordinating staff officer, the G-6 or S-6 provides signal subject matter expertise for the commander and other staff elements through the planning process. G-6 or S-6 planners should work closely with—

- The G-3 or S-3 to understand the commander's intent and concept of operations and ensure signal plans support the proposed course of action.
- The G-2 or S-2 to understand the threat and appropriate countermeasures.
- The cyberspace electromagnetic warfare officer (CEWO) to minimize the electromagnetic signatures of signal sites.
- The G-4 or S-4 to support sustainment, maintenance, and repair of signal systems.

## RADIO NETWORK PLANNING TOOLS

3-8. Tactical network planning tools enable Soldiers to operate without overburdening the network planners and managers with added complexities. These tools help planners develop, manage, and monitor tactical radio networks; share network plans; view network performance metrics; and provide situational awareness data across the assigned area. Planning tools used for tactical radio systems include—

- Joint Automated Communications-Electronics Operating Instructions System.

- Ruggedized Applications Platform-Tactical Radios.
- Automated Communications Engineering Software.
- Electromagnetic Warfare Planning and Management Tool.
- Systems Planning, Engineering, and Evaluation Device.
- Voice of America Coverage Analysis Program.
- Naval Research Laboratory Builder.

## **JOINT AUTOMATED COMMUNICATIONS-ELECTRONICS OPERATING INSTRUCTION SYSTEM**

3-9. The G-6 or S-6 staff's radio network planning ensures interoperability of RF networks, cryptographic key tag generation, transmission security key generation, signal operating instructions production, and radio network planning. Joint Automated Communications-Electronics Operating Instruction System and Automated Communications Engineering Software allow the G-6 or S-6 staff to perform pre-deployment and post-mission planning radio network planning.

## **RUGGEDIZED APPLICATIONS PLATFORM-TACTICAL RADIOS**

3-10. The Ruggedized Applications Platform-Tactical Radio is a host platform that combines multiple software applications into a single tactical network management tool that plans configures, loads, and monitors legacy waveforms and advanced networking waveforms. It supports secure but unclassified and classified radio networks. Atom, the primary Ruggedized Applications Platform-Tactical Radio software, is a radio planning and loading tool that provides simultaneous support for tactical radios and a collection of legacy and advanced waveforms.

## **AUTOMATED COMMUNICATIONS ENGINEERING SOFTWARE**

3-11. Automated Communications Engineering Software is a network planning software program that plans, creates, distributes, manages, and verifies cryptonets and key related information. Automated Communications Engineering Software allows users to perform fully automated cryptographic network planning; signal operating instructions; and electromagnetic protection planning, management, validation, generation, and distribution at the time and location needed.

3-12. The Automated Communications Engineering Software provides the capability to support black key packaging and distribution and provides fills for the electronic fill device, AN/PYQ-10, with secure network information through a direct cable connection or over-the-air rekey. Pre- and post-mission planning also consist of electromagnetic protection data and radio network engineering for secure communication.

3-13. The network planning functionality of Automated Communications Engineering Software incorporates cryptonet planning, key management, and key tag generation. The data for a given plan includes individual networks, which are assigned individual network members. Designated network members define the platforms, specific equipment fill locations and associated key tags, and keys with the equipment locations.

3-14. Network members download the equipment records, which include platform data, network data, and key tags associating the data with the required key. Similarly, the electromagnetic protection data and signal operating instructions generated by the Automated Communications Engineering Software workstation operator enable the data download to the electronic fill device. Mission requirements dictate whether the COMSEC account is resourced with an Automated Communications Engineering Software workstation.

## **ELECTROMAGNETIC WARFARE PLANNING AND MANAGEMENT TOOL**

3-15. The Electromagnetic Warfare Planning and Management Tool is a software application that uses computerized modeling and planning capabilities to assess and manage electromagnetic warfare asset in real-time. The Electromagnetic Warfare Planning and Management Tool supports spectrum planning by allowing Soldiers to sense emitter activity and sensor data within their electromagnetic operating environment. This tool provides commanders essential information and intelligence for analysis to develop a common operational picture while operating remotely. The Electromagnetic Warfare Planning and Management Tool integrates electromagnetic order of battle data into current and future simulations and command and control systems. This allows commanders to identify and visualize both threat and friendly force digital footprints and execute synchronized electromagnetic warfare, spectrum management, intelligence, and signal operations for successful offensive and defensive operations and planning.

### SYSTEMS PLANNING, ENGINEERING, AND EVALUATION DEVICE

3-16. The Systems Planning, Engineering, and Evaluation Device is a complete stand-alone, self-installing software package that provides the tools necessary to plan and analyze communications equipment. The software contains HF analysis, radio coverage analysis, point-to-point, and satellite planning tools, which allows planning in response to rapidly changing communications architectures.

3-17. The Systems Planning, Engineering, and Evaluation Device communications planner has three-dimensional mapping and mission planning capabilities, including frequency-dependent rejection interference analysis, enhanced jammer modeling, and effectiveness prediction. The software contains HF analysis, radio coverage analysis, point-to-point, and satellite planning tools, which allows planning in response to rapidly changing communications architectures. Communications planners load topographical information before each operation to generate reports, maps, and matrixes.

### VOICE OF AMERICA COVERAGE ANALYSIS PROGRAM

3-18. Voice of America Coverage Analysis Program software released to the public is downloadable from the Department of Commerce National Telecommunications Information Administration, Institute for Telecommunications Sciences to use as a HF prediction and analysis tool. Planners must input several parameters before the Voice of America Coverage Analysis Program can provide a propagation prediction determined by the method and the antennas used. Voice of America Coverage Analysis Program offers—

- Graphical user interface.
- Detailed point-to-point graphs.
- Signal-to-noise ratio.
- Required power gain.
- Signal power.
- Maximum usable frequency.
- Take-off and arrival angles.
- Point-to-point performance versus distance for any given parameters at one or all user assigned frequencies.
- Methods for antenna patterns.

### INTERACTIVE SCENARIO BUILDER (BUILDER)

3-19. The Interactive Scenario Builder (Builder) is a mission planning tool that provides a three-dimensional interactive representation, including RF propagation and geospatial and temporal situational awareness. The Interactive Scenario Builder (Builder) incorporates antenna and radar cross-section pattern data and meteorological, terrain, and countermeasure effects to model communications systems. This modeling helps calculate RF propagation and enhances the geospatial situational awareness in the commander's assigned area by visualizing National Geospatial-Intelligence Agency map products. The National Geospatial-Intelligence Agency map products include visual imagery and digital terrain elevation data. The National Geospatial-Intelligence Agency Builder can be used for pre-mission planning, near real-time situational awareness, and after-action reviews. The Interactive Scenario Builder's (Builder) interactive software is authorized to operate on Nonclassified Internet Protocol Router Network and SECRET Internet Protocol Router Network.

### COMMUNICATIONS SECURITY PLANNING

3-20. *Communications security* is protective measures to deny unauthorized access to information of value derived from use of communications systems (JP 6-0). Cryptographic systems and devices protect networks and the information being transmitted and give the command confidence in the security of transmissions. Leaders, network planners, and end users must guard against threat attempts to intercept and exploit U.S. operation plans through its networks and information services.

3-21. Automated Communications Engineering Software operators develop the COMSEC network overlay based on mission requirements or PACE plan. The network planner tailors Automated Communications Engineering Software and workstations as needed to correspond to planning requirements using the general-purpose planning module software application. Once tailored and complete, COMSEC account managers that maintain centralized accountability of cryptographic material and equipment, process the cryptonet plan

for delivery to the network planner and Automated Communications Engineering Software operators and subsequent distribution to the end user.

3-22. The Automated Communications Engineering Software includes a planning module for each network category. The categories of supported network plans include—

- Combat net radios.
- General-purpose networks.
- Adaptive networking wideband waveform.

3-23. The network planner finishes processing the cryptonet and provides the cryptonet plan, signal operating instructions, and loadset data to local end users. The COMSEC account manager generates or provisions and distributes the key to end users by loading keys into the users' fill devices. Users load the end cryptographic units using a fill device. Refer to ATP 6-02.75 for more information about COMSEC planning.

## KEYING MATERIAL

3-24. The COMSEC account manager is responsible for the receipt, custody, security, accountability, safeguarding, inventory, transfer, and destruction of COMSEC material. For a cryptonet to communicate securely, all members of the net must use compatible equipment and keying material. COMSEC account managers distribute and control keying material and manage routine updates so all authorized users—and only authorized users—have the correct key to communicate.

3-25. Commanders carry the responsibility of safeguarding and controlling COMSEC material in their units to ensure its continuous integrity and prevent access by unauthorized persons. The unit SOP includes risk assessments to establish and document security measures for COMSEC material and controlled cryptographic items. The commander appoints an account manager for the COMSEC account.

3-26. Safeguarding and controlling COMSEC material is also the responsibility of individuals in possession of the material. While COMSEC accounting is centralized, individual users hold keying material and devices at the lowest echelon that can maintain adequate physical security. End users protect and account for COMSEC material in their possession and report anything that could jeopardize its security or integrity. The unit SOP provides local procedures for safeguarding, controlling, and emergency destruction of COMSEC material. Refer to ATP 6-02.75 for additional information on techniques for COMSEC.

## KEY DISTRIBUTION

3-27. The Army uses electronic key generation and distribution as its primary means of transferring keying material from the COMSEC account in electronic form using a fill device. Encrypted key distribution allows secure delivery of a key to an end cryptographic unit and reduces the risk of compromise during transmission. Encrypted key distribution starts with the COMSEC account manager using the planning workstation to generate an encrypted key. COMSEC distribution takes place through—

- **Transfer.** The distribution of COMSEC material from one account to another. Transfers move accountability and responsibility for the transferred COMSEC material from the sending account to the receiving account.
- **Issue.** The distribution of one or more accountable COMSEC items from a COMSEC account to a local element. The issuing element retains accountability for the material. Issuing and filling keys are the two functional areas of key distribution between a COMSEC operating account and a local element.

## Initial Key Load

3-28. The stores, sends and receives mission data (hopsets, loadsets and signal operating instructions information). COMSEC account managers ensure loading and key distribution for various systems by first inputting the key into the electronic fill device. For further information on loading instructions for specific end cryptographic units, refer to the equipment technical manual and the radio and system operators handbook.

### **Over-the-Air Rekey**

3-29. The net control station may initiate over-the-air key distribution any time during the effective cryptographic period, or immediately before the end of the cryptographic period. Before transmitting a key, the net control station notifies all recipients over a secure communications link and outlines the details of key transmission.

### **EMERGENCY DESTRUCTION**

3-30. Allowing COMSEC materials to fall into enemy hands compromises secure communications. COMSEC compromise disrupts radio and data networks until operators and net control stations can conduct emergency cryptographic key supersession. If a keyed radio or a loaded electronic fill device falls into an enemy's hands, the unit SOP should assume the enemy has English-speaking Soldiers who can operate the radio and key loading device. SOPs should also assume the enemy is able to listen to U.S. secure frequency hopping network communications and can transmit on that same U.S. network if desired.

3-31. Unit SOPs must include procedures to evacuate COMSEC material and encryption devices or destroy them in place. By including emergency destruction battle drills in mission rehearsals, the risk of system compromise is mitigated, and forces are prepared to respond and defend during real-world operations. Refer to ATP 6-02.75 for more information about COMSEC operations.

### **EMISSION CONTROL PLANNING**

3-32. Commanders consult with the G-2 or S-2 and electromagnetic warfare personnel to identify threat, electromagnetic warfare, and signals intelligence capabilities. Additionally, electromagnetic warfare professionals and spectrum managers must evaluate the effectiveness of friendly force EMCON measures and recommend modifications or improvements. Units should implement and exercise EMCON plans before execution and conduct vulnerability analysis and assessments of friendly communications assets to understand the electromagnetic environment.

3-33. G-6 or S-6 planners collaborate with the CEWO during signal site planning and consider using electromagnetic masking and electromagnetic decoys to protect the locations of critical nodes. EMCON helps prevent threat electromagnetic warfare assets from locating and attacking friendly forces. Everyone in the organization must understand the importance of EMCON for it to be effective. See appendix A for additional techniques for EMCON.

3-34. Radio discipline is a basic and effective EMCON technique to protect against threat detection. Radio discipline can take the form of limiting transmissions, antenna masking, and the use of minimum required power. Additional techniques for EMCON include—

- Preplanning messages before transmitting. This minimizes the number of pauses in the transmission and decreases transmission time. It also ensures the conciseness of the message.
- Transmitting quickly and precisely. This reduces the need to repeat a radio transmission. Unnecessary repetition increases transmission time and the threat's opportunity to intercept U.S. transmissions and gain valuable information.
- Using equipment capable of data burst transmission. This is a significant advantage of TACSAT systems.
- Using an alternate means of communication. Soldiers use alternate means of communications, such as cable, wire, or messages to convey necessary directives and information.
- Using brevity codes and procedure words. This provides no security; however, its purpose is to shorten messages rather than conceal their content.

### **SPECTRUM PLANNING**

3-35. Spectrum planning coordinates frequency use before activation of emitters to mitigate or eliminate electromagnetic interference or other adverse effects by implementing friendly force recognition, anti-fratricide, and electromagnetic protection measures. Deconfliction ensures spectrum-dependent devices do not create unintended electromagnetic interference (frequency fratricide) with friendly communications, unmanned aircraft systems, weapon systems, or space-based GPS-enabled equipment providing precise PNT.

3-36. Spectrum planning considerations include—

- Transmitter and receiver locations.
- Antenna technical parameters and characteristics.
- Number of frequencies desired and frequency separation requirements.
- Nature of the operation (fixed, mobile land, mobile airborne, or maritime).
- Physical effects of the operational environment (ground, soil, humidity, and topology).
- Start and end dates for use.
- Joint restricted frequency list.

3-37. Spectrum managers use various tools to manage and plan the unit's spectrum needs and objectives. Examples include—

- Automated Communications Engineering Software.
- Electromagnetic Warfare Planning and Management Tool.
- Host-Nation Spectrum Worldwide Database Online.
- Joint Automated Communications-Electronics Operating Instructions System.
- Spectrum Situational Awareness System.
- Systems Planning, Engineering, and Evaluation Device.
- Global Electromagnetic Spectrum Information System.
- Spectrum XXI.

## PRIMARY, ALTERNATE, CONTINGENCY, AND EMERGENCY COMMUNICATION PLANS

3-38. Commanders depend on communications systems to exercise command and control of subordinate formations and coordinate mission plans with higher and adjacent units as well as supporting or supported joint forces, host-nations, and multinational forces. The PACE communication plan provides redundant communications if the primary means of communication fails and designates the order in which an element will move through redundant means of communication until contact is established with the desired distance elements.

3-39. The PACE communication plan applies to a specific mission or task, not a unit. It enables operational continuity during all operational phases ensuring maneuver elements can continue to communicate to perform mission objectives. Higher headquarters evaluates communication requirements with subordinate echelons and works with the G-6 or S-6 to develop an effective plan to mitigate the impact of degraded networks. By implementing a PACE communication plan, signal planners reduce communications disruption in congested or contested environments.

3-40. Planners identify appropriate PACE means of communication for each phase of the operation and publish them in Annex H (Signal) of the operation order. The PACE plan is implemented in accordance with command post battle drills.

3-41. If the commander determines that the primary means of communication is impacted, they take immediate action to mitigate the effects or initiate the PACE plan. Quickly changing to an alternate or contingency means of communications reduces disruption of command and control.

3-42. The unit must be capable of operating each means of communication in the PACE plan. If the designated equipment is non-mission capable, that capability should not be included in the PACE plan. Planners should ensure unit personnel are adequately trained on systems designated as alternate, contingency, or emergency means of communication. Planners should also consider the totality of information on the primary form of communication and assess whether that level of bandwidth can fulfill alternate, contingency, or emergency elements of the PACE and plan appropriately. When developing PACE plans, planners should ensure all affected elements have the equipment to execute the plan. Upon receipt of an order from a higher headquarters, the receiving unit evaluates the PACE plan for two essential elements:

- Availability of assets to execute the plan to higher.
- Ability to nest with the higher command's PACE plan.

3-43. If a unit cannot execute the full PACE plan to its higher command, it must inform the issuing headquarters with an assessment of shortfalls, gaps, and possible mitigation measures as part of mission analysis during the military decision-making process. Subordinate elements must be capable of providing

higher echelons with redundant means of communications support. Identified PACE capabilities must be diverse to ensure continued communication should any portion of the communication path or capability become non-mission capable or compromised. This allows forces to carry out mission objectives without impacting the other PACE capabilities.

3-44. Units should rehearse and validate the PACE plan during mission and communications rehearsals to ensure the alternate, contingency, and emergency means of communication are viable and that all personnel understand the triggers and can execute the plan as necessary. Table 3-1 shows an example PACE plan aligned with warfighting functions.

**Table 3-1. Example primary, alternate, contingency, and emergency communication plan by warfighting function**

| <b>Communications Means</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Command and Control</b> | <b>Intelligence</b> | <b>Fires</b>  | <b>Sustainment</b> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------|---------------|--------------------|
| Primary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | VHF (CMD net)              | DCGS-A              | AFATDS        | VHF (A&L)          |
| Alternate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | SC TACSAT                  | VHF                 | VHF (digital) | JBCP               |
| Contingency                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | JBCP                       | JBCP                | VHF (digital) | SC TACSAT (voice)  |
| Emergency                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | HF (chat)                  | SC TACSAT (voice)   | JBCP          | BCS3               |
| <b>Legend:</b><br>A&L                administrative and logistics<br>AFATDS          Advanced Field Artillery Tactical Data System<br>BCS3             Battle Command Sustainment and Support System<br>CMD              command<br>DCGS-A          Distributed Common Ground System-Army<br>HF                high frequency<br>net                network<br>JBCP             Joint Battle Command Platform<br>SC                single-channel<br>TACSAT          tactical satellite<br>VHF               very high frequency |                            |                     |               |                    |

## LINK GEOMETRY

3-45. G-6 or S-6 planners conduct terrain modeling and analysis for airborne, ground-based, and surface and subsurface spectrum-dependent devices to support tactical mission and operational planning. To increase team member survivability, spectrum managers ensure equipment is properly emplaced during planning. This maximizes linkage between communications and decrease susceptibility of detection from threat directional finding. Proper link geometry masks signals from threat direction-finding and jamming capabilities.

## SITE SIGNATURE CONTROL

3-46. Tactical radios and retransmission sites can be easily detected by threats from the air or through the EMS. To prevent this, soldiers use protective measures to conceal their activities and minimize the risk of being detected. This includes using camouflage and concealment, as well as adhering to noise and light discipline. These measures help to keep personnel safe and maintain their ability to act freely on the battlefield. Leaders combine multiple measures to mask signatures. These measures include—

- Camouflage and concealment.
- Noise and light discipline.
- Electromagnetic signature reduction.

## CONCEALMENT AND CAMOUFLAGE

3-47. Concealment offers protection from observation or surveillance and helps prevent threat detection. SOPs should include provisions to conceal personnel and equipment. Installing antennas as low as possible

on the backside of terrain features, and behind man-made obstacles helps conceal communications equipment while still permitting communication.

3-48. It is difficult to conceal large communication systems, but camouflage can make these systems harder to identify. Camouflage offers concealment and helps minimize the visual signature. Radar reflective camouflage netting partially mitigates the command post's electromagnetic signature. Using antenna multiplexers further reduces electromagnetic footprint by reducing the number of antennas needed.

## **NOISE AND LIGHT DISCIPLINE**

3-49. Generators, vehicles, and communications assemblages can produce significant noise revealing the location of command post and unit positions thereby increasing vulnerability. Operators reduce detection by minimizing and masking the noise from signal systems and disabling or turning equipment off when not in use. Units can also use available natural cover or place equipment against a wall or physical structure to minimize noise and avoid detection. Additional practices include limiting radio use and restricting unnecessary vehicle and foot movement and securing loose metal parts to prevent them from making noise in windy weather conditions. Operators can turn the speaker volume to the lowest setting to avoid location detection.

3-50. Excessive light makes tactical sites easy to locate at night. Units practice light discipline to minimize the probability of detection. Minimal use of flashlights or other electrical lights helps prevent visual detection. During site setup, Soldiers should cover or blacken reflective surfaces, such as bare metal surfaces or vehicle windshields when not in use. When using infrared light, leaders must consider the threat's night vision and infrared capabilities. Leaders should restrict unnecessary vehicle and foot movement once the signal site is set up.

## **ELECTROMAGNETIC SIGNATURE REDUCTION**

3-51. Soldiers use various spectrum-dependent devices that a peer threat can detect and target. These threats have advanced capabilities to locate the electromagnetic signatures of communications systems using ground-based or unmanned aerial electromagnetic warfare systems. To safeguard against these risks, leaders must minimize the unit's electromagnetic signature. This includes assessing and employing digital rather than analog command and control, implementing electromagnetic protection techniques, and establishing EMCON procedures.

## **SIGNAL OPERATING INSTRUCTIONS**

3-52. Signal operating instructions contain relevant instructions for radio communications. The signal operating instruction is a governing document for routine signal operations of a unit. It includes information on communications system priorities and general guidance for communications protocol. The signal operating instructions contain details on the unit's radio nets; assigning net control stations; and primary and alternate operating frequencies for each net. The signal operating instructions also contain guidance on authentication procedures and COMSEC.

## **HIGH FREQUENCY RADIO PLANNING**

3-53. HF radios provide beyond line of sight secure voice and data communications on two modes of HF propagation—ground wave and skywave. Planners consider that system adjustments are required to compensate for ionosphere and an ever-changing terrestrial environment, electromagnetic interference from other stations, atmospheric interference, and man-made noise. Determining the most favorable conditions is crucial to HF radio planning. Planners determine the optimal frequency, time of day, power, and antenna configuration to ensure the criteria align with the capabilities and assets of the unit. Listed at paragraph 3-16–3-18 are detailed descriptions of HF planning tools—

- Systems Planning, Engineering, and Evaluation Device.
- Voice of America Coverage Analysis Program.

3-54. HF radio data capabilities are limited by bandwidth, spectrum physics, and space and terrestrial weather. Planner awareness of the capabilities and limitations of HF radios will maximize availability. HF radio assets are most useful for long-range communications. Planners must understand the concepts of propagation, path loss, and digital signal processing. Antenna selection is also critical in HF radio planning.

To select the best antenna for HF radios, planners need to understand wavelength, frequency, resonance, and polarization. See appendix E for information on HF operations and appendix H for antenna selection techniques.

## **NARROWBAND (SINGLE-CHANNEL) TACTICAL SATELLITE PLANNING**

3-55. Narrowband (single-channel) TACSAT supports tactical communications worldwide by providing interoperability between legacy TACSAT radios, software-defined radios, and hardware-defined radio waveforms. The interoperability enables voice and limited data exchange for beyond line of sight tactical radio users. Planners use the Joint Integrated Satellite Communications Tool on SECRET Internet Protocol Router Network to prepare and submit satellite access requests for narrowband satellite communications service.

3-56. Planning considerations for narrowband (single-channel) TACSAT include establishing redundant and auxiliary communications and mitigation techniques to minimize interference and threat attacks on critical satellite resources. Additionally, planners should—

- Formulate communications plans that minimize RF footprint.
- Use modulation techniques that provide a low probability of signal detection or reception.
- Identify bandwidth requirements (lower data rates require less RF power).
- Practice EMCON (limit radio transmissions).
- Establish PACE plans to mitigate effects of a congested or contested environment.
- Locate antennas away from command posts to mitigate collateral damage if communications systems are destroyed.

3-57. Units establish and document valid service requirements in the Satellite Communications Database Management Tool prior to requesting satellite access in the Joint Integrated Satellite Communications Tool. Users submit connectivity requirements using the Joint Satellite Communications Database Management Tool. The combatant command forwards satellite requirements to the joint satellite communications panel administrator using the Joint Satellite Communications Database Management Tool. Following receipt and assessment by the Space Force, the request findings are returned to the joint satellite communications panel administrator. The communications planner at the division G-6 or other operations cell submits satellite access requests through corps headquarters to the theater army for verification once a unit receives a validated requirement for satellite access. The approved satellite access authorization permits operational access.

3-58. Planners should submit satellite access requests with sufficient time before the mission date to ensure the request can be validated and submitted to the supporting regional satellite communications support center 30 days before the requested access date. This timeline may be shortened if the validation authority identifies a satellite access request as an exception to policy, crisis, or contingency satellite access request. Planners use the Joint Integrated Satellite Communications Tool application on SECRET Internet Protocol Router Network to prepare and submit narrowband satellite access requests. When submitting a satellite access request, the requesting unit should ensure—

- Signal planners prepare a satellite access request.
- Unit forwards the satellite access request to the theater army satellite communications manager for validation; units in the continental United States submit requests to FORSCOM.
- Theater army or FORSCOM submits request to the combatant command for mission approval.
- The combatant command forwards validated satellite access requests to the supporting regional satellite communications support center for resourcing.

## **DEMAND-ASSIGNED MULTIPLE ACCESS**

3-59. Demand-assigned multiple access (DAMA) is a technique that matches user demands to available satellite bandwidth. Satellite channels are grouped as a bulk asset. DAMA assigns users variable time slots that match the radio operator's requirements. Since most users do not require constant bandwidth, this concept allows bandwidth sharing and results in greater flexibility in managing available satellite resources. The increase in networks or radio users supported using DAMA depends on the type of users. Regardless of format, DAMA generally increases satellite capability by a factor of 5–14 times over dedicated channel operation.

3-60. Extensive planning and coordination are required for establishing SATCOM. Planners should become familiar with—

- The SATCOM database.
- Requirements submission.
- Operating in degraded and denied environments.
- Satellite access requests.
- Access priorities.
- Apportionment.
- Access planning.
- Redundant communications procedures.
- Allocation process.
- After action reporting.
- Planning checklist.

## INTEGRATED WAVEFORM

3-61. Integrated waveform is an enhanced method of multiplexing radio networks on the same channel. This technique uses carrier phase modulation to allow more access using the same channel. Carrier phase modulation provides higher data throughput on UHF dedicated satellite channels in line of sight mode. The integrated wave form offers advantages over DAMA. The integrated waveform replaces DAMA to provide multiple users access to the SATCOM resources. Planning considerations for integrated waveform include—

- Communication classification level.
- Waveform flexibility.
- Mission partner collaboration.
- Common operational picture accounting.
- Range extension and scalability.
- Configuration interoperability.

## MOBILE USER OBJECTIVE SYSTEM

3-62. MUOS terminals are a combination radio and modem configuration. Using appropriate provisioning applications, unit planners ensure the radio and modem configurations are provisioned, programmed, and loaded to support operations upon deployment. Provisioning is the process whereby a terminal is registered with the MUOS network management system and given access to operate on the MUOS network in an approved configuration. Planners initiate important steps to register and obtain approval to operate on the MUOS network in an approved configuration. To provide on-the-move communications, planners initiate the pre-operations process which includes—

- Terminal capabilities planning.
  - Coordinate terminal requirements.
  - Coordinate network requirements.
- Access authorization process.
  - Coordinate terminal profiles with operational provisioning authority using the Joint Satellite Communications Management and Planning System.
  - Coordinate group service authorizations with the regional satellite communications support center using the Joint Integrated Satellite Communications Tool.
- COMSEC ordering process.
  - Order keying material.
  - Coordinate group service keying material.
  - MUOS keying material.
- COMSEC receipt process.
  - Coordinate terminal registration.
  - Coordinate keying material transfers.
  - Ensure the terminal's radio configuration files are built within the provisioning application and complete for transfer.
- Local provisioning process.
  - Ensure terminal accesses system using over-the-air file transfer.

- Ensure communications and network parameters are met and capabilities are fully functional.

3-63. MUOS group network planning requires the planner to coordinate terminal requirements with the communications element or spectrum manager and the narrowband communications and Internet Protocol structure with the unit's DODIN operations personnel. MUOS services are managed and processed by the regional satellite communications support centers and require completion of a satellite access request. Requests must be submitted according to access authorization policies and applicable enterprise tools and applications. Terminal provisioning requests and satellite access requests should be submitted no later than 90 days prior to mission start date. Unit SOPs should provide detailed descriptions for terminal provisioning request policies, access authorization, COMSEC keying material ordering policies, and satellite access request policies. Any compromise or loss of a terminal profile must be reported to the Global Narrowband Watch Office or a regional satellite communications support center.

3-64. TACSAT communications is a key component to many ground and aviation forces' communications plans. Maintaining capabilities that allow units to communicate with allied and joint forces is essential during transitioning from legacy TACSAT-based networks to MUOS in order to address capability gaps caused by new equipment fielding during network modernization. A method of bridging legacy satellite network users and MUOS users is done over a MUOS legacy gateway component which requires significant coordination with MUOS network control centers and end users. It is important for planners to ascertain communications requirements during transitional periods so forces operating at a distance can both maintain communication with lines of support and provide vital situational awareness information to commanders.

## DATA NETWORK PLANNING

3-65. Deployed units must be able access the needed data at the right time and at the right place. Data-centricity and interoperability, key aspects to network viability, create resiliency, eliminate silos, and enable greater interoperability with mission partners. Sharing common applications, services infrastructure, and transport layers enables data synchronization between echelons and across formations.

3-66. To plan effective data networks, network planners must understand the current threat situation and the electromagnetic order of battle. An *electromagnetic order of battle* is a subset of the overall order of battle that consists of the identification, strength, command structure, disposition, and operating parameters of the electromagnetic spectrum-dependent systems (JP 3-85).

3-67. Signal support plans must also account for threat electromagnetic warfare capabilities that potentially impact operations, disrupt communication, and degrade command and control. Collaboration between G-6 or S-6 planners, the G-2 or S-2 section, space operations officer, and the CEWO is essential to ensure adequate consideration is given to the effects of friendly communications when developing an electromagnetic protection plan. Network planners use automated planning and management tools to facilitate network planning. Additional considerations of network planning include—

- System and network availability.
- Frequency allocation.
- Internet Protocol planning.
- Mission partner interoperability.
- Frequency planning.

## INTEGRATED TACTICAL NETWORK PLANNING

3-68. Operational flexibility, interoperability, and security are essential components to ITN planning. Through common application standards, services infrastructure, and network transport, the ITN provides a broader range of capabilities to extend the data and voice communication when required to Army, allied forces, and mission partners. Commercial capabilities and cross-domain solutions integrated into the ITN create greater flexibility. The same versatility enables continued communications support and secure interaction between lower and higher echelons including interoperability with mission partners using a shared common operational environment.

3-69. ITN systems must provide maximum effectiveness while operating in highly dynamic environments, under challenging environmental conditions where peer threats exist. For further information on safeguarding ITN systems and subsystems, refer to MIL-STD-461G.

3-70. Planning is essential to maintain signal and data confidentiality, integrity, and availability and deny threat efforts to exploit network vulnerabilities. Key requirements for network modernization and ITN planning include—

- Equipment capable of supporting multiple mission modes including user interface.
- Visual tools that are customizable and enhance decision making and action.
- Equipment readiness and real-time monitoring of system health to enable predictive maintenance and reduce major component failures.
- Multipath transport systems for voice and data communications.
- Multi-level security classification across multiple security domains.
- Information systems that mediate and translate data to enhance situational and understanding.
- Relevant, multi-source data accessibility.
- Systems that allow flexibility using a system of systems architecture and standards or interoperability and rapid upgrades.
- Network transport compatibility and bandwidth aggregation that enable resiliency by leveraging available commercial, military terrestrial, aerial, and celestial capabilities.
- Remote and on-the-move voice and data communication in mobile, handheld, and mounted configurations.
- Spectrum supportability risk assessment conducted in accordance with DODI 4650.01 and AR 5-12 and submitted to the Army Spectrum management Office.
- Automated PACE prioritization and failover path selection to minimize disruption to tactical operations.
- Tactical radio network extension to augment or replace retransmission team operations.

3-71. To ensure ITN capabilities and network systems maintain compatibility, planners ensure spectrum supportability risk assessments are performed and appropriately submitted to the Army Spectrum Management Office to receive a spectrum supportability decision. Refer to ATP 6-02.70 for frequency request procedures.

## JOINT AND MULTINATIONAL INTEROPERABILITY NETWORK PLANNING

3-72. Building joint communications to support the joint force communications requires knowledge of the joint force organization, the commander's concept of operations, available communications resources and capabilities, and how they are employed (JP 6-0). Agreed-upon standards for information and data exchange ensure there is cohesive integration of the network architecture whether systems and capabilities are commercial-off-the-shelf or are transport network capabilities or systems belonging to mission partners.

3-73. The joint and coalition network section of the theater army G-6 plans and coordinates the management of joint functional computer systems—both hardware and software. The section integrates joint information systems with command post and sustainment information systems. The section develops the information management plan and maintains Global Command and Control System (Army and joint) servers to provide the common operational picture for the theater. The section represents the theater army on Army, joint, and multinational information management boards. The section also develops and maintains the concept of operations for multinational network information exchange, develops network architecture and support requirements for joint and multinational exercises, and plans implementation for new technologies.

## SECTION II – RETRANSMISSION PLANNING

3-74. Employing a retransmission site requires deliberate mission planning before deploying the team. Retransmission teams ensure required supplies, equipment, and site support are coordinated and available before leaving the staging area, command post, or forward operating base. Planners must coordinate for security and sustainment when deploying signal sites remotely from the supported unit. Team members assist in coordinated defense of retransmission sites but may require augmentation to adequately defend remote sites. Military police can provide additional security or serve as a response force to communication centers or nodes in the support area and along ground lines of communication, to include retransmission sites, when needed. Refer to ATP 3-39.30 for more information on security and mobility support.

## OVERVIEW

3-75. Essential to retransmission planning and survivability are support tasks performed by non-signal personnel such as quick reaction force, physical security, site defense, and casualty evacuation. Additional factors for retransmission planning include—

- Units supported.
- Number of nets or nodes required.
- Running estimates.
- Number of signal systems on hand and mission capable.
  - Capacity to support retransmission across the assigned area.
  - Radio coverage and capabilities of the radio systems being used.
- Compromise procedures.
- Contingency planning.
  - Planned retransmission locations throughout the mission, including alternate sites, with line of sight analysis.
  - Triggers for displacement to alternate locations.
  - Quick reaction force and relocation evacuation criteria.
  - Primary and alternate frequencies.
- Frequencies for emergency contact.
  - Medical evacuation.
  - Command.
  - Nearest units.
- Time required to be on station and retransmitting for supported units.
- Route and site security requirements.
- Ingress and egress routes.
- Whether unmanned aircraft systems or other aerial retransmission capabilities can meet the requirement.
- Plan to communicate with the parent unit (for example, contact the G-6 or S-6 every two hours with status reports).
- Resupply plan.
- Concealment of the site.
- Plans to mask or reduce the electromagnetic signature of the site.
- Reseed plan (vehicles, radios, and personnel).

## RETRANSMISSION PLANNING CONSIDERATIONS

3-76. Team chiefs, platoon sergeants, and platoon leaders should thoroughly inspect all retransmission systems and personnel before employing a remote retransmission site. Upon return, teams must recover their equipment and prepare for immediate follow-on missions.

3-77. Employing a retransmission team remotely from established command posts or other units requires additional planning and resources for site defense, logistics, and emergency evacuation. Planning considerations for retransmission include—

- Site analysis.
- Site selection.
- Site reconnaissance.
- Site security and defense.
- Sustainment and resupply.

3-78. Because retransmission teams often operate remotely from other elements. It is important for teams to track maneuver operations and anticipate emerging communications requirements. This awareness helps teams maintain fully operational networks supporting the commander's mission requirements.

## SITE ANALYSIS

3-79. The G-6 or S-6 should conduct a signal site analysis of the assigned area. Signal site analysis and selection is a collaborative effort between the G-6 or S-6 and the G-3 or S-3. A thorough site analysis enables

a meaningful course of action comparison during planning and provides flexibility when executing operations. The G-6 or S-6 analysis focuses on the ability of communications systems to provide coverage from the proposed site. The G-3 or S-3 analysis focuses on mobility, survivability, and sustainability.

3-80. If the mission changes and operations move to another area, planners should already know which areas they can support with effective communications. Signal planners consider the operating characteristics, limitations, and effective planning distance for each available communications asset. As planners define communications requirements, the number of retransmission sites needed to cover the assigned area may help determine supportability of proposed courses of action during the military decision-making process.

3-81. The G-6 or S-6 coordinates with the G-2 or S-2 for terrain analysis and the current threat estimate. Signal planners evaluate the terrain (including man-made features) at proposed retransmission sites to visualize how they can support the mission. Terrain and structures may make a proposed site unsuitable for retransmission operations. Planners consider the current threat estimate to prevent placing communications sites near known or anticipated enemy positions.

3-82. The G-6 or S-6 considers these facts and assumptions:

- Maintenance status of communications equipment and systems.
- Technical limitations of equipment—
  - Range.
  - Weather.
  - Bandwidth.
  - Frequencies.
- Areas of limited communications coverage.
- Higher headquarters support.
- Services.
- Personnel status.
- Effects of the operational environment on communications.
- Threat intelligence collection and electromagnetic warfare capabilities.
- COMSEC changes.
- Security and defense of remote retransmission sites.

## SITE SELECTION

3-83. Selecting a retransmission site location is best accomplished by conducting a thorough analysis of the potential site. If the situation does not allow for a site reconnaissance, leaders can use available planning resources such as line of sight analysis tools and satellite imagery. Leaders consider how well the site is concealed from major roads or other vantage points and how the site's natural terrain can support the mission. Planners should consider the enemy's most likely and most dangerous courses of action when evaluating proposed retransmission sites.

3-84. Based on the signal site analysis, the G-6 or S-6 recommends retransmission sites to support all proposed courses of action during the military decision-making process. The final site selected must be—

- Able to provide range extension.
- Accessible.
- Defensible—
  - Defensive plan.
  - Escape routes.
  - Cover.
  - Concealment.
- Logistically supportable.

## Range Extension

3-85. Extending the network to tactical units provides commanders increased situational awareness and enables communication with dispersed or deployed units. Signal teams employ tactical radio retransmission to increase the geographic area a commander can effectively control. Ground and aerial retransmission links

can extend the effective range of tactical radio networks to beyond line of sight distances in environments where satellite access is denied or degraded.

### **Accessibility**

3-86. The proposed site should be accessible with terrain flat enough to accommodate antenna emplacement. Favorable locations include ridgelines and hilltops away from roads, trails, and natural lines of drift that offer multiple avenues of escape.

### **Defensibility**

3-87. Because of their small size and limited defensive capabilities, retransmission teams require assistance from the supported unit to defend against enemy attack. Leaders must ensure the retransmission site is defensible and offers adequate escape routes to prevent overrun by an enemy force.

3-88. Retransmission sites supporting mobile defense should co-locate with other Army elements whenever possible. If the retransmission site operates remotely, the site requires security augmentation to survive a ground attack or level II and III threats. They must be prepared to survive enemy air; artillery; and chemical, biological, radiological, or nuclear hazards with limited external assistance. Refer to ATP 3-39.30 for more information on threat categories levels II and III.

3-89. Planners must coordinate for security and sustainment when employing signal sites remotely from the supported unit. Planners coordinate with the G-2 or S-2 for the current threat estimate and the G-3 or S-3 for security force arrangements when planning site defense. Members of the retransmission team assist in the coordinated defense of retransmission sites but require augmentation from the parent unit to adequately defend remote sites.

### **Logistical Supportability**

3-90. Mission-critical materials must be planned, and throughput must be coordinated to transport items to the site. Stockpiling of unit sustainment supplies (rations, water, and fuel) may also be required. When there are no pre-positioned caches, it is possible to deliver resupply packages to isolated personnel by aircraft. The need for this type of activity should be weighed against the risk to the isolated person and the forces involved, and the capability to mitigate that risk (JP 3-50).

3-91. Supply is essential for enhancing Soldiers' quality of life and provides the materiel required to accomplish the mission. Platoon leaders develop the sustainment plan by determining what is on hand so they can accurately project requirements. Company leaders must know what designated critical supplies the platoon has on hand. The platoon leader establishes priorities for delivery; however, combat demands that class I, III, and IX supplies and equipment take priority for signal platoons because they are critical to successful operations. Resupply operations are either planned or emergency. Refer to ATP 6-02.2 for more information about sustainment of retransmission sites.

## **CONTINGENCY PLANNING**

3-92. Leaders should ensure personnel understand the steps and can execute the contingency plan as necessary. Rehearsals should include crew drills such as requests for indirect fire support, contingency actions, actions on contact, and emergency destruction. Contingency plans include vehicle breakdowns, lost vehicles, and accidents.

## **ALTERNATE LOCATIONS**

3-93. Signal leaders must ensure planned retransmission sites will continue to support the unit in case of displacement to alternate and supplementary fighting positions (ATP 6-02.2). Signal plans should include movement triggers for dislocation to alternate site locations. Triggers for evacuation or displacement to alternate locations may include—

- Approaching enemy forces.
- Receiving indirect fires.
- Impending inclement weather.
- Imminent wildfires.

## COMPROMISE PROCEDURES

3-94. A COMSEC compromise occurs when COMSEC material is irretrievable or lost, information has been disclosed to unauthorized individuals, or a violation of the security policy for a system in which unauthorized intentional or unintentional disclosure, modification, or destruction may have occurred. Users experiencing compromised COMSEC should immediately stop transmission on the compromised system and notify the G-6 or S-6. At a minimum, operators should provide the following information to the G-6 or S-6:

- Unit experiencing compromise.
- Equipment type and nomenclature.
- Crypto segment or crypto period.
- Time and location of compromise.

3-95. All radio operators must be able to recognize an incident that may develop into a COMSEC compromise and report it to the appropriate security authorities. Undetected or unreported incidents or compromises are serious security vulnerabilities. Reportable COMSEC incidents include—

- Loss of accountability.
- Improper destruction.
- Possible tampering.
- Found on installation.
- Unsecured or unauthorized access.
- Improper storage of COMSEC material or equipment.
- Unauthorized use of keying material.
- Loss of COMSEC material.
- Loss of two-person integrity for top secret key material.

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## Appendix A

# Recognizing and Responding to Electromagnetic Attack

This appendix addresses tactics, techniques, and procedures to recognize and respond to threat effects and electromagnetic attack. Included are measures used to recognize and prevent threat activities and effects.

## THREAT TACTICS, TECHNIQUES, AND PROCEDURES

A-1. Information and its management, dissemination, and control have always been critical to the successful conduct of tactical missions. Given today's advancements in information and information systems technology, this importance is growing in scope, impact, and sophistication. The opposing force recognizes the unique opportunities that information warfare gives tactical commanders, and it continuously strives to incorporate information warfare activities in all tactical missions and battles (TC 7-100.2).

A-2. A peer threat's information warfare activities may include synchronized attack methods to create convergence of capabilities and their effects across multiple disciplines and domains to increase the probability of their success. Attacking across multiple domains and echelons, a peer threat will integrate electromagnetic attack, military deception, lethal fires, perception management, information attack, and computer warfare to deny U.S. and allied forces access to the EMS in a contested environment.

A-3. Understanding threat capabilities in the EMS is key to sound signal support plans. Attacks on friendly command and control nodes may combine electromagnetic attacks, other information warfare effects, and lethal fires to deny friendly forces the use of spectrum-dependent systems. To accomplish this goal, threat forces gather technical and combat information about friendly forces. As threat forces locate and identify friendly units, threat information warfare elements establish priorities to—

- Jam communications assets.
- Deceptively enter radio networks.
- Interfere with the normal flow of U.S. and allies' communications.

A-4. Commanders, their staffs, and equipment operators must train to recognize and react to peer threat information warfare tactics, techniques, and procedures if they are to continue communicating in a contested environment.

## MEASURES TO PREVENT THREAT EFFECTS

A-5. There are operational tactics, techniques, and procedures signal planners and operators can take to mitigate peer threat capabilities in the EMS. If a threat cannot detect friendly signals, they cannot geolocate or jam those signals.

### Limiting Electromagnetic Signature

A-6. To protect against a peer threat's ability to locate and target radio signals, the G-6 or S-6 must plan signal support in a way that limits the electromagnetic signature of the signal site. Measures to reduce the electromagnetic signature of command posts and communications sites include—

- Careful site selection for communications equipment.
- Employment of directional antennas.
- Operations using the lowest power required.
- Limiting radio transmissions.
- Using burst transmission to minimize transmission time.
- Using a random battle rhythm schedule.

A-7. Electromagnetic protection techniques can also help mask the electromagnetic signatures of command posts and communications sites. The CEWO assists the G-6 or S-6 in planning electromagnetic protection measures to reduce the command post's electromagnetic signature.

***Terrain Masking***

A-8. Terrain masking can effectively block radio signals from reaching threat direction-finding capabilities. Positioning communications systems with large terrain features or man-made structures between the communications system and the forward line of own troops effectively blocks a threat from detecting the signal.

***Camouflage Net Masking***

A-9. Radar reflective camouflage netting is an effective means of blocking unintended electromagnetic radiation from the rear and sides of directional antennas. Camouflage netting to the sides and back of a line of sight or SATCOM antenna ensures only the main beam of the antenna radiates. This main beam is highly directional; it is much harder to detect since the enemy would need to be directly in the transmission path.

***Line of Sight***

A-10. High-throughput line of sight radios can carry high bandwidth data over distances up to 25 miles, but the links need to be engineered to minimize the chance of detection, targeting, and jamming. If the line of sight path is parallel to the forward line of troops, a threat is less likely to detect the signal, and threat jammers will be unable to reach the antenna with a signal strong enough to jam the radio.

***Remote Antennas***

A-11. Large command posts and high-throughput communications systems emit a significant amount of electromagnetic energy. While planners and operators can mask some of this energy with careful siting, terrain masking, and directional antennas, some electromagnetic energy remains. Because peer threats target friendly command and control capabilities, anything near the communications system is at risk of destruction from lethal fires.

A-12. Commanders and signal planners should consider locating major communications assemblages as far from the supported command post as practical. Placing terrain features, man-made structures, or distance between communications systems and command posts provides the command post protection from lethal fires. Commanders and planners must consider the additional physical security and site defense requirements for a remote site during planning.

***Frequent Command Post Dislocation***

A-13. Despite all efforts to reduce and mask the electromagnetic signature of a command post, a peer threat is likely to locate it eventually. Moving a command post frequently reduces the chances of destruction. Frequent moves are especially important when operating within the range of enemy artillery.

A-14. Maintaining continuity during displacement of a command post or catastrophic loss requires designating alternate command posts and passing control between command posts. During training exercises, units must practice frequent command post dislocation and handoff between the main and tactical command posts.

**INTEGRATION WITH OTHER STAFF ELEMENTS**

A-15. The G-6 or S-6 cannot plan signal support alone. Other staff sections' functions and capabilities help inform sound, survivable signal support plans. Staff sections' activities during the military decision-making process reinforce each other's planning efforts.

A-16. The signal planner, spectrum manager, CEWO, space operations officer, and G-2 or S-2 personnel assess threats to friendly communications during the military decision-making process. Integrated planning counters threat attempts to exploit vulnerabilities of friendly communications systems. When creating an electromagnetic protection plan, the CEWO and signal planners consider deployment, employment, replacement, and concealment of communications systems.

## Cyberspace Electromagnetic Activities Section

A-17. The CEMA section from brigade to corps coordinates and synchronizes cyberspace and electromagnetic warfare operations into the scheme of maneuver. This section includes the CEWO, the spectrum manager, the electromagnetic warfare technician, and electromagnetic warfare noncommissioned officers. The CEMA section is key to the collaboration of cyberspace and electromagnetic warfare operations. The CEWO is the cyberspace planner and understands the operations and missions of the unit and the commander's intent.

A-18. The CEMA section participates in the planning and targeting process and leads the CEMA working group to support the military decision-making process. The cyberspace planner submits requests for effects provided by non-organic resources. Other key staff members who assist in CEMA include the G-2 or S-2, G-3 or S-3, and G-6 or S-6. The CEMA section—

- Plans, requests, and synchronizes effects in cyberspace and the EMS to support freedom of maneuver.
- Coordinates with higher headquarters staff to integrate and synchronize information collection efforts to support cyberspace and electromagnetic warfare operations.
- Synchronizes cyberspace and electromagnetic warfare effects requests with organic targeting capabilities.
- Prepares and submits effect requests using the cyber effects request format or electromagnetic attack request format.
- Develops, maintains, and disseminates cyberspace and EMS input for the common operational picture.
- Deconflicts spectrum use for electromagnetic warfare capabilities with the G-6 or S-6 spectrum manager.
- Provide recommendations on commander's critical information requirements.

## G-6 or S-6

A-19. The G-6 or S-6 section prepares annex H (Signal) to the operation plan or order, with appropriate cross-references to the electromagnetic warfare portions of annex C (Operations), operations security portion of annex E (Protection), the military deception plan, and the signal operating instructions. The G-6 or S-6 integrates and coordinates DODIN operations and spectrum management operations through close collaboration with the CEMA section during planning and operations. The G-6 or S-6 also—

- Plans redundant means of communications for those systems most vulnerable to jamming.
- Provides COMSEC to protect systems most vulnerable to threat information collection.
- Plans measures to protect critical friendly frequencies from intentional and unintentional electromagnetic interference.
- Enforces proper use of radio, electromagnetic protection, and transmission security procedures.
- Performs spectrum management, including designating alternate frequencies for radio networks and modeling the electromagnetic environment.
- Publishes and updates signal operating instructions.
- Disseminates the joint restricted frequency list—taboo, protected, and guarded frequencies.
- Implements cybersecurity on the network and automated information systems.

## G-3 or S-3

A-20. The G-3 or S-3 is the staff lead in planning and the conduct of Army operations. During the planning and execution of electromagnetic warfare and signal support, the G-3 or S-3—

- Exercises staff responsibility for electromagnetic protection.
- Includes electromagnetic support and electromagnetic attack scenarios in all command post and field training exercises.
- Evaluates electromagnetic protection techniques employed.
- Includes electromagnetic protection in unit training.

## **Cyber Electromagnetic Warfare Officer**

A-21. The CEWO plans, coordinates, and directs the execution of electromagnetic warfare activities. The CEWO is also the lead for the CEMA section. The CEWO—

- Serves as subject matter expert on threat electromagnetic warfare capabilities and electromagnetic warfare rules of engagement.
- Leads the CEMA working group.
- Submits information requirements to the G-2 or S-2 to support electromagnetic warfare planning and assessment.
- Supports the G-2 or S-2 during intelligence preparation of the operational environment.
- Coordinates with the G-6 or S-6 to plan, implement, and assess friendly electromagnetic protection measures.
- Coordinates with the G-6 or S-6 spectrum manager to ensure planned electromagnetic warfare activities do not interfere with friendly communications.
- Deconflicts electromagnetic warfare operations with the CEMA spectrum manager.
- Collaborates with the G-2 or S-2 to synchronize and deconflict electromagnetic warfare operations with intelligence activities.

## **Space Operations Officer**

A-22. The primary task of the space operations officer to advise the supported commander of current space assessment from planning through mission execution. Their responsibility includes providing commanders insight on effective space-related planning, operational support, and space capabilities. These tasks require an expert knowledge of space-based systems and situational awareness of the space domain and its operations to include an understanding of the process for acquiring space support.

A-23. Space operations officers integrate space-based capabilities and effects in conjunction with the support staff and designated senior officer responsible for coordinating space requirements. They assist in identifying requirements based on available capabilities and needed materials for a unit to fulfill their mission. A space operations officer's role includes providing space-related support and expertise on space capabilities. Additional responsibilities include—

- Providing information on space-based capabilities to enhance the commander's situational awareness and common operational picture.
  - Recommending mitigation actions to the commander and staff elements.
  - Analyzing space and terrestrial weather impacts to mission.
  - Coordinating planning activities with internal and external organizations.
- Assisting the G-6 in maximizing SATCOM support to operations.
  - Understanding SATCOM architecture.
  - Assessing impact of loss of SATCOM operations and recommend issue solutions.
  - Integrating space capabilities and synchronize effect.
- Assessing GPS satellite technical capabilities.
  - Assessing requirements and gaps consistent with space technology resolutions.
  - Analyzing the impacts of navigation warfare and ensure PNT considerations are integrated into plans and operations.
- Recommending ways to deny threat use of all satellite-based navigation systems, in coordination with G-2.
  - Assessing timing and impacts of multiplexing of communications in coordinate with G-6.
- Coordinating with the CEMA section, G-2 or S-2, and G-6 or S-6 for space-based environmental monitoring and terrestrial weather impacts to operations.
  - Assessing space-based and terrestrial weather and their effects on forecasting accuracy.
  - Analyzing weather impacts to assets including sensors, electro-optical, multispectral imagery, and infrared detection and data collection capabilities.

## **G-2 or S-2**

A-24. The G-2 or S-2 provides current intelligence estimates during the military decision-making process, and updated estimates during operations. The intelligence estimate includes—

- Current enemy tactics, techniques, and procedures.
- Enemy locations in relation to friendly forces.
- The enemy's electromagnetic order of battle—enemy capabilities that could be used to deny friendly use of the EMS.
- Known cyberspace threats to inform cybersecurity and defensive cyberspace operations efforts.

## RECOGNIZING AND RESPONDING TO THREAT EFFECTS AND ELECTROMAGNETIC INTERFERENCE

A-25. Because peer threats consider friendly command and control capabilities to be high priority targets, they have developed capabilities to deny friendly use of the EMS for communications. Soldiers and leaders must learn to identify and respond to threat effects in the EMS.

### Electromagnetic Jamming

A-26. *Electromagnetic jamming* is the deliberate radiation, reradiation, or reflection of electromagnetic energy for the purpose of preventing or reducing an enemy's effective use of the electromagnetic spectrum, with the intent of degrading or neutralizing the enemy's combat capability (JP 3-85).

A-27. Jammers operate against receivers, not transmitters. Two modes of jamming are spot and barrage jamming. Spot jamming concentrates power on one channel or frequency. Barrage jamming is power spread over several frequencies or channels at the same time. It is important to recognize jamming, but it can be difficult to detect.

### Communications Jamming

A-28. Radio operators must learn to recognize and react to electromagnetic jamming. This is not always an easy task since interference can be either internal or external. Other sources having nothing to do with jamming may cause electromagnetic interference. Unintentional electromagnetic interference may be caused by—

- Other radios (friendly and enemy).
- Other electronic, electrical, or electromechanical equipment.
- Atmospheric conditions.
- Equipment malfunction.

A-29. Radio operators must train to differentiate internal and external interference quickly. Electromagnetic jamming most commonly affects single-channel radio systems. These radios include HF, VHF, and UHF radios. Jamming effects may be obvious or subtle. Obvious jamming is normally simple to detect. When experiencing jamming, it is more important to recognize and overcome the incident than to identify it formally.

A-30. Subtle jamming is less obvious because subtle jamming signals produce no sound from the receivers. Although everything may appear normal to the radio operator, the receiver cannot receive an incoming friendly signal. Often, users assume their radios are malfunctioning instead of recognizing subtle jamming. Table A-1 on page 46 lists common jamming signals. Refer to GTA 13-01-001 for detailed information on electromagnetic attack tactics, techniques, and procedures. Additional information is listed at the Cyber Lessons and Best Practices website.

Table A-1. Common jamming signals

| <b>Signal</b>    | <b>Description</b>                                                                                                                                                                                                                                                                                                   |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Random Noise     | Synthetic radio noise. Random noise is indiscriminate in amplitude and frequency. It is similar to normal background noise and can degrade all types of signals. Operators often mistake random noise jamming for receiver or atmospheric noise and fail to take appropriate electromagnetic protection actions.     |
| Stepped Tones    | Tones transmitted in increasing and decreasing pitch. They resemble the sound of bagpipes. Stepped tones are effective against single-channel amplitude modulation or frequency modulation voice circuits.                                                                                                           |
| Spark            | Bursts of short duration and high intensity; they are repeated at a rapid rate. This signal is effective in disrupting all types of radio communications. Spark jamming is easy to produce and an effective jamming signal.                                                                                          |
| Gulls            | Quickly rising and slowly falling variable radio frequency. The effect produced is similar to the cry of a seagull. Gulls produce a nuisance effect and are very effective against voice radio communications.                                                                                                       |
| Random Pulse     | Pulses of varying amplitude, duration, and rate. Pulses disrupt radar and various data transmission systems.                                                                                                                                                                                                         |
| Recorded Sounds  | Any audible sound, especially of a variable nature. Recorded sounds can distract radio operators and disrupt communications. Music, screams, applause, whistles, machinery noise, and laughter are examples.                                                                                                         |
| Wobbler          | A single frequency, modulated by a low and slowly varying tone. The result is a howling sound that causes a nuisance effect on voice radio communications.                                                                                                                                                           |
| Preamble Jamming | A tone resembling the synchronization preamble of the speech security equipment broadcast over the operating frequency of secure radio sets. Results in all radios being locked in the receive mode. Preamble jamming is especially effective when employed against radio networks that use speech security devices. |

### **Preventive Measures**

A-31. Measures operators and planners can use to reduce susceptibility to jamming include—

- Minimize radio transmissions (optimal radio transmissions are six seconds or less).
- Use electromagnetic counter-countermeasures, such as frequency hopping.
- Train using radio silence.
- Use low power setting on radios for normal operations to reduce the probability of detection.
- Use terrain masking to reduce the probability of detection and block potential sources of jamming.

### **Indicators**

A-32. The threat strives to perfect and use new and more confusing forms of jamming, which requires radio operators to be increasingly alert to the possibility of jamming. Training and experience allow operators to determine whether a particular signal is a jamming signal. During operations, radio operators should remain alert to possible jamming indicators. Observable indications of jamming include—

- Apparently random noise or static over voice channels.
- Recorded sounds—messages or music—over voice channels.
- No answer to a radio transmission.

## Reaction

A-33. Communications jamming requires prompt corrective action to restore critical communications capabilities. Possible reactions to jamming include—

- **Continuing to operate.** Enemy jamming usually involves a period of jamming followed by a brief listening period. Operator activity during this short period indicates to threats whether their jamming efforts were successful. Continuing to operate normally gives the enemy no indication of success or failure. If the enemy hears discussion of the problem on the air, or radio the operation terminates, the enemy may assume their jamming is effective. Operators should never terminate operation of a radio network unless ordered to do so. Operators should be careful not to disclose adverse effects of enemy jamming. This means normal operations should continue even when degraded by jamming.
- **Increasing transmitter power output.** Using low power for normal operations minimizes the chance of detection. Once the enemy begins jamming the radios, the risk of detection becomes secondary to the radio delivering required communications. Higher radio power may overcome the enemy's jamming signal but increases the risk of detection by enemy direction finding capabilities.
- **Improving the signal-to-jamming ratio.** The signal-to-jamming ratio is the relative strength of the desired signal to the jamming signal at the receiver. If the desired signal is much stronger than the jamming signal, the jamming does not significantly degrade communications. To improve the signal-to-jamming ratio operators and signal leaders can consider the following—
  - **Adjusting or changing the antenna.** When jamming occurs, the radio operator should adjust the antenna to receive the maximum incoming signal strength. Depending on the antenna, some methods include reorienting the antenna, changing antenna polarization at all stations, or installing an antenna with a greater range.
  - **Establishing a retransmission site.** A retransmission site can increase the effective range and power of a signal between radio stations without increasing transmit power.
  - **Relocating the antenna.** Operators may use terrain masking to block the incoming jamming signal. This may require relocating the antenna and associated radio setup to several hundred meters from its previous location.
- **Changing frequencies.** If a communications network cannot overcome enemy jamming, the commander may direct forces using an alternate or spare frequency. Preplanned and well-coordinated actions are required for practical dummy stations to continue to operate on the jammed frequency to mask the change to an alternate frequency. During a jamming incident, it may be difficult to coordinate a frequency change. All radio operators require knowledge of when, and under what circumstances, they should switch to a backup frequency. If the frequency change is not smooth, the enemy may discover what is happening and try to degrade communications on the new frequency.
- **Executing the PACE plan.** Quickly changing to the alternate or contingency means of communications reduces disruption of the command and control system.
- **Using signals intelligence or electromagnetic warfare capabilities to locate the jamming signal.** Leveraging signals intelligence or electromagnetic warfare capabilities requires coordination and collaboration with the G-2 or S-2 and the CEWO.

A-34. If any of the corrective actions taken mitigate the enemy jamming, operators should continue operation of the network and submit a joint spectrum interference resolution (JSIR) report to higher headquarters. JSIR reports document a history of problems and help identify possible causes for subsequent interference. Maintaining a historical record of interference helps develop countermeasures to future jamming incidents. Refer to ATP 6-02.70 for more information about JSIR reporting.

## Positioning, Navigation, and Timing Jamming

A-35. A typical brigade combat team has over 2,500 PNT-enabled devices and over 250 SATCOM-enabled devices that are used to conduct precision fires, movement, maneuver, communications, protection, command and control, and other mission requirements. Peer threats have capabilities to contest the ground, space, and link segments of satellite-based PNT. These attacks may have significant impacts across all warfighting functions and many weapon platforms.

A-36. Electromagnetic jamming of PNT satellite capabilities affects not only communications, but also many other capabilities in tactical formations. Systems affected include—

- Communications systems.
- Friendly force tracking.
- Navigation.
- Reconnaissance.
- Radar systems.
- Precision guided munitions.

### *Preventive Measures*

A-37. Measures to reduce susceptibility to, and mitigate the effects of, PNT jamming include—

- Using only encrypted PNT systems.
- Employing antenna masking.
- Using terrain masking.
- Training and maintaining the ability to navigate using a map and compass.

### *Indicators*

A-38. User indications that an enemy may be jamming PNT signals include—

- Loss of satellite signal.
- Red GPS icon on the network management system.
- Loss of timing or incorrect time displayed on equipment.
- Inaccurate map display or imprecise position or geolocation indicator data.
- Jamming environment warning message.

### *Reaction*

A-39. Because of the diverse and widespread effects of enemy GPS jamming, a prompt, coordinated response is necessary. Operators of all affected systems should—

- **Navigate using map and compass.** While this does not restore system timing and situational awareness displays, navigation using a map and compass cannot be jammed.
- **Increase distance between affected systems and jammer.** If the jammer operates from a known location, increased distance or terrain masking may mitigate interference.
- **Use signals intelligence or electromagnetic warfare capabilities to locate the jamming signal.** Leveraging signals intelligence or electromagnetic warfare capabilities requires coordination and collaboration with the G-2 or S-2 and the CEWO.
- **Report jamming to higher headquarters.** Submitting a JSIR report to higher headquarters documents a history of problems and helps identify possible causes for subsequent interference.

## **Satellite Communications Jamming**

A-40. Expeditionary forces rely heavily on SATCOM capabilities for beyond line of sight network transport. Systems and capabilities affected by SATCOM jamming include—

- Friendly force tracking.
- Wideband SATCOM.
- TACSAT radios.
- Intelligence reporting systems.

### *Preventive Measures*

A-41. Operational and employment measures to prevent SATCOM jamming include—

- Minimizing transmissions on narrowband (single-channel) TACSAT radios (radio transmissions six seconds or less is optimal).
- Using terrain masking.
- Using camouflage net masking.

### Indicators

A-42. Possible operator indications of satellite jamming include—

- Seemingly random noise or static on narrowband (single-channel) TACSAT radios.
- Recorded sounds, such as messages or music, over narrowband (single-channel) TACSAT radios.
- No answer to transmission.
- Red satellite icon on network management system display.
- Loss of data from the satellite.
- Low signal-to-noise indicated on wideband satellite terminal.

### Reaction

A-43. The reactive measures listed here apply mostly to narrowband (single-channel) TACSAT systems. When a narrowband (single-channel) TACSAT radio operator recognizes a jamming attempt, they may—

- **Increase radio transmit power.** Only increase power on wideband SATCOM terminals if directed by the satellite controller.
- **Change to a preapproved alternate frequency.** Minimizes electromagnetic footprint of communications systems operating within the EMS environment.
- **Execute the PACE plan.** Quickly changing to the alternate or contingency means of communications reduces disruption of the command and control system.
- **Use signals intelligence or electromagnetic warfare capabilities to locate the jamming signal.** Leveraging signals intelligence or electromagnetic warfare capabilities requires coordination and collaboration with the G-2 or S-2 or the CEWO.
- **Report jamming to higher headquarters.** The higher headquarters' frequency manager and CEWO can correlate reports from units across the assigned area to isolate enemy jammers and plan countermeasures, including nominating targets for lethal fires.
- **Use line of sight systems for network transport.** Units cannot communicate beyond line of sight, or through significant physical obstacles.

## ELECTROMAGNETIC INTERFERENCE

A-44. With the increased use of the EMS and complex, sophisticated radio systems and electronic devices, there is increased vulnerability to electromagnetic interference. *Electromagnetic interference* is any electromagnetic disturbance, induced intentionally or unintentionally, that interrupts, obstructs, or otherwise degrades or limits the effective performance of electromagnetic spectrum-dependent systems and electrical equipment (JP 3-85).

## TYPES OF ELECTROMAGNETIC INTERFERENCE

A-45. Electromagnetic interference can be caused by threat, neutral, friendly, or natural sources. Electromagnetic interference from natural sources, or natural electromagnetic interference, includes geographic electromagnetic phenomena, atmospheric noise (thunderstorms), and galactic noise (stars). Electromagnetic interference from man-made sources includes power lines, power generators, alarm systems, electrified railroads, and fluorescent lights.

A-46. Since electromagnetic interference prevents successful radio transmission, units must recognize potential internal and external sources of electromagnetic interference, both intentional and unintentional. They must then take appropriate action to minimize its effects on friendly communications.

### Environmental

A-47. Weather data and other naturally occurring phenomena are part of the information required by commanders and staffs when planning, preparing, executing, and assessing combat operations. Naturally occurring subsurface and surface phenomena create hazards that may endanger personnel or cause electromagnetic interference. Advanced knowledge of electromagnetic interference resulting from weather or environmental factors allows ground forces to avoid adverse conditions while taking advantage of other

conditions that enhance operations. Natural weather or space environmental factors that can affect signal transmission include—

- Solar radio bursts.
- Noise storms.
- Ionospheric scintillation.
- Aurora activity.
- Solar flares.

A-48. The Air Force Space Weather Operations Center issues real-time and forecast space weather data addressing natural electromagnetic interference from solar radio bursts or noise storms, ionospheric scintillation, aurora activity, or solar flares. To establish the presence of solar electromagnetic interference, planners determine whether the link line of sight is aligned with the sun and whether the side lobes of the antenna intersect the sun's radio frequency output. For ionospheric scintillation, planners determine whether communication links intersect a region of the ionospheric scintillation.

A-49. Planners analyze weather data, identify potential weather effects, and assess the impact of weather on systems, and provide vital information to optimally employ communications assets. Planners and signal personnel should consider local weather conditions when assessing electromagnetic interference causes and effects as well as Space Force-provided data, alerts, and warnings on meteorological, oceanographic, and space environment factors that could affect military operations.

## **Cosite**

A-50. Cosite interference is the effect of unwanted energy, due to emissions, radiation, or induction, on reception in a radio communications system. This could cause system performance degradation, misinterpretation, or loss of information. Several options are available to mitigate cosite interference, but there are no comprehensive solutions. The user must decide if an option is applicable to the tactical situation and take the appropriate action to resolve cosite interference.

A-51. Some equipment systems are not as critical as others. The G-6 or S-6 recommends a system priority list that ensures the transmission of mission-critical information. During interference events, the G-6 or S-6 should be prepared to shut down less critical systems to preserve the functionality of mission-critical systems. To reduce cosite interference—

- **Transmit at the lowest power level when possible.** This allows co-located SINCGARS and other transmission systems to operate with minimal interference at the receivers. This option may be unacceptable due to the significant transmission range reduction of the SINCGARS.
- **Locate antennas remotely and transmit from the command post at low power to a full power retransmission system.** Certain critical command post networks would then be able to maintain their high-power advantage.
- **Separate antennas as much as possible when in the same or nearby frequency ranges.** The greater the separation between the transmitting and receiving antennas, the less interference encountered. Tilting the tops of the transmitting and receiving antennas away from each other can enhance vertically polarized ground wave communications. Tilt angles from 15–30 degrees provide the best results.

## **JOINT SPECTRUM INTERFERENCE RESOLUTION**

A-52. The JSIR report addresses electromagnetic interference incidents, electromagnetic attack, and directed energy affecting the DOD. The objective is to report and help resolve electromagnetic attack and persistent, recurring electromagnetic interference. Units must report all electromagnetic interference disturbances regardless of the severity, intensity, or duration. Users encountering electromagnetic interference submit reports of suspected hostile electromagnetic interference using the secure JSIR-Online Portal. Users should not delay submitting reports if information is not readily available. They should use follow-up reports to provide additional information as it becomes available.

A-53. Table A-2 on page 51 shows the minimum information requirements for JSIR reporting. The message subject line should indicate whether the report is initial.

**Table A-2. Joint spectrum interference resolution information requirements**

| <i>Item Number</i> | <i>Data Input</i>                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                  | Frequencies affected by the interference.                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2                  | Locations of systems experiencing the interference.                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3                  | The affected system name, nomenclature, and manufacturer (with model number), or other system description. If available, include the equipment characteristics of the victim receiver, as bandwidth, antenna type, and antenna size.                                                                                                                                                                                                                        |
| 4                  | The operating mode of the affected system. If applicable, include —frequency agile, pulse Doppler, search, and upper and lower sidebands.                                                                                                                                                                                                                                                                                                                   |
| 5                  | The characteristics of the interference (noise, pulsed, continuous, intermittent, frequency, or bandwidth).                                                                                                                                                                                                                                                                                                                                                 |
| 6                  | The description of the interference effects on victim performance (reduced range, false targets, reduced intelligibility, or data errors).                                                                                                                                                                                                                                                                                                                  |
| 7                  | Dates and times the interference occurred. Indicate whether the duration of the interference is continuous or intermittent. Indicate whether the approximate repetition rate of the interference, and whether the amplitude of the interference is varying or constant. Indicate if the interference is occurring at a regular or irregular time of day, and if the occurrence of the interference coincides with any ongoing local activity.               |
| 8                  | The location of possible interference sources (coordinates or line of bearing, if known; otherwise, state the location as unknown).                                                                                                                                                                                                                                                                                                                         |
| 9                  | A listing of other units affected by the interference (if known) and their location or distance and bearing from the reporting site.                                                                                                                                                                                                                                                                                                                        |
| 10                 | A clear and concise narrative summary of the interference, and any local actions that have been taken to resolve the problem. The operator is encouraged to provide any other information, based on observation or estimation, pertinent in the technical or operational analysis of the incident. Identify whether the information furnished is based on actual observation and measurement or estimated. Avoid using Army or program jargon and acronyms. |
| 11                 | Other message traffic that relates to the interference problem reported, including the message date-time group, originator, action addressees, and subject line.                                                                                                                                                                                                                                                                                            |
| 12                 | Whether the problem has been identified or is resolved.                                                                                                                                                                                                                                                                                                                                                                                                     |
| 13                 | Whether joint spectrum interference resolution technical assistance is desired or anticipated.                                                                                                                                                                                                                                                                                                                                                              |
| 14                 | Point of contact information, including name, unit, and contact phone numbers.                                                                                                                                                                                                                                                                                                                                                                              |

A-54. It is critical for users to attempt to resolve electromagnetic interference problems at the lowest possible level before submitting JSIR reports to higher headquarters. The unit experiencing the electromagnetic interference incident forwards the initial JSIR report through the chain of command to the unit operations center.

A-55. Electronic report submissions are recorded using the joint spectrum management system or Spectrum XXI programs. The sender classifies the report by evaluating the security sensitivity of the electromagnetic interference on the affected system, and by considering the classification of the text comments.

A-56. Spectrum managers assign precedence to the JSIR report consistent with the urgency of the reported situation. Routine or priority precedence is used for JSIR reports, unless the organization originating the report believes the incident is hazardous to military operations. Table A-3 on page 52 is an example security classification guide used in JSIR reporting.

A-57. Each Army unit submits reports through its chain of command, up to the theater army, the combatant command, and to the United States Army Communications-Electronics Services Office. Users submit

information copies of incident reports to the Joint Spectrum Center for inclusion in the JSIR database. Refer to CJCSM 3320.02E for additional information on JSIR reporting procedures.

**Table A-3. Joint spectrum interference resolution security classification guide**

| <b>Information Revealing</b>                                                                                                                                                       | <b>Security Classification</b>                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| The specific identification of an unfriendly platform or location, by country or coordinates, as the source of interference or electromagnetic attack.                             | Secret.                                                                            |
| Specific susceptibility or vulnerability of U.S. electromagnetic equipment and systems.                                                                                            | Secret.                                                                            |
| Parametric data of classified U.S. electromagnetic equipment.                                                                                                                      | In accordance with the classification guide of the affected equipment.             |
| Suspected interference from unidentified sources while operating in or near hostile countries.                                                                                     | Secret.                                                                            |
| Interference to U.S. electromagnetic equipment and systems caused by electromagnetic attack exercises in foreign nations.                                                          | Confidential.                                                                      |
| Suspected interference from friendly sources.                                                                                                                                      | Unclassified or secret (if the report reveals a specific equipment vulnerability). |
| Information referring to joint spectrum interference resolution reporting stating that joint spectrum interference reporting analyses are a function of the Joint Spectrum Center. | Unclassified.                                                                      |

## INTERFERENCE RESPONSE

A-58. To avoid future electromagnetic interference, units must respond in a consistent, methodical manner. One way to reduce the operational impact of electromagnetic interference is to incorporate preventive measures that identify and mitigate electromagnetic interference effects. By instituting best practices and lessons learned, units can also reduce future interference that disrupts communication and minimizes the negative impact of electromagnetic interference on operations. The electromagnetic interference battle drill example at table A-4 provides appropriate responses to electromagnetic interference. Refer to CJCSM 3320.02E for information about electromagnetic interference resolution.

**Table A-4. Electromagnetic interference battle drill**

| <b>Step</b>                                                                                                       | <b>Description</b>                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                                                                                                                 | Follow equipment troubleshooting steps (verify frequency, cable, antenna connections, and communications security). If interference continues, follow steps 2 through 6. |
| 2                                                                                                                 | Determine start and stop times or duration of interference.                                                                                                              |
| 3                                                                                                                 | Identify interference effect (interfering voice, noise, or static).                                                                                                      |
| 4                                                                                                                 | Identify other emitters in the assigned area.                                                                                                                            |
| 5                                                                                                                 | Check adjacent and nearby units for similar problems.                                                                                                                    |
| 6                                                                                                                 | Prepare and submit a joint spectrum interference resolution report through the G-6 or S-6 spectrum manager.                                                              |
| <b>Legend:</b><br>G-6      assistant chief of staff, signal<br>S-6      battalion or brigade signal staff officer |                                                                                                                                                                          |

A-59. Figure A-1 on page 53 depicts a logical flow diagram for instances when an Army unit is the victim of interference in a tactical operation. Figure A-2 on page 54 shows a flow diagram for interference when the Army unit is the source of the interference.

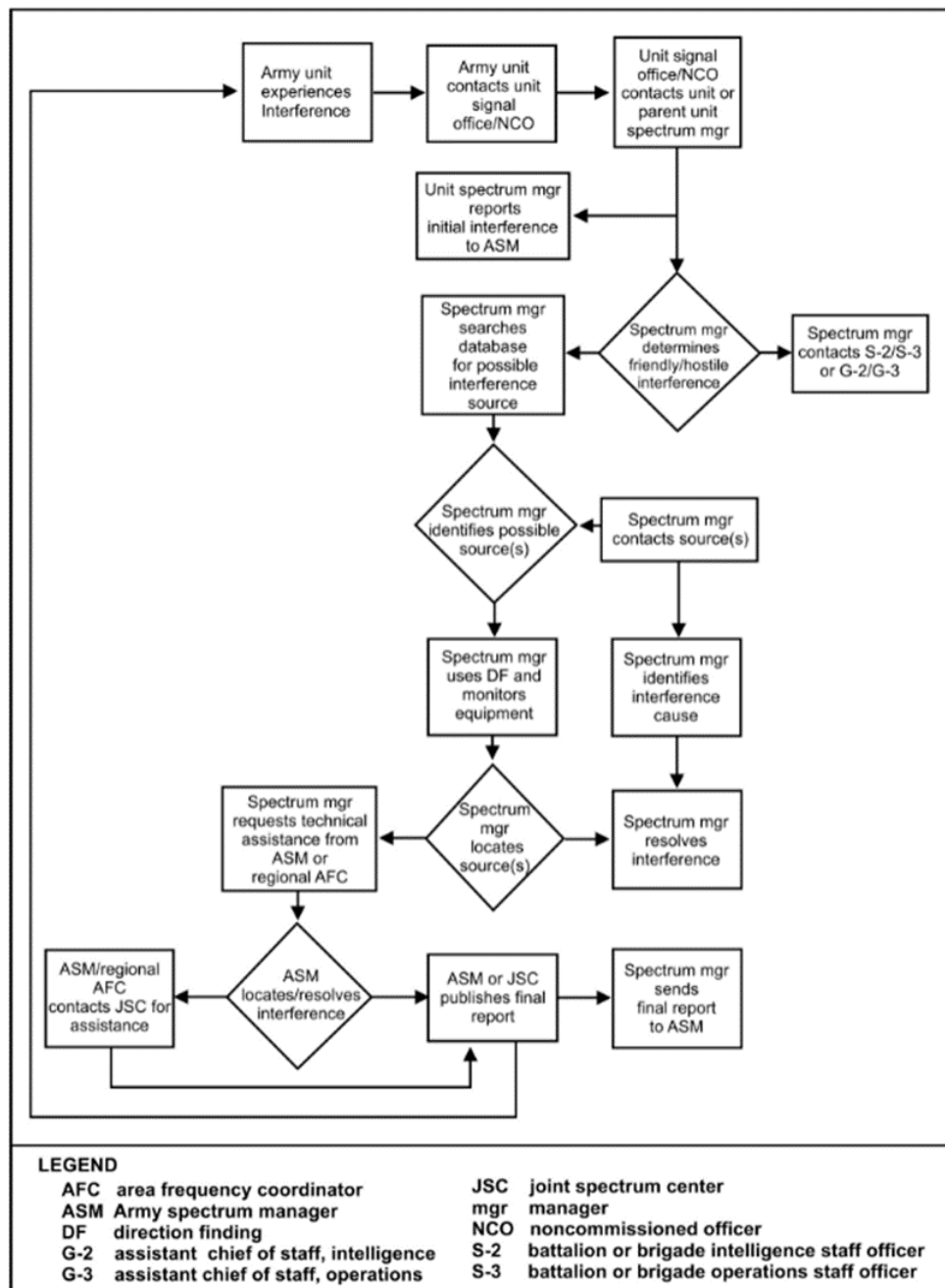


Figure A-1. Interference resolution (Army victim)

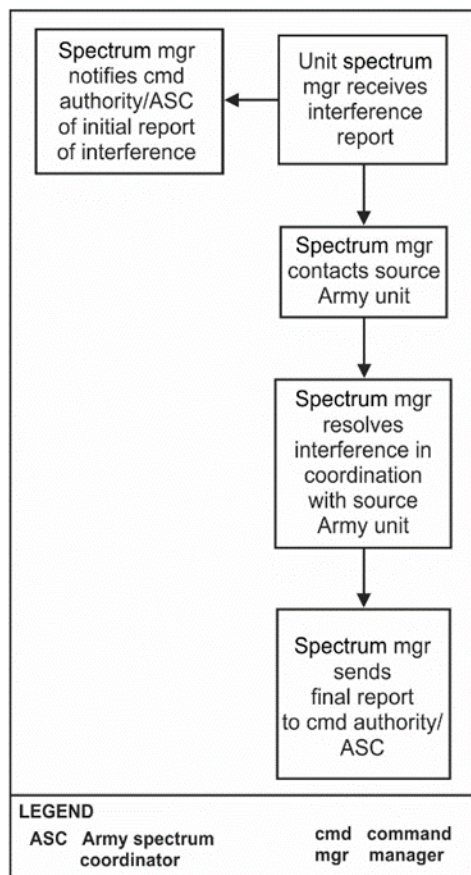


Figure A-2. Interference resolution (Army source)

## EMISSION CONTROL

A-60. *Emission control* is the selective and controlled use of electromagnetic, acoustic, or other emitters to optimize command and control capabilities while minimizing, for operations security: a. detection by enemy sensors, b. mutual interference among friendly systems, and/or c. enemy interference with the ability to execute a military deception plan (JP 3-85).

### EMISSION CONTROL CONDITIONS

A-61. EMCON prevents the threat from discovering and attacking the locations of friendly forces with electromagnetic warfare or lethal fires. It is important to understand the general categories and status criteria for EMCON conditions. Based on the tactical situation or operational environment, the commander can dictate the appropriate EMCON condition. During operations, commanders should consider EMCON condition 3 (amber) as the baseline condition. Units should routinely execute transitions to EMCON 4 (red) and EMCON 5 (black) during conflict.

A-62. Table A-5 on page 55 captures the five EMCON conditions and the general descriptive criteria associated with each condition. The EMCON emitter matrix establishes the operational criteria for emitters at each EMCON condition and works in conjunction with the PACE plan. Every emitter within the unit should be included and the corresponding criteria described. The CEWO assists the G-3 or S-3 and the G-6 or S-6 in determining operating criteria for each type of emitter for each EMCON condition. Refer to FM 3-12 for staff responsibilities including information on planning, coordination, and execution of electromagnetic protection activities.

**Table A-5. Emission control conditions**

| EMCON Condition                                                                                                                                                                      | Description                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EMCON 1</b><br>Green                                                                                                                                                              | Describes a situation where there is <u>no apparent hostile activity</u> against friendly emitter operations. Operational performance of all spectrum-dependent systems is monitored; password and encryption systems are used as a layer of protection.                                                                                                                |
| <b>EMCON 2</b><br>Yellow                                                                                                                                                             | Describes an <u>increased risk of attack</u> after detection. Increased monitoring of all EMS activities is mandated, and all Department of Defense end users must make sure their systems are secure, encrypted, power levels monitored, and transmissions limited. EMS usage may be restricted to certain emitters. Rehearsal for elevated EMCON conditions is ideal. |
| <b>EMCON 3</b><br>Amber                                                                                                                                                              | Describes when a <u>risk has been identified</u> . Electromagnetic counter-countermeasures (encryption/frequency hop/directional antennas) on important systems are a priority, and the CEWO's alertness is increased. All unencrypted systems are disconnected.                                                                                                        |
| <b>EMCON 4</b><br>Red                                                                                                                                                                | Describes when an <u>attack has taken place</u> , but the EMCON system is not at its highest alertness. Non-essential emitters may be taken offline, alternate or contingency means of communication may be implemented, and modifications are made to standard lower EMCON configurations (power levels and antenna types).                                            |
| <b>EMCON 5</b><br>Black                                                                                                                                                              | Describes when <u>attacks are taking place</u> within the EMS. The most restrictive electromagnetic protection methods are enforced. Any compromised systems are isolated from the rest of the network.                                                                                                                                                                 |
| <b>LEGEND</b><br>CEWO                      cyber electromagnetic warfare officer<br>EMCON                    emission control<br>EMS                        electromagnetic spectrum |                                                                                                                                                                                                                                                                                                                                                                         |

A-63. Table A-6 illustrates the use of the EMCON conditions and shows how they escalate for the SINGARS Advanced System Improvement Program (ASIP) radio.

**Table A-6. Example emitter criteria table**

| Emitter Criteria                                                                                                                     | Emitter Criteria by EMCON Condition              |                                    |                                                |                                                                                                                              |                    |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------|
|                                                                                                                                      | EMCON 1<br>(Green)                               | EMCON 2<br>(Yellow)                | EMCON 3<br>(Amber)                             | EMCON 4<br>(Red)                                                                                                             | EMCON 5<br>(Black) |
| <b>SINGARS</b>                                                                                                                       | Use system as prescribed in the technical manual | Must encrypt systems for operation | Must encrypt and operate in frequency hop mode | Must encrypt and operate in frequency hop mode. Do not exceed 10 watts transmit power. Use directional antennas if available | DO NOT USE         |
| <b>LEGEND</b><br>EMCON                    emission control<br>SINGARS                single-channel ground and airborne radio system |                                                  |                                    |                                                |                                                                                                                              |                    |

**EMISSION CONTROL TECHNIQUES**

A-64. Users of RF-enabled devices adhere to EMCON techniques to prevent unintentional transmission of signals. These signals, if intercepted and analyzed, could compromise the mission. Appropriate countermeasures should be applied along with applicable operations security guidelines for that device or radio to prevent exploitation of compromising emanations.

## Appendix B

# Retransmission Team Training and Certification

Retransmission team certification consists of training and testing team members to perform critical individual and collective tasks and drills to an established standard. Section I provides detailed information to implement signal assessment tables I–IV, section II covers team training, and section III provides battle drills in which the team must be proficient.

### SECTION I – Signal Assessment Tables

B-1. Signal assessment tables provide a tiered approach to collective signal training. The tables provide a crawl, walk, run, training methodology from individual through team, section, platoon, and higher collective tasks. Tables I–IV provide a training framework for individual and team tasks. The individual and collective training tasks within this section comply with training developed and approved by the Army Training and Development Capability office. Refer to FM 7-0 or the Army Training Network Website, for more information on building effective unit training plans.

### TABLE I – INDIVIDUAL TRAINING

B-2. During table I training, team chiefs train Soldiers to conduct the individual tasks that support the team’s collective tasks. Leaders train Soldiers on equipment maintenance during table I training. Team chiefs train their teams using table I and determine when individuals are prepared for table II certification. Table I training includes personnel and equipment training.

#### PERSONNEL TRAINING

B-3. Military training improves the capabilities of Soldiers in their respective roles. Team chiefs ensure Soldiers complete training in the following—

- Role-based cyberspace workforce training.
- Individual task training.
- Preventive maintenance.
- AR 350-1 training requirements.

#### Role-Based Cyberspace Workforce Training

B-4. Cyberspace workforce training unifies the overall cyberspace workforce and establishes specific workforce elements for information technology, cybersecurity, cyberspace effects, intelligence, and enablers to align and manage the cyberspace workforce. During table I training, Soldiers complete cyberspace workforce training based on their respective roles. The minimum cyberspace workforce training for general purpose users is the DOD cyber awareness challenge.

#### Individual Task Training

B-5. Team chiefs document individual task training in the Digital Training Management System and the Soldier’s individual training record. Leaders monitor Soldier training and individual task completion to certify proficiency in technical, tactical, and leadership skills. Individual task training includes common task training and military occupational specialty task training.

#### *Common Tasks*

B-6. Team chiefs enforce common tasks taught to Soldiers during initial training. These tasks are considered critical for the Soldiers’ survival. Retransmission teams must be proficient in all common warrior tasks and battle drills. Warrior tasks and battle drills enhance a Soldier’s readiness. Warrior tasks are a collection of individual Soldier skills known to be critical to a Soldier’s survival. Refer to STP 21-1-SMCT for specific tasks. Examples of common tasks include—

- Weapons training.
- Tactical communications.
- Urban operations.
- Tactical combat casualty care—
  - First aid.
  - Self-aid buddy care.

### ***Military Occupational Specialty Tasks***

B-7. Team chiefs ensure Soldiers receive training specific to their military role, including the use of any specialty equipment. Soldiers perform and reinforce military occupational specialty skills during unit training events. Individual tasks for retransmission teams include—

- Task 113-25U-1003, Operate Radio Retransmission Station.
- Task 113-611-1000, Select a Radio Retransmission Site.
- Task 113-25U-3004, Direct Retransmission Operations.
- Task 113-596-1006, Install Antenna Group OE-254.
- Task 113-100-0002, Install COM 201 Antenna.
- Task 113-000-1001, Operate the Radio Retransmission Station using SINCGARS Family of Radios.
- Task 113-2kw-0001, Operate Generator Set 2 kilowatts.
- Task 113-2kw-0000, Perform Field-Level Maintenance on Generator Set 2 kilowatts.

### **Preventive Maintenance**

B-8. Preventive maintenance checks and services identify and correct faults and ensure continued operation of the equipment. Soldiers train to conduct preventive maintenance specified in the equipment technical manual.

### **AR 350-1 Training Requirements**

B-9. AR 350-1 training requirements cover unit and individual training. Team leaders monitor training throughout the year and ensure Soldiers complete Army training. Examples of AR 350-1 training includes—

- Anti-Terrorism Level 1.
- Combating trafficking in person.
- Controlled unclassified information.

### **EQUIPMENT TRAINING**

B-10. Units conduct individual equipment training and qualification to certify Soldiers on their assigned equipment. Equipment training includes—

- Supply discipline.
- Operator maintenance.
- Driver training.
- Weapons training.
- COMSEC.

### **Supply Discipline**

B-11. Team chiefs enforce supply discipline by maintaining equipment accountability of all classes of supplies. Team chiefs must enforce programs to ensure resources are used without fraud, waste, and abuse.

### **Operator Maintenance**

B-12. Operators perform preventive maintenance checks and services on their assigned equipment. Based on the mission, operators diagnose faults and replace parts and subcomponents from available spares. Team chiefs ensure Soldiers conduct operator level preventive maintenance according to the technical manual maintenance schedule.

## Driver Training

B-13. Driver training qualifies Soldiers to operate the team's assigned vehicle. Retransmission teams are expected to conduct operations at night, including tactical movement to and displacement from retransmission sites. Soldiers must be able to operate vehicles at night during blackout conditions and adverse weather conditions. Soldiers must be proficient in the use of night observation devices. Driver training includes—

- Driver responsibilities.
- Laws and regulations.
- Basic operating procedures.
- Emergency response procedures.
- Accident causes and reporting.
- Hazards and safety measures.
- Operation of vehicles with trailers attached.
- Operation of vehicles on steep slopes.
- Operation of government vehicles off post.
- Transportation of personnel.
- Maintenance responsibilities and forms.
- Convoy operations.
- Driving using night observation devices.
- Hands-on training.
- Performance road test.

## Weapons Training

B-14. Units conduct individual weapons training and qualification to certify Soldiers on their assigned weapons. Soldiers who fail to qualify will participate in retraining and qualification.

## Communications Security

B-15. Soldiers train to use and protect COMSEC devices and keying material. For more information about COMSEC, refer to ATP 6-02.75. COMSEC training includes—

- Key loading.
- Emergency destruction.
- Over-the-air rekeying.
- Incident reporting.
- Material accountability.

## TABLE II – INDIVIDUAL SKILLS TEST

B-16. During table II skill testing, a staff sergeant or above certifies a Soldier's proficiency in the individual skills trained during table I training. Soldiers certify on personnel and equipment tasks.

B-17. In table II, Soldiers will take an individual skills test that certifies their competence on the supporting tasks necessary to install, operate, and maintain their assigned system. Soldiers must complete a physical fitness test, qualify on assigned weapon, complete role-based cyberspace workforce training, and recommended by their team chief as proficient in their individual supporting tasks.

## PERSONNEL CERTIFICATION

B-18. Soldiers will be trained to conduct all supporting individual tasks in the system they are assigned. They will be trained and resourced for other individual certifications required to perform their tasks to include—

- Role-based cyberspace workforce certification.
- Individual task proficiency.
- Physical fitness test qualification.

## **Role-Based Cyberspace Workforce Certification**

B-19. The certifying leader verifies cyberspace workforce certification through the Army Training and Certification Tracking System Website. The certifying leader documents completion according to the unit SOP.

## **Proficiency in Individual Tasks**

B-20. The certifying leader evaluates Soldiers' proficiency in individual common and military occupational specialty tasks. Proficiency is achieved when Soldiers perform a training task to standard.

### ***Common Tasks***

B-21. The Soldier's manual of common tasks provides information to plan, prepare, train, and evaluate individual training of warrior tasks. Refer to STP 21-1-SMCT for training and evaluation outlines for common tasks.

### ***Military Occupational Specialty Tasks***

B-22. Certifying leaders verify Soldiers' proficiency in the individual tasks that support retransmission operations. The evaluation checklists are available on the Army Training Network Website.

## **Physical Fitness Test**

B-23. Certifying leaders verify the Soldier has passed a physical fitness test through the unit training noncommissioned officer. Soldiers who fail their physical fitness test are subject to administrative action and retest.

## **EQUIPMENT CERTIFICATION**

B-24. Equipment certification provides a guide to the unit ensuring Soldiers meet proficiency standards. Records of certification are retained for training management purposes. Certifying leaders verify the following tasks—

- Weapons qualification.
- Vehicle licensing.
- Equipment accountability.

## **Weapons Qualification**

B-25. Soldiers qualify on weapons assigned to their team. Certifying leaders verify weapons qualification through the company training records.

## **Vehicle Licensing**

B-26. Retransmission teams require at least two licensed drivers for the team vehicle. Certifying leaders verify valid licenses through inspection.

## **Equipment Accountability**

B-27. Table II certification includes verifying accountability of assigned equipment. Certifying leaders conduct hand receipt inspections and equipment layouts to verify equipment accountability.

## **TABLE III – TEAM TRAINING**

B-28. After individual Soldiers complete table II certification, the team can begin training collective tasks and drills. Teams train and rehearse collective tasks and battle drills based on their assigned equipment and mission.

B-29. Table III is the crawl phase of the overall team certification. During table III training, teams train according to the unit SOP and the training and evaluation outlines for the team collective tasks and battle

drills. Teams progress from daytime iterations under static conditions to nighttime iterations under dynamic conditions. As the team masters the collective tasks, they conduct some training events in mission oriented protective posture 4. Table III consists of personnel and equipment training. The section sergeant conducts table III training and recommends when teams are ready to progress to table IV certification.

## **PERSONNEL TRAINING**

B-30. Team training improves the capabilities of Soldiers in their respective team roles. Section sergeants ensure Soldiers complete training in the following.

- Table II certification.
- SOP.
- Drills.
- Training reinforcement.

### **Table II Certification**

B-31. Table II individual certification is a prerequisite for table III team training. Leaders verify all team members have completed table II certification before beginning table III training.

### **Standard Operating Procedures**

B-32. SOPs use past lessons and best practices to make the unit more effective. Team members review the SOP to ensure they are familiar with unit-specific requirements.

### **Drills**

B-33. Teams build proficiency by conducting drills that focus on collective skills. Drills for retransmission teams consist of—

- Warrior battle drills.
- Retransmission team drills.
- Command post battle drills.
- Battle drills.

#### ***Warrior Battle Drills***

B-34. Retransmission teams must be proficient in all common warrior battle drills. Battle drills are group skills designed to teach a unit to react and survive in common combat situations. Refer to STP 21-1-SMCT for a complete list of warrior battle drills with their supporting individual warrior tasks. Team chiefs must also be proficient in skill level 2 drills in STP 21-24-SMCT. Examples of warrior battle drills important to a retransmission team include—

- React to contact.
- Establish security at the halt.
- Perform tactical combat casualty care.
- React to ambush—
  - Near.
  - Far.

#### ***Retransmission Team Drills***

B-35. Team drills are an effective mechanism to validate and sustain team training. Teams should train and conduct team drills according to the unit's combined arms training strategy or SOP. The unit should implement team certification standards to support the unit's mission and incorporate these standards into their operational planning.

#### ***Command Post Battle Drills***

B-36. Command post battle drills ensure all team members understand a technique or a specific set of procedures. Examples of command post battle drills for a retransmission team include—

- Site defense.
- Emergency destruction.
- React to jamming.

### ***Battle Drills***

B-37. Battle drills are collective actions performed without the application of a deliberate decision-making process. A battle drill is initiated on a cue, such as an enemy action or a leader's command. A battle drill provides a reflexive response to a given stimulus. Battle drills require minimal leader orders to accomplish. Synchronization and reaction speed are enhanced when battle drills are identified, and the required reactions are defined in the unit SOP and rehearsed during training.

### **Training Reinforcement**

B-38. Training on the collective tasks reinforces individual task training. Reinforcement training of collective tasks should be performed to standard through continual training.

### **EQUIPMENT TRAINING**

B-39. Section sergeants conduct equipment training on the team's assigned equipment. Equipment training includes—

- Team collective tasks.
- Supply discipline.
- Operator maintenance.
- Load plan complete.

### **TEAM COLLECTIVE TASKS**

B-40. A collective task requires participation from each team member acting in concert to accomplish the team's assigned mission. Retransmission teams train to achieve proficiency in collective task 11-CW-7017. A complete training and evaluation outline is available on the Army Training Network Website.

### **Supply Discipline**

B-41. Team leaders must adhere to supply discipline procedures and conduct training to Soldiers. Supply discipline training includes—

- Proper use of equipment.
- Safeguarding government property.
- Inventory techniques.
- Hand receipts procedures.

### **Operator Maintenance**

B-42. Signal teams and operators should perform preventive maintenance checks and services on their assigned equipment. Based on the mission, teams and operators diagnose faults and replace parts and subcomponents from available spares. Leaders should ensure operator level preventive maintenance is conducted according to technical manual maintenance schedules.

### **Load Plan Complete**

B-43. Vehicle load plans should be based on equipment that is needed first and readily accessible. Section sergeants inspect load plans to ensure teams carry what is necessary for the mission. The standardization of load plans allows the platoon leader and platoon sergeant to quickly check accountability of equipment and standards.

## TABLE IV – TEAM CERTIFICATION

B-44. Soldiers must complete table III training before conducting table IV team certification. During table IV team certification, a sergeant first class or above certifies team proficiency in personnel and equipment tasks.

B-45. During table IV certification, teams are evaluated on the execution of collective tasks and drills. The certification begins with site occupation and consists of day and night iterations. During certification, the team must complete deployment readiness exercise level I, preventive maintenance checks and services, and submit DA Form 5988-E (*Equipment Maintenance and Inspection Worksheet [EGA]*) on the assigned system.

### PERSONNEL TASKS

B-46. Team certification improves the capabilities of Soldiers in their respective team roles. Section sergeants document Soldiers' proficiency in—

- Unit-specific requirements.
- Day and night performance of tasks and drills.
- Minimal distractions.
- Deployment readiness exercise level 1.

### Unit-Specific Requirements

B-47. Sections complete certification using training and evaluation outlines or unit SOP. A sergeant first class or above certifies teams. Examples of unit-specific training requirements include—

- Unit-specified time standards.
- Location.
- Mission oriented protective posture conditions.
- Operational environment.
- Reporting requirements.

### Day and Night Iterations

B-48. Teams perform tasks in day and night iterations according to unit-specific requirements for certification listed in table IV. Sergeants first class or above certify teams performing day and night iterations.

### Minimal Distractions

B-49. Teams train and certify on tasks with minimal distractions. Leaders safeguard their organization's approved training from external influences that may disrupt training.

### Deployment Readiness Exercise

B-50. Unit commanders conduct deployment readiness exercises. Leaders ensure movement requests and equipment documentation is complete. Deployment readiness exercise level I evaluates a unit's ability to—

- Alert and assemble the unit.
- Conduct Soldier readiness tasks.
- Verify appropriate deployment certification.
- Verify appointment orders.

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**Note.** Refer to AR 525-93 and AR 600-8-101 for details on conducting deployment readiness exercises.

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### EQUIPMENT TASKS

B-51. Certifying leaders verify the team's proficiency in equipment-related tasks. Equipment certification includes—

- Team collective tasks.

- Maintenance and service verification.
- Hand receipt verification.
- Load plan verified.

**Team Collective Tasks**

B-52. Certifying leaders verify retransmission teams are proficient in their team collective tasks and drills. Each task is evaluated based on the training and evaluation outline. Collective task proficiency is rated as—

- Trained.
- Practice.
- Untrained.

**Maintenance and Service Verification**

B-53. Unit maintenance personnel verify all annotated faults on the DA-5988-E and order any parts required to resolve faults. Leaders monitor status of ordered parts by verifying entry into the Standard Army Maintenance System-Enhanced.

**Hand Receipt Verification**

B-54. During table IV certification, hand receipt holders verify accountability for assigned equipment. Certifying leaders annotate discrepancies and notify unit supply personnel of shortages. Unit supply personnel review all sub-hand receipts for accuracy before signing.

**Load Plan Stressed**

B-55. During table IV certification, certifying leaders verify team load plans developed during table III training. Correctly developed load plans enable teams to quickly acquire needed items during certification. Figure B-1 on page 65 illustrates signal assessment tables I–IV.

|                                                                                                                                                                                                           |  | Table I<br>Individual Training                                                                                                                                                                                                                                                   | Table II<br>Individual Skills Test                                                                                                                                                        | Table III<br>Team Training                                                                                                                                                                                                    | Table IV<br>Team Certification                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                           |  | Trained by Team Chief                                                                                                                                                                                                                                                            | Certified by SSG or Above                                                                                                                                                                 | Trained by SSG or Above                                                                                                                                                                                                       | Certified by SFC or Above                                                                                                                                                                                                               |
| PERSONNEL                                                                                                                                                                                                 |  | <ul style="list-style-type: none"> <li>• Role-based cyberspace workforce training</li> <li>• Individual task training against team systems</li> <li>• Train and conduct preventive maintenance</li> <li>• Army and unit individual training requirements IAW AR 350-1</li> </ul> | <ul style="list-style-type: none"> <li>• Role-based cyberspace workforce certification</li> <li>• Demonstrate proficiency in individual tasks</li> <li>• Physical fitness test</li> </ul> | <ul style="list-style-type: none"> <li>• Team members certified in table II</li> <li>• SOP known by entire team</li> <li>• Battle drills under static conditions (day and night)</li> <li>• Training reinforcement</li> </ul> | <ul style="list-style-type: none"> <li>• Team meets drill time standards</li> <li>• Day and night iterations</li> <li>• Certified with minimal distractions to system tasks</li> <li>• Deployment readiness exercise level I</li> </ul> |
|                                                                                                                                                                                                           |  | INDIVIDUAL                                                                                                                                                                                                                                                                       |                                                                                                                                                                                           | SYSTEM                                                                                                                                                                                                                        |                                                                                                                                                                                                                                         |
| EQUIPMENT                                                                                                                                                                                                 |  | <ul style="list-style-type: none"> <li>• Supply discipline</li> <li>• Operator maintenance</li> <li>• Driver training</li> <li>• Weapons training</li> <li>• COMSEC training</li> </ul>                                                                                          | <ul style="list-style-type: none"> <li>• Weapons qualification</li> <li>• Vehicle licensing</li> <li>• Equipment accountability</li> </ul>                                                | <ul style="list-style-type: none"> <li>• Supply discipline</li> <li>• Operator/crew maintenance</li> <li>• Load plan complete</li> </ul>                                                                                      | <ul style="list-style-type: none"> <li>• Mechanic verifies maintenance</li> <li>• Hand receipt verified</li> <li>• Services verified</li> <li>• Load plan stressed</li> </ul>                                                           |
| <b>Legend:</b><br>COMSEC    communications security    SOP    standard operating procedure<br>IAW        in accordance with                      SSG    staff sergeant<br>SFC        sergeant first class |  |                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                           |                                                                                                                                                                                                                               |                                                                                                                                                                                                                                         |

Figure B-1. Signal assessment tables I–IV

## SECTION III – BATTLE DRILLS

B-56. A battle drill ensures all participants understand a technique or a specific set of procedures. Throughout unit training and preparation for missions, retransmission teams rehearse battle drills. These rehearsals do not require a complete operation order to execute. Leaders place priority on drills for actions they anticipate occurring during the operation. Refer to FM 6-0 for more information about battle drills. See section II for abbreviated drill outlines for retransmission team battle drills.

B-57. To ensure survivability, retransmission teams should be proficient in the following squad battle drills:

- Battle Drill 1: React to Direct Fire Contact while Dismounted—Squad (07-SQD-D9501).
- Battle Drill 2: Break Contact—Squad (07-SQD-D9505).
- Battle Drill 3: React to Ambush (Near)—Squad (07-SQD-D9502).
- Battle Drill 4: React to Indirect Fire while Mounted.

**Note.** The following are abbreviated drill outlines. The full battle drills with task measures and evaluation standards are available on the Army Training Network Website.

## SAFETY

B-58. In a training environment, leaders must perform a risk assessment before each training event in accordance with ATP 5-19. Leaders complete the current DD Form 2977 (*Deliberate Risk Assessment Worksheet*) during the planning and completion of each task and sub-task by assessing the risk factors and the mission variables of METT-TC (I).

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**Note.** Leaders must monitor personnel for potential heat injury during simulated chemical, biological, radiological, and nuclear conditions while wearing individual mission-oriented protective posture equipment. Local policies and procedures must be followed if there is an increased heat category to avoid heat-related injury. Leaders consider the mission-oriented protective posture work-rest cycles and water replacement guidelines in accordance with current chemical, biological, radiological, and nuclear doctrine. Refer to ATP 3-11.32 for more information on chemical, biological, radiological, or nuclear protection.

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## **BATTLE DRILL 1: REACT TO DIRECT FIRE CONTACT WHILE DISMOUNTED—SQUAD (07-SQD-D9501)**

### **Conditions**

The squad is conducting operations in a live training environment independently or as part of a platoon or larger force. The squad is dismounted. While stationary or moving, the enemy engages the squad with direct fire. Some iterations of this task should be performed in mission oriented protective posture 4.

### **Cue**

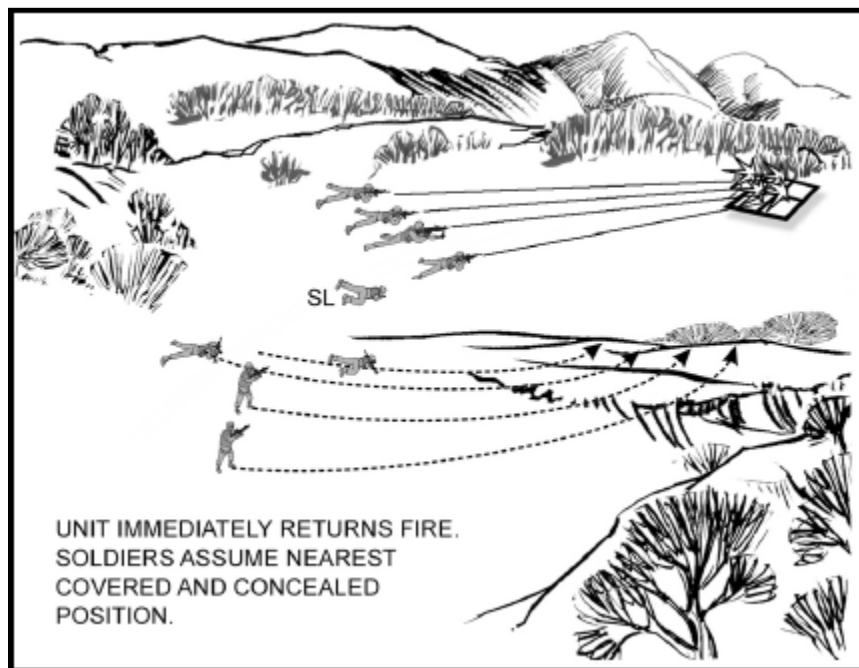
The drill begins when the enemy initiates direct fire contact.

### **Standards**

The squad reacts to direct fire contact while dismounted according to ATP 3-21.8. The team in contact returns fire immediately and seeks cover. The team in contact locates the enemy and places well-aimed fire on known enemy positions. Leaders point out enemy positions and identify the types of weapons, such as small arms and light machine guns. The team not in contact assumes the nearest covered and concealed position. The squad leader reports the contact to the platoon leader.

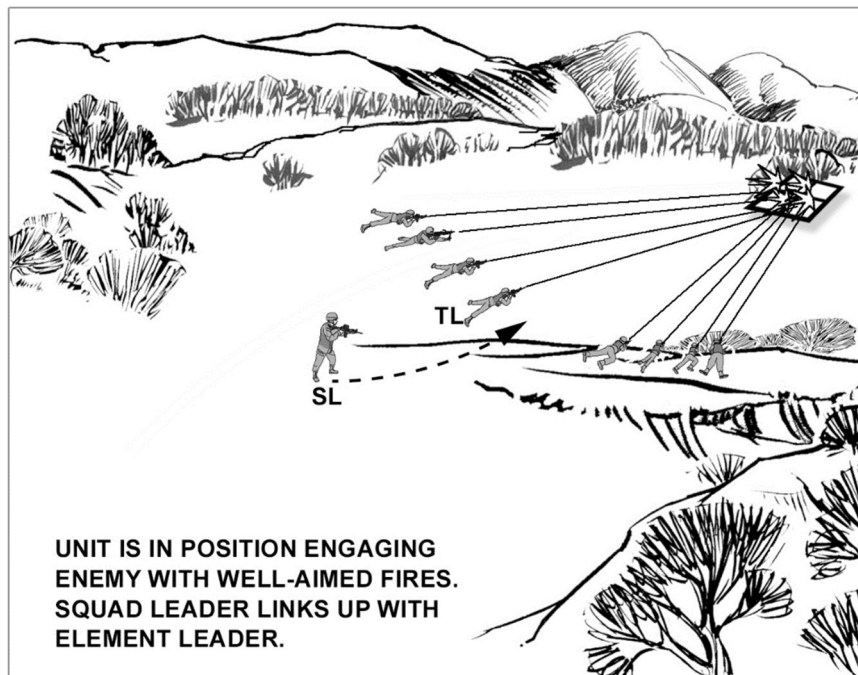
### **Task Steps and Performance Measures (Asterisks Indicate Leader Performance Steps)**

1. The team in contact immediately returns well-aimed fire on known enemy positions.
2. The team not in contact assumes the nearest covered and concealed position (see figure B-2 on page 67).



**Figure B-2. Covered and concealed position**

- \* 3. Team leaders engage known enemy positions with well-aimed fire and report information to the squad leader.
- \* 4. Team leaders control the fire of their teams by using standard fire commands (initial and supplemental) containing the following information:
  - a. Alert.
  - b. Weapon or ammunition (optional).
  - c. Target description.
  - d. Direction.
  - e. Range.
  - f. Method.
  - g. Control (optional).
  - h. Execution.
  - i. Termination.
- 5. Soldiers maintain visual or vocal contact with their team leader and the other Soldiers on their left or right (if applicable).
- 6. Soldiers maintain contact with the team leader and indicate the location of the enemy positions.
- \*7. Leaders visually or vocally check the status of their personnel.
- \*8. Team leaders maintain visual contact with the squad leader.
- 9. The squad leader moves up to the team in contact and links up with its leader (see figure B-3 on page 68).



**Figure B-3. Squad leader links up with the team leader**

- \* 10. The squad leader determines whether or not the squad can gain and maintain suppressive fires with the team already in contact (based on the volume and accuracy of enemy fires against the team in contact).
- \* 11. The squad leader confirms the commander's criteria to disengage and determines whether or not the squad must move out of the engagement area.
- \* 12. The squad leader assesses the situation and identifies—
  - a. The location of the enemy position and obstacles.
  - b. The size of the enemy force engaging the team in contact. (The number of enemy automatic weapons, the presence of any vehicles, and the employment of indirect fires are indicators of enemy strength.)
  - c. Vulnerable flanks.
  - d. Covered and concealed flanking routes to the enemy positions.
- \* 13. The squad leader decides whether to conduct an assault, bypass (if authorized by the platoon leader), or break contact.
- \* 14. The squad leader reports the situation to the platoon leader and begins to maneuver the squad.

## **Battle Drill 2: BREAK CONTACT—SQUAD (07-SQD-D9505) )**

### **Conditions**

The squad is conducting operations in a live training environment independently or as part of a platoon or larger force. The squad is moving as part of a larger force, conducting a movement to contact or an attack. Following direct fire contact with the enemy, the squad leader decides to break contact. Some iterations of this task should be performed in mission oriented protective posture 4.

### **Cue**

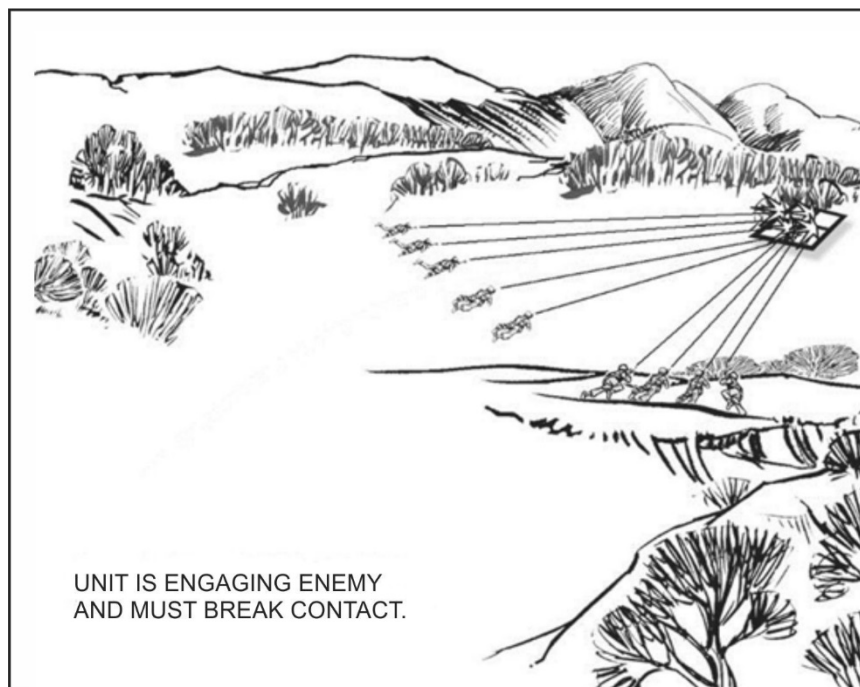
The drill begins when the squad receives direct fire contact from the enemy.

## Standards

The squad breaks contact according to ATP 3-21.8. Using fire and movement, the squad continues to move until the enemy cannot observe or place fire on them. The squad leader reports the contact to the platoon leader.

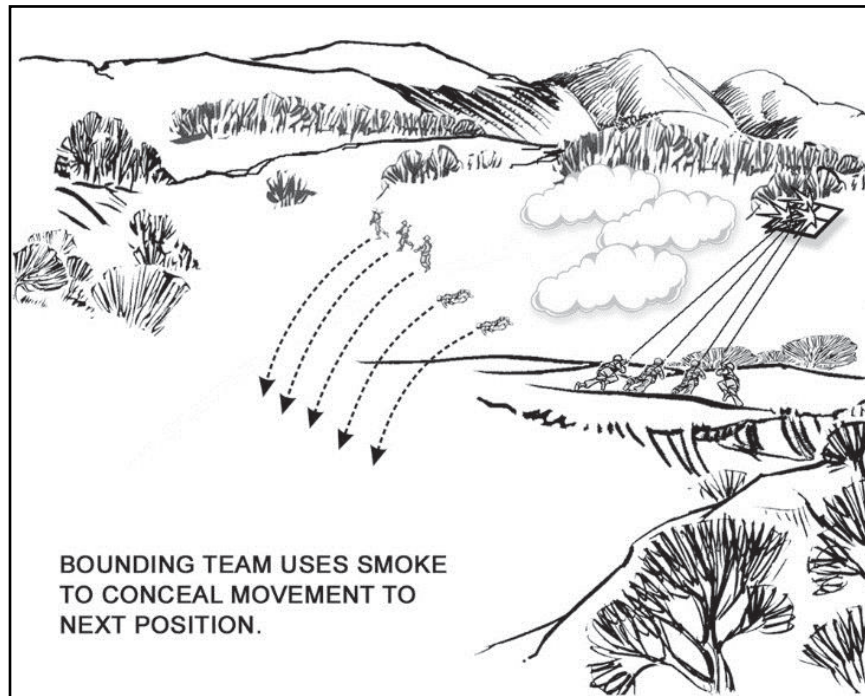
### Task Steps and Performance Measures (Asterisks Indicate Leader Performance Steps)

- \* 1. While receiving direct fire from the enemy or on orders, the squad leader directs the squad to break contact.
- \* 2. The squad leader directs one team to suppress by fire to support the disengagement of the remainder of the squad.
- \* 3. The squad leader orders a distance and direction, terrain feature, or last rally point of the movement of the team in contact.
- \* 4. The squad leader employs direct fires to suppress enemy positions (see figure B-4).



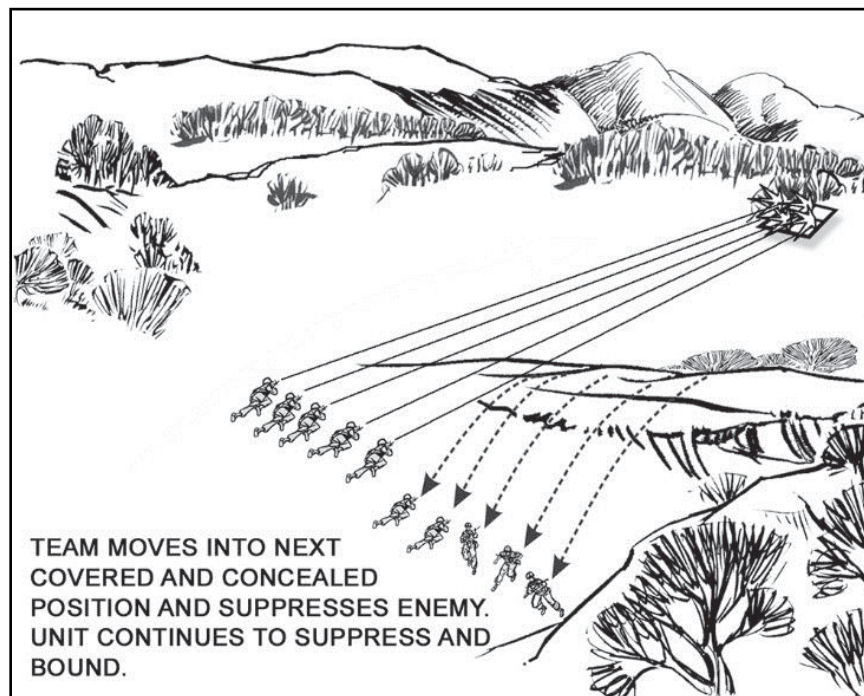
**Figure B-4. Employing direct and indirect fires to suppress enemy**

- 5. The moving team moves to occupy the overwatch position, employing smoke M203 or M320, smoke grenades, indirect fires, and other options to screen movement. If necessary, employ fragmentation and concussion grenades to facilitate breaking contact.
- 6. The base-of-fire team continues to suppress the enemy.
- 7. The moving team occupies their overwatch position and engages enemy positions (see figure B-5 on page 70).



**Figure B-5. Moving element occupies overwatch and engages enemy**

8. The squad leader directs the base-of-fire team to move to its next covered and concealed position. Based on the terrain and the volume and accuracy of the enemy's fire, the moving team may need to use fire and movement techniques (see figure B-6).



**Figure B-6. Fire and movement technique**

9. The squad continues to move away from the enemy until—

- a. It breaks contact (the squad must continue to suppress the enemy as it breaks contact).
  - b. It passes through a higher-level support-by-fire position.
  - c. Its teams are in the assigned positions to conduct the next mission.
- \* 10. The squad leader moves the squad onto an azimuth or alternate route away from enemy forces.

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*Note.* The squad leader should change the unit's direction of movement once contact is broken. This reduces the ability of the enemy to place effective indirect fire on the squad.

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11. Teams and Soldiers that become divided stay together and move to the last designated rally point.
- \* 12. Leaders account for Soldiers, report the situation to the platoon leader, reorganize as necessary, and continue the mission.

### **BATTLE DRILL 3: REACT TO AMBUSH (NEAR)—SQUAD (07-SQD-D9502) )**

#### **Conditions**

The squad is conducting operations in a live training environment independently or as part of a platoon or larger force. The squad is moving tactically either dismounted or mounted. The enemy initiates contact with direct fire within hand grenade range from covered and concealed positions. All or part of the squad is receiving accurate enemy direct fire. Indirect fire is available. Some iterations of this task should be performed in mission oriented protective posture 4.

#### **Cue**

This drill begins when the enemy initiates ambush within hand grenade range.

#### **Standards**

The squad reacts to an ambush according to ATP 3-21.8.

#### **Dismounted**

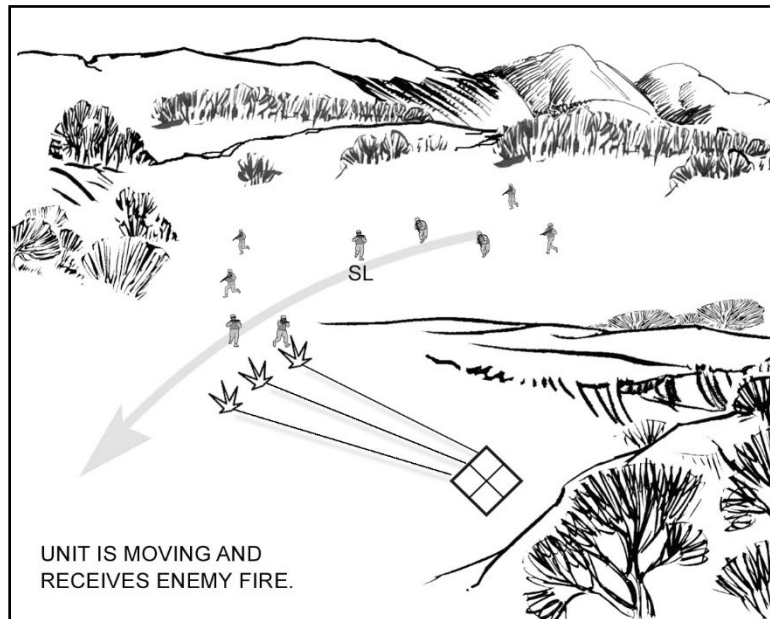
Soldiers in the kill zone immediately return fire on known or suspected enemy positions and assault through the kill zone. Soldiers not in the kill zone locate and place well-aimed suppressive fire on the enemy. The squad assaults through the kill zone and destroys the enemy.

#### **Mounted**

Vehicle gunners immediately return fire on known or suspected enemy positions as the squad continues to move out of the kill zone. Soldiers on disabled vehicles in the kill zone dismount, occupy covered positions, and engage the enemy with accurate fire. Vehicle gunners and Soldiers outside the kill zone suppress the enemy. The squad assaults through the kill zone and destroys the enemy. The squad leader reports the contact to the platoon leader.

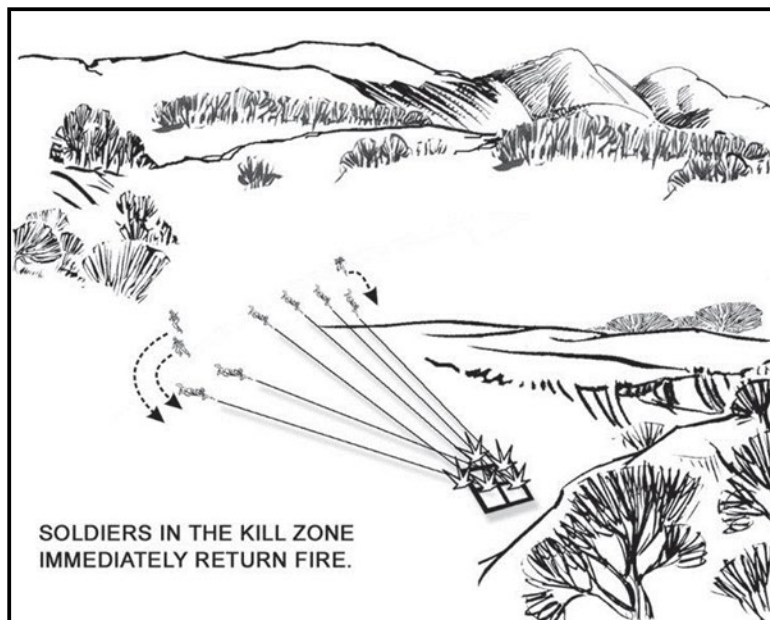
#### **Task Steps and Performance Measures (Asterisks Indicate Leader Performance Steps)**

1. The squad is moving dismounted (see figure B-7 on page 72), receives direct fire from the enemy, and takes the following actions.
  - a. Soldiers in the kill zone execute one of the following two actions:
    - (1) Return fire immediately. If cover is not available, immediately assault through the kill zone without order or signal.



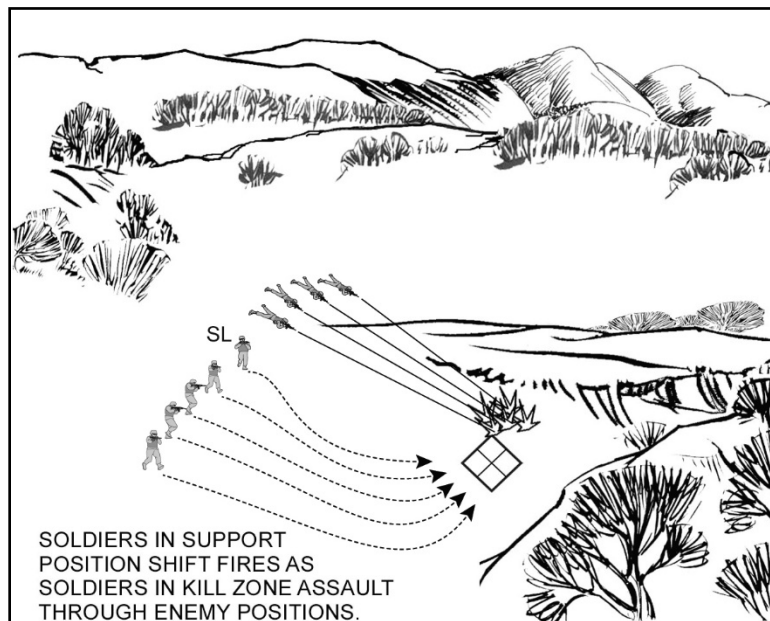
**Figure B-7. Squad receives enemy fire**

(2) Return fire immediately. If cover is available, without order or signal, occupy the nearest covered position and throw obscurant smoke grenades (see figure B-8).



**Figure B-8. Squad immediately returns fire**

- a. The Soldiers in the kill zone assault through the ambush using fire and movement.
- b. The Soldiers not in the kill zone identify the enemy location, place well-aimed suppressive fire on the enemy's position, and shift fire as Soldiers assault the objective.
- c. The Soldiers assault through and destroy the enemy position (see figure B-9 on page 73).



**Figure B-9. Squad assaults through enemy position**

- d. The squad leader reports the contact to the platoon leader.
2. The squad is moving mounted, receives direct fire from the enemy, and takes the following actions:
  - a. The vehicle gunners in the kill zone immediately return fire and deploy vehicle obscurant smoke, while moving out of the kill zone.
  - b. Soldiers in disabled vehicles in the kill zone immediately obscure themselves from the enemy with smoke, dismount if possible, seek covered positions, and return fire.
  - c. The vehicle gunners and Soldiers outside of the kill zone identify the enemy positions, place well-aimed suppressive fire on the enemy, and shift fire as Soldiers assault the objective.
  - \* d. The squad leader calls for and adjusts indirect fire and requests close air support according to the mission variables of METT-TC (I).
  - e. The Soldiers in the kill zone assault through the ambush and destroy the enemy.
  - \* f. The squad leader reports the contact to the platoon leader.

#### **BATTLE DRILL 4: REACT TO INDIRECT FIRE WHILE DISMOUNTED—SQUAD (07-03-D9504)**

##### **Conditions**

The squad is conducting operations in a live training environment independently or as part of a platoon or larger force. The squad is dismounted. While stationary or moving, a member of the squad alerts INCOMING, or a round impacts nearby. Some iterations of this task should be performed in mission oriented protective posture 4.

##### **Cue**

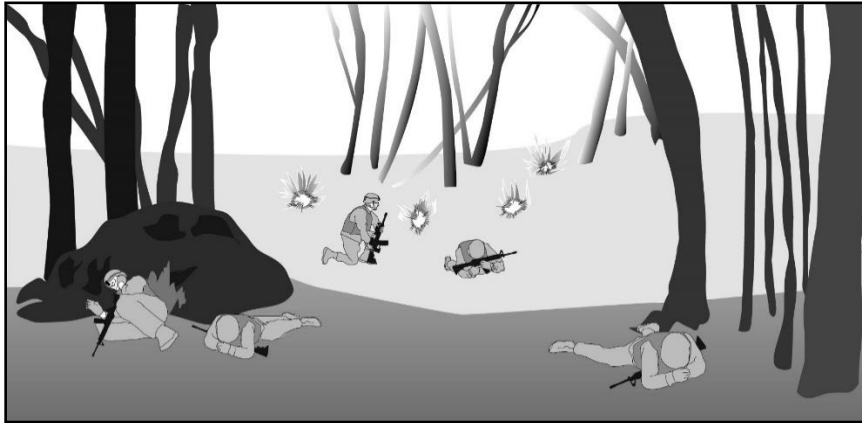
The drill begins when any squad member alerts INCOMING, or a round impacts nearby.

##### **Standards**

The squad reacts to indirect fire while dismounted according to ATP 3-21.8. Soldiers immediately seek the best available cover. The squad moves out of the area to the designated rally point after the impacts. The squad leader reports the contact to the platoon leader.

### **Task Steps and Performance Measures**

1. A member of the squad hears or observes artillery impacting near the squad and alerts the squad with the announcement INCOMING!
2. Soldiers immediately assume the prone position or immediately move to available cover during initial impacts (see figure B-10).



**Figure B-10. Soldiers reacting to indirect fire**

- \*3. The squad leader orders the squad to move to a rally point by giving a direction and distance.
4. Soldiers move rapidly in the given direction and distance to the designated rally point and reestablish security after the impacts.
- \*5. Leaders regain accountability of their Soldiers, weapons, and equipment at the rally point and treat casualties as required.
- \*6. The squad leader reports the action to the platoon leader.

## **Appendix C**

# **Combatant Commander Communications Team**

This appendix discusses combatant commander communications team personnel supporting the combatant commander and theater army commander. It describes the team's role and responsibilities to provide secure, reliable communications support.

## **MISSION**

C-1. The combatant commander communications team provides communications support using secure frequency modulation radios and narrowband (single-channel) TACSAT to combatant commanders and theater army commanders.

## **ROLES AND RESPONSIBILITIES**

C-2. The combatant commander communications team is comprised of—

- Network management technician.
- Information systems chief.
- Senior combatant commander information technology noncommissioned officer.
- Combatant commander information technology noncommissioned officer.
- Senior combatant commander communications noncommissioned officer.
- Combatant commander communications noncommissioned officer.
- Transmission systems operations noncommissioned officer.

## **NETWORK MANAGEMENT TECHNICIAN**

C-3. The network management technician's responsibilities are—

- Supervises and manages the tactical networking environment and administers the local area network and radio system for the combatant commander or theater army commander's location.
- Plans, administers, manages, maintains, operates, integrates, secures, and troubleshoots commercial off-the-shelf communications and automated information systems and radio systems.
- Leads the team and manages the training of personnel on the installation, administration, management, maintenance, operation, integration, securing, and troubleshooting of commercial off-the-shelf, intranets, radio systems, and video teleconferencing systems.
- Performs system integration and administration.
- Implements the cybersecurity program to protect and defend information, computers, and networks from disruption, denial of service, degradation, or destruction.
- Develops policy recommendations and advises commanders and staffs on planning, installing, administering, managing, maintaining, operating, integrating, and securing commercial off-the-shelf communications, radio systems, and video teleconferencing systems on Army, joint, combined, and multinational networks.

## **INFORMATION SYSTEMS CHIEF**

C-4. The information systems chief's responsibilities are—

- Assumes the duties of the network management technician, when required.
- Serves as the principal information systems noncommissioned officer for the combatant commander communications team.
- Supervises, plans, coordinates, and directs the employment, operation, management, and unit-level maintenance of multi-functional and multi-user information processing systems in mobile and fixed facilities.
- Provides technical and tactical advice to the command and staff concerning all aspects of information processing system operations, maintenance, and logistical support.

- Supervises the installation, operation, configuration, preventive maintenance checks and services, and software upgrades on COMSEC devices.
- Conducts briefings on the status, relationship, and interface of information processing systems in an area of interest.
- Supervises or prepares technical studies, evaluations, reports, correspondence, and records of multi-functional and multi-user information processing systems.
- Plans, organizes, and conducts technical inspections.
- Supervises development of the information systems plan, information management plan, and the information management master plan.
- Reviews, consolidates, and forwards final written input for the continuity of operations plan.
- Develops and enforces policy and procedure for facility operations security and physical security in accordance with regulations and policies.
- Prepares or supervises the preparation of technical studies, evaluations, reports, correspondence, software programs, program editing, debugging and associated functions.
- Maintains records pertaining to information system operations.

### **SENIOR COMBATANT COMMANDER INFORMATION TECHNOLOGY NONCOMMISSIONED OFFICER**

C-5. The senior combatant commander information technology noncommissioned officer's responsibilities are—

- Plans, supervises, coordinates, and provides technical assistance for the installation, operation, systems analyst functions, unit-level maintenance, and management of combatant commander communications team communications and information processing systems to support the combatant commander or theater army commander.
- Installs, operates, and maintains communications and automated information systems to support the combatant commander or theater army commander.
- Supervises and implements classified document control policies, procedures, standards, and inspections.
- Develops, directs, and supervises training programs to ensure Soldier proficiency and career development.
- Organizes work schedules and ensures compliance with directives and policies on operations security, signal security, COMSEC and physical security.
- Prepares or supervises the preparation of technical studies, evaluations, reports, correspondence, and records on information system operations.
- Briefs staff and operations personnel on matters regarding the team's communications and information systems.

### **COMBATANT COMMANDER INFORMATION TECHNOLOGY NONCOMMISSIONED OFFICER**

C-6. The combatant commander information technology noncommissioned officer's responsibilities are—

- Installs, operates, and maintains standard Army and commercial off-the-shelf communications and automated information systems equipment to support the combatant commander or theater army commander.
- Compiles reports to support information systems operations.
- Performs system studies using established techniques to develop new or revised system applications and programs.
- Analyzes telecommunications and information management requirements and requests logistical support and coordinates system integration.
- Ensures required spare parts, supplies, and operating essentials are maintained.
- Performs maintenance management and administrative duties related to facility operations, maintenance, security, and personnel.
- Requisitions, receives, stores, issues, destroys, and accounts for COMSEC equipment and keying material on a hand receipt from the property book officer or COMSEC account manager, including the over-the-air key.

**SENIOR COMBATANT COMMANDER COMMUNICATIONS NONCOMMISSIONED OFFICER**

- C-7. The senior combatant commander communications noncommissioned officer's responsibilities are—
- Assumes supervisory responsibility for Soldiers on the combatant commander communications team.
  - Supervises, plans, and executes the installation, operation, and maintenance of signal support systems, including local area networks, wide-area networks, and routers; satellite radio communications systems; and network integration using radio, wire, and commercial off-the-shelf automated information systems.
  - Installs, operates, and maintains the standard Army and commercial off-the-shelf communications and automated information systems to support the combatant command or theater army commander.
  - Develops and implements signal unit-level maintenance programs.
  - Supervises signal training and provides technical advice and assistance to the combatant command or theater army commander communications team chief.
  - Coordinates external signal support.
  - Plans and requests logistics support for unit-level operations and maintenance.

**COMBATANT COMMANDER COMMUNICATIONS NONCOMMISSIONED OFFICER**

- C-8. The combatant commander communications noncommissioned officer's responsibilities are—
- Assumes responsibility for installing, operating, and maintaining a standard for the Army and commercial off-the-shelf communications and automated information systems to support the combatant command or theater army commander.
  - Provides technical assistance and unit-level training for automation, communication, and user owned and operated automated telecommunications computer systems.
  - Provides local area networks and routers, signal communications, electronic equipment, and satellite radio communications equipment.
  - Prepares maintenance and supply requests for unit-level signal support.

**TRANSMISSION SYSTEMS OPERATIONS NONCOMMISSIONED OFFICER**

- C-9. The transmission system operations noncommissioned officer's responsibilities are—
- Plans, installs, operates, and maintains communications and automated information systems to support the combatant command or theater army commander.
  - Provides the combatant commander communications team chief assistance and advises on communications systems planning, satellite access requests, propagation, spectrum management, and maintenance management for the standard Army and commercial off-the-shelf communications and automated information systems used to support the combatant command or theater army commander.

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## Appendix D

# Waveforms and Waveform Applications

This appendix discusses the waveforms and waveform applications used in tactical radio networks. Tactical radios provide the means to pass voice and data across the assigned area.

### WAVEFORM OVERVIEW

D-1. A waveform is the representation of a signal that consists of the frequency, modulation type, message format, and transmission system. The term waveform refers to a known set of characteristics, for example, frequency bands, modulation techniques, message standards, and transmission systems. In tactical radio system usage, the term waveform describes the entire set of radio functions that occur from the user input to the RF output.

D-2. Waveforms of the tactical networking environment process voice and essential data elements. Data elements within the tactical networking environment are centered on friendly forces, situational awareness, and transport capabilities to enable communications in disconnected, intermittent, and limited bandwidth environments. The spectrum and bandwidth availability are limited, and only critical functions operate within this portion of the tactical networking environment.

### Wave Modulation

D-3. Frequency modulation and amplitude modulation (AM) transmitters produce RF carriers. The carrier is a wave of constant amplitude, frequency, and phase that modulates by changing its amplitude, frequency, or phase. Modulation is superimposing information, voice, or coded signals on a carrier. Figure D-1 shows different wave shapes.

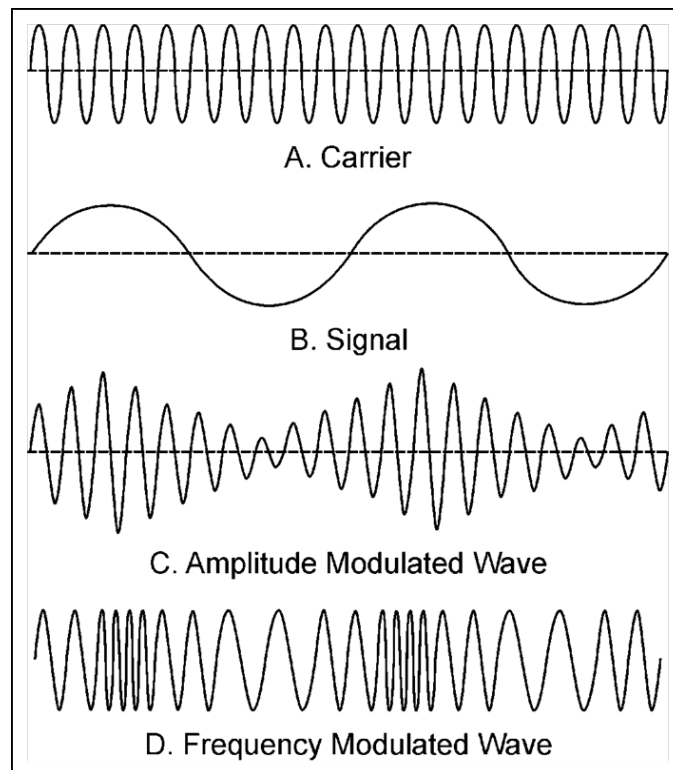
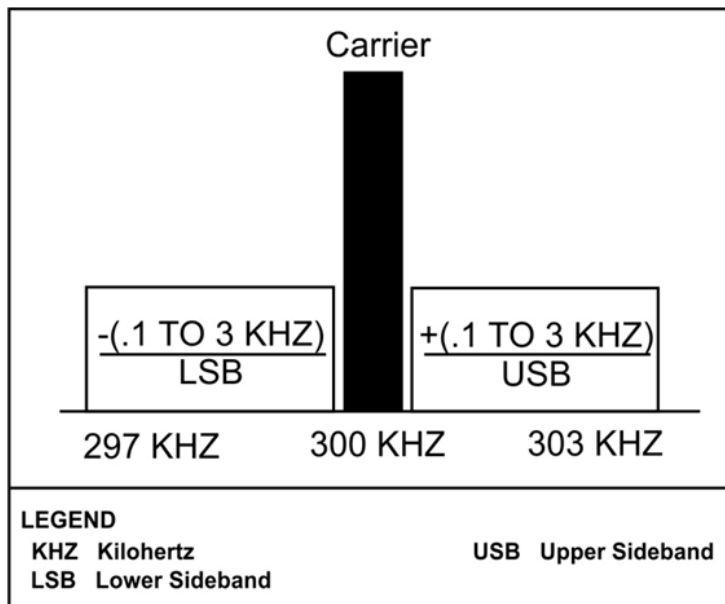


Figure D-1. Wave shapes

## Amplitude Modulation

D-4. AM is the variation of the RF power output of a transmitter at an audio rate. The RF energy increases and decreases in power, according to the audio frequencies superimposed on the carrier signal.

D-5. When the RF carrier modulates by complex tones, such as speech, each separate frequency component of the modulating signal produces its own upper and lower sideband frequencies. The upper sideband contains the sum of the RF and audio frequency signals, and the lower sideband includes the difference between the RF and frequency signals. Figure D-2 shows an AM carrier.



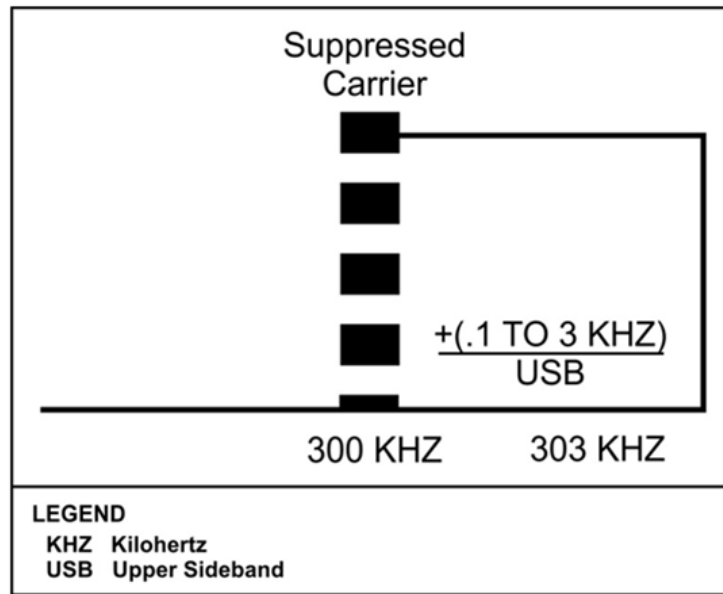
**Figure D-2. Amplitude modulation carrier**

D-6. The space occupied by a carrier and its associated sidebands in the RF spectrum is a channel. In AM, the width of the channel (bandwidth) is equal to twice the highest modulating frequency. For example, if a band of frequencies modulates a 5,000 kHz (5 MHz) carrier ranging from 0.2–5 kHz (200–5,000 cycles), the upper sideband extends from 5000.2–5005 kHz. The lower sideband extends from 4,999.8–4,995 kHz. The bandwidth is the difference between 5,005 Hz–4,995 kHz, a total of 10 kHz.

D-7. Radio transmitters operating in the medium and HF bands generally use AM. The information of an AM signal exists solely in the sidebands.

## Single-Sideband

D-8. Each sideband contains all the information needed for communication. Both sidebands occur within the modulation circuitry of the single-sideband radio set; the transmitter suppresses the carrier on one sideband before any signal transmits. Figure D-3 on page 81 shows a single-sideband carrier.



**Figure D-3. Single-sideband carrier**

D-9. The upper sideband is higher in frequency than the carrier, and the lower sideband is lower in frequency. Planners use either an upper or a lower sideband for communication and make appropriate adjustment to the transmitter and the receiver to the same sideband.

D-10. Most Army single-sideband equipment operates in the upper sideband mode. The transmission of only one sideband leaves open that portion of the RF spectrum normally occupied by the other sideband of an AM signal. This allows more RF carriers within a given frequency range. Single-sideband transmission is used in applications to—

- Obtain greater reliability.
- Limit size and weight of equipment.
- Increase effective output without increasing antenna voltage.
- Operate a large number of radio sets without heterodyne interference (whistles and squeals) from RF carriers.
- Operate over longer ranges without signal degradation due to selective fading.

## WAVEFORM APPLICATIONS

D-11. The Network Operations Management System provides a mechanism to plan, configure, and load waveform applications onto the radio platforms. The waveform application is composed of all current and future software-defined waveform applications that provide voice and data across the transport layer of the network. Waveform applications are peer-to-peer programs that facilitate the exchange of application data across the spectrum of radio networks. Knowledge of waveform applications and characteristics is essential for network planners as they develop their network architecture to meet their commander's command and control requirements.

### Mobile User Objective System Waveform

D-12. The MUOS is a UHF SATCOM waveform that enables satellite network connectivity and communications services for mounted and dismounted units to support beyond line of sight communications capability at brigade and below.

D-13. MUOS is primarily intended for mobile user's aerial, maritime, and ground vehicles, and dismounted soldiers. MUOS extends voice and data communications beyond line of sight. The MUOS waveform operates

as a global cellular service provider to support Soldiers with modern cell phone-like capabilities. It adopts a commercial third generation wideband code division multiple access mobile phone architecture for use in the military UHF SATCOM frequency spectrum using four geosynchronous satellites in place of cell towers covering the globe. By operating in the UHF band, 300–320 MHz, to transmit user to base and 360–380 MHz to receive base to user, a lower frequency band is required than that which is used by conventional terrestrial cellular networks. MUOS provides Soldiers with the tactical ability to communicate in disadvantaged environments, such as heavily forested regions where the forest canopy would unacceptably attenuate higher frequency signals. The radio uses the MUOS waveform when operating on one of four 5 MHz channels on one of sixteen beams on one of four geosynchronous satellites.

D-14. MUOS waveform provides military point-to-point and communication users with precedence-based and pre-emptive access to voice and data up to 64 kilobits per second that span the globe. Each MUOS terminal has a 10-digit Defense Switched Network assigned phone number when provisioned. Users in the field establish connections on demand within seconds, freeing resources for other users. The multichannel, multifunction tactical radios that support the MUOS waveform technology are the AN/PRC-158 and AN/PRC-162.

### **Integrated Waveform**

D-15. SATCOM integrated waveform is an enhanced method of multiplexing radio networks on the same channel. An integrated waveform uses carrier phase modulation to allow more access using the same channel. Carrier phase modulation implemented in radios provides higher data throughput on the UHF dedicated satellite channels in line of sight mode. An integrated waveform is an augmentation to time division multiple access services; its flexible waveform structure allows communication access tailoring based upon operational requirements. Narrowband (single-channel) TACSAT-capable radios that support integrated waveform technology are the AN/PSC-5C/D, AN/PRC-117F/G, AN/PRC-148, AN/PRC-158, AN/PRC-162, and AN/PRC-163. Integrated waveforms—

- Support data rates up to 19.2 kilobits per second.
- Provide up to 14 networks operating at 2400 bits per second each.
- Support narrowband voice operations with mixed excitation linear prediction.

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**Note.** The Army plans to divest its inventory of the AN/PSC family of radios(replacing them with the AN/PRC-158 or AN/PRC-162) and the AN/PRC-117F/G radios by fiscal year 28 to support cryptographic modernization.

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### **SATURN**

D-16. The Second Generation Anti-jam Tactical Ultrahigh Frequency Radio for North Atlantic Treaty Organization (SATURN) is an Air Force sponsored waveform with fast frequency hopping capabilities that replaces the legacy Havequick waveforms. SATURN is the current solution to counter threats from precision signal intercepting capabilities and jamming efforts. SATURN is designed for military airborne operations as defined by STANAG 4372. Although there are similarities between the legacy Havequick waveform, significant differences exists between the performance of the digital waveform and the Havequick analog waveform. SATURN's frequency hopping capabilities outperform Havequick waveforms allowing users to define up to 40 frequencies as opposed to 16 frequencies allowed by Havequick. Additional capabilities are—

- Supports improved data rates, synchronous and asynchronous.
- Allows automatic transmission time updates and synchronization.
- Provides cease reception and cease transmission capabilities (receiver and transmitter break-in).
- Supports contested environments including air-to-ground, air-to-air, command and control voice line-of-sight communications.

## **Appendix E**

# **High Frequency Radio Networks**

This appendix describes HF radio networks and capabilities. It also discusses establishing HF networks at echelons corps through brigade. This appendix concludes with factors to consider for successful HF network operations.

## **OVERVIEW**

E-1. Units use HF networks when unit dispersal exceeds the planning range for VHF radios. Commanders routinely establish a HF command network as a secondary means of controlling operations. The primary advantage of HF radio is its capability to provide long-range, over-the-horizon communication. Additionally, HF radio networks—

- Provide enhanced voice and data connectivity using ALE.
- Extend beyond line of sight and point-to-point communications.
- Provide lower probability of transmission interception.
- Provide secure and reliable data and voice encryption.
- Allow synchronous collaboration using tactical chat applications.
- Maximize secure voice availability with mixed excitation linear prediction.
- Supplement VHF systems.

E-2. HF radios' longer range overcomes terrain limitations and is useful where line of sight cannot be achieved. HF is the only tactical communications asset that may achieve long-range communications independent of terrestrial or satellite relays. HF communications may be either voice or secure data.

## **ESTABLISHING HIGH FREQUENCY RADIO NETWORK COMMUNICATION**

E-3. HF radio networks provide commanders reliable, secure, fast, and flexible communications exchange with higher and subordinate echelons. Units from battalion to theater establish HF radios and radio networks to enable communication during offensive or defensive operations.

E-4. All echelons employ HF networks to support the command network. Each echelon typically establishes sustainment networks and operations and intelligence command networks to support fires and reconnaissance operations. Radio stations are divided into nets according to the function that each station serves within the command. The station in charge of maintaining the net control station is usually a station in a superior unit communicating with stations in one or more subordinate units.

## **HIGH FREQUENCY RADIO NETWORK**

E-5. HF communications networks are easily established, reconfigured, and maintained for Soldiers requiring long-distance, wide-area, fixed or on-the-move, ground, and ground-to-air communications. HF communications networks are established at large command post that lack the mobility required for tactical enabling tasks. The corps, division, and brigade employ HF radios when there are terrain and distance challenges that disrupt or degrade single-channel VHF communications. HF radio, terrestrial beyond line of sight systems require an understanding of antenna design and awareness of the capabilities needed to support requirements. When determining site locations, planners work with G-3 or S-3 to minimize adverse effects that terrain and structures have on a unit's communications capabilities.

## **Corps**

E-6. The G-6 coordinates with higher, adjacent, and subordinate echelons to ensure adequate network support. The corps G-6 and the corps signal, intelligence, and sustainment company provide HF radio support for the corps headquarters and the various companies in the corps headquarters. These systems are mostly

user-owned and operated systems with the corps signal, intelligence, and sustainment company responsible for network control.

### **Division**

E-7. The division G-6 and the division signal, intelligence, and sustainment company provide HF radio support for the division command posts. The division G-6 and the division signal, intelligence, and sustainment company provide HF radio support for the division headquarters and the various companies in the division headquarters.

### **Brigade Combat Team**

E-8. The brigade signal company provides HF radio support to the brigade command posts. The brigade's organic signal assets support continuous operations to sustain the brigade communications network. The AN/PRC-150 and AN/PRC-160 are both manpack radios that can be integrated into a mount for operation within a tactical operations center, or vehicle mounted.

#### ***Brigade Combat Team S-6***

E-9. The brigade employs traditional HF networks to support command networks. Brigade combat teams typically establish command, administrative and logistics, and operations and intelligence networks to support fires and reconnaissance operations. A typical brigade has enough HF radios to establish command networks down to the company and lower levels as needed. The brigade S-6 collaborates with the higher headquarters J-6 or G-6, assigned or attached battalion S-6 staffs, and adjacent units to ensure effective communications throughout the brigade assigned area.

#### ***Brigade Signal Company***

E-10. HF radios enable brigade elements to coordinate operations with higher, lower, and adjacent headquarters throughout their assigned area. The brigade signal company extends voice and data retransmission capabilities for its tactical radio network.

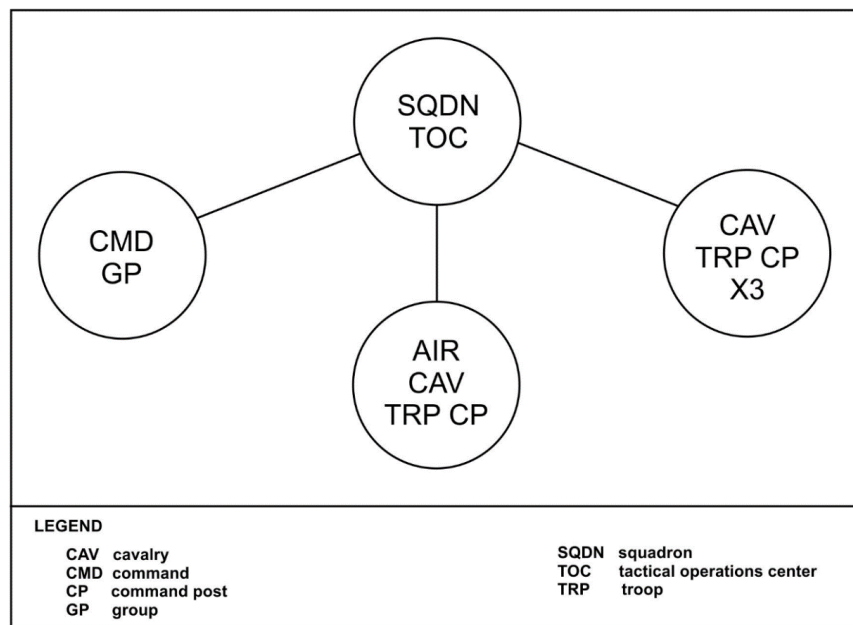
E-11. The brigade signal company uses HF radios as a redundant means of communications when terrain or distance challenge line of sight radio capabilities. The brigade signal company supports both the brigade and subordinate battalion command posts.

#### ***Maneuver Battalion S-6***

E-12. The maneuver battalion S-6 employs traditional HF networks to support command networks. The maneuver battalion S-6 typically establishes command, administrative and logistics, and operations and intelligence networks to support fires and reconnaissance operations. A typical maneuver battalion has enough HF radios to establish command networks down to the company as needed. The maneuver battalion collaborates with the higher headquarters and adjacent units to ensure effective communications throughout the assigned area.

### **DATA NETWORK**

E-13. Combat aviation brigades and air cavalry units use HF data networks to provide long-range, beyond line of sight communications. Cavalry squadrons and troops use the low power HF data network for their command networks when the distance is not an issue. The same is true for divisional and regimental cavalry. Figure E-1 on page 85 shows a typical cavalry squadron HF data network.



**Figure E-1. Example of a cavalry squadron high frequency data network**

## HIGH FREQUENCY RADIO CONSIDERATIONS

E-14. HF radio equipment is capable of both long- and short-range secure voice and data communications. Data communications capability is typically limited to rates of 2.4 kilobits per second. Data transmission requires modems specifically designed for operation in the HF range.

E-15. Reliable HF communication requires a detailed analysis of the terrain and availability of appropriate frequencies. HF performance factors include—

- The sun's position in relation to transmitter and receiver stations.
  - Lower frequencies are optimal during times of darkness.
  - Higher frequencies are optimal during times of sunlight.
- Weather seasons.
  - Lower frequencies are optimal during winter weather conditions.
  - Higher frequencies are optimal during summer weather conditions.
- Solar activity.
  - HF band propagation is optimal during the solar maximum (maximum sunspot activity period).
  - HF band propagation is extremely limited during the solar minimum (minimum sunspot activity period).
- Local electromagnetic interference from man-made equipment, which can greatly reduce the available frequencies in the HF band.

E-16. Successful data communications over the HF range depends on several factors including equipment position, proper equipment grounding, and types of antennas used. Other considerations are the tactical employment of the radio equipment, path assessment and analysis, and frequency planning and assignment. When commercial data terminal equipment is used, signal teams and operators employing HF radios need to be aware of radio interference and potential shock hazards that can easily affect unprotected data terminal equipment. Whenever possible, HF radios should be remoted from data terminal equipment.

E-17. The primary limiting factors when using HF radios are frequency allocation and management and bandwidth availability. Frequency allocation and management is concerned with frequency, time of day, time of year, and location. The ability to reflect HF radio waves off the ionosphere to a distant location is in a constant state of flux because of ionospheric activity. Therefore, planners consider environmental limitations associated with HF radios including characteristics of ground wave and skywave propagation.

E-18. The maximum ground wave range for most tactical HF communications is about 20–30 kilometers (12–22 miles) for manpack equipment and 80–100 kilometers (50–62 miles) for high-power vehicular and van equipment. The range may be decreased by heavy vegetation, mountainous terrain, or dry desert soil. Beyond this range, it is necessary to communicate by skywave.

E-19. Skywave frequencies are a very important consideration, as those above a certain value will not bend back to earth but will transmit through the ionosphere into outer space. On the other hand, lower frequencies are noisier and become absorbed by the ionosphere. The reflective nature of the ionosphere will change when sunlight hits it each day. As a result, at least two frequencies are usually required during a 24-hour period: a lower nighttime frequency and a higher daytime frequency.

E-20. Another consideration in using HF communications is how to compensate for skip zones, a phenomenon where no signal is received from a particular transmitter for a particular frequency. Skip zones are formed when the nearest point at which a skywave is received is beyond the furthest point at which a ground wave is received. Multipath interference occurs when both the skywave and the ground wave signals from the transmitter arrive at different times at the receiver. By using an antenna with a high radiation take-off angle, HF radio waves can be bounced off the ionosphere and come back to earth closer than they can with more commonly used antennas. This can cause the skip zone to disappear if the waves do not penetrate the ionosphere.

E-21. The use of high radiation take-off angle or near-vertical incident skywave communications using a horizontally polarized antenna is useful for HF radio transmission since signals can be transmitted from moving vehicles. Near-vertical incident skywave communications require a horizontally polarized antenna and are done over frequencies from 2–12 MHz. The limit of the effective range of near-vertical incident skywave communications is usually about 300 miles. With proper application, near-vertical incident skywave communication provides reliable, continuous communications beyond the range of HF ground wave and VHF and UHF line of sight.

E-22. The site selection for the antenna is another important consideration. Usually, the tactical situation determines the position of the antenna. The ideal setting is a clear, flat area with no trees, fences, power lines, or mountains. Situations often require constructing an antenna on an irregular or less than optimal site. This does not mean the antenna will not work. It means that the site affects the antenna's pattern and function.

## Antenna Types

E-23. HF antennas vary in size, shape, and capability. HF antennas are generally longer and require more clearance space to assemble and establish radio communications. Before selecting an antenna, operators should examine the available construction materials. For instance, horizontal dipoles require at least two supports to erect. A third (center) support is necessary for frequencies 5 MHz or lower. Operators should not construct a dipole antenna without the necessary support structure. If the support structures are unavailable, the operator should select another antenna type. Planners and operators examine the proposed antenna site to determine whether the antenna can fulfill the mission requirements.

E-24. Radio waves in the 3–30 MHz frequency range can reflect and return to Earth by the ionosphere with predictable regularity. To optimize the probability of a successful skywave communications link, operators select the frequency, and take-off angle most appropriate for the time of day transmission is to take place. Various large conducting objects, and the Earth's surface, modify an antenna's radiation pattern. Sometimes nearby scattering objects may modify the antenna's pattern favorable by concentrating more power toward the receiving antenna.

E-25. Selecting the right antenna for a HF radio circuit is very important. When selecting a HF antenna, planners consider the type of propagation desired. Ground wave propagation requires low take-off angle and vertically polarized antennas. The whip antenna included with most radio sets provides good omnidirectional ground wave radiation. Selecting an antenna for skywave propagation is very complex. Listed below are considerations for selecting antennas for skywave propagation—

- **Determining the circuit range distance required for the take-off angle.** A circuit distance of 966 kilometers (600 miles) requires a take-off angle of approximately 25 degrees during the day and 40 degrees at night.

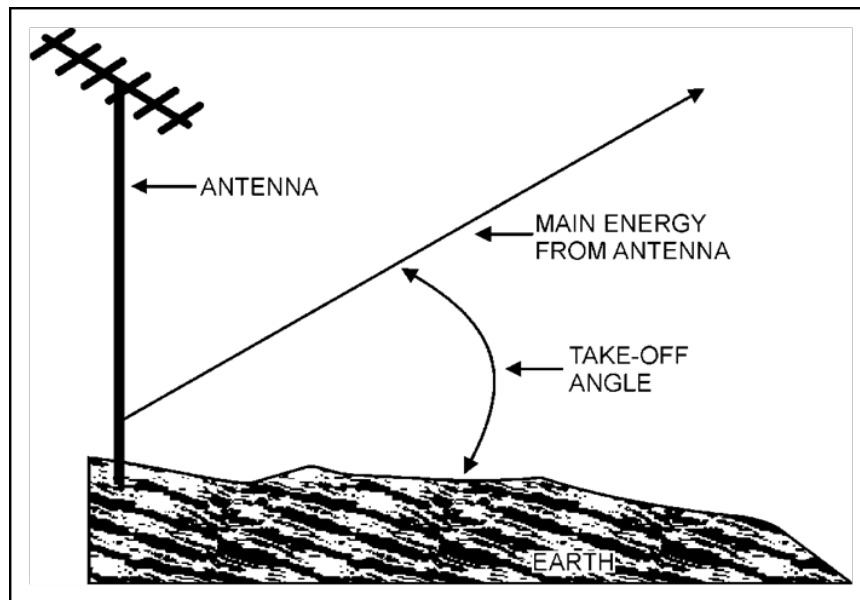
- **Selecting a high gain antenna.** The optimal range for propagation predictions is 25–40 degrees. This prediction indicates the take-off angles required.
- **Determining the required coverage.** A point-to-point circuit uses either a bidirectional or directional antenna. The receiving station location dictates the antenna choice. See table E-1 for the take-off angles versus distance.

*Note.* A radio circuit with mobile vehicle stations (or several stations located away from the transmitter) requires an omnidirectional antenna.

**Table E-1. Antenna take-off angle versus distance**

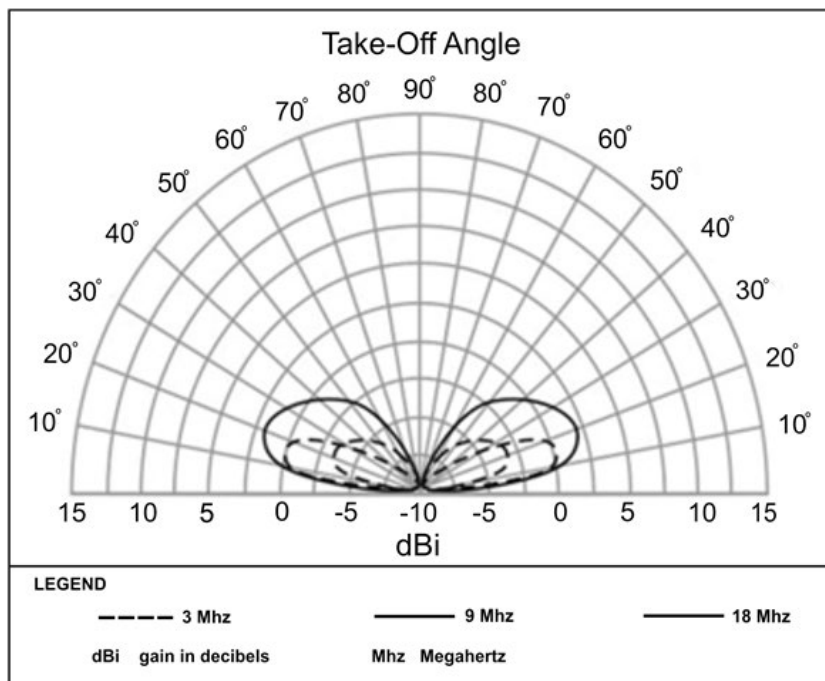
| <i>Take off Angle (Degrees)</i> | <i>Distance</i>          |              |                            |              |
|---------------------------------|--------------------------|--------------|----------------------------|--------------|
|                                 | <i>F2 Region Daytime</i> |              | <i>F2 Region Nighttime</i> |              |
|                                 | <i>Kilometers</i>        | <i>Miles</i> | <i>Kilometers</i>          | <i>Miles</i> |
| 0                               | 3,220                    | 2,000        | 4,508                      | 2,800        |
| 5                               | 2,415                    | 1,500        | 3,703                      | 2,300        |
| 10                              | 1,932                    | 1,200        | 2,898                      | 1,800        |
| 15                              | 1,450                    | 900          | 2,254                      | 1,400        |
| 20                              | 1,127                    | 700          | 1,771                      | 1,100        |
| 25                              | 966                      | 600          | 1,610                      | 1,000        |
| 30                              | 725                      | 450          | 1,328                      | 825          |
| 35                              | 644                      | 400          | 1,127                      | 700          |
| 40                              | 564                      | 350          | 966                        | 600          |
| 45                              | 443                      | 275          | 805                        | 500          |
| 50                              | 403                      | 250          | 685                        | 425          |
| 60                              | 258                      | 160          | 443                        | 275          |
| 70                              | 153                      | 95           | 290                        | 180          |
| 80                              | 80                       | 50           | 145                        | 90           |
| 90                              | 0                        | 0            | 0                          | 0            |

E-26. The antenna's take-off angle is the angle above the horizon that an antenna radiates the largest amount of energy. The take-off angle of an HF communications antenna can determine whether a circuit is thriving or not. HF skywave antennas are designed to support exact take-off angles, depending on the circuit distance. Operators use high take-off angles for short-range communications and use low take-off angles for long-range communications. Figure E-2 on page 88 depicts an example of an antenna take-off angle.



**Figure E-2. Antenna take-off angle**

E-27. Figure E-3 shows the vertical antenna pattern for the 32-foot vertical whip antenna. The numbers along the outer ring 90, 80, and 70 degrees represent the angle above the Earth—90 degrees would be straight up, and 0 degrees would be parallel to the ground. Along the bottom of the pattern are numbers from -10 (at the center) to +15 at the edges. These numbers represent decibels over a theoretical isotropic radiator.



**Figure E-3. 32-foot vertical whip antenna pattern**

E-28. Using the example at figure E-3, signal leaders can determine the antenna gain at a particular frequency and take-off angle and locate the desired take-off angle on the plot. The plotted center line represents the pattern of the desired frequency. The gain appears at the bottom scale. To find the gain of a 32-foot vertical

whip at 9 MHz with a 20-degree take-off angle, locate 20 degrees along the outer scale and follow this line to the 9 MHz pattern line. The intersecting line represents the 9 MHz pattern line. The resulting antenna gain in the example appears where the intersecting vertical and horizontal lines meet at the bottom scale or slightly less than 2.5 decibels isotropic. The gain of the 32-foot vertical whip at 9 MHz and 20 degrees is 2 decibels above effective isotropic power.

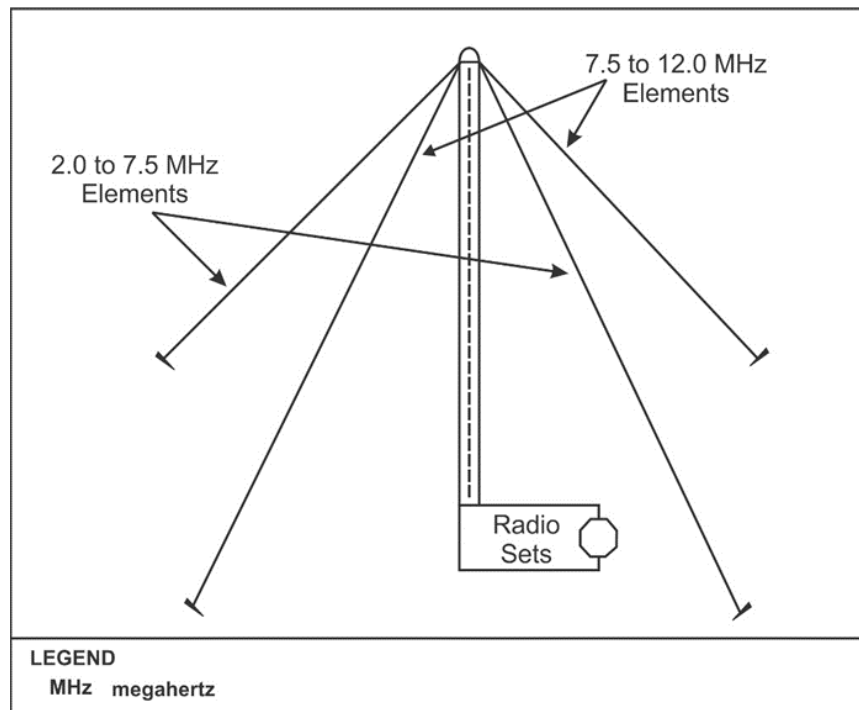
## Antenna Selection

E-29. Table E-2 shows a HF antenna selection matrix planners can use to determine the appropriate antenna for a circuit. If the proposed circuit requires a short-range, omnidirectional, wideband antenna, the HF antenna selection matrix shows the only antenna that meets all the criteria is the AS-2259/GR.

**Table E-2. High frequency antenna selection matrix**

| Antenna               | Use         |                 |                      |                     | Directivity     |               |             | Pol        |          | B/W  |        |
|-----------------------|-------------|-----------------|----------------------|---------------------|-----------------|---------------|-------------|------------|----------|------|--------|
|                       | Ground wave | Skywave         |                      |                     | Omnidirectional | Bidirectional | Directional | Horizontal | Vertical | Wide | Narrow |
|                       |             | Short—0–500 mi. | Medium—500–1,200 mi. | Long—over 1,200 mi. |                 |               |             |            |          |      |        |
| AS-2259/GR            |             | X               |                      |                     | X               |               |             |            |          | X    |        |
| Vertical whip         | X           |                 |                      |                     | X               |               |             |            | X        | X    |        |
| Half-wave dipole      |             | X               | X                    |                     |                 | X             |             | X          |          |      | X      |
| Long wire             | X           |                 | X                    | X                   |                 | X             | X           |            | X        | X    |        |
| Inverted L            | X           | X               | X                    |                     | X               | X             |             | X          | X        |      | X      |
| Sloping V             | X           |                 | X                    | X                   |                 |               | X           | X          |          | X    |        |
| Vertical half rhombic | X           |                 | X                    | X                   |                 |               | X           |            | X        | X    |        |
| <b>Legend:</b>        |             |                 |                      |                     |                 |               |             |            |          |      |        |
| B/W                   |             | bandwidth       |                      |                     |                 |               |             |            |          |      |        |
| mi                    |             | miles           |                      |                     |                 |               |             |            |          |      |        |
| Pol                   |             | polarization    |                      |                     |                 |               |             |            |          |      |        |

E-30. The near-vertical incidence skywave antenna, (AS-2259/GR) is a lightweight sloping dipole omnidirectional antenna. The AS-2259/GR antenna provides high-angle radiation, near-vertical incidence to permit short-range skywave propagation to support HF communications up to 800 kilometers (500 miles). The frequency range of the antenna is 2.0–12.0 MHz. The maximum RF power capacity is 1,000 watts. Figure E-4 on page 90 shows a near-vertical incidence skywave antenna, AS-2259/GR.



**Figure E-4. Near-vertical incidence skywave antenna, AS-2259/G**

#### ***Harris RF-1944, Inverted Vee High Frequency Antenna***

E-31. The Harris RF-1944 Inverted Vee HF antenna is a lightweight, broadband dipole commercial off-the-shelf antenna that provides radiation patterns to support HF skywave communications. The antenna's primary use is for its ALE and frequency hopping capabilities. Other capabilities of the RF-1944 antenna are—

- Horizontal polarization.
- 0–804.7 kilometer (0–500 miles) radiation pattern. This range is ideal for HF skywave communications.
- 1.6–30 MHz frequency range.
- Up to 20 watts power and 50 ohm input impedance.
- Gain—
  - 16 decibels at 2 MHz.
  - 2 decibels at 30 MHz.
- Weight—
  - 9 pounds.

E-32. The RF-1944 antenna does not include a mast. The primary components are a balanced to unbalanced transformer, two radiation elements with integral terminating loads, two ground stakes, a coaxial cable, a weighted throwing line, and a carrying bag. The RF-1944 antenna's light-weight design makes it easily transportable in mobile, tactical environments.

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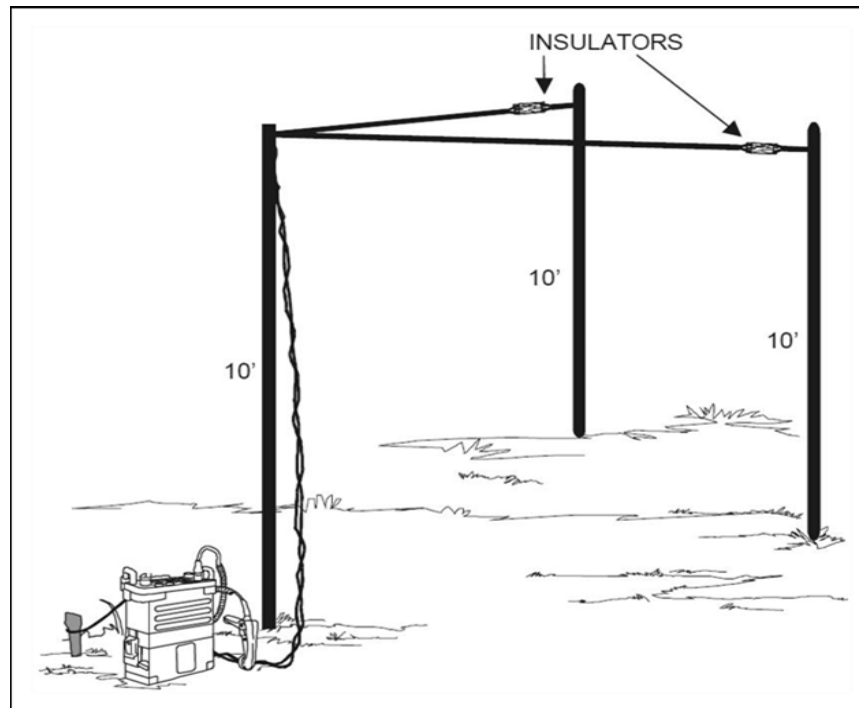
**Note.** A balanced-to-unbalanced transformer is a device used to couple a balanced device or line to an unbalanced device or line.

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#### ***V Antenna***

E-33. The V antenna (see figure E-5 on page 91) is a medium- to long-range, broadband skywave antenna. The V antenna provides point-to-point communications to ranges exceeding 4,000 kilometers (2,500 miles). The V antenna consists of two wires arranged to form a V, with its ends at the apex (where the legs come

together) attached to a transmission line. Radiation lobes off each wire combine to increase gain in the direction of an imaginary line bisecting the apex angle; the pattern is bidirectional. Adding 300 ohm terminating resistors to the far end of each leg makes the pattern unidirectional (in the direction away from the apex angle).



**Figure E-5. V antenna**

E-34. The angle between the legs varies with the length of the legs to achieve maximum performance. Table E-3 shows the angle and the length of the legs. When using the V antenna with more than one frequency or wavelength, operators should use an apex angle that is midway between the extreme angles as determined in table E-3.

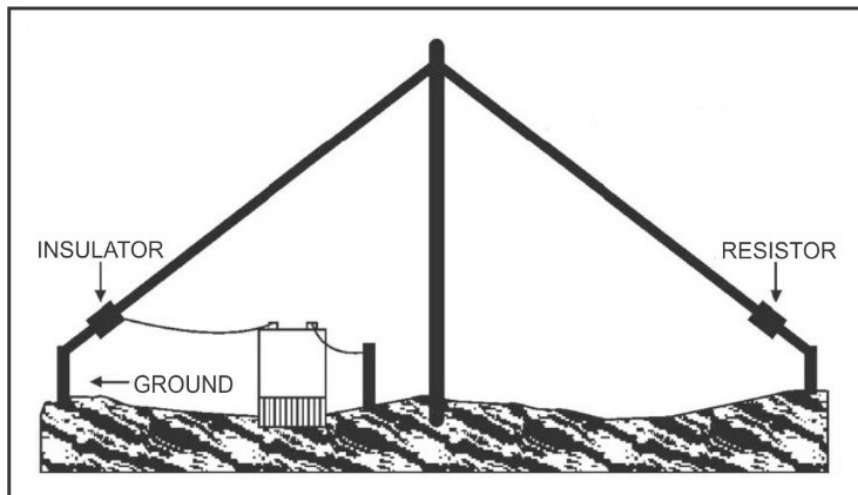
**Table E-3. Leg angle for V antenna**

| <b>Antenna Length<br/>(Wavelength)</b> | <b>Optimum Apex Angle<br/>(Degrees)</b> |
|----------------------------------------|-----------------------------------------|
| 1                                      | 90                                      |
| 2                                      | 70                                      |
| 3                                      | 58                                      |
| 4                                      | 50                                      |
| 6                                      | 40                                      |
| 8                                      | 35                                      |
| 10                                     | 33                                      |

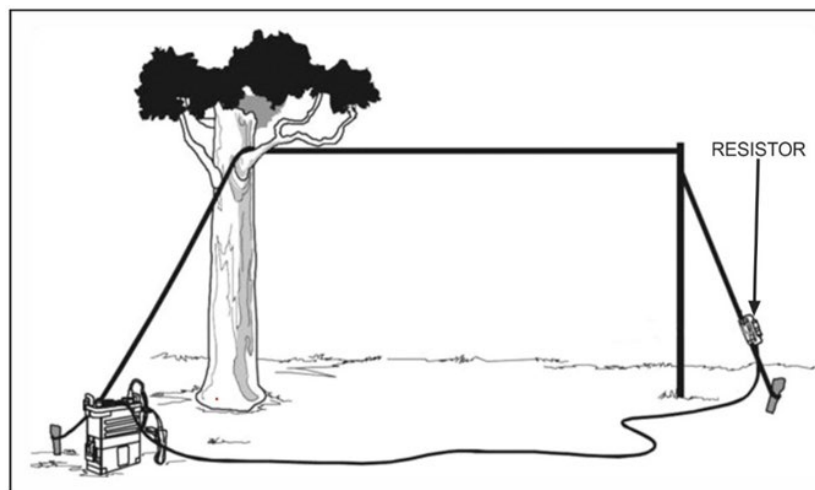
#### ***Vertical Half-Rhombic Antenna and Long-Wire Antennas***

E-35. The vertical half-rhombic antenna and the long-wire antenna are two field expedient directional antennas. The long wire antenna directive pattern radiates in the horizontal and vertical planes. The vertical half-rhombic antenna radiates to the front and back of the sloping wires if resistors are not used. These antennas consist of a single wire, preferably two or more wavelengths long, supported on poles at the height of 3–7 meters (10–20 feet) above the ground. The antennas operate satisfactorily as low as 1 meter (3.2 feet) above the ground. Resistors on the far end of the wire connect to ground through a non-inductive resistor of

500–600 ohms. To avoid resistor burnout from the output power of the transmitter, leaders ensure personnel use a resistor rated at least one-half the wattage output of the transmitter, good ground with multiple ground rods, or a counterpoise at both ends of the antenna. Figure E-6 is an example of a vertical half-rhombic antenna. Figure E-7 is an example of the long wire antenna.



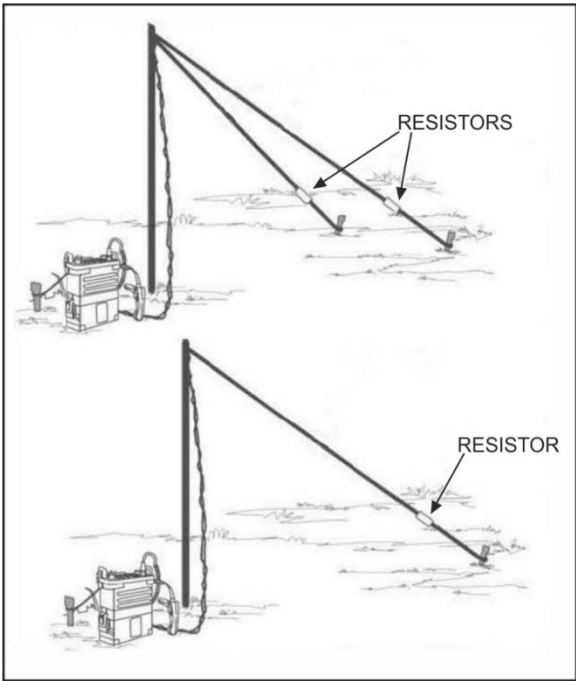
**Figure E-6. Vertical half-rhombic antenna**



**Figure E-7. Long-wire antenna**

### ***Sloping V Antenna***

E-36. The sloping V antenna is a field expedient directional antenna with legs sloping downward from its apex. To ensure that the antenna radiates in only one direction, add non-inductive terminating resistors from the end of each leg, not at the apex to ground. The resistors should be approximately 500 ohms and have a power rating at least one-half that of the output power of the transmitter used. Without the resistors, the antenna radiates bidirectionally, to the front and back. A balanced transmission line feeds the antenna. Figure E-8 on page 93 shows an example of a sloping V antenna.



**Figure E-8. Sloping-V antenna**

***Inverted L Antenna***

E-37. The inverted L is a combination antenna made up of vertical and horizontal wire sections. It provides radiation when using no resistors from the vertical element for ground wave propagation, and high-angle radiation from the horizontal element for short-range skywave propagation, 0–400 kilometers (0–250 miles). The classic inverted L has a quarter-wave vertical section and a half-wave horizontal section.

E-38. Table E-4 shows the frequency and length of the horizontal element. Reasonable performance for short-range skywave circuits is obtained with a vertical height of 11–12 meters (35–40 feet). Figure E-9 on page 94 shows an example inverted L antenna.

**Table E-4. Frequency and inverted L horizontal element length**

| <i>Operating Frequency</i> | <i>Length of Horizontal Element</i> |
|----------------------------|-------------------------------------|
| 5.0–7.0 MHz                | 24.3 meters (80 feet)               |
| 3.5–6.0 MHz                | 30.4 meters (100 feet)              |
| 2.5–4.0 MHz                | 45.7 meters (150 feet)              |
| <b>Legend:</b>             |                                     |
| MHz                        | megahertz                           |

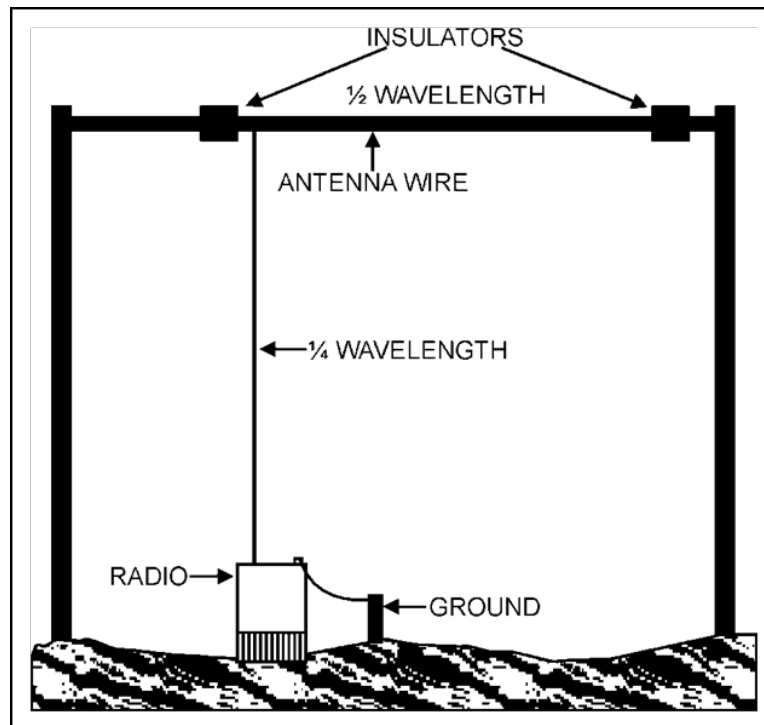


Figure E-9. Inverted L antenna

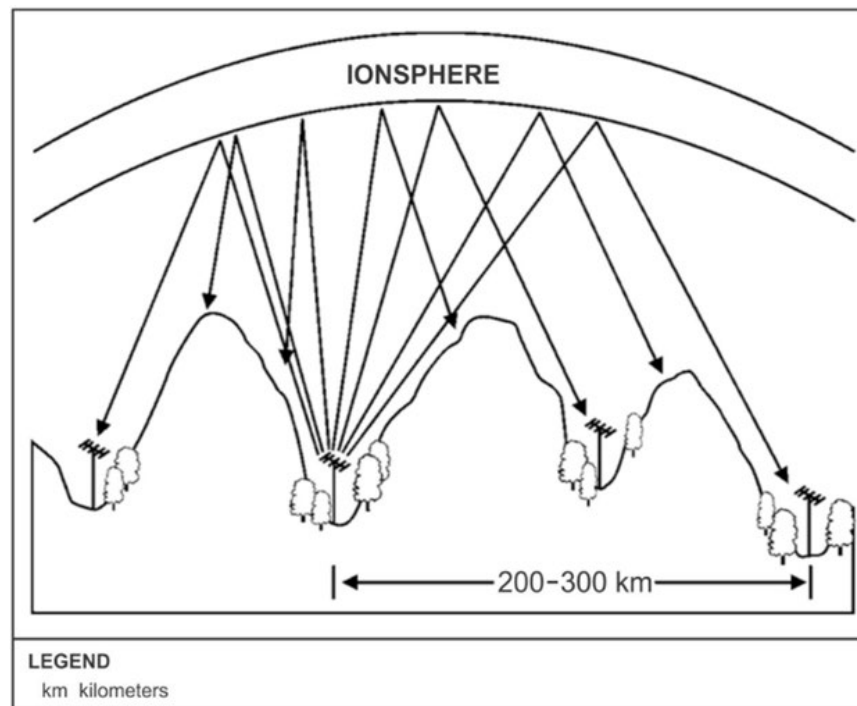
#### *Near Vertical Incidence Skywave Propagation*

E-39. HF radio and the near-vertical incidence skywave mode play an essential role in establishing reliable communications. HF communications typically deploy first and are frequently the only means of communicating with fast-moving or widely separated units. Communications planners, commanders, and operators must be familiar with near-vertical incidence skywave techniques and their applications and shortcomings.

E-40. Near-vertical incidence skywave propagation uses antennas with high-angle radiation and low operating frequencies. Just as the proper selection of antenna can increase the reliability of a long-range circuit, the same holds true for short-range communications.

E-41. Near-vertical incidence skywave propagation uses a high take-off angle of 60–90 degrees to radiate the signal upward. The signal reflects from the ionosphere and returns to Earth in a circular pattern around the transmitter. In near-vertical incidence skywave propagation, there is an area between the maximum ground wave distance and the shortest skywave distances where no communication is possible. Communications are continuous out to several hundred kilometers from the transmitter.

E-42. The nearly vertical angle of radiation also requires lower frequencies. Generally, near-vertical incidence skywave propagation uses frequencies up to 8 MHz. The steep up-and-down propagation of the signal gives the radio operator the ability to communicate over nearby ridgelines, mountains, and dense vegetation. A valley location may give the radio operator terrain shielding from hostile intercept or protect the circuit from ground wave and long wave electromagnetic interference. Antennas used for near-vertical incidence skywave propagation require high take-off angle radiation with very little ground wave radiation. Figure E-10 on page 95 is an example of near-vertical incidence skywave propagation.



**Figure E-10. Near-vertical incidence skywave propagation**

E-43. Using the HF antenna selection matrix in table E-2 on page 89, the AS-2259/GR and the half-wave dipole are the only antennas listed that meet the requirements of near-vertical incidence skywave propagation. While the inverted V and inverted L have high angle radiation, they can also have strong ground wave radiation that could interfere with the close-in, near-vertical incidence skywave communications.

E-44. Disadvantages of near-vertical incidence skywave propagation include—

- Increased potential for electromagnetic interference—where near-vertical incidence skywave and ground wave signals are present, the ground wave can cause destructive electromagnetic interference.
- Ground wave radiation suppression—proper antenna selection suppresses ground wave radiation, minimizes electromagnetic interference effects, and maximizes the amount of energy output while in the near-vertical incidence skywave mode.

E-45. Advantages of using near-vertical incidence skywave propagation in a tactical environment include—

- Skip zone-free omnidirectional communication.
- Terrain does not affect the loss of signal. This gives a more consistent received signal level over the operational range instead of one that varies widely with distance.
- Operators can operate from protected, dug-in positions. Commanders do not have to control the high ground for HF communication.
- Orientation, such as doublets and inverted antennas are not as critical.

E-46. Advantages of using near-vertical incidence skywave when operating against an electromagnetic warfare-capable threat include—

- **Lower probability of geolocation.** Near-vertical incidence skywave energy from above at very steep angles reduces direction finding from nearby but makes beyond ground wave range more difficult. Successful near vertical incidence skywave is achieved with very low power HF sets.
- **Communications are more laborious to jam.** Ground wave jammers are subject to path loss. Terrain features attenuate a ground wave jammer without degrading the desired communication path. This forces the jammer to move very close to the target or transmit power. Either tactic makes jamming more difficult.

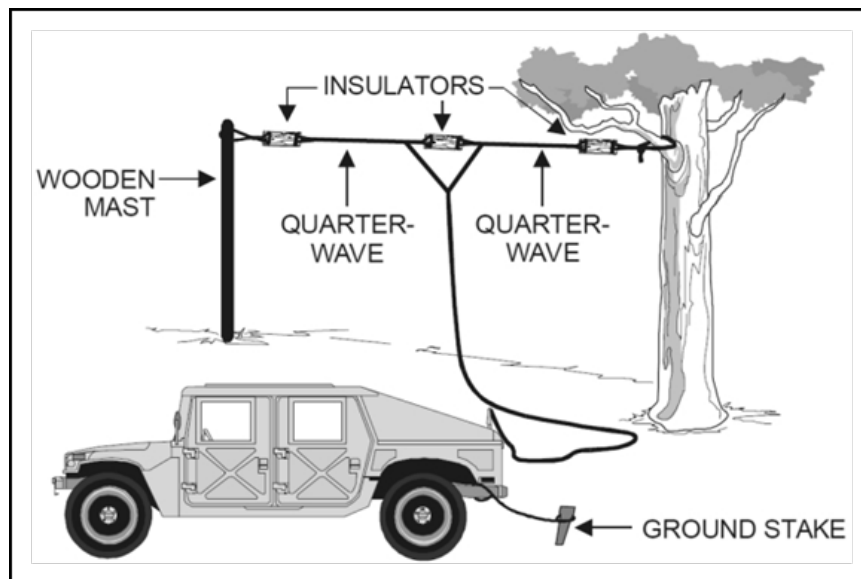
### Field Expedient Antennas

E-47. Field expedient antennas are generally used for HF communication. Construction material is what differentiates field expedient antennas used for HF and VHF or UHF field applications. Users may have to construct an emergency antenna from common items. Plastic and glass, including plastic spoons, buttons, bottlenecks, and plastic bags, can be effective insulators. Wood and rope also act as insulators, although they are less effective than plastic and glass.

### High Frequency Antennas Usable at Very High Frequency and Ultrahigh Frequency

E-48. Simple vertical half-wave dipole (doublet) and quarter wave monopole antennas are effective for omnidirectional transmission and reception over short-range distances. Longer distances typically use rhombic antennas made of wire similar in design to HF versions to provide an advantage at frequencies as high as 1 GHz.

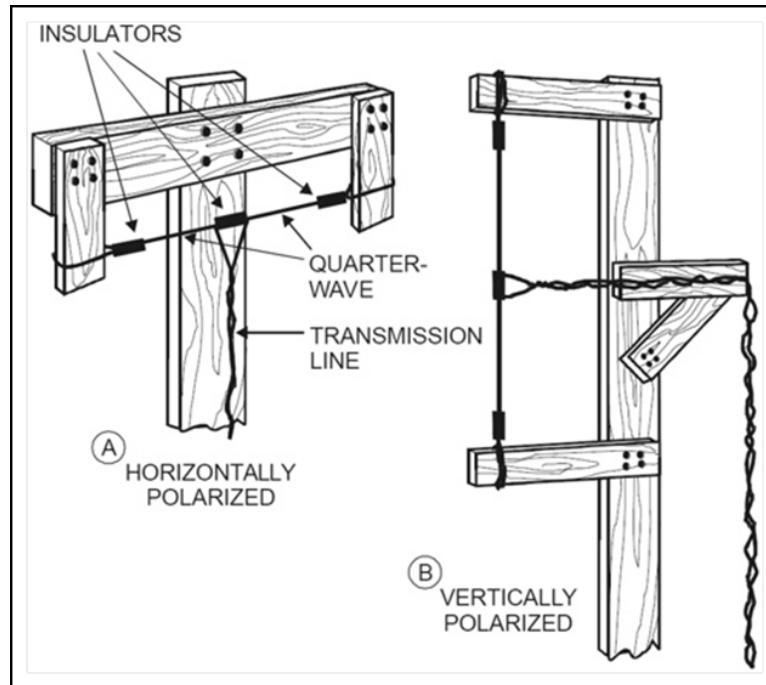
E-49. The dipole (doublet) antenna, also considered a center fed antenna, is a half-wave antenna consisting of two-quarter wavelength sections on each side of the center. Figure E-11 is an example of the improvised dipole (doublet) antenna used with VHF and UHF radios.



**Figure E-11. Half-wave dipole (doublet) antenna**

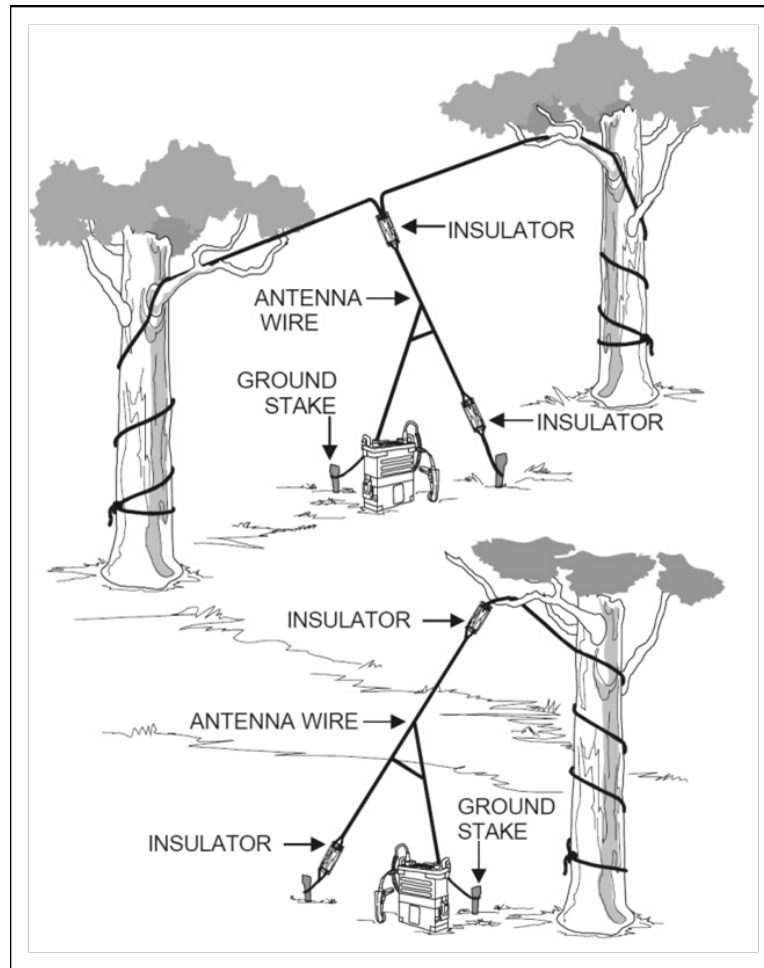
E-50. A transmission line conducts electrical energy from one point to another and transfers the output of a transmitter to an antenna. It is possible to connect an antenna directly to a transmitter. In a vehicular installation, for example, the antenna is mounted outside, and the transmitter is inside the vehicle.

E-51. Operators may use pieces of wood to support center-fed half-wave frequency modulation antennas. These antennas rotate to any position to obtain the best performance. If the antenna is vertical, the transmission line should be positioned horizontally from the antenna or a distance equal to at least one-half of the antenna's length, before dropping down to the radio set. Figure E-12 on page 97 shows examples of horizontal (A) and vertical (B) center-fed half-wave antennas.



**Figure E-12. Center-fed half-wave antenna**

E-52. Figure E-13 on page 98 is an example of an improvised vertical half-wave antenna. VHF and UHF radios primarily use this technique. An improvised vertical half-wave antenna is effective in heavily wooded areas to increase the range of portable radios. The top guy wire can connect to a limb or pass over the limb and connect to the tree trunk or a stake.



**Figure E-13. Improvised vertical half-wave antennas**

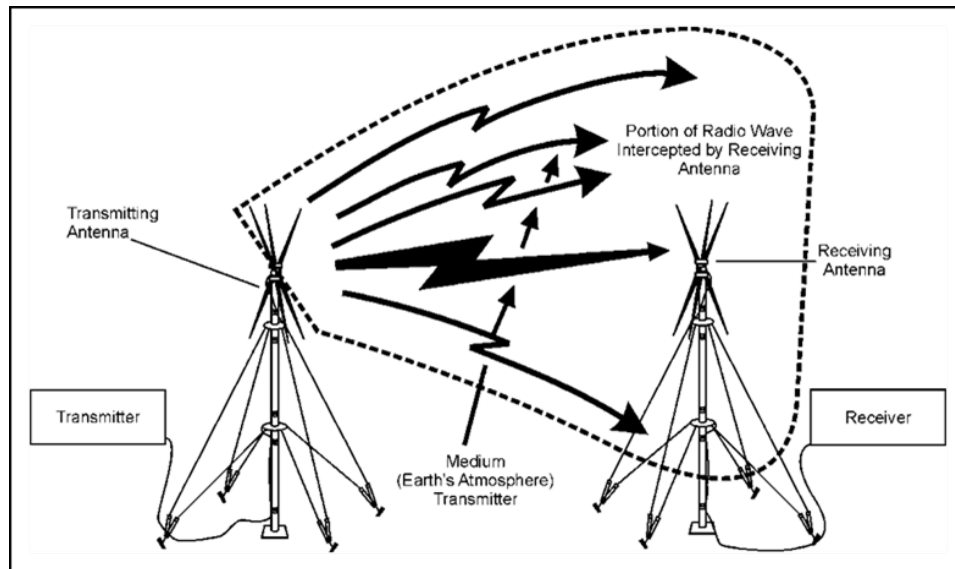
## HIGH FREQUENCY ANTENNA PLACEMENT

E-53. During operations, units are not always able to locate their fixed and mobile radio assets in ideal positions. Planners should comply with as many of the following criteria as possible to gain the best technical advantages for the tactical situation—

- Use of ground radials and ground stakes under a vertical antenna to improve antenna efficiency and lower take-off angles for better ground wave communications.
- Placement of vertical antennas on higher ground, if possible, to enhance ground wave communications.
- Avoiding placement of vertical antennas behind metal fencing that shield ground wave signals.
- Avoiding placement of a vertical antenna close to vertical conducting structures such as masts, light poles, trees, or metal buildings. Antennas need to be at least one wavelength from conducting structures to eliminate major pattern distortions and antenna impedance changes by induced current and reflections.
- Location of antennas as far from the radio as practical to reduce electromagnetic interference effects between the radio the radio and antenna.

E-54. Units that operate in less than ideal positions may need to communicate using simplex operation. Simplex operation, or one-way-reversible, consists of sending and receiving radio signals on one antenna. HF radios normally use simplex operation. Units use two antennas during duplex operation: one for transmitting and one for receiving. The transmitter generates a radio signal; a transmission line delivers the signal from the transmitter to the antenna.

E-55. The transmitting antenna sends the radio signal into the atmosphere toward the receiving antenna, which intercepts the signal and sends it through a transmission line to the receiver. The receiver processes the radio signal and is supported by an AN/UXC-10 facsimile. Figure E-14 is an example of a typical transmitter and receiver connection.



**Figure E-14. Transmitter and receiver connection example**

E-56. The function of all antennas depends on whether it is transmitting or receiving. A transmitting antenna transforms the output RF signal in the form of an alternating electrical current produced by a radio transmitter and RF output power. The signal is then transmitted into an electromagnetic field. The receiving antenna reverses this process; it transforms the electromagnetic field into electrical energy delivered to a radio receiver.

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## Appendix F

# Julian Day, Synchronization Time, and Time Conversion

Accurate time is essential for SINCGARS radios to operate in the frequency-hopping mode. This appendix addresses the Julian day, synchronization time, and ZULU time. A time variance greater than plus or minus four seconds can disrupt the SINCGAR's frequency-hopping capability.

## JULIAN DAY

F-1. The SINCGARS uses a special two-digit form of the Julian day as part of the synchronization time. The two-digit Julian day begins with 01 on 1 January and continues through 00, repeating as necessary to cover the entire year.

F-2. Table F-1 on page 102 shows the two-digit Julian day calendar for a regular year. Table F-2 on page 103 shows the two-digit Julian day calendar for a leap year. The regular year, two-digit Julian calendar terminates on 65 (or 66 for the leap year). Net control station personnel and operators change the Julian day to 01 on 1 January each year. Resetting the Julian day can be accomplished by—

- Sending an electronic remote refill.
- Reloading the time from a Defense Advanced Global Positioning System Receiver.
- Changing the date in the radio using the radio transceiver (RT) keypad.

Table F-1. Two-digit Julian day calendar (regular year)

| <i>Julian Day Calendar (Regular Year)</i> |            |            |            |            |            |            |            |            |            |            |            |            |
|-------------------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| <i>Day</i>                                | <i>JAN</i> | <i>FEB</i> | <i>MAR</i> | <i>APR</i> | <i>MAY</i> | <i>JUN</i> | <i>JUL</i> | <i>AUG</i> | <i>SEP</i> | <i>OCT</i> | <i>NOV</i> | <i>DEC</i> |
| 1                                         | 01         | 32         | 60         | 91         | 21         | 52         | 82         | 13         | 44         | 74         | 05         | 35         |
| 2                                         | 02         | 33         | 61         | 92         | 22         | 53         | 83         | 14         | 45         | 75         | 06         | 36         |
| 3                                         | 03         | 34         | 62         | 93         | 23         | 54         | 84         | 15         | 46         | 76         | 07         | 37         |
| 4                                         | 04         | 35         | 63         | 94         | 24         | 55         | 85         | 16         | 47         | 77         | 08         | 38         |
| 5                                         | 05         | 36         | 64         | 95         | 25         | 56         | 86         | 17         | 48         | 78         | 09         | 39         |
| 6                                         | 06         | 37         | 65         | 96         | 26         | 57         | 87         | 18         | 49         | 79         | 10         | 40         |
| 7                                         | 07         | 38         | 66         | 97         | 27         | 58         | 88         | 19         | 50         | 80         | 11         | 41         |
| 8                                         | 08         | 39         | 67         | 98         | 28         | 59         | 89         | 20         | 51         | 81         | 12         | 42         |
| 9                                         | 09         | 40         | 68         | 99         | 29         | 60         | 90         | 21         | 52         | 82         | 13         | 43         |
| 10                                        | 10         | 41         | 69         | 00         | 30         | 61         | 91         | 22         | 53         | 83         | 14         | 44         |
| 11                                        | 11         | 42         | 70         | 01         | 31         | 62         | 92         | 23         | 54         | 84         | 15         | 45         |
| 12                                        | 12         | 43         | 71         | 02         | 32         | 63         | 93         | 24         | 55         | 85         | 16         | 46         |
| 13                                        | 13         | 44         | 72         | 03         | 33         | 64         | 94         | 25         | 56         | 86         | 17         | 47         |
| 14                                        | 14         | 45         | 73         | 04         | 34         | 65         | 95         | 26         | 57         | 87         | 18         | 48         |
| 15                                        | 15         | 46         | 74         | 05         | 35         | 66         | 96         | 27         | 58         | 88         | 19         | 49         |
| 16                                        | 16         | 47         | 75         | 06         | 36         | 67         | 97         | 28         | 59         | 89         | 20         | 50         |
| 17                                        | 17         | 48         | 76         | 07         | 37         | 68         | 98         | 29         | 60         | 90         | 21         | 51         |
| 18                                        | 18         | 49         | 77         | 08         | 38         | 69         | 99         | 30         | 61         | 91         | 22         | 52         |
| 19                                        | 19         | 50         | 78         | 09         | 39         | 70         | 00         | 31         | 62         | 92         | 23         | 53         |
| 20                                        | 20         | 51         | 79         | 10         | 40         | 71         | 01         | 32         | 63         | 93         | 24         | 54         |
| 21                                        | 21         | 52         | 80         | 11         | 41         | 72         | 02         | 33         | 64         | 94         | 25         | 55         |
| 22                                        | 22         | 53         | 81         | 12         | 42         | 73         | 03         | 34         | 65         | 95         | 26         | 56         |
| 23                                        | 23         | 54         | 82         | 13         | 43         | 74         | 04         | 35         | 66         | 96         | 27         | 57         |
| 24                                        | 24         | 55         | 83         | 14         | 44         | 75         | 05         | 36         | 67         | 97         | 28         | 58         |
| 25                                        | 25         | 56         | 84         | 15         | 45         | 76         | 06         | 37         | 68         | 98         | 29         | 59         |
| 26                                        | 26         | 57         | 85         | 16         | 46         | 77         | 07         | 38         | 69         | 99         | 30         | 60         |
| 27                                        | 27         | 58         | 86         | 17         | 47         | 78         | 08         | 39         | 70         | 00         | 31         | 61         |
| 28                                        | 28         | 59         | 87         | 18         | 48         | 79         | 09         | 40         | 71         | 01         | 32         | 62         |
| 29                                        | 29         |            | 88         | 19         | 49         | 80         | 10         | 41         | 72         | 02         | 33         | 63         |
| 30                                        | 30         |            | 89         | 20         | 50         | 81         | 11         | 42         | 73         | 03         | 34         | 64         |
| 31                                        | 31         |            | 90         |            | 51         |            | 12         | 43         |            | 04         |            | 65         |

Table F-2. Two-digit Julian day calendar (leap year)

| <i>Julian Day Calendar (Leap Year)</i> |            |            |            |            |            |            |            |            |            |            |            |            |
|----------------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| <i>Day</i>                             | <i>JAN</i> | <i>FEB</i> | <i>MAR</i> | <i>APR</i> | <i>MAY</i> | <i>JUN</i> | <i>JUL</i> | <i>AUG</i> | <i>SEP</i> | <i>OCT</i> | <i>NOV</i> | <i>DEC</i> |
| <b>1</b>                               | 01         | 32         | 61         | 92         | 22         | 53         | 83         | 14         | 45         | 75         | 06         | 36         |
| <b>2</b>                               | 02         | 33         | 62         | 93         | 23         | 54         | 84         | 15         | 46         | 76         | 07         | 37         |
| <b>3</b>                               | 03         | 34         | 63         | 94         | 24         | 55         | 85         | 16         | 47         | 77         | 08         | 38         |
| <b>4</b>                               | 04         | 35         | 64         | 95         | 25         | 56         | 86         | 17         | 48         | 78         | 09         | 39         |
| <b>5</b>                               | 05         | 36         | 65         | 96         | 26         | 57         | 87         | 18         | 49         | 79         | 10         | 40         |
| <b>6</b>                               | 06         | 37         | 66         | 97         | 27         | 58         | 88         | 19         | 50         | 80         | 11         | 41         |
| <b>7</b>                               | 07         | 38         | 67         | 98         | 28         | 59         | 89         | 20         | 51         | 81         | 12         | 42         |
| <b>8</b>                               | 08         | 39         | 68         | 99         | 29         | 60         | 90         | 21         | 52         | 82         | 13         | 43         |
| <b>9</b>                               | 09         | 40         | 69         | 00         | 30         | 61         | 91         | 22         | 53         | 83         | 14         | 44         |
| <b>10</b>                              | 10         | 41         | 70         | 01         | 31         | 62         | 92         | 23         | 54         | 84         | 15         | 45         |
| <b>11</b>                              | 11         | 42         | 71         | 02         | 32         | 63         | 93         | 24         | 55         | 85         | 16         | 46         |
| <b>12</b>                              | 12         | 43         | 72         | 03         | 33         | 64         | 94         | 25         | 56         | 86         | 17         | 47         |
| <b>13</b>                              | 13         | 44         | 73         | 04         | 34         | 65         | 95         | 26         | 57         | 87         | 18         | 48         |
| <b>14</b>                              | 14         | 45         | 74         | 05         | 35         | 66         | 96         | 27         | 58         | 88         | 19         | 49         |
| <b>15</b>                              | 15         | 46         | 75         | 06         | 36         | 67         | 97         | 28         | 59         | 89         | 20         | 50         |
| <b>16</b>                              | 16         | 47         | 76         | 07         | 37         | 68         | 98         | 29         | 60         | 90         | 21         | 51         |
| <b>17</b>                              | 17         | 48         | 77         | 08         | 38         | 69         | 99         | 30         | 61         | 91         | 22         | 52         |
| <b>18</b>                              | 18         | 49         | 78         | 09         | 39         | 70         | 00         | 31         | 62         | 92         | 23         | 53         |
| <b>19</b>                              | 19         | 50         | 79         | 10         | 40         | 71         | 01         | 32         | 63         | 93         | 24         | 54         |
| <b>20</b>                              | 20         | 51         | 80         | 11         | 41         | 72         | 02         | 33         | 64         | 94         | 25         | 55         |
| <b>21</b>                              | 21         | 52         | 81         | 12         | 42         | 73         | 03         | 34         | 65         | 95         | 26         | 56         |
| <b>22</b>                              | 22         | 53         | 82         | 13         | 43         | 74         | 04         | 35         | 66         | 96         | 27         | 57         |
| <b>23</b>                              | 23         | 54         | 83         | 14         | 44         | 75         | 05         | 36         | 67         | 97         | 28         | 58         |
| <b>24</b>                              | 24         | 55         | 84         | 15         | 45         | 76         | 06         | 37         | 68         | 98         | 29         | 59         |
| <b>25</b>                              | 25         | 56         | 85         | 16         | 46         | 77         | 07         | 38         | 69         | 99         | 30         | 60         |
| <b>26</b>                              | 26         | 57         | 86         | 17         | 47         | 78         | 08         | 39         | 70         | 00         | 31         | 61         |
| <b>27</b>                              | 27         | 58         | 87         | 18         | 48         | 79         | 09         | 40         | 71         | 01         | 32         | 62         |
| <b>28</b>                              | 28         | 59         | 88         | 19         | 49         | 80         | 10         | 41         | 72         | 02         | 33         | 63         |
| <b>29</b>                              | 29         | 60         | 89         | 20         | 50         | 81         | 11         | 42         | 73         | 03         | 34         | 64         |
| <b>30</b>                              | 30         |            | 90         | 21         | 51         | 82         | 12         | 43         | 74         | 04         | 35         | 65         |
| <b>31</b>                              | 31         |            | 91         |            | 52         |            | 13         | 44         |            | 05         |            | 66         |

## SYNCHRONIZATION TIME

F-3. The SINCGARS radio uses seven internal clocks to maintain proper synchronization time: There is a base clock and a clock for each of the six frequency hopping channels. Manual and cue settings display the base clock time.

F-4. The Defense Advanced Global Positioning System Receiver provides units a ready source of precision timing. Opening networks on GPS time and updating net control station RT synchronization time to GPS time daily keeps all networks within the plus or minus four-second window required for frequency hopping communications.

## ZULU TIME

F-5. ZULU time remains in synchronization with the United States Naval Observatory master atomic clock. An alternative is to use the time from a Defense Advanced Global Positioning System Receiver for tracking at least one satellite. The net control station should update and verify net time daily or according to the unit SOP.

F-6. There are 25 world time zones from -12 through +12 Coordinated Universal Time (ZULU time). Each is 15 degrees longitude, as measured East and West from the Prime Meridian. Table F-3 shows the world time zones and time zone conversion.

**Table F-3. Example of world time zone conversion (standard time)**

| Y                                               | X                                | W    | V    | U    | T    | S    | R      | Q    | P    | O            | N    | Z    | A                            | B    | C    | D    | E    | F    | G    | H    | I    | K    | L    | M     |  |
|-------------------------------------------------|----------------------------------|------|------|------|------|------|--------|------|------|--------------|------|------|------------------------------|------|------|------|------|------|------|------|------|------|------|-------|--|
| Civilian Time Zones                             |                                  |      |      |      |      |      |        |      |      |              |      |      |                              |      |      |      |      |      |      |      |      |      |      |       |  |
| IDLW                                            | NT                               | HST  | ASDT | PST  | MST  | CST  | EST    | AST  | NST  | AT           | WAT  | UTC  | CET                          | EET  | BT   | ZP4  | ZP5  | ZP6  | WAST | CCT  | JST  | GST  | SBT  | IDL F |  |
| 1200                                            | 1300                             | 1400 | 1500 | 1600 | 1700 | 1800 | 1900** | 2000 | 2100 | 2200         | 2300 | 2400 | 0100                         | 0200 | 0300 | 0400 | 0500 | 0600 | 0700 | 0800 | 0900 | 1000 | 1100 | 1200* |  |
| Standard Time=Universal Time + Value from Table |                                  |      |      |      |      |      |        |      |      |              |      |      |                              |      |      |      |      |      |      |      |      |      |      |       |  |
| Z                                               | 0                                |      | E    |      | +5   |      | K      |      | +10  |              | P    |      | -3                           |      | U    |      | -8   |      |      |      |      |      |      |       |  |
| A                                               | +1                               |      | F    |      | +6   |      | L      |      | +11  |              | Q    |      | -4                           |      | V    |      | -9   |      |      |      |      |      |      |       |  |
| B                                               | +2                               |      | G    |      | +7   |      | M      |      | +12  |              | R    |      | -5                           |      | W    |      | -10  |      |      |      |      |      |      |       |  |
| C                                               | +3                               |      | H    |      | +8   |      | N      |      | -1   |              | S    |      | -6                           |      | X    |      | -11  |      |      |      |      |      |      |       |  |
| D                                               | +4                               |      | I    |      | +9   |      | O      |      | -2   |              | T    |      | -7                           |      | Y    |      | -12  |      |      |      |      |      |      |       |  |
| *=Today                                         |                                  |      |      |      |      |      |        |      |      | **=Yesterday |      |      |                              |      |      |      |      |      |      |      |      |      |      |       |  |
| Legend:                                         |                                  |      |      |      |      |      |        |      |      |              |      |      |                              |      |      |      |      |      |      |      |      |      |      |       |  |
| ASDT                                            | Alaska Standard Time             |      |      |      |      |      |        |      |      |              |      | HST  | Hawaii Standard Time         |      |      |      |      |      |      |      |      |      |      |       |  |
| AST                                             | Atlantic Standard Time           |      |      |      |      |      |        |      |      |              |      | IDLE | International Date Line East |      |      |      |      |      |      |      |      |      |      |       |  |
| AT                                              | Azores Time                      |      |      |      |      |      |        |      |      |              |      | IDLW | International Date Line West |      |      |      |      |      |      |      |      |      |      |       |  |
| AWST                                            | Australian Western Standard Time |      |      |      |      |      |        |      |      |              |      | JST  | Japan Standard Time          |      |      |      |      |      |      |      |      |      |      |       |  |
| BT                                              | Baghdad Time                     |      |      |      |      |      |        |      |      |              |      | MST  | Mountain Standard Time       |      |      |      |      |      |      |      |      |      |      |       |  |
| CET                                             | Central European Time            |      |      |      |      |      |        |      |      |              |      | NST  | Newfoundland Standard Time   |      |      |      |      |      |      |      |      |      |      |       |  |
| CCT                                             | China Coast Time                 |      |      |      |      |      |        |      |      |              |      | NT   | Nome Time                    |      |      |      |      |      |      |      |      |      |      |       |  |
| CST                                             | Central Standard Time            |      |      |      |      |      |        |      |      |              |      | PST  | Pacific Standard Time        |      |      |      |      |      |      |      |      |      |      |       |  |
| EET                                             | Eastern European Time            |      |      |      |      |      |        |      |      |              |      | SBT  | Solomon Island Time          |      |      |      |      |      |      |      |      |      |      |       |  |
| EST                                             | Eastern Standard Time            |      |      |      |      |      |        |      |      |              |      | UTC  | Coordinated Universal Time   |      |      |      |      |      |      |      |      |      |      |       |  |
| GST                                             | Guam Standard Time               |      |      |      |      |      |        |      |      |              |      | WAST | West Africa Summer Time      |      |      |      |      |      |      |      |      |      |      |       |  |
| NST                                             | Newfoundland Standard Time       |      |      |      |      |      |        |      |      |              |      | WAT  | West Africa Time             |      |      |      |      |      |      |      |      |      |      |       |  |

F-7. When ZULU time is 12:00, the opposite time zone is 00:00 and indicates a date change. By convention, the area to the left of the dashed line is the following day, while the area to the right is the preceding day. Figure F-1 on page 105 shows a world time zone map.

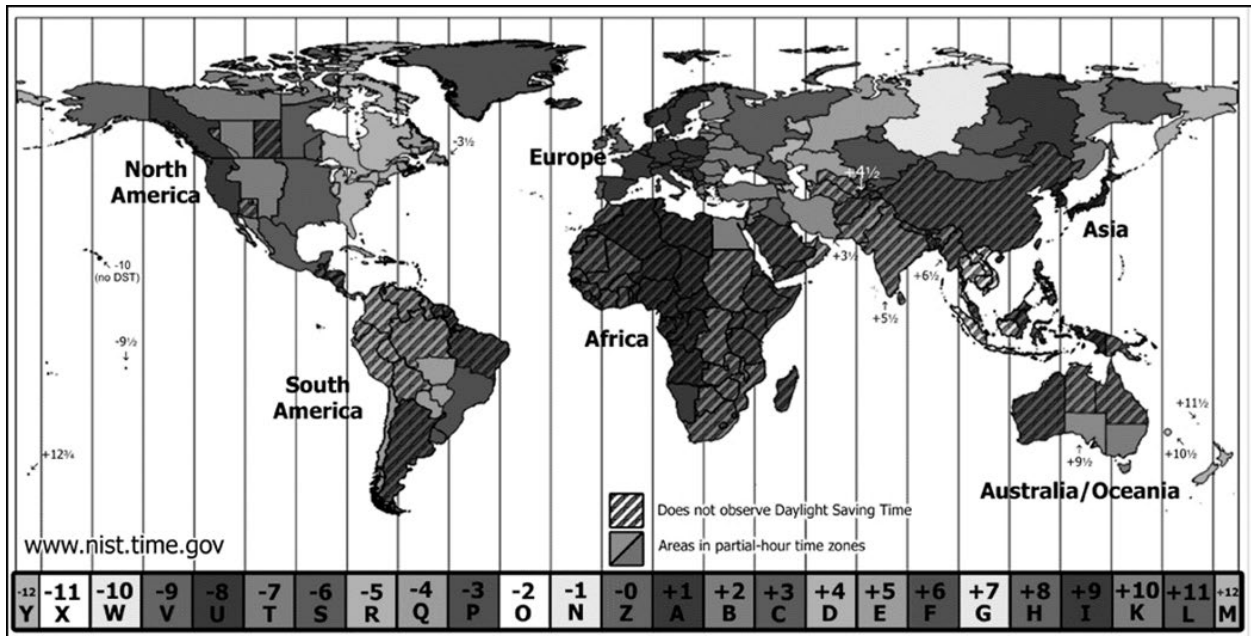


Figure F-1. World time zone map

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## **Appendix G**

# **Communications in Adverse Conditions**

The appendix addresses radio operations in cold weather, jungle, urban, desert, mountain areas, and nuclear areas. Communications in austere, unusual environments require special consideration.

### **OPERATIONS IN COLD WEATHER**

G-1. Single-channel radio equipment has certain capabilities and limitations that require careful consideration when operating in extremely cold weather. Despite significant limitations, the radio is still the normal means of communication in such environments.

G-2. An important attribute of a radio in cold weather operations is versatility. Vehicle-mounted radios can move to almost any location where there is a command headquarters. Smaller, manpack radios move to any area accessible by aircraft or on foot.

G-3. Electromagnetic interference by ionospheric disturbances limits radio communications in extremely cold areas. These disturbances, known as ionospheric storms, have a degrading effect on skywave propagation. Ionospheric storms and aurora activity can cause complete failure of radio communications. Static may block out some frequencies completely for extended periods during storm activity. Fading caused by changes in the density and height of the ionosphere can also occur and may last from several minutes to several weeks. These occurrences are difficult to predict; but when they do occur, the spectrum manager needs to assign alternate frequencies.

### **TECHNIQUES FOR BETTER COMMUNICATIONS IN COLD WEATHER**

G-4. Radio operators implement preventive measures to reduce the adverse effects of extremely cold weather conditions when possible. To avoid cold weather-related malfunctions, communications equipment should be stored inside vehicles. Storing equipment inside vehicles provides added protection from severe weather conditions and potentially avoids some of the grounding and antenna installation problems experienced when operating equipment in cold weather environments.

G-5. It is difficult to establish a good electrical ground in extremely cold areas because of permafrost and deep snow. The conductivity of frozen ground is often too low to provide good ground wave propagation. To improve ground wave operation, radio operators use a counterpoise to offset the degrading effects of poor electrical ground conductivity. To prevent snow from covering an antenna, operators should install a counterpoise; and if possible, install the antenna high enough above the ground to prevent snow from covering the antenna.

G-6. In general, antenna installation in arctic conditions presents no serious difficulties. Antenna installation may take longer due to adverse working conditions. Tips for installing antennas in extremely cold areas include—

- Handle the mast sections and antenna cables carefully since they become brittle in very low temperatures.
- Construct antennas overhead to prevent damage from heavy snow and frost.
- Use nylon rope if available, in preference to cotton or hemp, because nylon ropes do not readily absorb moisture, and are less likely to freeze and break.
- Use extra wires, supports, and anchor stakes to strengthen antennas and to withstand heavy ice and wind loading.

G-7. Some radios, generally older generation radios adjusted to a frequency in a relatively warm place, may drift off frequency when exposed to extreme cold. Low battery voltage can also cause frequency drift. When possible, operators should allow a radio to warm up for several minutes before placing it into operation. Extreme cold tends to lower the output voltage of a dry battery; operators can use body heat to elevate the battery temperature. This technique minimizes frequency drift.

G-8. Northern regions sometime experience flakes or pellets of electrically charged snow. When these particles strike the antenna, the resulting electrical discharge causes a high-pitched static roar that can affect frequencies. To overcome this static, operators cover antenna elements with polystyrene tape or shellac.

## MAINTENANCE IN COLD WEATHER

G-9. The maintenance of radio equipment in extreme cold presents many difficulties. Snow potentially freezes dials and knobs and can get into the wiring, causing shorts and ground faults. Radio sets should be protected from blowing snow. Cold weather can damage or reduce cord and cable flexibility. Operators avoid additional damage by handling these items with care. Operators should winterize communications equipment and power units according to technical manual instructions. Batteries, power units, and microphones require routine maintenance for cold weather operations. Table G-1 is a list of winterization techniques. Refer to the equipment technical manual for additional instructions on winterization.

**Table G-1. Winterization techniques**

| <i>Item</i>                                      | <i>Technique</i>                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Power Units                                      | Operators safeguard generators as much as possible from the weather by using shields and protective covering.                                                                                                                                                                                                                                                                                                                                        |
| Batteries                                        | Maintain moderate temperatures to extend battery life during cold temperatures.                                                                                                                                                                                                                                                                                                                                                                      |
| Shock Damage                                     | Check shock mounts frequently and change them as required to avoid damage to radio equipment.                                                                                                                                                                                                                                                                                                                                                        |
| Winterization                                    | Replace normal lubricants with the recommended arctic lubricants and add a light coat of silicon compound on antenna mast connections to prevent connective wiring from freezing together and becoming hard to dismantle.                                                                                                                                                                                                                            |
| Microphones                                      | Prevent moisture from entering the perforated cover plate of the microphone using standard covers, if available, or improvise a suitable cover using rubber or cellophane membranes, or rayon or nylon cloths.                                                                                                                                                                                                                                       |
| Condensation Management and Corrosion Prevention | Wrap the equipment in a blanket or parka to ensure that it warms gradually to reduce sweating before moving cold equipment into a heated area. Equipment must be thoroughly dry before taking it back into the cold air, or the moisture freeze. Sweating occurs when cold equipment suddenly encounters warm air and moisture condenses on the equipment parts.                                                                                     |
| Vehicle-Mounted Radios                           | Observe proper starting procedures. Allow time for the vehicle's heater to warm the radio sufficiently to ensure that any frost collected on or inside the radio has a chance to thaw or stabilize (approximately 15 minutes). Charge the batteries before increasing engine revolutions per minute. Turn off radios to avoid an excessive power surge. A vehicle operated at a low idle with radios, heater, and lights on may drain the batteries. |

## OPERATIONS IN JUNGLE AREAS

G-10. Limitations on radio communications in jungle areas stem from the climate and the density of jungle growth. The hot and humid climate increases the maintenance demands. Thick jungle growth acts as a vertically polarized absorbing screen for RF energy that reduces transmission range. Therefore, increased emphasis on maintenance and antenna site selection is inherently important when operating in jungle areas.

G-11. Radio communications in jungle areas requires careful planning. Dense jungle growth, heavy rains, and hilly terrain significantly reduce the range of radio transmission. Trees and underbrush absorb VHF and UHF radio energy. In addition to the ordinary free space loss between transmitting and receiving antennas, a

radio wave passing through a forest undergoes additional loss. This extra loss increases rapidly as the transmission frequency increases. Near the ground, with antenna heights of less than 3 meters (9.8 feet), vertical polarization is preferred. If possible, operators should elevate the receiving and transmitting antennas as much as 10–20 meters (32.8–65.6 feet). Horizontal polarization is preferable to vertical polarization. Considerable reduction in total path loss results if either the transmitting or receiving antenna placement occurs above tree level.

G-12. Planners should carefully consider the capabilities and limitations of tactical radios in a jungle environment. Their mobility and various configurations are primary advantages of tactical radios in jungle areas.

## TECHNIQUES FOR BETTER COMMUNICATIONS IN JUNGLE AREAS

G-13. Antenna site selection is the main challenge while establishing radio communications in jungle areas. Techniques to improve communication in the jungle include—

- Placing antennas as high as possible in clearings on the edge farthest from the distant station.
- Keeping antenna cables and connectors off the ground to lessen the effects of moisture, fungus, and insects. This also applies to power and telephone cables.
- Using complete antenna systems, such as broadband and dipoles. Complete antenna systems are more effective than fractional wavelength whip antennas.
- Clearing vegetation from antenna sites. If an antenna touches any foliage, especially wet foliage, grounding of the signal occurs.
- Using horizontally polarized antennas in preference to vertically polarized antennas because vegetation, particularly when wet, acts as a vertically polarized screen and absorbs much of a vertically polarized signal.

## MAINTENANCE IN JUNGLE AREAS

G-14. Maintenance of radio sets in tropical climates is more difficult than in temperate climates due to excessive moisture and fungus. High relative humidity causes condensation to form on the equipment and encourages the growth of fungus. Radio operators and maintenance personnel should check equipment technical manuals for any special maintenance requirements. Techniques for improving maintenance in jungle areas include—

- Keeping the equipment as dry as possible and in lighted areas to retard fungal growth.
- Keeping air vents clear of obstructions to ensure that air can properly circulate for cooling and drying of the equipment.
- Using moisture and fungus proofing paint, tape, or silicone grease to protect equipment after repairs, or to protect damaged or scratched painted surfaces.

## OPERATIONS IN URBAN AREAS

G-15. Radio communications in urbanized terrain poses special challenges. During urban combat operations, the communications situation is considerably different from the situation faced by civil government or cell phone users. Military factors include—

- Restriction of operation to the frequency range of common military radios (2–512 MHz).
- Limits on the output power of military radio equipment.
- Limited number of available repeater assets, if any.
- Limited access to good repeater locations due to threat actions.
- Need to communicate to both outside street locations and inside structures.
- Lack of standard compact antenna systems useful for urban combat.
- Severe restrictions on the movements of system users.
- Lack of manpower required to cover multiple signal sites can easily exceed available resources.
- Problems with buildings or other man-made structures blocking transmission paths.
- Poor electrical conductivity of paved surfaces.
- Commercial power line interference.
- Distorted radio wave propagation in built-up areas.

G-16. VHF radios have their effectiveness reduced in urban areas. The operating frequencies and power output of these radios require line of sight between antennas. Line of sight at street level is not always possible in built-up areas. AM HF radios are less affected by the line of sight challenges because operating frequencies are lower, and power output is greater. HF radios are generally not organic to the small units that conduct urban clearing operations. Retransmission, if available, may overcome this limitation.

## **TECHNIQUES FOR BETTER COMMUNICATIONS IN URBAN AREAS**

G-17. Depending on requirements, units should use organic retransmission teams to extend radio range in urban environments. When available, aerial retransmission platforms can provide effective radio communications in urban environments. Units should hide or blend radio antennas in with the surroundings to ensure they will not be landmarks for the threat to locate. Water towers, commercial antennas, and steeples can conceal military antennas. Signal elements executing operations in challenging urban environments leverage the barrage relay TSM technology of the ITN that offers reliable, redundant communications paths, and improves transmission completion rates. The technology simplifies network configurations and is particularly beneficial for on-the-move operations.

G-18. Signal personnel install wire while friendly forces are in static positions and use existing telephone poles to raise wire lines above the streets. They use ditches, culverts, and tunnels to conceal the wire below the streets. If proper precautions are not taken, tracked and wheeled vehicles could sever lines and disrupt communications.

G-19. Messengers provide security and flexibility. Planners should carefully select messenger routes to avoid pockets of enemy resistance. Units should vary routes and time schedules to avoid establishing a predictable pattern. Pyrotechnics, smoke, and marker panels are also excellent means for communicating, but they require coordination and understanding by air and ground forces. The noise of combat in built-up areas makes it difficult to use sound signals effectively.

G-20. The possible seizure or retention of established communications facilities must be included in planning. Units must make every effort to prevent damage or destruction of these facilities. The local telephone system is already in place and tailored to the city or town. Forces can use local telephone systems to provide immediate access to wire communications with overhead and buried cable. This procedure helps overcome the challenges encountered with radios and provides a cable system less susceptible to combat damage.

G-21. Local media such as newspapers, radio stations, and television stations, provide communication with the local populace after the level of combat declines. Intact police or taxi communications facilities are also possible radio systems, tailored to the city, with retransmission facilities already in place.

G-22. Signal elements and leaders can park radio-equipped vehicles inside of buildings or garages for cover and concealment. It is important to dismount radio equipment, when possible, and install it inside the basements of buildings, if available. To increase noise absorption and provide concealment, operators should position generators against buildings or under sheds with adequate ventilation.

G-23. Another important consideration for urban combat is raw power. The more power used, the more path loss and the deeper the signals penetrate structures. Common tactical VHF manpack radios like SINCGARS have a maximum output power of four watts. The AN/PRC-150 HF radio has a maximum output power of 20 watts. This means there is 7 decibels more signal power to overcome losses caused by the path, path obstructions, inefficient antennas, and other signal attenuating factors. The extra power improves the radio capabilities. The following are examples of power relationships—

- 4 watts = 36 decibels above one milliwatt (dBm).
- 20 watts = 43 dBm.
- 50 watts = 47 dBm.
- 150 watts = 52 dBm.
- 400 watts = 56 dBm.

G-24. The decibel is a logarithmic unit used to describe a ratio. The ratio may be power, voltage, intensity, or several other factors; but in this case, it is transmitter power in watts. Radio operators can calculate the difference of two power levels by taking a logarithm of log ten of their power ratio. For example, if the ratio of power is two—meaning one radio transmitter is double the power of the other—the difference is 3 decibels. Every 3 decibels gained through a more efficient antenna system or reduced transmission line loss is the

equivalent of doubling the transmitter power. The important point is that adjustments to antenna systems or operational frequencies to make an antenna more efficient can often produce far more effective signal power than simply increasing the transmitter power. More effective power may be necessary to overcome path loss for the near-vertical incidence skywave and ground wave systems. If the radio is already transmitting at maximum power, adjustments to the antenna systems or frequencies are the only way to compensate for path loss and improve signal penetration in an urban environment.

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**Note.** It is important to remember that in many situations the power required to operate a radio may not need to be the maximum power. Operators should use only the power necessary to establish communication.

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G-25. Communication between two radio stations requires that the combined transmitter power, transmitting antenna gain, receiving antenna gain, and receiver performance overcome the path loss between stations. A base station typically uses a higher performance receiver and a more efficient antenna, and can compensate for a low power outstation radio, like the manpack radios and inefficient antennas used by forward troops. Communication between low-power outstations is much more difficult and may even require retransmission through a more efficient base station.

G-26. In urban operations, small HF radios like the AN/PRC-150, are extremely portable, but are antenna- and power-challenged based on location. Obtaining a high degree of portable near-vertical incidence skywave effect when needed can be done by physically reorienting standard vertical manpack or vehicle (whip) antennas to the horizontal plane. Direct (surface wave) signals are simpler to generate and use inside structures produced from the same antenna by leaving the antenna vertical.

G-27. Because of their longer wavelengths, lower frequency HF signals (2–30 MHz) naturally penetrate urban structures deeper than signals on higher (shorter wavelength) frequencies. How deep the penetration depends on exact frequency, signal power level, antenna efficiency, and the makeup of the urban structures in the path.

G-28. It is not uncommon for a small ground penetrating radar transmitter operating in the HF range to penetrate over 30.4 meters (100 feet) into common kinds of earth while the same power radar on a higher frequency penetrates much less. If the radio operator is using a common VHF military radio operating at 30 MHz (lowest frequency for SINCGARS) replaces it with a HF radio, such as the AN/PRC-150 operating at 5 MHz; the path loss drops by 20 decibels because of the way that longer wavelength lower frequency signals propagate. In this case, lowering the frequency is the equivalent to increasing the power of the transmitter by a factor of almost seven.

## MAINTENANCE IN URBAN AREAS

G-29. The physical aspects of urban infrastructure and the need to protect communications systems presents unique operational challenges in urban areas. Units should be aware of potential for restrictions to communication ranges with non-vehicular systems due to line of sight limitations, urban structures, dispersion, and threat jamming. To alleviate electromagnetic challenges associated with urban environments, operators can implement redundant communications such as HF radios, satellite, and retransmission. To prevent personnel loss, equipment damage and degraded communications, operators should accomplish maintenance and equipment recovery expeditiously to secure themselves and their equipment. Techniques for improving maintenance in urban areas include—

- Using camouflage and concealment to prevent signal detection, reduces communications footprint, and matches the local urban terrain.
  - Park radio-equipped vehicles inside buildings for cover and concealment.
  - Dismount radio equipment and install it inside buildings (in basement, if available).
  - Place generators against buildings or under sheds to decrease noise and provide concealment.
- Keeping equipment clear of obstructions that impede direct signal transmission to reduce interference.
- Provide adequate ventilation to prevent heat buildup and subsequent failure of generators and communications equipment.

- Training and use of redundant platforms and systems to include retransmission and relay sites and equipment.
- Remote radios by positioning antennas in upper stories or in adjacent buildings based on remote capabilities.

## **OPERATIONS IN DESERT AREAS**

G-30. Radios are usually the primary means of communication in the desert. Radios employed in desert climate and terrain provide the highly mobile means of communications required for widely dispersed forces. However, desert terrain provides poor electrical ground, and counterpoises are needed to improve operation.

G-31. Dust and extreme heat are two of the biggest problems involved in desert operations. Temperatures may vary from 58 degrees Celsius (136 degrees Fahrenheit) in the summer to -46 degrees Celsius (-51 degrees Fahrenheit) in the winter. Extreme heat can damage generators, wire, and communications equipment and increase personnel risk to serious heat-related illnesses or injuries.

G-32. Dust and sand particles damage equipment. Some combat net radios have ventilating ports and channels that may clog with dust. Operators should check and clean these ports regularly to prevent overheating.

G-33. In a desert environment, operators should ground equipment by burying ground plates in the sand and pouring salt solutions on the ground plates. Daily preventive maintenance checks and services should include cleaning generators and air filters to prevent equipment damage.

## **TECHNIQUES FOR BETTER COMMUNICATIONS IN DESERT AREAS**

G-34. Operators should cut or adjust antennas to the length of the operating frequency and emplace antennas in the required direction. Approximate azimuth produced by guesswork is not sufficient. A whip antenna relies on the capacitor effect between itself and the ground for efficient propagation. The electrical ground may be very poor and degrade the antenna performance by as much as one-third if the surface soil lacks moisture, which is normally the case in the desert.

G-35. If a ground-mounted antenna is not fitted with a counterpoise, operators should dampen the ground around it using any fluid available. See appendix H for more information on a counterpoise. Vehicle mounted antennas are more efficient if the main mass structure of the vehicle is forward of the antenna and is oriented toward the distant station.

G-36. Operators should keep radios cool and clean in accordance with technical manual procedures. Units should operate radios in a shaded or ventilated area, and at low power whenever possible. A flat sheet of wood, cardboard, or a vehicle's canvas top over the top of the radio can create man-made shade to keep radios cool. Leaving a space between the wood, cardboard, and the radio, helps to cool the radio by causing air to circulate in the shaded area between the radio and the wood. Carefully covering hot radios with a damp cloth without blocking ventilation outlets allows moisture evaporation from the cloth to cool the radio.

G-37. Desert terrain can cause excessive signal attenuation, making planning ranges shorter. Desert operations require dispersion; yet the environment is likely to degrade the transmission range of radios, particularly secure VHF frequency modulation radios. The greatest degradation is most likely to occur during the hottest part of the day, from approximately 1200–1700.

G-38. If, during the hottest time of day, radio stations begin to lose contact, units should exercise their PACE plan. Techniques may include—

- Using retransmission capabilities—including airborne retransmission. Aerial retransmission aircraft should remain at least 4,000 meters behind the line of contact. Planners create retransmission plans to best support the scheme of maneuver.
- Deploying an unemployed vehicle with a radio as a retransmission station.
- Using alternative radio links, such as VHF multichannel telephones, a higher power level, or HF single-sideband voice.

G-39. After dark, a rapid temperature drop creates heat inversion. This inversion can disrupt radio communication until the atmosphere stabilizes.

G-40. Wired connections are impractical during high-tempo military operations. Wire may be of some value in static defensive situations. When possible, installers and operators should bury wire and cables deep in the soft sand to prevent heat damage to cable insulation and prevent vehicle or foot traffic damage.

G-41. Operators should protect computers and removable storage media from dust and sand. Covering computers and disks with plastic bags may reduce damage. Covering computers and radios for an extended period may cause condensation inside these components, resulting in equipment damage or data loss. Canned compressed air is a useful tool to clean keyboards and other components.

## **MAINTENANCE IN DESERT AREAS**

G-42. Wind-blown sand and grit can damage electrical wire insulation over time. Tape can help protect cables from sand damage. Sand may also work into parts of items such as spaghetti cord plugs, either preventing electrical contact or making it impossible to join the plugs together. A soft brush, such as an old toothbrush, can clean cords and plugs before connection.

G-43. Static electricity is prevalent in the desert. Many factors cause static electricity. One factor is wind-blown dust particles. Extremely low humidity contributes to static discharges between charged particles. Poor grounding practices exacerbate the problem. Operators should tape all sharp edges and tips of antennas to reduce wind-caused static discharges and the accompanying noise. When operating from a fixed position, operators should install a proper earth ground. Since static-caused noise diminishes with an increase in frequency, operators should use the highest frequency that is available and authorized. Refer to TC 6-02.6 for more information about grounding techniques.

## **OPERATIONS IN MOUNTAINOUS TERRAIN**

G-44. Radio operation in mountainous terrain shares some of the same challenges as in cold weather areas. Mobility is difficult in mountainous terrain, and it can be difficult to find a level area to establish a communications site.

### **TECHNIQUES FOR BETTER COMMUNICATIONS IN MOUNTAINOUS TERRAIN**

G-45. Line of sight paths are more difficult to establish in mountainous terrain. Operating in mountainous terrain may require additional retransmission assets. Signal plans should include the use of retransmission to improve communications range. Antenna positioning is critical in mountainous terrain; moving an antenna even a short distance can drastically affect reception.

## **MAINTENANCE IN MOUNTAINOUS TERRAIN**

G-46. Generators and communications equipment need level ground for proper operation. It may be difficult to drive ground rods and guy wire stakes into rocky ground. An alternate grounding method may be necessary. Rocky soil provides poor ground conductivity. Adding salt solutions improves electrical flow. Refer to TC 6-02.6 for more information about grounding equipment in rocky soil.

## **OPERATIONS IN NUCLEAR ENVIRONMENTS**

G-47. A nuclear environment adversely affects sensitive radio equipment and components. Operators should take measures to protect signal equipment and ensure equipment survivability and availability for future use. The ionization of the atmosphere by a nuclear explosion will degrade communications because of static and the disruption of the ionosphere.

### **TECHNIQUES FOR BETTER COMMUNICATION IN A NUCLEAR ENVIRONMENT**

G-48. Electronic equipment is very susceptible to electromagnetic pulse. *Electromagnetic pulse* is a strong burst of electromagnetic radiation caused by a nuclear explosion, energy weapon, or by natural phenomenon, that may couple with electrical or electronic systems to produce damaging current and voltage surges (JP 3-85). Electromagnetic pulse radiates an intense electromagnetic field. This electromagnetic energy affects unprotected communications equipment, causing disruption and destruction of delicate circuitry and components. The residual ionized cloud also disrupts radio transmissions. The electromagnetic field from an

electromagnetic pulse affects communications over all frequency ranges, but lower frequencies recover from the effects much sooner than the highest frequencies.

G-49. A high-altitude electromagnetic pulse can disable electronic systems as far as 6,000 kilometers (3,720 miles) from the site of the detonation. An electromagnetic pulse can also cause severe disruption and sometimes damage when other weapon effects are absent. The enemy may explode a high-yield nuclear weapon above the atmosphere to knock out narrowband (single-channel) TACSAT without causing any other significant damage. The range of an electromagnetic pulse shortens if the weapon detonates at a lower altitude.

G-50. A typical high-altitude electromagnetic pulse could be up to 1,000 times more intense than a radar beam. A radar beam has enough power to cause biological damage to the human body, including blindness or sterilization. The electromagnetic pulse is broad spectrum and extends from extremely low frequency into the extremely ultrahigh frequency band. A likely electromagnetic pulse effect would be temporary disruption of communications. This can occur even without permanent damage. This delay could give an enemy enough of an advantage to change the outcome of a battle.

G-51. Narrowband (single-channel) TACSAT radios incorporate built-in features and techniques to counter electromagnetic pulse effects. Shielding can further reduce the effects of electromagnetic pulse; Whenever possible, operators should connect cable shields to grounding systems and keep cable length as short as possible. Units assess the probable direction of the nuclear blast, consider available shielding, then select unit and equipment locations to reduce exposure.

G-52. Electronic equipment not in use should remain disconnected and stored in sealed shelters or other shielded enclosures to protect against electromagnetic pulse. This reduces the likelihood of the simultaneous damage to all equipment and provides a source of backup components to reinstall affected systems. If communications equipment cannot be disconnected, the unit should consider acquiring additional communications equipment to provide operational spares in the event of a nuclear hazard.

## **MAINTENANCE IN A NUCLEAR ENVIRONMENT**

G-53. Installers and operators should shield and properly ground wire and cable and keep cable length as short as possible. Operators should connect cable shields to the grounding system, whenever possible. Effective grounding can reduce electromagnetic pulse effects. Most tactical radios with fully-enclosed metal cases provide adequate protection if external connectors are disconnected. Placing radios in vehicles, vans, and underground shelters may enhance protection.

## Appendix H

# Antenna Selection and Techniques

The antenna is the key component in establishing reliable communications to support operations. This chapter addresses antenna techniques, concepts, terms, ground effects, antenna length, types of antennas, and examples of antenna field repairs.

### SECTION I – OVERVIEW

H-1. All radios, whether transmitting or receiving, require an antenna. The antenna is a key component in establishing reliable communications. The selection, proper installation, and positioning of antennas around command posts is critical. The G-6 or S-6 staff ensures that antennas are more than twice the distance of their height from infrastructure such as power lines. For safety, units avoid placing antennas near sleep areas, tents, and vehicle parking areas. Selection factors to consider include the operational environment, means of employment, planning range, and asset availability.

H-2. When selecting an antenna, the operator should consider—

- Radio type.
- Required coverage.
- Take-off angle versus distance.
- Frequency range or band.

H-3. Identifying the specific conditions in the operational environment provides additional information necessary for determining what type of communications system and antennas to install. Tactical radio networks require antennas that are—

- Reliable.
- Robust.
- Capable of interconnection between networks of the same type.
- Capable of interconnection between networks that are dissimilar.

H-4. Antennas can be a significant hazard. Planners and installers must assess and manage antenna assembly, emplacement, electrical discharge, and physical hazards to maintain safe conditions. Radio planners should assess the following when establishing a tactical radio network:

- Antenna location.
- Antenna selection.
- Environment and terrain conditions.
- Mode of transmission.
- Threat situation.
- Frequency band.
- Antenna masking.
- Electromagnetic warfare system deconfliction.
- Signals intelligence system deconfliction.

## ANTENNA FUNDAMENTALS

H-5. Antennas transmit and receive electromagnetic waves, referred to as radio waves. An understanding of antenna concepts and terms is critical to selecting the right antenna. Planners must have a fundamental knowledge of propagation, transmission, and waveform properties to develop reliable communications systems that meet the commander's requirements.

### FORMING A RADIO WAVE

H-6. Alternating current flows through a conductor. If the length of the conductor is short in comparison to the wavelength, the electric and magnetic fields generally die out within one or two wavelengths.

Lengthening the conductor increases the intensity of the field. Thus, an ever-increasing amount of energy escapes into space.

## RADIATION

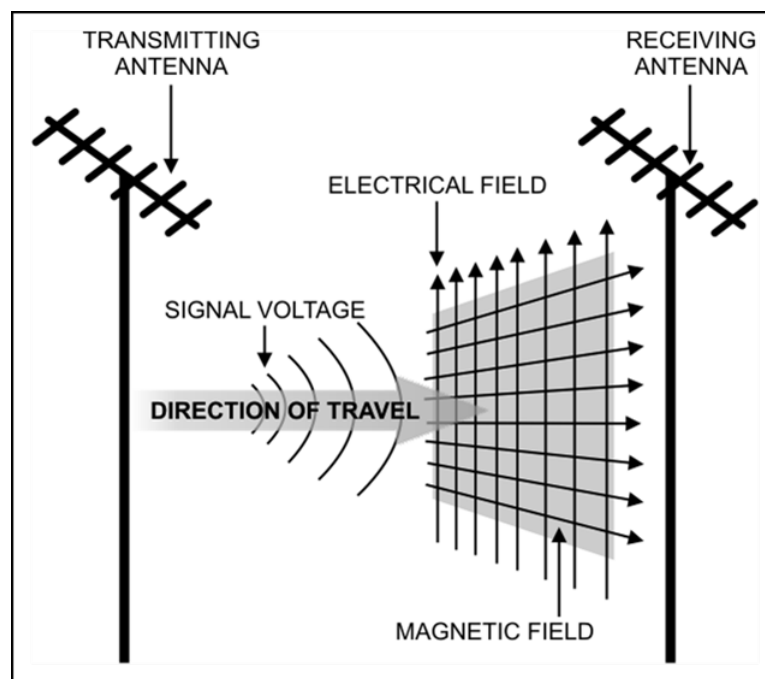
H-7. The alternating flow of electrons impressed on the bottom end of a wire creates electromagnetic energy. A wire connected to a transmitter and properly grounded oscillates electrically, causing the wave to convert the transmitter power into a radio wave. The electrons travel upward on the wire to the top, where they have no place to go, and bounce back toward the lower end. As the electrons reach the lower end in phase, they are in step with the radio energy applied by the transmitter.

H-8. The energy of their motion is strongly reinforced as the electrons travel upward along the wire. This regenerative process sustains the oscillation. The wire is resonant at the frequency at which the source of energy is alternating. The energy stored at any location along the wire is equal to the product of the voltage and the current at that point. The electric current reaches its maximum near the bottom end of the wire.

## RADIATION FIELDS

H-9. RF power delivered to an antenna creates two fields: an induction field, which associates with the stored energy; and a radiation field oriented at a right angle. At the antenna, the intensities of these fields are large and are proportional to the RF power delivered to the antenna.

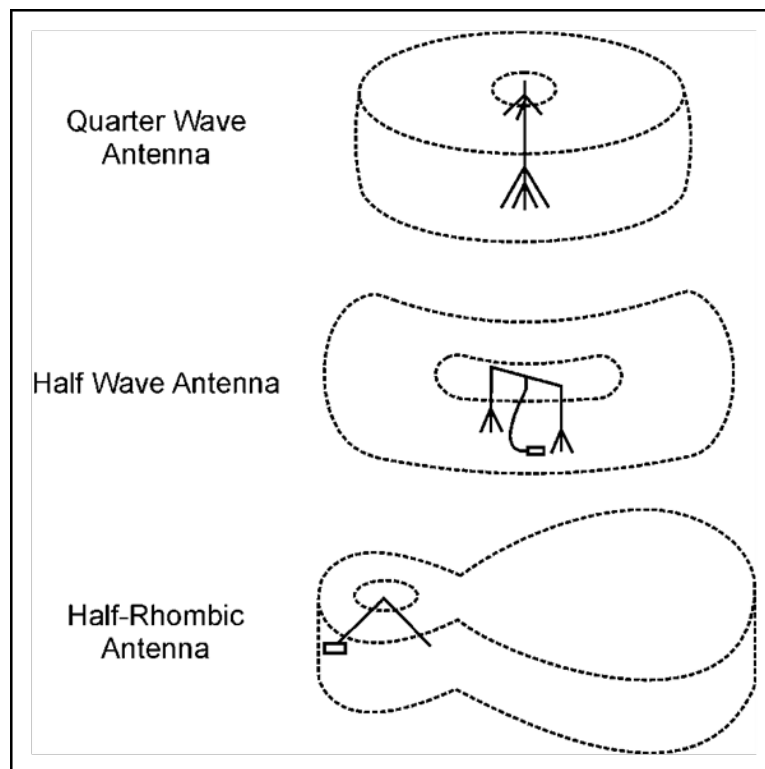
H-10. The electric and magnetic field components radiating from an antenna form an electromagnetic field that transmits electromagnetic energy through free space. A radio wave is a moving electromagnetic field that has velocity in the direction of travel. Figure H-1 is a visual representation of the components of electromagnetic waves.



**Figure H-1. Components of electromagnetic waves**

## RADIATION PATTERNS

H-11. The radiation pattern is a graphical depiction of the relative field strength transmitted from or received by an antenna. The full- or solid-radiation pattern appears as a three-dimensional figure that looks somewhat like a doughnut with a transmitting antenna in the center. Figure H-2 on page 117 is an example of solid antenna radiation patterns. The top figure shows the radiation pattern of a quarter-wave vertical antenna; the middle figure shows the radiation pattern of a half-wave horizontal antenna located one-half wavelength above the ground; and the bottom figure shows the radiation pattern of a vertical half-rhombic antenna.



**Figure H-2. Solid radiation patterns**

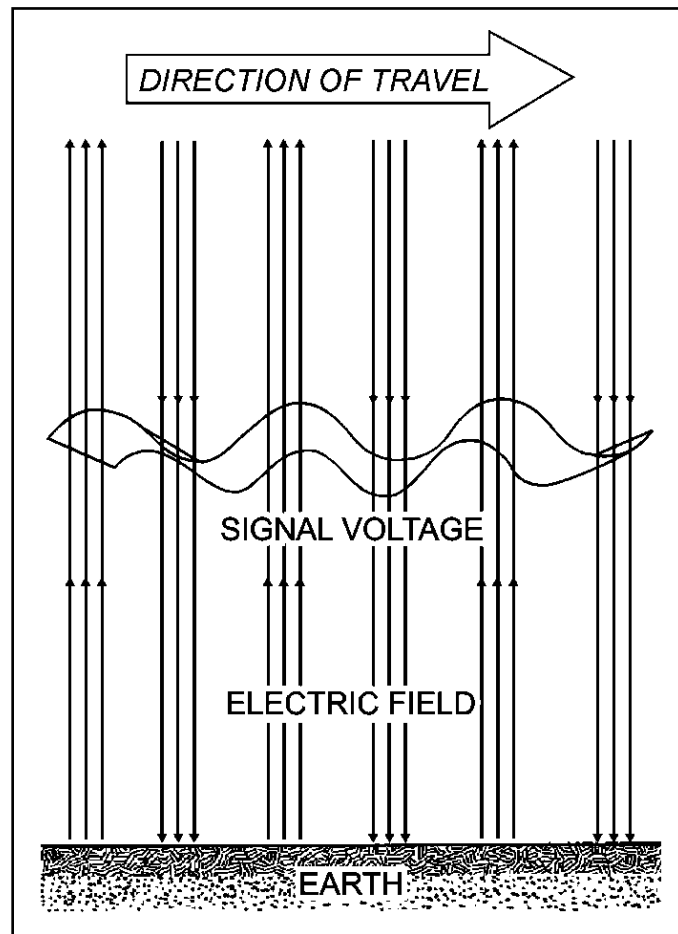
## **POLARIZATION**

H-12. The direction of the lines of force making up the electric field determines the polarization of a radio wave. Polarization can be vertical, horizontal, or elliptical. When a single-wire antenna is used to extract energy from a passing radio wave, maximum pickup result when the antenna is oriented in the same direction as the electric field. Correct polarization ensures the antenna lies in the same direction as the electric field component.

H-13. Horizontal or vertical polarization is satisfactory for VHF or UHF signals. The signal maintains the original polarization produced at the transmitting antenna maintains as the wave travels to the receiving antenna.

## **Vertical Polarization**

H-14. In a vertically polarized wave, the lines of electric force are at right angles to the surface of the Earth. Figure H-3 on page 118 illustrates a vertically polarized wave. Operators should use a vertical antenna for the efficient reception of vertically polarized waves.



**Figure H-3. Vertically polarized wave**

H-15. Vertical polarization is necessary for ground wave transmission at medium and low frequencies. Vertical lines of force are perpendicular to the ground, and the radio wave can travel a considerable distance along the ground with minimal signal loss.

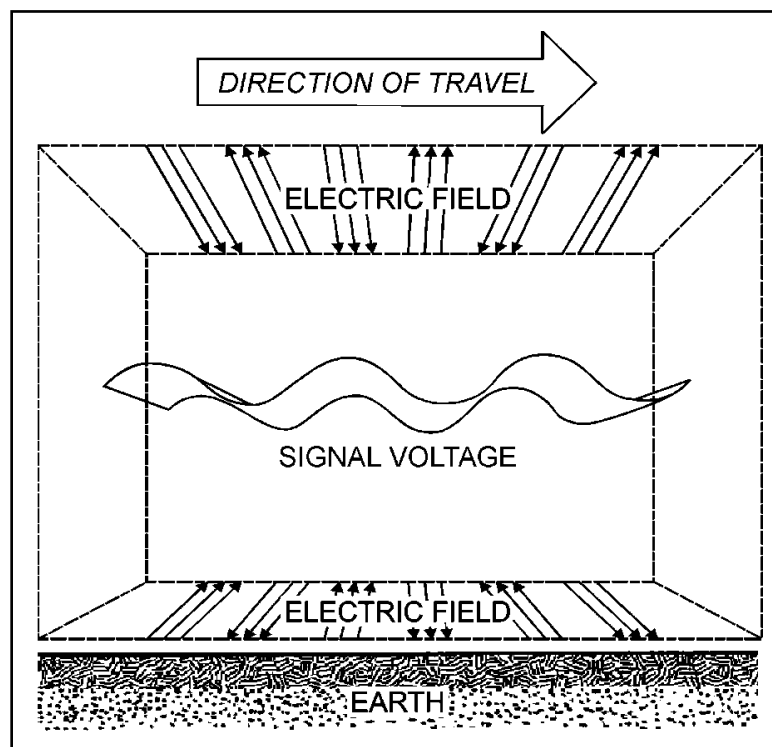
H-16. Vertical polarization provides a stronger received signal at frequencies up to approximately 50 MHz when antenna heights are limited to 3.05 meters (10 feet) over land, as in a vehicular installation.

H-17. Reflections from aircraft flying over the transmission path have a lesser effect on vertically polarized radiation. This factor is advantageous in areas with heavy aircraft traffic.

H-18. Using vertical polarization results in a lesser production and reception of electromagnetic interference from the strong VHF and UHF transmissions used in television and commercial radio broadcasts. This factor is important when locating an antenna in an urban area that has television or radio broadcast stations.

### Horizontal Polarization

H-19. In a horizontally polarized wave, the lines of electric force are parallel to the surface of the Earth. It is essential to use a horizontal antenna to receive horizontally polarized waves. Figure H-4 on page 119 is an example of a horizontal polarized wave.



**Figure H-4. Horizontally polarized wave**

### ***Skywave Transmission***

H-20. With skywave transmission at higher frequencies, it makes little difference whether horizontal or vertical polarization is used. The skywave, after reflection by the ionosphere, arrives at the receiving antenna elliptically polarized. Operators should mount the transmitting and receiving antennas either horizontally or vertically. Horizontal antennas are preferred since they radiate at high angles and have inherently directional properties. A simple horizontal half-wave antenna is bidirectional. This characteristic is useful when minimizing electromagnetic interference from specific directions and masking signals from the threat.

### ***Advantages***

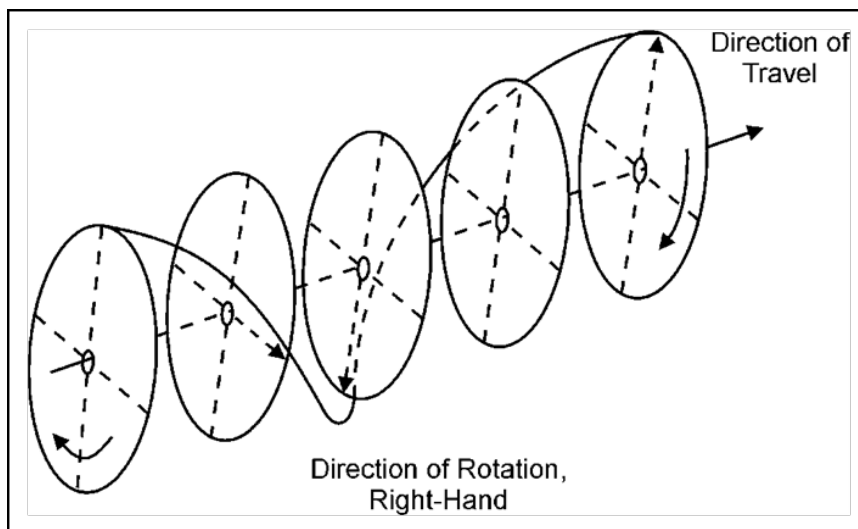
H-21. Although either horizontally or vertically polarized antennas can be used for HF communications, horizontally polarized antennas are preferred because of certain advantages. One advantage is that vertically polarized interference signals, such as those produced by automobile ignition systems and electrical appliances, are minimized by horizontal polarization. In addition, less absorption of radiated energy by buildings or wiring occurs when these antennas are used. Another advantage is that support structures for these antennas are of a more convenient size than those for vertically polarized antennas.

H-22. Horizontal antennas are less likely to pick up man-made interference. Horizontally polarized waves, especially at frequencies above 100 MHz, suffer lower losses when antennas are located near dense forests. Small changes in antenna location do not cause significant variations in the field intensity of horizontally polarized waves.

### ***Elliptical Polarization***

H-23. With elliptical polarization, the electromagnetic field rotates as the radio waves travel through space. Satellites and SATCOM ground terminals use a form of elliptical polarization called circular polarization. Circular polarization describes a wave where the plane of polarization rotates through 360 degrees as it progresses forward. Depending on the phase relationship between the electrical and magnetic fields, this causes rotation either in one direction or the other. Figure H-5 on page 120 is an example of circular

polarization. The rotation can be clockwise (right-hand circular polarization) or counterclockwise (left hand circular polarization). Circular polarization occurs when equal magnitudes of vertically and horizontally polarized waves combine with a phase difference of 90 degrees.



**Figure H-5. Circular polarization**

## DIRECTIONALITY

H-24. Vertical transmitting antennas radiate equally in horizontal directions; vertical receiving antennas accept radio signals equally from all horizontal directions. Stations operating on the same or nearby frequencies may interfere with the desired signal, making reception difficult or impossible. To improve the reception of the desired signal, operators use directional antennas.

H-25. Horizontal half-wave antennas accept radio signals from all directions. The most robust reception occurs from a direction perpendicular to the antenna, while the weakest reception occurs from the direction of the ends of the antenna. When eliminating or reducing interfering signals, it is beneficial to change the antenna installation to ensure that the end of the antenna points directly at the interfering station.

## RESONANCE

H-26. Resonance occurs when an object vibrates (or resonates) at its natural frequency. Resonance can also be achieved by either physically matching the length of the antenna to the wavelength or electrically matching the length of the antenna to the wavelength.

H-27. In a resonant antenna, all radio signals feed to the antenna that radiates. When different frequencies are produced, the sound is louder at its resonant frequency than at all other frequencies. A non-resonant antenna effectively radiates a broad range of frequencies with less efficiency. When using a resonant antenna for a radio circuit, it is important to build a separate antenna for each frequency used.

## RECEPTION

H-28. Electrons in the path of radio waves influence other electrons in the path of radio waves. For example, as a high frequency wave enters the ionosphere, it reflects or refracts back to the Earth by the action of free electrons. When the radio wave encounters the wire or metallic conductors of the receiving antenna, the radio wave's electric field causes the electrons in the antenna to oscillate in step with the wave. The movement of these electrons within the antenna generates a small alternating electrical current, which the radio receiver detects.

H-29. When radio waves encounter electrons that are free to move under the influence of the wave's electric field, the free electrons oscillate in sympathy with the wave. This generates the electric current, which then creates waves of its own called reflected or scattered waves. This process is electromagnetic scattering. Materials that are good electrical conductors reflect or scatter RF energy. Since a receiving antenna is a good

conductor, it too scatters energy. Only a portion of the energy that contacts the antenna converts into received electrical power; the wire reradiates a sizable portion of the total power.

H-30. If an antenna is in a congested urban environment or inside a building, many objects may scatter or reradiate the energy and interfere with reception. For example, the electrical wiring inside a building can strongly reradiate RF energy. If a receiving antenna is positioned near electrical wiring, the reflected energy can cancel the incoming RF signal. When this condition exists, it is beneficial to move the receiving antenna away from electrical wiring where the reflected and direct signals may reinforce rather than cancel each other. Refer to TC 9-64 for more information about radio wave propagation.

## **RECIPROCITY**

H-31. Reciprocity refers to the various properties of an antenna that apply equally, regardless of whether the antenna is used for transmitting or receiving. For example, the more efficient an antenna is for transmitting, the more efficient it is for receiving the same frequency. The directive properties of a given antenna are the same whether used for transmission or reception.

H-32. If using the transmitting antenna as a receiving antenna; the antenna receives best in the same directions in which it produces maximum radiation, at right angles to the axis of the antenna. The minimum signal is received from a transmitter located in line with the antenna wire.

## **IMPEDANCE**

H-33. How much power an antenna radiates depends on the amount of current applied to it. Maximum power emits when there is maximum current flowing. Maximum current flows when the impedance is minimized or when the antenna is resonated so its impedance is pure resistance. The impedance of an antenna is equal to the ratio of the voltage to the current at the point on the antenna where the feed points connect. If the feed point is located at a maximum voltage point, the impedance is as much as 500–10,000 ohms.

H-34. The input impedance of an antenna depends on the conductivity or impedance of the ground. For example, if the ground is a simple stake driven about 1 meter (3.2 feet) into soil of average conductivity, the impedance of the monopole may be 2–3 times the quoted values. Resistance occurs at a point on the antenna circuit where the current is high, and a significant amount of transmitter power dissipates as heat into the ground rather than radiated as intended. It is essential to use Earth, artificial ground, or a counterpoise connection when using a vertical whip or monopole antenna.

## **BANDWIDTH**

H-35. Bandwidth describes how much data is transmitted over a connection in a specified amount of time. In the radio communications process, information changes from speech or writing to a low frequency signal used to modulate a much higher frequency radio signal. Natural laws govern and limit signal transmission. The more words per minute, the higher the modulation frequency needed; which generates a wider bandwidth signal. To transmit and receive the necessary information, the antenna bandwidth must be equal to or wider than the signal bandwidth; otherwise, it limits the signal frequencies causing voice and data to be unintelligible. If the bandwidth is too wide the signal may become degraded due to additional voices and signal to noise ratio.

H-36. Antenna bandwidth describes the range of frequencies over which the antenna can adequately radiate or receive energy. The bandwidth of an antenna reflects the frequency range over which it performs within certain specified limits with respect to impedance matching, gain, and radiation pattern characteristics.

## **ANTENNA DIRECTIVITY AND GAIN**

H-37. The gain of an antenna depends on its design. Transmitting antennas are designed for high efficiency in radiating energy, and receiving antennas are designed for high efficiency in receiving RF energy.

### **Directional Antennas**

H-38. On many radio circuits, transmission is desired between a single transmitter and one receiving station. Directional antennas radiate most of their RF energy in one direction. Directional receiving antennas increase the effective gain in the favored direction and reduce the reception of unwanted noise in signals from the

other directions. Transmitting and receiving antennas should have small energy losses and should be as efficient as radiators and receptors.

### Omnidirectional Antennas

H-39. When employed in forward combat areas, omnidirectional antennas transmit and receive signals equally in all directions. This characteristic provides a similarly strong signal to intended receivers and threat electromagnetic sensors.

## GROUND ANTENNA THEORY

H-40. The ground is an excellent conductor for medium and low frequencies and acts as a large mirror for the radiated energy. As a result, the ground reflects a significant amount of energy that radiates downward from an antenna mounted over it. It is important that the ground have as high conductivity as possible. This reduces ground loss and provides the best possible reflecting surface for the downward-radiated energy from the antenna. Thus, a quarter-wave antenna erected vertically, with its lower end connected electrically to the ground, behaves like a half-wave antenna. Under these conditions, the vertical antenna quarter wavelength and the ground create the half wavelength. The ground portrays the quarter wavelength of radiated energy that reflects on completing the half wavelength. At higher frequencies, artificial grounds constructed of large metal surfaces are common and provide better wave propagation. Figure H-6 is an example of a grounded quarter-wave antenna.

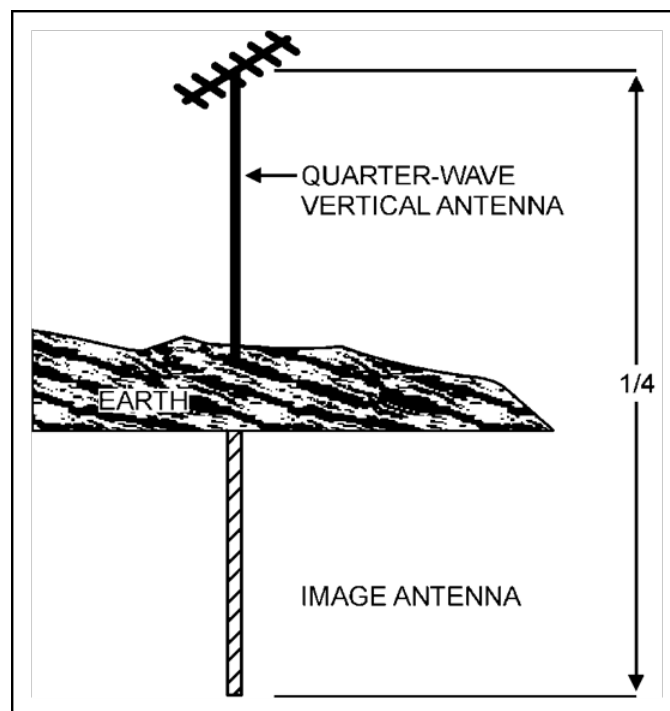


Figure H-6. Grounded quarter-wave antenna

### TYPES OF GROUND

H-41. Operators should establish the ground connection to introduce the least resistance to ground. At higher frequencies, artificial grounds constructed of large metal surfaces are standard.

H-42. Ground connections take many forms, depending on the type of installation and acceptable signal loss. In many simple field installations, one or more metal rods driven into the soil make an adequate ground connection. Operators connect ground leads to existing grounded devices to compensate for unsatisfactory ground. Ground connections typically consist of metal structures or underground pipe systems. In an emergency, a ground connection can be created by forcing one or more metal rods or stakes into the soil.

## SOIL CONDITIONS

H-43. Favorable, less favorable, or unfavorable are the categories used to describe soil ground conditions. When erecting an antenna on soil with low conductivity, the soil can be treated using a variety of grounding techniques to reduce resistance, depending on the specific category and corresponding soil condition.

### Favorable Soil Conditions

H-44. For simple installations, operators can fabricate a single ground rod in the field from a pipe or conduit. It is essential to establish a low resistance connection between the ground wire and the ground rod. It is important to clean the rod. Cleaning can be done by scraping and sanding the ground rod at the desired connection point and installing a clean ground clamp. Operators should solder a ground wire to adjoin the clamp and cover the wire with tape to prevent an increase in resistance caused by oxidation.

### Unfavorable Soil Conditions

H-45. Unfavorable soil conditions occur when there is a high resistance of the soil or a large, buried ground system is not practical. A counterpoise or a ground screen can replace the direct ground connection in unfavorable soil.

#### Counterpoise

H-46. A counterpoise is a device made of wire, erected a short distance above the ground and insulated from it. A counterpoise can replace the usual direct ground connection with high-resistant soil. The counterpoise should be equal in size to, or larger than, the size of the antenna. Figure H-7 is an example of a counterpoise.

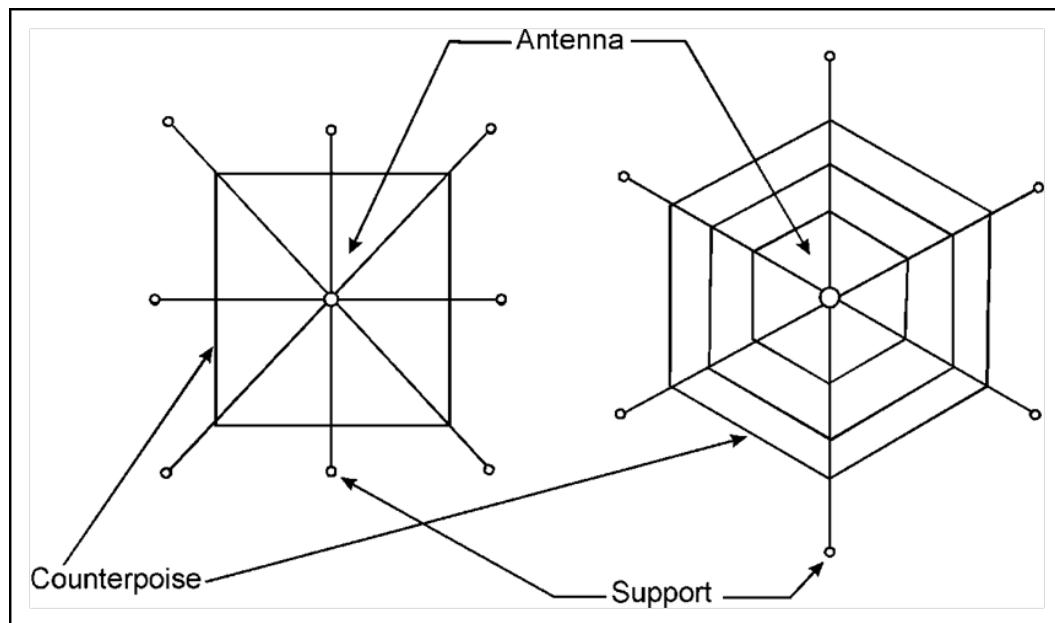


Figure H-7. Wire counterpoise

#### Ground Screen

H-47. A ground screen consists of a large area of metal mesh or screen laid on the surface of the ground under the antenna. There are two advantages to using ground screens. First, the ground screen reduces ground absorption losses that occur when an antenna is installed over an area with poor soil conductivity. Secondly, the height of the antenna can be set accurately. Thus, the radiation resistance of the antenna is determined more accurately.

## **GROUND EFFECTS**

H-48. The ground alters an antenna's radiation pattern. The ground also affects the electrical characteristics of antennas, specifically those mounted relatively close to the ground wavelength. For example, medium and HF antennas have radiation patterns that are entirely different from free-space patterns when elevated above the ground by only a fraction of a wavelength.

## **TYPES OF ANTENNA**

H-49. Tactical antenna design requires that the antennas be rugged and permit mobility with the least possible sacrifice of efficiency. Typically, antennas are mounted using either a single mast or suspended between sets of masts. All tactical antennas must be easy to install. Small antennas can be mounted on the helmet of the radio operator. Large antennas must be accessible to dismantle, pack, and transport.

H-50. The best kinds of wire for antennas are copper and aluminum. In an emergency, operators use any available wire. The exact length of most antennas is critical. An expedient antenna should be the same length as the antenna it replaces.

## **HERTZ ANTENNA**

H-51. A Hertz antenna—also known as a doublet, dipole, an ungrounded, or a half-wave antenna—mounts in a vertical, horizontal, or slanting position and is generally used at frequencies above 2 MHz. With Hertz antennas, the wavelength to which any wire electrically tunes depends directly upon its physical length. The Hertz antenna is center-fed, and its total wire length is equal to approximately half the wavelength of the transmitted signal.

## **MARCONI ANTENNA**

H-52. A Marconi antenna is a quarter-wave antenna with one end usually grounded through the output of the transmitter or the coupling coil at the end of the feed line, which is required for the antenna to resonate. Positioned perpendicular to the Earth, a Marconi antenna is generally useful at lower frequencies. When used on vehicles or aircraft, Marconi antennas operate at high frequencies; and the vehicle or aircraft serves as the ground for the antenna.

H-53. The main advantage of the Marconi antenna over the Hertz antenna is that, for any given frequency, the Marconi antenna is physically much shorter. This is particularly important in all field and vehicular radio installations. Typical Marconi antennas include the inverted L and the whip.

## **ANTENNA LENGTH**

H-54. An antenna has a physical and electrical length; the two are never the same. The reduced velocity of a wave on the antenna, and a capacitive effect—known as end-effect—make the antenna effectively longer electrically than physically. Contributing factors include the ratio of the diameter of the antenna to its length and the capacitive effect of terminal equipment.

H-55. To calculate the physical length of an antenna, operators apply a correction factor of 0.95 for frequencies between 3 and 50 MHz. Table H-1 on page 125 illustrates the antenna length calculation for a half-wave antenna.

Table H-1. Antenna length calculation

|                                                                                                                                                                                                                               |                                  |                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------------------------|
| <b><i>The formula below calculates the half-wavelength and uses a correction factor of 0.95 for frequencies between 3 and 50 MHz. The same formula calculates the height above ground for HF wire antennas.</i></b>           |                                  |                         |
| Length (meters)                                                                                                                                                                                                               | =150 X 0.95/frequency in MHz     | =142.5/frequency in MHz |
| Length (feet)                                                                                                                                                                                                                 | =492 X 0.95/frequency in MHz     | =468/frequency in MHz   |
| The length of a long wire antenna (one wavelength or longer) for harmonic operation is calculated by using the following formula:                                                                                             |                                  |                         |
| Length (meters)                                                                                                                                                                                                               | =150 X (N-0.05)/frequency in MHz |                         |
| Length (feet)                                                                                                                                                                                                                 | =492 X (N-0.05)/frequency in MHz |                         |
| Where N equals the number of half wavelengths in the total length of the antenna.<br>For example, if the number of half wavelengths is 3 and the frequency in MHz is 7, then:<br>Length (meters)=150(N-0.05)/frequency in MHz |                                  |                         |
| =150(3-0.05)/7                                                                                                                                                                                                                | =150 X 2.95/7                    |                         |
| <b>Legend:</b>                                                                                                                                                                                                                |                                  |                         |
| HF                                                                                                                                                                                                                            | high frequency                   |                         |
| MHz                                                                                                                                                                                                                           | megahertz                        |                         |
| <b>Note.</b> For HF antennas, a half wavelength in meters is 143/ <i>f</i> where <i>f</i> is the frequency in MHz. If the frequency is 30 MHz, the wavelength is 5 meters. Often a center-fed half-wave dipole is used.       |                                  |                         |

## ANTENNA ORIENTATION AND LOCATION CONSIDERATIONS

H-56. The orientation of an antenna is critical. Determining the position of an antenna concerning the points of the compass can make the difference between a marginal and good radio circuit. If the azimuth of the radio path is not provided, the azimuth should be determined by the best available means. The accuracy required in determining the azimuth of the path depends on the radiation pattern of the directional antenna.

H-57. If an antenna has a wide beamwidth is vast (for example, a 90-degree angle between half-power points) a 10-degree azimuth error is of little consequence. In transportable operation, the rhombic and V antennas may have such a narrow beam as to require great accuracy in azimuth determination. If a line of a known azimuth is not available, the antenna with the correct azimuth should be erected at the site. A magnetic compass determines the best direction of the path. Figure H-8 on page 126 is an example of a beamwidth measured on relative field strength and corresponding power patterns.

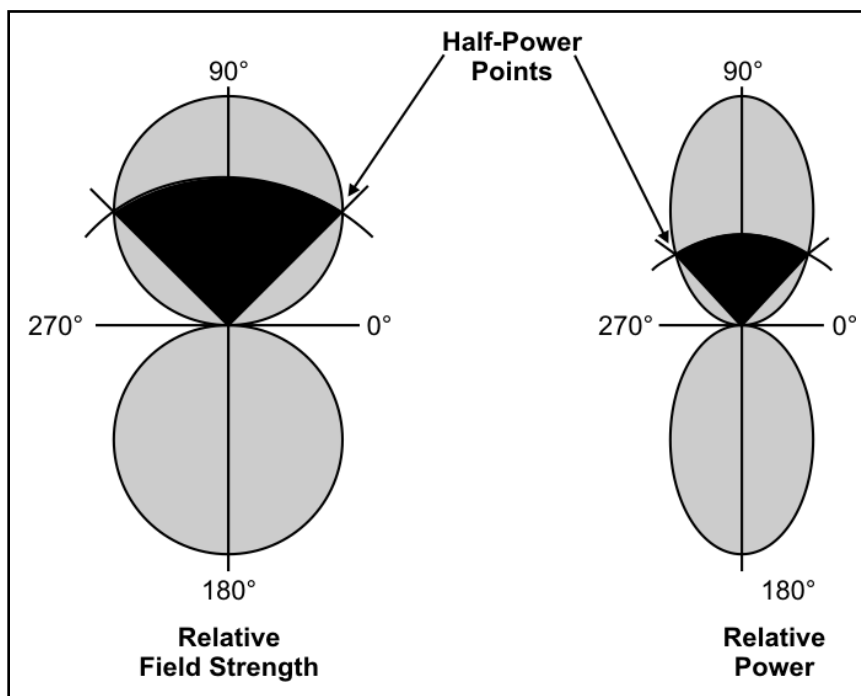


Figure H-8. Beamwidth

H-58. Figure H-9 is an example of a declination diagram. The diagram shows the relationship between the three north points: magnetic north, grid north, and true north. These points are represented on the bottom margin of a topographic map by a declination diagram. It is important to understand the difference between the three north points and understand how to calculate the three north points. Magnetic azimuth is measured using magnetic instruments such as a lensatic compass or an M2 pocket transit, while a grid azimuth is plotted on a map between two points. The points are joined together by a straight line and a protractor is used to measure the angle between grid north and drawn line.

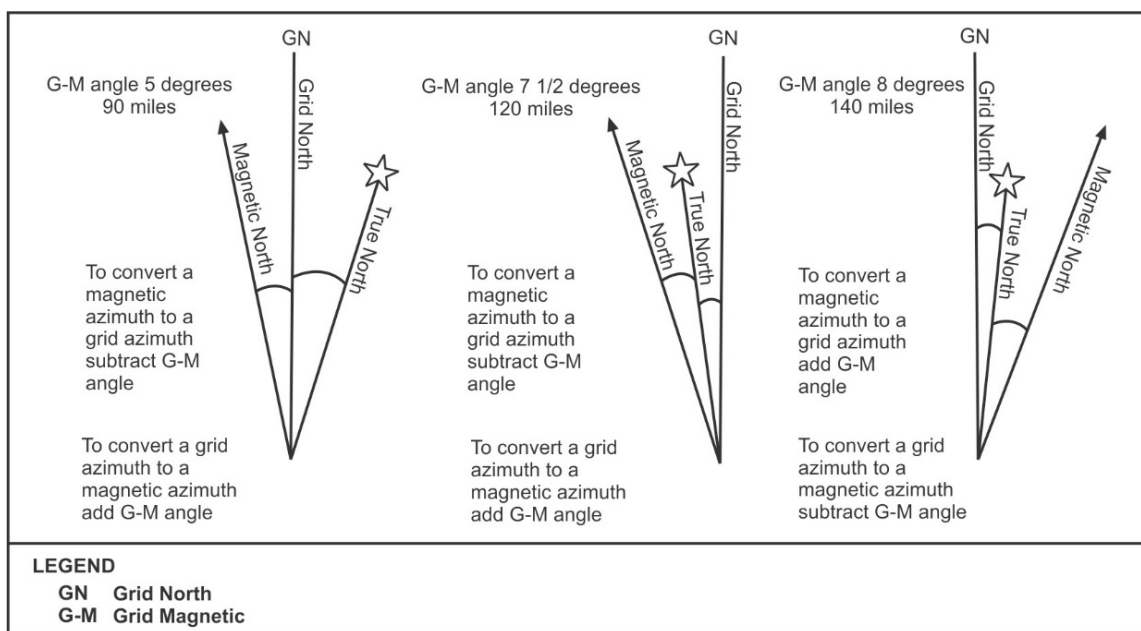


Figure H-9. Example of a declination diagram

## ULTRAHIGH FREQUENCY ANTENNA

H-59. UHF antennas are much shorter and need less space than HF antennas. HF antennas are not used, in general, with VHF and UHF since there are tradeoff considerations between antenna placement and radio performance. Whip antennas are parts of the radio system components. Examples of radio sets with the antennas included are the AN/PRC-119, AN/PRC-148, AN/PRC-152, AN/PRC-150, AN/PRC-158, AN/PRC-162, AN/PRC-163, and AN/PRC-117 series. HF radios may use whip antennas when in VHF (frequency modulation mode); however, the antennas used are usually larger antennas, except for those explained in the matrix shown at table E-2 on page 89. Base station and vehicular antennas for UHF ranges include OE-254, COM-201, COM-201B, OE-303, and vehicular whip (3 meters long), COM-202.

## VERY HIGH FREQUENCY ANTENNAS

H-60. VHF propagation is line of sight in nature and is restricted to shorter-range communication. Atmospheric and man-made influences can adversely impact or limit signal transmission. Heavy vegetation and mountainous terrain are factors that can also reduce the expected signal range. Maximum line of sight distance depends on the height of an antenna above the ground. The higher the antenna, the greater the line of sight distance.

H-61. Due to VHF propagation characteristics as signals travel through the air, obstructions—such as a mountain between the two antennas or dense tree lines—can block or reduce the signal and prevent effective communication. Heavy vegetation weakens radio signals considerably. Therefore, planners must understand the full spectrum of radio communications systems including the effect of antenna types on communication and select and locate antennas to obtain the maximum line of sight range. In vegetation, the antenna should be positioned away from trees that are in the direction of the signal and erected as high as possible. Specific propagation influences that affect VHF transmission and antenna selection include, direct, refracted, reflected, and diffracted rays. Other antenna properties that may improve VHF antenna radiation (omnidirectional, bidirectional, or directional) and efficiency include proper take-off angle, gain, ground screens, and ground planes. See detailed antenna descriptions and properties in section III of this appendix.

H-62. A relationship exists between antenna length and the wavelength of the transmit frequency. This relationship is important when determining the appropriate antenna for a specific frequency or range of frequencies. The wavelength of a radio signal is the distance traveled in the time it takes to complete one wave cycle.

## MULTIBAND ANTENNA

H-63. The Nextgen multiband antenna operates with advanced networking waveform and SINCGARS waveforms to provide transmission reliability for radio operations spanning from lower-lowerband, upper-lowerband, and VHF frequencies. The Nextgen multiband antennas are rugged, and of high quality and durable construction with tri-band design and high isolation between ports for VHF, 30–88 MHz; UHF, 225–450 MHz; and L-band ranges 1200–2110. The UHF dipole and L-band antenna elements are located high up in the whip for maximum range. A ground plane is required for the VHF element. Additional features include—

- Designed for operation on all military platforms.
- Suitable for operation on shelters, mounted on masts or in other permanent installations.
- Includes North Atlantic Treaty Organization flange base.
- Optional GPS installed in the antenna base for L1 and L2 configurations.

## IMPROVEMENT OF MARGINAL COMMUNICATION

H-64. Under certain situations, it may not be possible to orient directional antennas to the correct azimuth of the desired radio path, possibly causing marginal communications. To improve marginal communications, operators should—

- Check, tighten, and tape cable couplings and connections.
- Check antenna adjustment for the proper operating frequency (if possible).
- Change the antenna height.
- Move the antenna a short distance away, and in different locations, from its original location.

- Separate transmitters from receiving equipment, if possible.
- Separate transmitters from power fields.
- Ensure transmission lines do not cross power lines.

H-65. An improvised antenna may change the performance of a radio set. An operator can use a distant station to test whether an antenna is operating correctly. If the signal received from this station is strong, the antenna is operating satisfactorily. If the signal is weak, operators should adjust the height and length of the antenna and transmission line to receive the strongest signal at a given setting on the volume control of the receiver.

H-66. Impedance matching a load to its source is an important consideration in transmission systems. If the load and source are mismatched, part of the power reflects along the transmission line toward the source. This prevents maximum power transfer and can be responsible for erroneous measurements of other parameters. It may also cause circuit damage in high-power applications.

H-67. The power reflected from the load interferes with the forward incident power, creating standing waves of voltages and current to exist along the line. Standing wave maximum-to-minimum ratio directly relates to the impedance mismatch of the load. Therefore, the standing wave ratio provides the means of determining impedance mismatch.

H-68. After site selection and proper antenna orientation, the signal level at the receiver is proportional to the strength of the transmitted signal. Operators can obtain a stronger signal using a high-gain antenna. Using a high quality transmission line as short as possible and adequately matched at both ends reduces signal loss between the antenna and the equipment.

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**Note.** Excessive signal strength may result in threat interception and electromagnetic interference, or interference with adjacent frequencies.

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## **SECTION II – PROPAGATION THEORY**

H-69. The behavior and frequency of radio waves as they travel govern the level and quality of the signal received. Radio waves are also affected by the medium through which they travel and the objects in their path. This section describes three types of propagation—line of sight, ground wave, and skywave and lists additional factors affecting the way in which radio signals or radio waves propagate.

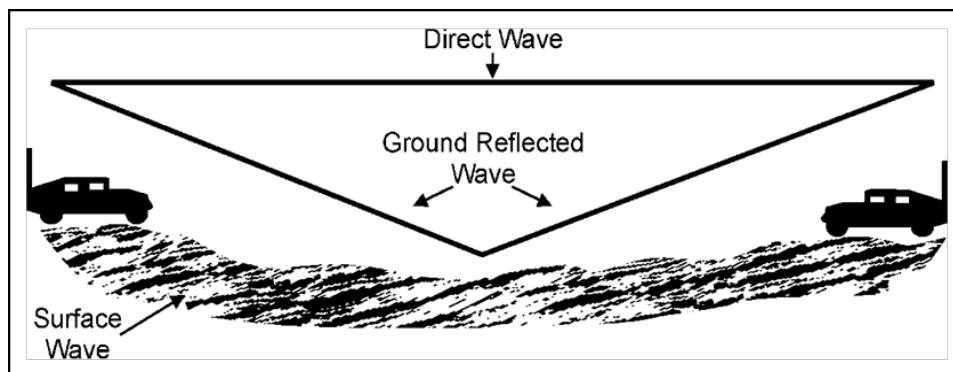
### **LINE OF SIGHT**

H-70. When planning and establishing radio networks, what is between two antennas is an important consideration. Line of sight is the path between two antennas. Line of sight can have the following characteristics—

- No obstacles reside between the two antennas.
- Partial obstructions, such as trees between the two antennas.
- Full obstructions exist between the two antennas.

### **GROUND WAVE**

H-71. Ground wave propagation does not depend on waves that refract from the ionosphere skywaves. The electrical characteristics of the Earth and the amount of diffraction affect ground wave propagation. The strength of the ground wave at the receiver depends on the transmitter power output and frequency, the shape and conductivity of the Earth along the transmission path, and the local weather conditions. Figure H-10 on page 129 shows an example of possible routes for ground waves.



**Figure H-10. Possible routes for ground waves**

### DIRECT WAVE

H-72. The direct wave travels directly from the transmitting antenna to the receiving antenna. The direct part of the wave is limited to the line of sight distance between the transmitting and receiving antennas, and the small distance added by atmospheric refraction and diffraction of the wave around the curvature of the Earth. Increasing the height of the transmitting or receiving antenna, or both, can extend the antenna range.

### GROUND-REFLECTED WAVE

H-73. The ground wave reaches the receiving antenna after reflecting from the surface of the Earth. Cancellation of the radio signal can occur when the ground reflected component and the direct wave component arrive at the receiving antenna at the same time and are 180 degrees out of phase.

### SURFACE WAVE

H-74. The surface wave follows the Earth's curvature. The Earth's conductivity and dielectric constant affect the surface wave.

### CHARACTERISTICS OF GROUND WAVES

H-75. Various frequencies determine which wave component prevails along a given signal path. For example, when the Earth's conductivity is high, and the frequency of a radiated signal is low, the surface wave is the predominant component. For frequencies below 10 MHz, the surface wave is sometimes the predominant component. Above 10 MHz, losses sustained by the surface wave component are significant that the other components, direct and skywave become predominant.

H-76. At frequencies of 30–300 kHz, ground losses are very small, so the surface wave component follows the Earth's curvature. Radio operators use the Earth's curvature for long-distance communication, provided the radio operator has enough power from the transmitter. Use 300 kHz–3 MHz frequencies for long distance communication over seawater and for medium-distance communication over land.

H-77. At the HF range (3–30 MHz), ground conductivity is extremely important, especially above 10 MHz where the dielectric constant or conductivity of the Earth's surface determines how much signal absorption occurs. In general, the signal is strongest at lower frequencies when the surface over which it travels has a high dielectric constant and conductivity.

H-78. The dielectric constant or Earth's surface conductivity determines how much of the surface wave signal energy absorbed or lost. The Earth's surface conductivity is generally poor. Table H-2 on page 130 shows a comparison of the conductivity of varying surface conditions.

**Table H-2. Surface conductivity**

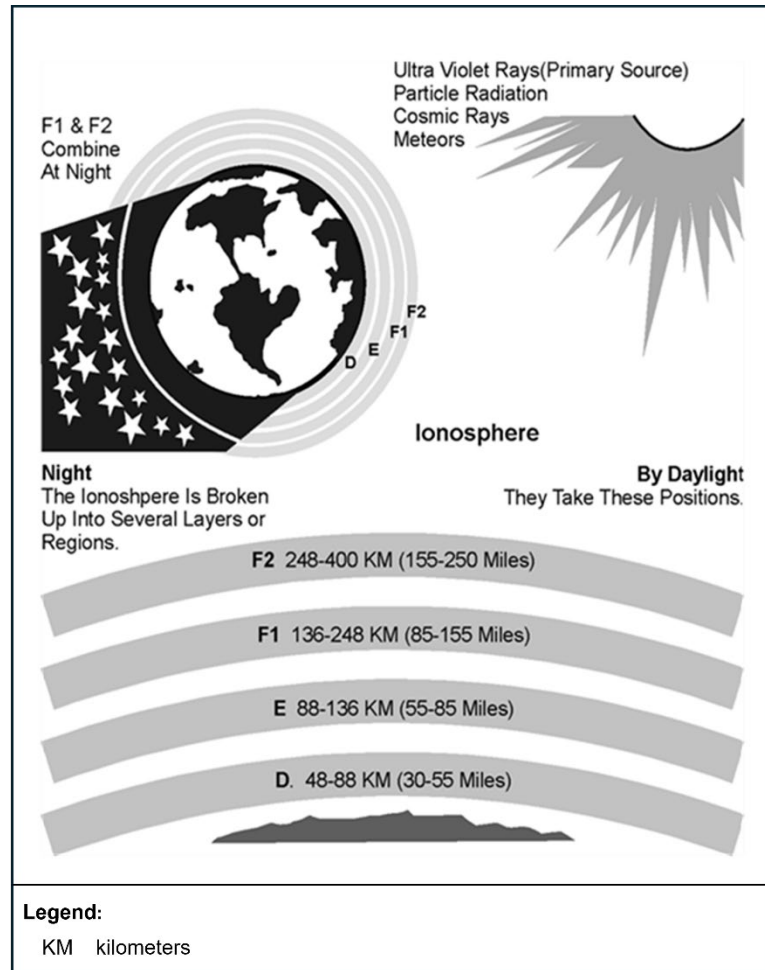
| <b>Surface Type</b>       | <b>Relative Conductivity</b> |
|---------------------------|------------------------------|
| Large body of fresh water | Very good                    |
| Ocean or sea water        | Good                         |
| Flat or hilly loamy soil  | Fair                         |
| Rocky terrain             | Poor                         |
| Desert                    | Poor                         |
| Jungle                    | Very poor                    |

## SKYWAVE

H-79. Radio communications that use skywave propagation depend on the ionosphere to provide the signal path between the transmitting and receiving antennas. The ionosphere has four distinct layers. These regions are labeled D, E, F1, and F2, in the order of increasing heights and decreasing molecular densities. During the day, when the rays of the sun direct toward that portion of the atmosphere, all four regions may be present. During the night, the F1 and F2 regions seem to merge into a single F layer, while the D and E regions fade out. The actual number of regions, their height above the Earth, and their relative intensity of ionization vary continuously. Table H-3 provides a description of the ionosphere regions. Figure H-11 on page 131 shows the average layer distribution of the ionosphere.

**Table H-3. Ionosphere layers**

| <b>Region</b>                          | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D Region                               | Exists only during daylight hours and has little effect in bending the paths of HF radio waves. The main effect of the D region is to attenuate HF waves when the transmission path is in sunlit regions.                                                                                                                                                                                                                                                                                                                        |
| E Region                               | Used during the day for HF radio transmission over intermediate distances (less than 2,400 kilometers [1,500 miles]). At night, the intensity of the E region decreases, and it becomes useless for radio transmission.                                                                                                                                                                                                                                                                                                          |
| F Region                               | Exists at heights up to 380 kilometers (240 miles) above the Earth and is ionized all the time. It has two well-defined layers (F1 and F2) during the day and one layer (F) during the night. At night, the F region remains at the height of about 260 kilometers (170 miles) and is useful for long-range radio communications (over 2,400 kilometers [1,500 miles]). The F2 layer is the most useful of all layers for long-range radio communications, although its degree of ionization varies appreciably from day to day. |
| <b>Legend:</b><br>HF    high frequency |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



**Figure H-11. Average layer distribution of the ionosphere**

H-80. The movement of the Earth around the sun and changes in the sun's activity contribute to ionospheric variations. These variations are either regular, predictable, or irregular, which occurs from the abnormal behavior of the sun. Table H-4 shows the periodic variations of the ionosphere.

**Table H-4. Regular variations of the ionosphere**

| <i><b>Variation</b></i> | <i><b>Description</b></i>                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------|
| Daily                   | Caused by the rotation of the Earth.                                                                         |
| Seasonal                | Caused by the north and south progression of the sun.                                                        |
| 27-day                  | Caused by the rotation of the sun on its axis.                                                               |
| 11-year                 | Caused by the sunspot activity cycle going from maximum through minimum back to maximum levels of intensity. |

H-81. Planners should anticipate the status of the four regular variations. Irregular variations have a degrading effect, at times blanking out communications, and cannot be controlled or compensated for. Table H-5 on page 132 lists some irregular variations of the ionosphere.

**Table H-5. Irregular variations of the ionosphere**

| <b>Variation</b>                                   | <b>Description</b>                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sporadic E                                         | When excessively ionized, the E layer often blanks out the reflections back from the higher layers. It can also cause unexpected propagation of signals hundreds of miles beyond the normal range. This effect can occur at any time.                                                                                   |
| Sudden Ionospheric Disturbance                     | Coincides with a bright solar eruption and causes abnormal ionization of the D layer. This effect causes total absorption of all frequencies above approximately 1 MHz. It can occur without warning during daylight hours and can last from a few minutes to several hours. When it occurs, receivers seem to go dead. |
| Ionospheric Storms                                 | During these storms, skywave reception above approximately 1.5 MHz shows low intensity and is subject to a type of rapid blasting and fading called flutter fading. May last from several hours to several days, and usually extend over the entire Earth.                                                              |
| <b>Legend:</b><br>MHz                    megahertz |                                                                                                                                                                                                                                                                                                                         |

H-82. Sunspots generate bursts of radiation that cause high levels of ionization. More sunspots equate to greater ionization. During periods of low sunspot activity, frequencies above 20 MHz tend to be unusable because the E and F layer's weak ionization reflects the signal back to Earth. At the peak of the sunspot cycle, it is unusual to have worldwide propagation on frequencies above 30 MHz.

H-83. Primarily, the ionization density of each layer determines the range of long-distance radio transmissions, the higher the frequency, the greater the ionization density required to reflect radio waves back to Earth. The upper E and F regions reflect the higher frequencies because they are the most highly ionized. The D region, which is the least ionized, does not reflect frequencies above approximately 500 kHz. In each ionized region, there is an upper frequency limit known as critical frequency, which radio waves sent vertically upward reflect back to Earth.

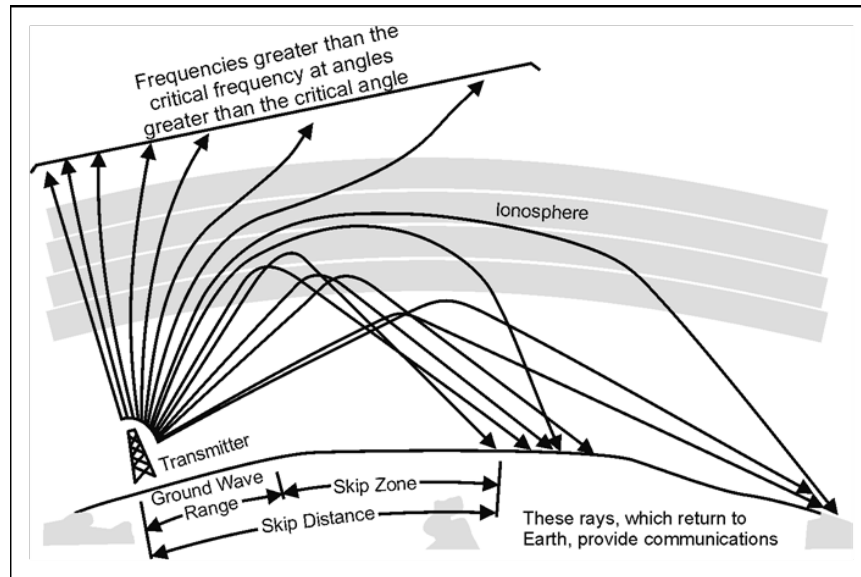
H-84. Radio waves directed vertically at frequencies higher than the critical frequency pass through the ionized layer of the Earth's atmosphere into outer space. All radio waves directed vertically into the ionosphere at frequencies lower than the critical frequency reflect back to Earth.

H-85. Generally, radio waves used in communications directed toward the ionosphere at some oblique angle called the angle of incidence. Radio waves at frequencies above the critical frequency reflect to Earth if transmitted at angles of incidence smaller than the critical angle, called the critical angle. At the critical angle, and at all angles larger than the critical angle, the radio waves pass through the ionosphere if the frequency is higher than the critical frequency.

## TRANSMISSION PATHS

H-86. Skywave propagation refers to those types of radio transmissions that depend on the ionosphere to provide signal paths between transmitters and receivers. Skip distance refers to the distance from the transmitting antenna to the location, where the skywaves first return to Earth. The skip distance depends upon the angle of incidence, the operating frequency, and the height and density of the ionosphere.

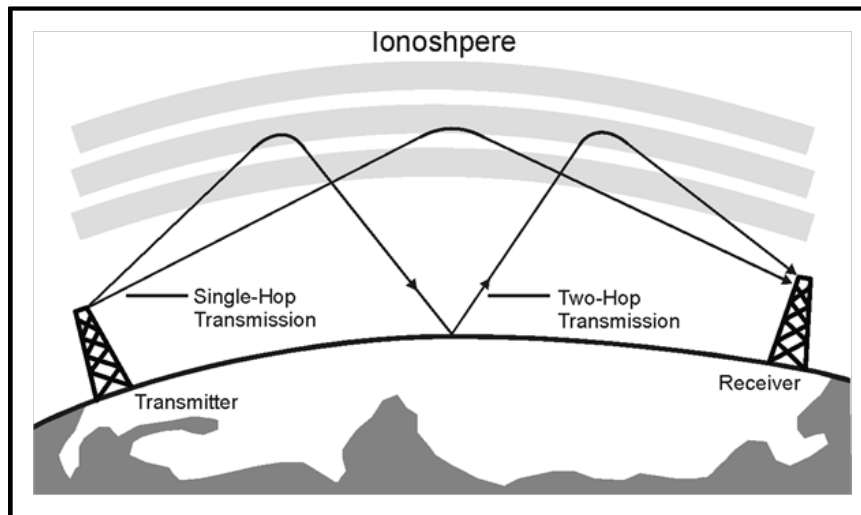
H-87. The antenna height and the operating frequency affect the angles at which transmitted radio waves strike and penetrate the ionosphere and then return to Earth. Control this angle of incidence to obtain the desired area of coverage. Lowering the antenna height increases the angle of transmission. This provides broad and even signal patterns in an area the size of a typical corps. Near-vertical incidence skywave uses near-vertical transmission paths. Raising the antenna height lowers the angle of incidence. Figure H-12 on page 133 is an example of skywave transmission paths.



**Figure H-12. Skywave transmission path**

H-88. Lowering the angle of incidence can produce a skip zone in which does not allow the receipt of a usable signal. This bounds the area by the outer edge of usable ground wave propagation and the point nearest the antenna at which the skywave returns to Earth. Low angles of incidence make long-distance communications possible.

H-89. When a transmitted wave reflects back to the surface of the Earth, the Earth absorbs part of its energy. The remainder of its energy is reflected into the ionosphere and back to Earth. This means of transmission (alternately reflecting the radio wave between the ionosphere and the Earth) referred to as hops, enables the reception of radio waves at great distances from the point of origin. Figure H-13 is an example of skywave transmission hop paths.



**Figure H-13. Skywave transmission hop path**

## Fading

H-90. Fading is the periodic increase and decrease of received signal strength. Fading occurs from a radio signal received over a long distance path in the HF range. There is little common knowledge of how to reduce or eliminate fading effects. Fading associated with skywave paths is a significant detriment to reliable

communication. Those responsible for communications circuits rely on raising the transmitter power or increasing antenna gain to overcome fading. Sections often do not work and seldom improve reliability. Only when the signal level fades below the background noise level for an appreciable fraction of time will an increase in transmitter power or antenna gain yield an overall circuit improvement. Choosing the correct frequency and using transmitting and receiving equipment properly ensure a reliable and robust receive signal, even when low power is used.

### **Maximum Usable Frequency and Lowest Usable Frequency**

H-91. The maximum usable frequency is the maximum frequency at which a radio wave returns to Earth at a given distance when using a given ionized layer and a transmitting antenna with a fixed angle of radiation. The monthly median of the daily highest frequency predicted for skywave transmission over a path in a specific hour of the day. The maximum usable frequency is always higher than the critical frequency because the angle of incidence is less than 90 degrees.

H-92. If the distance between the transmitter and the receiver increases, the maximum usable frequency also increases. Radio waves lose some of their energy through absorption by the D region, and a portion of the E region of the ionosphere, on specific transmission frequencies. The total absorption is less, and communications more satisfactory, when using higher frequencies up to the level of the maximum usable frequency.

H-93. The absorption rate is most significant for frequencies ranging from approximately 500 kHz–2 MHz during the day. During the night, the absorption rate decreases for all frequencies. As the frequency of transmission over any skywave path increases from low to high, a frequency occurs at which the received signal overrides the level of atmospheric and other radio noise interference. This is the lowest usable frequency because frequencies lower than these are too weak for useful communications. The lowest usable frequency also depends on the power output of the transmitter and the transmission distance. When the lowest usable frequency is greater than the maximum usable frequency, no skywave transmission is possible. The spectrum manager uses Spectrum XXI to identify optimum frequency groupings.

### **OTHER FACTORS AFFECTING PROPAGATION**

H-94. In the VHF and UHF ranges (30–300 MHz and beyond), the presence of an object may produce strong reflections that arrive at the receiving antenna in a way that cancels the desired signal and renders communication ineffective.

H-95. Planners should choose receiver locations away from airfields due to possible adverse electromagnetic interference from signals bouncing off the aircraft. Planners and operators should avoid locating transmitters and receivers near airfields or near the midpoint of the propagation path of frequencies above 20 MHz.

H-96. Many other factors may affect the propagation of a radio wave. Hills, mountains, buildings, water towers, tall fences, and even other antenna can affect the condition and reliability of a propagation path. The conductivity of the local ground or body of water can significantly alter the strength of the transmitted or received signal. Energy radiation from the Sun's surface also dramatically affects conditions in the ionosphere and alters the characteristics of long-distance propagation at 2–30 MHz.

### **Path Loss**

H-97. Radio waves become weaker as they travel outwards from the transmitter. Path loss refers to the ratio of the received power to the transmit power. Line of sight paths at VHF and UHF require relatively little power since the total path loss at the radio horizon is only about 25 decibels greater than the path loss over the same distance in free space (absence of ground). This additional loss results from some energy reflected from the ground, canceling part of the direct wave energy. This is unavoidable in almost every practical case. The total path loss for a line of sight path above average terrain varies with the following factors:

- Total path loss between transmitting and receiving antenna terminals.
- Frequency.
- Distance.
- Transmitting antenna gain.
- Receiving antenna gain.

- Reflected Waves.

H-98. It is often possible to communicate beyond the normal line of sight distance by exploiting the reflection from a tall building, nearby mountain, or water tower. If the transmitting and receiving antennas can see the top portion of a structure or hill readily, it may be possible to achieve efficient communications by directing both antennas toward the point of maximum reflection. If the reflecting object is very large in terms of a wavelength, the path loss, including the reflection, can be very low.

H-99. If a structure or hill exists adjacent to a line of sight path, reflected energy may either add to or subtract from the energy arriving from the direct path. If the reflected energy arrives at the receiving antenna with the same strength as the direct signal but has the opposite phase, the signals cancel, and communication is impossible. Under the same condition exists, if both signals arrive in phase they add and double the effective signal strength. These two conditions represent destructive and constructive combinations of the reflected and direct waves.

H-100. Reflection from the ground at the common midpoint between the receiving and transmitting antennas may also arrive constructively or destructively. Generally, in the VHF and UHF range, the reflected wave is out of phase (destructive) concerning the direct wave at vertical angles less than a few degrees above the horizon. Since the ground is not a perfect conductor, the amplitude of the reflected wave seldom approaches that of the direct wave. Thus, even though the two arrive out of phase, complete cancellation does not occur. Improvements may result from using vertical polarization rather than horizontal polarization over line of sight paths because there tends to be less phase difference between direct and reflected waves. The difference is usually less than 10 decibels, in favor of vertical polarization.

## Diffraction

H-101. Unlike a ship passing beyond the visual horizon, a radio wave does not fade out completely when it reaches the radio horizon. A small amount of radio energy travels beyond the radio horizon by a process called diffraction. Diffraction also occurs when a light source held near an opaque object, casts a shadow on a surface behind it. Near the edge of the shadow, a narrow band displays which is neither completely light nor dark. The transition from total light to total darkness does not occur abruptly but changes smoothly as the light diffracts.

H-102. A radio wave passing over either the curved surface of the Earth or a mountain ridge behaves in much the same fashion as a light wave. For example, people living in a valley below a high, sharp, mountain ridge can often receive a television station located many miles below on the other side. Mountain ridges diffract and bend television station waves downward in the direction of the town. The energy decays very rapidly as the angle of propagation departs from the straight line of sight path. Typically, a diffracted signal may undergo a reduction of 30–40 decibels by a bend of only 1.5 meters (5 feet) by a mountain ridge. The actual amount of diffracted signal depends on the shape of the surface, the frequency, the diffraction angle, and many other factors.

## Refraction

H-103. Refraction is the bending of a wave as it passes through air layers of different density. In semitropical regions, a layer of air 5–100 meters (16–328 feet) thick with distinctive characteristics may form close to the ground, usually the result of a temperature inversion. For example, on an unusually warm day after a period of rain, the Sun may heat up the ground and create a layer of warm, moist air. After sunset, the air a few meters above the ground cools very rapidly while the moisture in the air close to the ground serves as a blanket for the remaining heat. After a few hours, a significant difference in temperature may exist between the air near the ground and the air at the height of 10–20 meters (32.8–65.6 feet), resulting in a marked difference in air pressure. Thus, the air near the ground is considerably denser than the air higher up. This condition may exist over an area of several hundred square kilometers or an extended area of land near a seacoast. When an air mass forms, it usually remains stable until dawn, when the ground begins to cool, and the temperature inversion ends.

H-104. When a VHF or UHF radio wave launches in air mass, it may bend or become trapped and forced to follow the inversion layer. This layer then acts as a duct between the transmitting antenna and a distant receiving site. The effects of ducting frequently occur during the year in locations where television or VHF

modulation stations received over paths of several hundred kilometers. The total path loss within a duct is usually very low and may exceed the free space loss by only a few decibels.

H-105. It is possible to communicate over long distances utilizing tropospheric scatter. At altitudes of a few kilometers, the air mass has a varying temperature, pressure, and moisture content. Small fluctuations in tropospheric characteristics at high altitude create blobs. In a blob, the temperature, pressure, and humidity are different from the surrounding air. If the difference is large enough, it may modify the refractive index at VHF and UHF. A random distribution of these blobs exists at various altitudes. If a high-power transmitter greater than 1 kilowatt and a high-gain (10 decibels or more) antenna are used, enough energy may be scattered from these blobs down to the receiver to make reliable communication possible over several hundred kilometers. Communications circuits employing this mode of propagation must use susceptible receivers and the form of diversity to reduce the effects of the rapid and deep fading. Scatter propagation is usually limited to path distances of less than about 500 kilometers (311 miles).

## Noise

H-106. Noise consists of all undesired radio signals, man-made or natural. Noise masks and degrades useful information reception. The radio signal's strength is of little importance if the signal power is greater than the received noise power. The signal to noise ratio is an important factor in a receiving system. Increasing receiver amplification cannot improve the signal to noise ratio since signal and noise amplifies and signal to noise ratio remains unchanged. Typically, receivers have more than enough amplification.

H-107. Natural noise has two principal sources: thunderstorms (atmospheric noise) and stars (galactic noise). Both sources generate sharp pulses of electromagnetic energy over all frequencies. The pulses propagate according to the same laws as man-made signals and receiving systems must accept them and the desired signal. Atmospheric noise is dominant from 0–5 MHz; galactic noise is most important at higher frequencies. Low frequency transmitters must generate very strong signals to overcome noise. Strong signals and strong noise mean that the receiving antenna does not have to be large to collect a usable signal. A 1.5-meter (4.9-foot) tuned whip antenna adequately delivers all the signals received at frequencies below 1 MHz.

H-108. Man-made noise appears wherever electric power is used. Each source is small, but there are so many that together they can completely overwhelm a weak signal that would be above the natural noise in rural areas. Man-made noise is troublesome when the receiving antenna is near the source, but being near the source produces exploitable noise wave characteristics. Waves near a source tend to be vertically polarized. A horizontally polarized receiving antenna generally receives less noise than a vertically polarized antenna.

H-109. Any conductors near the source, including the antenna, transmission line, and equipment cases, induce man-made noise currents. If the antenna and transmission lines should balance with respect to the ground, then the noise voltages will balance and cancel with respect to the receiver input terminals zero voltage across terminals and this noise will not receive. Near perfect balance is difficult to achieve, but any balance may help.

H-110. Other ways to avoid man-made noise are to locate troublesome sources and turn them off or move the receiving system away from them. Moving at least one kilometer (.6 miles) away from a busy street or highway significantly reduces noise. Broadband receiving antennas are convenient because they do not require tuning to each working frequency, sometimes a narrowband antenna can make the difference between communicating and not communicating. The HF band is now so crowded with users that electromagnetic interference and noise, not signal strength, are the main reasons for poor communications. A narrowband antenna rejects strong interfering signals near the desired frequency and helps maintain good communication.

## SECTION III – VERY HIGH FREQUENCY AND ULTRAHIGH FREQUENCY ANTENNA TECHNIQUES

### ANTENNA TYPE

H-111. Army forces employ a wide array of VHF and UHF antennas to expand communications over a more extended range and transmit communications signals to more users. Because VHF and UHF antennas are small, it is possible to use multiple radiating elements to form arrays, which provide a considerable gain in a

direction(s). An array is an arrangement of antenna elements, usually, dipoles used to control the direction in which most of the antenna power radiates.

## ANTENNA SELECTION

H-112. The VHF portion of the radio spectrum extends from 30–300 MHz and the UHF range reaches from 300–3,000 MHz (0–3 GHz). Both frequency ranges are extremely useful for short-range communications—less than 50 kilometers (31 miles). This includes point-to-point, mobile, air-to-ground, and general-purpose communications. Wavelengths at these frequency ranges are considerably shorter.

H-113. In the VHF and UHF portion of the spectrum, there are sub-frequency bands for specific uses, such as VHF aircraft band, UHF aircraft band, and public communications.

## ANTENNA SELECTION PROCEDURE

H-114. VHF and UHF antennas, such as the whip antenna, can be mounted on vehicles allowing units to extend communication over an extended range. Planners also consider weather during antenna selection and employment. High winds may adversely impact large antennas that comprise some signal systems degrading communication during a mission. When selecting VHF or UHF antennas, planners should consider—

- Propagation.
- Desired coverage and characteristics.
- Construction material.
- Site selection.
- Power and power supply.
- Polarization.
- Gain.
- Directivity.

## POLARIZATION

H-115. In many countries, frequency modulation and television broadcast in the VHF range using horizontal polarization. One reason is that it reduces ignition interference, which is mainly vertically polarized. Mobile communications often use vertical polarization or two reasons. First, the vehicle antenna installation has physical limitations, and second, there is no interruption in the reception or transmission as the vehicle changes heading.

H-116. Using directional antennas and horizontal polarization, when possible, reduces man-made noise interference in urban locations. Choose horizontal polarization only where an antenna height of many wavelengths is possible. Ground reflections tend to cancel horizontally polarized waves at low angles. Use only vertically polarized antennas when the antenna is less than 10 meters (33 feet) above the ground, or when desiring omnidirectional radiation or reception.

## GAIN AND DIRECTIVITY

H-117. VHF and UHF above 30 MHz antenna gain are extremely important for several reasons. Assuming the same antenna gain and propagation path, the received signal strength drops as the frequency is increased. At VHF and UHF, more of the received signal is lost in the transmission line than is lost at HF. A 10–20 decibel loss is uncommon in a 30-meter (98.4 feet) length of the coaxial line at 450 MHz.

H-118. At frequencies below 30 MHz, system sensitivity is usually limited by receive noise rather than by noise external to the antenna. Generally, wider modulation or signal bandwidths are employed in VHF and UHF transmissions rather than in the HF range. Since system noise power is directly proportional to bandwidth, additional antenna gain is necessary to preserve a usable signal to noise ratio.

H-119. VHF and UHF antenna gain and directivity aids security by restricting how much power radiated in unwanted directions. Receiver sensitivity is generally poorer at VHF and UHF except for high quality state-of-the-art receivers. Obstructions buildings, trees, hills may seriously decrease the signal strength available to the receiving antenna because VHF and UHF signals travel a straight line of sight path.

H-120. Obtaining communications reliability over difficult VHF and UHF propagation paths requires considerable attention to the design of high-gain directive antenna arrays. Unlike HF communications, the

shorter VHF and UHF wavelengths support walkie-talkie transceivers and simple mobile transmissions units. Communicating or receiving with such devices over distances beyond 1 or 2 kilometers requires maximum antenna gain at the base station or fixed end of the link.

H-121. An array provides directivity the ability to concentrate radiated energy into a beam directed toward the intended receiver. Arrays of resonant elements, half-wave dipoles, constructed of rigid metal rods or tubing or copper foil laid out or pasted on a flat non-conducting surface. Directing power helps to increase the range of the communications path and tends to decrease the likelihood of the interception of jamming from hostile radio stations. Highly directive antennas place an added burden on the radio operator to ensure that the antenna is properly oriented.

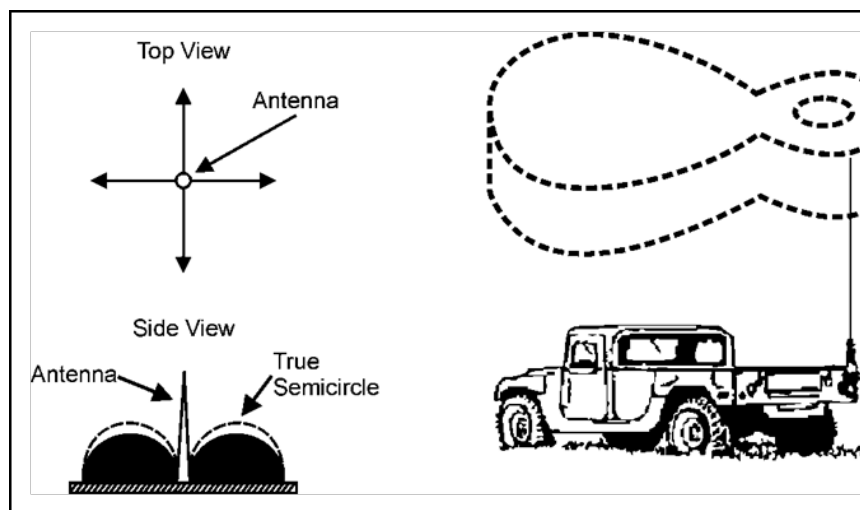
## WHIP ANTENNA

H-122. Whip antennas for VHF tactical radio sets are usually 4.5 meters (15 feet) long. A HF vehicular whip antenna has a planning range of 400–4,000 kilometers (250–2,500 miles).

H-123. Lightweight portable frequency modulation radios use two whip antennas: 0.9-meter (2.9-foot) semi-rigid steel tape antenna, and a 3-meter (10-foot) multi-section whip antenna. These antennas are shorter than a quarter wavelength to ensure they are a practical length. A quarter wavelength antenna for a 5.0 MHz radio would be over 14 meters (46 feet) long. An antenna tuning unit, either built into the radio set or supplied with it, compensates for the missing length of the antenna. The tuning unit varies the electrical length of the antenna to accommodate a range of frequencies.

H-124. Whip antennas on tactical radio sets radiate equally in all directions on the horizontal plane. Since stations in a radio net lie in random directions and change their positions frequently, the radiation pattern is ideal for tactical communications.

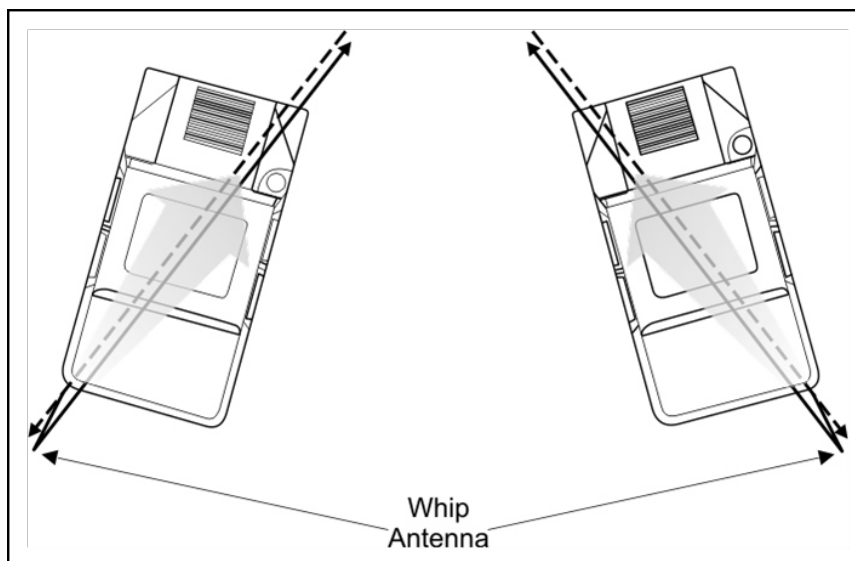
H-125. When a whip antenna mounts on a vehicle, the metal of the vehicle affects the operation of the antenna. Thus, the direction in which the vehicle is facing may also affect transmission and reception, particularly of distant or weak signals. Tactical whip antennas are electrically short, vertical, base loaded types, fed with a non-resonant coaxial cable of about 52 ohms impedance. Figure H-14 is an example of a whip antenna.



**Figure H-14. Whip antenna**

H-126. To attain efficiency comparable to that of a half-wave antenna, operators should position the height of the tactical whip antenna to a quarter wavelength antenna. This is not always possible; therefore, operators should use the loaded whip instead. The loading increases the electrical length of the vertical radiator to a quarter wavelength. The ground, counterpoise, or any conducting surface large enough, supplies the missing quarter-wavelength of the antenna.

H-127. Figure H-15 shows the best direction for whip antennas mounted on vehicles. A vehicle with a whip antenna mounted on the left rear side of the vehicle transmits its strongest signal in a line running from the antenna through the right front side of the vehicle. Similarly, an antenna mounted on the right rear side of the vehicle radiates its strongest signal in a direction toward the left front side. In some cases, the best reception from signals traveling in the direction shown by the dashed arrows on the figure. In some cases, the best path for transmission can be determined by driving the vehicle in a small circle until the best position is located. Normally, the best direction for receiving from a distant station is also the best direction for transmitting to the same station.



**Figure H-15. Whip antennas mounted on a vehicle**

H-128. A whip antenna mounted on a vehicle may require leaving the antenna fully extended for instant use while the vehicle is in motion. The base-mounted insulator of the whip is fitted with a coil spring attached to a mounting bracket on the vehicle. The spring base allows horizontal tie down of the whip antenna when the vehicle is in motion, and when driving under low bridges or obstructions. Even in the vertical position, if the antenna hits an obstruction, the whip usually will not break because the spring base absorbs most of the shock.

H-129. The energy leaving a whip antenna travels downward and the ground reflects the energy with little loss. Raising the whip antenna increases the transmitting and receiving range. When raised, the efficiency of a whip antenna decreases because it is further from the ground. Therefore, when using a whip antenna at the top of a mast, the operator should supply an elevated substitute for the ground plane.

### **DANGER**

**When fully extended and in motion, drivers must avoid antenna contact with overhead power lines. Death or serious injury can result if a vehicular antenna strikes a high-voltage transmission line. When tied down, operators should ensure the antenna tip protector is in place.**

## **BROADBAND OMNIDIRECTIONAL ANTENNA**

H-130. The OE-254 is a broadband omnidirectional antenna that contains a balanced to the unbalanced transformer and supports frequency hopping. Table H-6 on page 140 shows the planning ranges for the OE-254 antenna. The OE-254 antenna—

- Operates in the 30–88 MHz range without any physical adjustments.

- Has an input impedance of 50 ohms unbalanced with an average voltage standing wave ratio of 3:1 or less, at RF power levels up to 350 watts.
- Can be assembled and set up by one individual.
- Meets the broadband and power handling requirements of the frequency hopping multiplexer.

**Table H-6. OE-254 planning ranges**

| <i>Terrain</i>                | <i>High Power</i>          | <i>Low Power (Nominal Conditions)</i> |
|-------------------------------|----------------------------|---------------------------------------|
| <b>OE-254 to OE-254</b>       |                            |                                       |
| Average Terrain               | 57.9 kilometers (36 miles) | 19.3 kilometers (12 miles)            |
| Difficult Terrain             | 48.3 kilometers (30 miles) |                                       |
| <b>OE-254 to Vehicle Whip</b> |                            |                                       |
| Average Terrain               | 48.3 kilometers (30 miles) | 12.9 kilometers (8 miles)             |
| Difficult Terrain             | 40.3 kilometers (25 miles) |                                       |

## COM 201B ANTENNA

H-131. The COM 201B antenna is a commercial, VHF and UHF, vertically polarized, omnidirectional antenna. Its unique design supports quick employment to support different applications. The COM 201B antenna has a tripod leg structure that allows mounting the antenna directly on the ground or on a standard communications mast. The COM 201B tripod legs provide the ground wave radiation of the signal transmitted and are necessary for proper operations. The COM 201B supports quick assembly and disassembly for transport. This makes it ideal for situations where there is not enough time to erect an OE-254 antenna.

H-132. The COM 201B antenna's ease of operation makes it ideal for use as a field expedient antenna or mounting to a vehicle when requiring more elevation. The eye fitting at the top of the antenna facilitates suspending it from buildings or trees when a mast is not available and when greater height is needed. The COM 201B antenna has the following characteristics and capabilities—

- Provides range extending from 30–88 MHz.
- Operates vertically polarized with no requirement for active tuning.
- Uses input impedance of 50 ohms unbalanced with an average voltage standing wave ratio of 2.5:1, at RF power levels up to 200 watts.
- Supports maximum power towards the horizon with a typical antenna gain of zero.
- Uses minimal personnel to assemble and erect and can be assembled by one Soldier.
- Uses compact assemblages that allow it to be stored in a space less than 36 inches long by 10 inches in diameter.

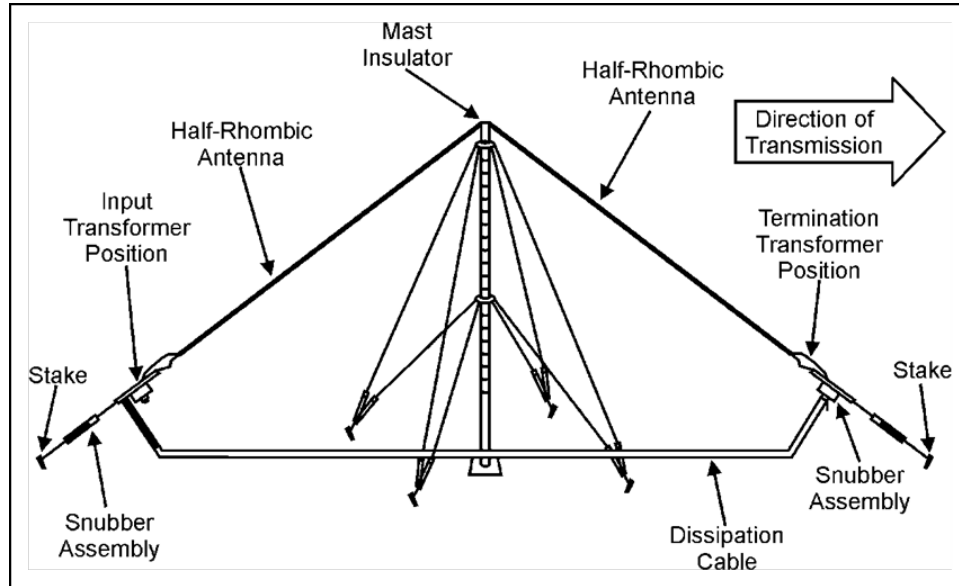
## QUICK ERECT ANTENNA MAST

H-133. The quick erect antenna mast, AB 1386/U elevates tactical radio antennas to a maximum height of 10 meters (33 feet) for more reliable communication over extended ranges. The quick erect antenna mast uses the same antenna elements and RF cable as the OE-254 antenna. The mast can deploy and operate in a ground or vehicular (wheeled or tracked) configuration. The antenna erects in less than 8 minutes with 2 Soldiers and in 15 minutes with one Soldier.

## VERY HIGH FREQUENCY HALF-RHOMBIC ANTENNA

H-134. The VHF half-rhombic antenna is a high gain, lightweight, directional antenna that operates over the frequency range of 30–88 MHz. The antenna and ancillary equipment, guy wires, stakes, tools, and mast sections fit in a carrying bag for manpack or vehicular transportation. The OE-303, VHF half-rhombic antenna is a vertically polarized antenna that, when used with VHF radios, extends the range of transmission considerably, and provides some electromagnetic protection. The half-rhombic antenna, when properly employed, decreases VHF radio susceptibility to jamming and enhances the communications ranges of radios. This effect occurs by directing the maximum signal strength in the direction of the desired friendly unit.

H-135. The planning range for the OE-303 is equivalent to the planning range of the OE-254. The OE-303 antenna is used with the AB-1244 mast assembly, consisting of 12 tubular mast sections—five lower-mast sections, one mast transition adapter, five upper-mast sections, and antenna adapter, a mast base assembly, and different ancillary equipment. Stabilization of the erected mast assembly uses a two-level, four-way guying system. Figure H-16 shows an example of the OE-303 VHF half-rhombic antenna.



**Figure H-16. OE-303 very high frequency half-rhombic antenna**

- The OE-303 antenna handles RF power levels up to 200 watts. It matches nominal 50-ohm impedance with a voltage standing wave ratio of no more than 2:1, over the entire frequency range of the antenna. The antenna has the following characteristics and capabilities:
  - Erectable in a space of 53.3 meters (175 feet) diameter, or less, depending upon the frequency.
  - Mountable on any structure at least 15.2 meters (50 feet) high.
  - Azimuth directional change within 1 minute.
  - Transported by manpack or tactical vehicle when fitted into a package.
  - Operation with the four-port frequency hopping multiplexer.

H-136. The OE-303 antenna supports special applications and is a task assigned as required. Its primary use is on command and intelligence networks to a higher headquarters. Units that habitually operate over extended distances from parent units and special task units use the OE-303 half-rhombic antenna.

## FIELD EXPEDIENT ANTENNAS

H-137. Field expedient antennas are generally used for HF communication. Field expedient antennas that can be used for VHF include the half-rhombic, half wave dipole, and improvised vertical half-wave.

## DIRECTIONAL ANTENNAS

H-138. Operators should use directional antennas whenever possible to minimize the probability of detection by threat electromagnetic sensors. This may require the prefabrication of VHF directional antennas since these are not available in the current Army inventory. Operators can change antenna polarization on systems where distance is not an issue. A horizontally polarized ground wave has less signal loss than a vertically polarized ground wave if antenna heights exceed treetop levels or other horizontal energy absorbers.

## MAST ASSEMBLIES

H-139. If possible, Operators should stack antennas in the null space of another vertical antenna. The radiation pattern of a vertical antenna has a deep energy void directly overhead 90 degrees. Figure H-17

shows possible antenna stacks. Operators should configure the mast assemblies to mount two OE-254 broadband antennas using vertical separation.

H-140. Both dual-antenna mast assemblies must provide at least 12 decibels or greater antenna isolation (at 30 MHz) over that obtained using the same distance horizontal separation. Operators can take advantage of the lateral wave propagation of vertical antennas. Energy transference is negligible on a receiving antenna in this null space. Early fabrication of mounting devices may be required to achieve antenna stacking.

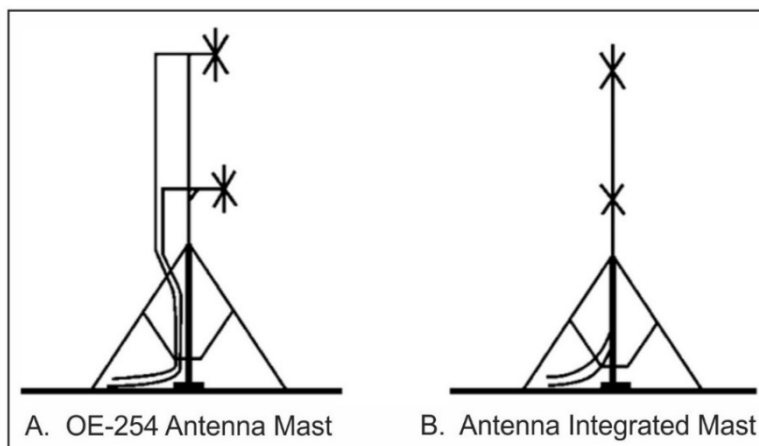


Figure H-17. Possible antenna stacks

## GROUNDING

H-141. Operators must properly ground electronic equipment in the command post to electromagnetic interference-and reduce electrical shock hazards. Another option is to counterpoise the antenna. The wires used in the counterpoise should be either a half wavelength or a full wavelength long for best results. Greater directional gain can be achieved by orienting the counterpoise wires in the direction of the receiving antenna.

## SECTION IV – SATELLITE COMMUNICATIONS ANTENNAS

### ANTENNA CHARACTERISTICS

H-142. The AN/PSC-5 family of radios are the most widely used narrowband (single-channel) TACSAT radios. The AN/PSC-5 provides a line of sight communications with the AS-3566 antenna and long-range SATCOM with the AS-3567 and AS-3568 antennas.

**Note.** The Army plans to divest its inventory of the AN/PSC-5 family of radios replacing them with the AN/PRC-158 and AN/PRC 162 by fiscal year 28 to support cryptographic modernization.

H-143. Low, medium, and high gain are distinct features of the AS-3566, AS-3567, and AS-3568 antennas. Additional characteristics to consider when selecting antennas for SATCOM include—

#### LOW GAIN ANTENNA

H-144. Antenna characteristics for the AS-3566 are—

- Frequency range (line of sight)—30–400 MHz.
- DAMA—225–400 MHz.
- Non-DAMA—225–400 MHz.
- Polarization—directional.
- Power capability—determined by terminating resistor.
- Azimuth (bearing)—directional.

## MEDIUM GAIN ANTENNA

- H-145. Antenna characteristics for the AS-3567 are—
- Frequency range—225–399.995 MHz.
  - Beamwidth—85 degrees.
  - Orientation—Directional.
  - Elevation—0–90 degrees.
  - Input impedance—50 ohms.
  - Voltage standing wave ratio—1.5:1.
  - Gain—6 decibels (225–318 MHz); 5 decibels (318–399.995 MHz).

## HIGH GAIN ANTENNA

- H-146. Antenna characteristics for the AS-3568 are—
- Frequency range—240–400 MHz.
  - Beamwidth—77 degrees.
  - Orientation—Directional.
  - Elevation—0–90 degrees.
  - Azimuth—180 degrees.
  - Input impedance—50 ohms.
  - Voltage standing wave ratio—1.5:1.
  - Gain—8 decibels (240–318 MHz) ; 6 decibels (318–400 MHz).
  - Power—up to 140 watts.

## ANTENNA SELECTION

H-147. An important consideration in addressing line of sight equipment is the antenna elevation with respect to the path terrain. Planners should choose sites that exploit natural elevations.

H-148. An important consideration in siting over-the-horizon systems is the antenna horizon screening angles at the terminals. As the horizon angle increases, the transmission loss increases, resulting in a weaker signal.

H-149. The effect of the horizon on transmission loss is significant. Except where the consideration of one or more other factors outweighs the effect of horizon angles, the site with the most negative angle should be the first choice. If no sites with negative angles exist, the site with the smallest positive angle should be the first choice.

H-150. The horizon angle can be determined by using a transit to survey each site and sighting along the circuit path. The on-site survey determines the visual horizon angle. Though the radio horizon angle is slightly different from the visual horizon angle, the difference is generally insignificant.

H-151. During the survey, planners measure the horizon angle between the tangent at the proposed location of the antenna and a direct line of sight to the horizon. The tangent line is a right angle to a plumb line at the antenna site. If the line of sight to the horizon is below the tangent line, the horizon angle is negative.

H-152. Trees, buildings, hills, or the curvature of the Earth can partially block UHF signals (obstruction loss). To avoid signal loss due to obstruction requires a clearance between the direct line of sight and the terrain. Planners use path profile plots to determine if adequate clearance in line-of-sight systems.

H-153. SATCOM sets weakening or producing distorted signals may result from operating near steel bridges, water towers, power lines, or power units. The presence of congested air-traffic conditions on the proximity of microwave equipment can result in significant signal fading.

## SECTION V – FIELD EXPEDIENT ANTENNA REPAIR

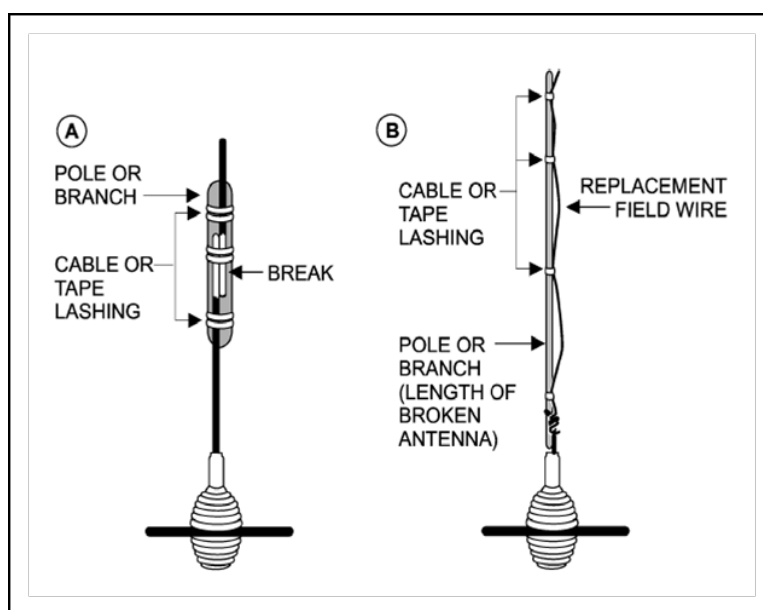
H-154. Broken or damaged antennas may cause poor communications or communications failure. If a spare antenna is available, the operator should replace the damaged antenna. When a spare is not available, the user may have to construct an emergency field expedient antenna.

H-155. Knowing the structural makeup of an antenna aids in assessing its condition and determining what material is required to reconstruct faulty or malfunctioning parts. The following paragraphs provide examples of field repair techniques for tactical radio antennas.

## WHIP ANTENNA

H-156. If a whip antenna is broken into two sections, an operator can temporarily repair the antenna by rejoining the sections. The operator must remove the paint at the join and clean the sections to ensure a good electrical connection. After placing the sections together, the operator secures them with a pole or branch, and then lashes them together with bare wire or tape above and below the break. Figure H-18 (example A) shows a field repair for a broken whip antenna using a pole or branch.

H-157. Severely damaged whip antennas may be temporarily repaired with a length of field wire the same length as the original antenna. The operator removes the insulation from the lower end of the field wire, twist the conductors together, inserts them into the antenna base connector, and secures them with a wooden block. The operator should use either a pole or a tree branch the same length as the broken antenna to support the antenna wire. Figure H-18 (example B) shows a field repair for a broken whip antenna using field wire.

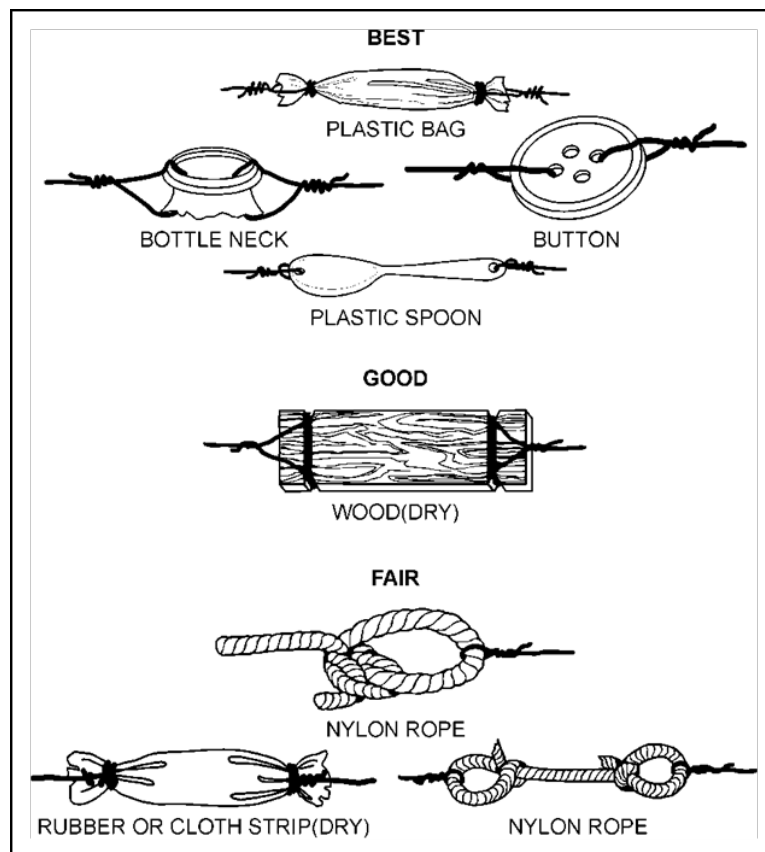


**Figure H-18. Field repair of broken whip antennas**

H-158. Emergency repair of a wire antenna may involve the repair or replacement of the wire used as the antenna or transmission line. It may also require the repair or replacement of the assembly used to support the antenna. When one or more antenna wires are broken, reconnecting the broken wires can repair the antenna. To do this, the operator lowers the antenna to the ground, cleans the ends of the wires, and twists the wires together. When possible, the operator should solder the connection and reassemble.

H-159. Damaged antenna supports may require using a substitute for the damaged support and if properly insulated may consist of any material of adequate strength. Radiating elements not correctly insulated may short field antennas to ground and be ineffective.

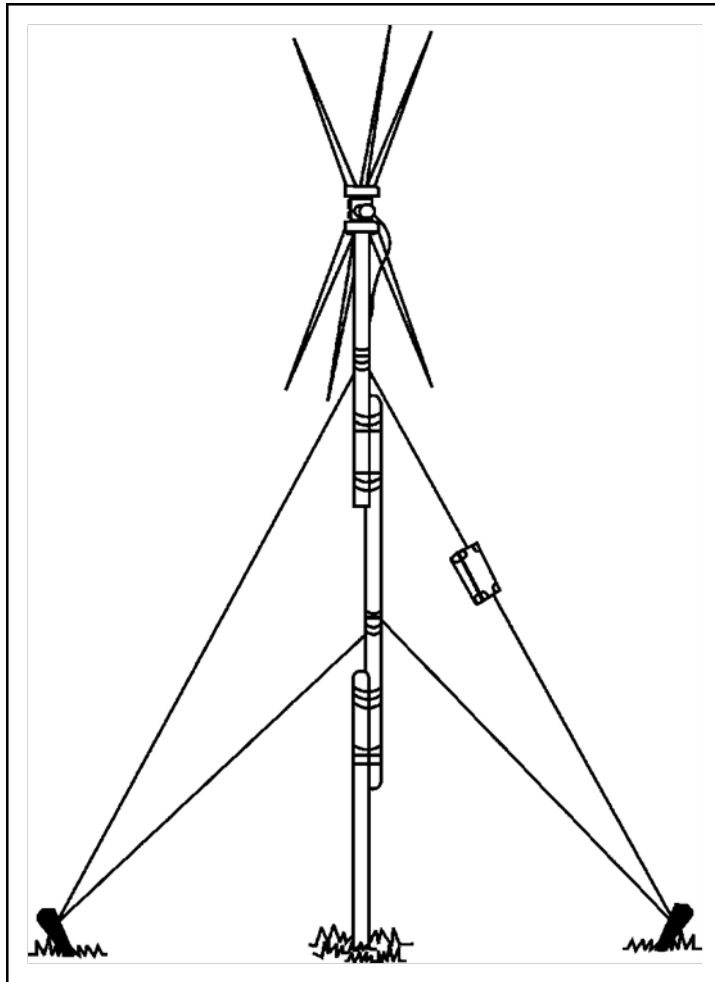
H-160. Operators may use many common items as expedient field insulators. Plastic or glass, including plastic spoons, buttons, bottlenecks, and plastic bags are the best insulator. Wood and rope also act as insulators, though they are less effective than plastic or glass. The antenna wire should only touch the antenna terminal and be electronically isolated from all other objects other than the supporting insulator. Figure H-19 on page 145 shows examples of field expedient antenna insulators.



**Figure H-19. Examples of field expedient antenna insulator**

## ANTENNA GUYS

H-161. Antenna guys stabilize the supports for an antenna. They are usually made of wire, manila rope, or nylon rope. An operator can repair a broken guy rope by tying the broken ends together. If the rope is too short after tying, the operator should add another piece of rope or a piece of dry wood or cloth to lengthen it. Broken guy wire can replace another piece of wire. To ensure guy wires do not affect the operation of the antenna, the operator should cut the wire into several short lengths and connect the pieces with insulators. Figure H-20 on page 146 shows an example of antenna guy lines and mast repaired with wood.



**Figure H-20. Antenna guy lines and masts repaired with wood**

#### **ANTENNA MASTS**

H-162. If an antenna mast is broken, it may be replaced with another mast of the same length. When long poles are not available as replacements, an operator can overlap and lash shorter poles together with rope or wire to provide a pole of the required length.

## Appendix I

# Tactical Radio Platforms

This appendix addresses the tactical radio platforms and associated waveforms forces employ at echelons corps and below across the range of military operations.

### SECTION I – OVERVIEW

I-1. Army tactical radio platforms traditionally supported a triad of voice and data communications. This triad included SINCGARS, HF, and narrowband (single-channel) TACSAT. The advent of software-defined radios and technological advances such as networking waveforms allow tactical radios to perform multiple communications functions, enable data services down to the user level, and offer far greater flexibility and resilience on the battlefield. Emerging technology allows tactical network operations using a mix of legacy radios and software-defined radios as well as the integration of radios on multiple platforms and components. These radio platform components are a combination of the hardware, which includes antenna, batteries, end-user devices, and the software design inherent in the radio operating system. This diverse asset conglomerate extends communications options and provides greater operational flexibility to suit the commander's requirements.

### SINGLE-CHANNEL GROUND AND AIRBORNE RADIO SYSTEM

I-2. The SINCGARS remains the primary FM voice communications for dismounted and mounted operations as well as joint air-ground coordination. Since its initial fielding in 1986, SINCGARS continues to provide the predominant line of sight radio capability in the DOD, serving as a mainstay for command and control. SINCGARS operates in the 30–88 MHz range providing a frequency hopping capability extending over 2,000 separate channels. SINCGARS supports manpack, handheld, vehicle mounted, and aviation platform mounted systems. Improvements in second and emerging generations of SINCGARS waveforms on software-defined radios will ensure that SINCGARS will remain a significant communications capability.

I-3. The SINCGARS system improvement program and ASIP increased the performance of the SINCGARS family of radios employing RT-1523. SINCGARS ASIP increases capabilities to support the tactical networking environment, specifically improved data capability, manpower and personnel integration requirement compliance, and flexible interfaces with other systems. SINCGARS ASIP incorporates waveform improvements, optimizations, and enhancements including—

- ASIP waveform updates.
- Noisy channel avoidance algorithm optimizations.
- Time of day enhancement.
- End of message enhancement.

I-4. All ASIP radios are capable of remote operation controlled by another ASIP radio up to 4 kilometers (2.4 miles) away, using a twisted pair wire, typically WD-1 or WF-16. To remote a radio, operators use an external two-wire adapter to interface between the radio and the wires. Performance of the remote control feature occurs between the dismounted RT and the vehicular amplifier adapter, or between two dismounted RTs. Another host controller can control the ASIP radio using the external control interface when the ASIP integrates as part of a more extensive system.

I-5. ASIP extends the functionality of the SINCGARS internet controller card and ancillary equipment. The internet controller card is in the right-hand side of the vehicular amplifier adapter and only needed when the SINCGARS system is operating in packet mode.

I-6. The vehicular amplifier adapter mounted internet controller card is the predominant communications router for the tactical maneuver platforms participating in a SINCGARS enabled tactical networking environment. The internet controller card uses commercial Internet Protocol services to deliver unicast and multicast data packets. The internet controller card has improved microprocessor with increased memory buffer size and ethernet interface. This feature allows connection of multiple internet controller cards for sharing and dissemination of information in a local area network configuration. ASIP components include—

- SINCGARS internet controller card.
  - Supports data flow and permits a sufficient flow of information.
  - Enables sharing of voice and data over the same network via the packet mode.
  - Generates messages and Internet Protocol addresses of intended receiver from the host computer.
- SINCGARS ancillary equipment.
  - Allows intravehicular and external remote radio control.
  - Provides securable remote control away from the radio.

I-7. Remote control devices, data fill devices, variable storage transfer devices, and the vehicular intercommunications system are the main categories of ancillary equipment associated with SINCGARS. Remote control devices include—

- Intravehicular remote-control unit.
  - Operates with an integrated COMSEC radio.
  - Operates with non-integrated COMSEC radio.
- Securable remote-control unit.
  - Remotes a single radio up to four kilometers (2.4 miles).
  - Provides an intercommunication function from the radio to the terminal unit and vice versa.

I-8. COMSEC and data adapter devices attach directly to the securable remote-control unit for secure communications over the transmission line and optimal interface with digital data terminals. The securable remote-control unit replaced the AN/GRA-39.

## BEYOND LINE OF SIGHT AND SATELLITE-ENABLED PLATFORMS

I-9. Beyond line of sight and satellite capabilities facilitate secure, reliable information exchange and is an indispensable element for supporting dispersed troops that encounter infrastructure or environmental challenges. Along with familiarity with system features and requirements, Signal personnel and teams should understand key propagation concepts including environmental factors and their effect on transmission before employing beyond line of sight communications. These concepts have a profound impact on antenna choice and site selection and can severely degrade connectivity when not taken into consideration. Also essential is knowledge on specialized satellite terminals and communications equipment such as high-gain antennas, powerful transmitters, and sensitive receivers.

I-10. To effectively employ space-based capabilities, signal planners must understand system capabilities and how to coordinate access. Users must also adhere to international and national regulatory policies for the use of RF and satellite communications.

## HIGH FREQUENCY SINGLE-SIDEBAND

I-11. High frequency (3–30 MHz) radios predominantly support ground forces for both line of sight and beyond line of sight communications using radios with advanced single-sideband and ALE functionality. Army HF use is growing in part due to spectrum availability, updates in Army communications architectures, and improved signaling capability of modernized HF tactical radios. Some Army and joint aircraft still operate legacy HF systems, while the use of wideband HF continues to evolve for data networks and shows significant potential for Army use. HF single-sideband radios provide a terrestrial beyond line of sight tactical radio capability often employed as a backup system to counter jamming or overcome distance limitations without relying on SATCOM. HF radio wave propagation requires an understanding of the radio's power capabilities, and the frequency used on the radio. The planning distances often dictates which type of antenna to use (whip, near-vertical incidence skywave, or doublet).

## SATELLITE COMMUNICATIONS

I-12. U.S. dependence on SATCOM is well recognized by threats who continue to direct their training and assets against U.S. capabilities in efforts to degrade systems and mission objectives. Advanced communications capabilities are available through the combination of hardware, such as a receiver, with a software interface capability that allows satellite signal reception and processing to include real-time position, velocity, navigation, reporting, and timing information.

I-13. Leaders and planners at all levels will realize even greater mission success as knowledge of satellite-based communications capabilities and the importance of their application are emphasized. This section describes enhanced capabilities in the Army's inventory that aid in providing secure communications and situational awareness for tactical radio operations.

## BLUE FORCE TRACKER

I-14. Blue Force Tracker is a GPS-enabled system that provides commanders and Soldiers with location information about friendly military forces. Using ground-based line of sight radios in operational environments with obstructive terrain can cause network limitations and hinder a commanders' ability to communicate and provide and maintain situational awareness. To overcome these network limitations, units employ Blue Force Tracker. Typical employment of Blue Force Tracker systems consists of—

- A computer to display location information.
- A satellite terminal.
- An antenna to transmit location information.
- A GPS receiver to determine position.
- Software to send and receive orders.
- Mapping software, usually as a geographic information system that plots the Blue Force Tracker device on a map.

I-15. The system displays the location of the host vehicle on the computer's terrain-map display, the locations of other friendly platforms in blue, and the threat in red in their respective locations. Blue Force Tracker sends and receives text and imagery messages and has a mechanism for reporting the locations of threat forces and other battlefield conditions, for example, the location of minefields, battlefield obstacles, and damaged bridges.

I-16. Additional capability in some Blue Force Tracker devices include route planning tools. By entering grid coordinates, the Blue Force Tracker becomes both the map and compass for motorized units. With proximity warnings enabled, the vehicle crew can maintain awareness as they approach critical or turn points.

I-17. Blue Force Tracker monitors the location and progress of friendly and threat forces and sends those coordinates to a central location, typically command posts. There the data are consolidated into a common operational picture, and sent to headquarters element, other forces in theater, or back out to other military units for situational awareness. The system also allows users to input or update operational graphics, obstacles, engineer reconnaissance, and threat forces information. Once uploaded, this information is sent to higher headquarters or mailed to other subscribers of that user's list, or other Blue Force Tracker users within the subscription system.

## JOINT BATTLE COMMAND PLATFORM

I-18. The Joint Battle Command Platform is one of the Army's friendly force tracking systems that equips Soldiers with a faster satellite network, secure data encryption and advanced logistics. Joint Battle Command Platform includes an intuitive interface with features like touch-to-zoom maps and drag-and-drop icons. Joint Battle Command Platform is interoperable with the Nett Warrior handheld device. Joint Battle Command Platform builds on Blue Force Tracker capabilities.

## JOINT CAPABILITIES RELEASE

I-19. The Joint Capabilities Release is the next generation of battlefield situational awareness and command and control systems featuring enhanced capabilities. The Joint Capabilities Release tactical network has greater bandwidth, allowing the movement of more information to more users within seconds rather than in minutes. The Joint Capabilities Release system upgrade includes Blue Force Tracker 2, a high-tech, high-speed force-tracking SATCOM network. The Blue Force Tracker 2 is approximately ten times faster than the legacy Blue Force Tracker system. Employment of the Joint Capabilities Release system with the Blue Force Tracker 2 transceiver and network upgrade enables friendly position updates in seconds.

## NETWORK MODERNIZATION

I-20. Network modernization is a critical enabler to shaping future Army forces and underpins the Army's modernization priorities. It aligns with core tenets of a multidomain-capable force by—

- Establishing a unified network to enable multidomain operations.
- Implementing security and survivability to allow commander's freedom of action in cyberspace.
- Reforming processes and policies to improve performance and affordability.
- Providing network sustainment for a sustainable enterprise and tactical network infrastructure.

I-21. As the Army evolves its operating concept to meet future requirements, central to its vision is the need for a unified network that extends the range and increases the speed of operations for armed conflicts involving the U.S. forces, its partners, and allies. The ITN enables readiness and will allow multidomain-capable forces to integrate and operate seamlessly and simultaneously across multiple domains and environments, geographics, and warfighting functions and to converge capabilities at the point of need. The ITN includes applications, devices, infrastructure, and network transport to support Army, joint, and coalition forces operating in congested and contested environments.

## INTEGRATED TACTICAL NETWORK

I-22. Future U.S. engagements with peer threats require network mobility and continual connectivity. The ITN allows multiple voice and data transport pathways that provide network resiliency and alternate transport capability options while stationary or on-the-move. The network's ability to provide automatic failover facilitates planners in implementing PACE plan without interaction from the user or manual switchover to select redundant communications means.

I-23. The ITN is the tactical portion of the unified network, and combines existing tactical network capabilities with emerging commercial capabilities and systems, including non-developmental items. The ITN includes voice and data networks and gateways; tactical cross-domain solutions; small aperture satellite terminals; expeditionary servers; variable height antennas, commercial phone technology; and small satellite terminals with additional capabilities enabling transmission between different formats. ITN supports air-ground interoperability and enables future technology insertions.

I-24. ITN contains a suite of high-capacity software-defined tactical radio voice and data communications systems that operate using advanced networking waveforms. Most often employed at brigade and below formations, these radios enable command and control between dispersed forces, offering line of sight and beyond line-of-sight communications over extended ranges.

I-25. Tactical radios are critical to coordinate with combat enablers such as fires, joint effects, casualty evacuation, and air-ground coordination in real-time, in disparate conditions, at the time of need.

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**Note.** Only Army formations with the modernized tactical radios (handheld, manpack and small form fit; GEN (generation) II Manpack; and Leader Radios) are scheduled to receive the ITN capability upgrades. The AN/PRC-117G, AN/PRC-152A, AN/PRC-154C, and GEN I Manpack (AN/PRC-155) legacy radios are not cryptographic compliant, nor are they scheduled for ITN modernization based on current fielding guidelines.

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## DEVICES, KITS, AND ANCILLARY EQUIPMENT

I-26. The integration of military and commercial capabilities that support multiple platforms creates near-continuous data sharing between higher and lower echelons providing leaders with a common operational picture that is both accurate and timely. These capabilities, along with additional supporting devices and kits, are critical to maneuver elements' ability to communicate in a contested environment.

## Nett Warrior End-User Device

I-27. The Nett Warrior end-user device is a handheld Android-based application device that provides users simple data services such as situational awareness, automatic position location information, and tactical chat. Nett Warrior allows Soldiers to transition from vehicular platforms to dismounted operations and provides leaders a data environment supporting platoon and company operations. The Nett Warrior end-user device—

- Supports National Security Agency-certified (type-1) security architectures and secure but secure but unclassified networks.
- Allows data categorization in accordance with classification to enable greater information sharing with multinational mission partners.
- Enables PACE communications in a contested or congested environment with secure data transmission to coalition forces in an unclassified environment.
- Simplifies network training, set-up, and employment.
- Enables use of alternate transport, including 4G/LTE.
- Interoperates with legacy network systems to enable command and control.
- Provides robust, reliable voice and data services immune to network volatility associated with highly dynamic networks.
- Establishes connection for mobile wireless hotspot devices.
- Interfaces with Ruggedized Applications Platform-Tactical Radio's software for mission planning

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**Note.** The AN/PRC-154A does not meet cryptographic modernization requirements as mandated by the DOD and the National Security Agency and is being replaced by the AN/PRC-148C/D or AN/PRC-163 Leader Radio.

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## LEGACY AND ENDURING RADIO PLATFORMS

I-28. Most legacy and enduring radio platform are non-digital and operate on an analog infrastructure. Legacy and enduring radio platforms offer consistency of design, reliability, availability, and maintainability at the expense of reduced performance over time and rigidity in the network design of each radio platform type. The primary value of the legacy and enduring radio platforms is immediate availability to meet mission needs and operators' familiarity and level of training with existing radios. SINCGARS provides the predominant voice capability and offers an electromagnetic counter-countermeasures capability using frequency hopping on any of 2,320 frequencies from 30–88 MHz, with a 25 kHz channel separation. However, legacy SINCGARS lacks adequate electromagnetic counter-countermeasures to counter emerging threat techniques to defeat its first- and second-generation frequency hopping modes. Threats now have an increased ability to detect and intercept SINCGARS voice nets, break transmission security schemes, exploit SINCGARS signals, and profile network stations. Part of this problem exists from the fact that SINCGARS lacks multi-band operation and does not employ spectrum outside of its original 30–88 MHz range. Table I-1 on pages 152 through 164 provides a summary of tactical radios and radio specifications.

Table I-1. Tactical radio specifications

| Nomenclature/<br>Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Frequency Bands |      |                      |      |                        |      | Frequency<br>Ranges     | CryptoMod<br>Compliant | Select<br>Features,<br>Waveforms,<br>and Modes of<br>Operation                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|----------------------|------|------------------------|------|-------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | HF<br>2–30 MHz  |      | VHF<br>30–300<br>MHz |      | UHF<br>300–3000<br>MHz |      |                         |                        |                                                                                                                                                                                                                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | LOS             | BLOS | LOS                  | BLOS | LOS                    | BLOS |                         |                        |                                                                                                                                                                                                                                  |
| AN/ARC-231<br>Series<br><br>Multiband<br>Aviation Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 |      | X                    | X    | X                      | X    | 30.00–<br>512.00<br>MHz | No                     | Voice and data<br>radio capable of<br>operating<br>VHF/UHF LOS<br>in AM or FM<br>Mode, ATC,<br>maritime,<br>Havequick I/II<br>and TACSAT/<br>DAMA;<br>improved ARC-<br>231D will<br>support MUOS<br>and SATURN.<br>(AM/FM Modes) |
| AN/ARC-220<br><br>HF Aviation<br>Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | X               | X    |                      |      |                        |      | 2.00–29.99<br>MHz       | No                     | HF SSB single-<br>channel<br>primarily for air-<br>to-air and air-to-<br>ground voice<br>communications;<br>operates 2G/3G<br>ALE.<br>(Mode: SSB)                                                                                |
| <b>Legend:</b><br>AM                    amplitude modulation<br>ATC                  air traffic control<br>BLOS                beyond line of sight<br>CryptoMod        cryptographic modernization<br>DAMA              demand-assigned multiple access<br>ECCM                electromagnetic counter-countermeasures<br>FM                    frequency modulation<br>GHz                  gigahertz<br>HF                    high frequency<br>LOS                  line of sight<br>MHz                megahertz<br>MIL-STD            military standard<br>MUOS                mobile user objective system<br>SATURN            Second Generation Anti-jam Tactical Ultrahigh Frequency for North Atlantic Treaty Organization<br><br>SINCGARS        single-channel ground and airborne radio system<br>SSB                    single-sideband<br>TACSAT            tactical satellite<br>UHF                    ultrahigh frequency<br>VHF                    very high frequency |                 |      |                      |      |                        |      |                         |                        |                                                                                                                                                                                                                                  |

Table I-1. Tactical radio specification (continued)

| Nomenclature/<br>Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Frequency Bands |      |                   |      |                     |      |                                     |                     |                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|-------------------|------|---------------------|------|-------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | HF<br>2–30 MHz  |      | VHF<br>30–300 MHz |      | UHF<br>300–3000 MHz |      | Frequency Ranges                    | CryptoMod Compliant | Select Features, Waveforms, and Modes of Operation                                                                                                                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | LOS             | BLOS | LOS               | BLOS | LOS                 | BLOS |                                     |                     |                                                                                                                                                                                      |
| AN/PRC-117F<br><br>Multiband Manpack and Vehicle Mountable Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |      | X                 | X    | X                   | X    | 30.00–512.00 MHz and up to 2.00 GHz | No                  | Voice and data radio capable of operating SINGARS ECCM, VHF/UHF LOS in both AM/FM, Havequick I/II, ECCM, and TACSAT, MIL-STD-188-181C, MIL-STD-188-182B, MIL-STD-188-183B. (FM Mode) |
| <b>Legend:</b><br>AM                    amplitude modulation<br>ATC                  air traffic control<br>BLOS                beyond line of sight<br>CryptoMod        cryptographic modernization<br>DAMA              demand-assigned multiple access<br>ECCM                electromagnetic counter-countermeasures<br>FM                   frequency modulation<br>GHz                  gigahertz<br>HF                    high frequency<br>LOS                  line of sight<br>MHz                  megahertz<br>MIL-STD            military standard<br>MUOS                mobile user objective system<br>SATURN             Second Generation Anti-jam Tactical Ultrahigh Frequency for North Atlantic Treaty Organization<br><br>SINGARS           single-channel ground and airborne radio system<br>SSB                   single-sideband<br>TACSAT             tactical satellite<br>UHF                   ultrahigh frequency<br>VHF                   very high frequency |                 |      |                   |      |                     |      |                                     |                     |                                                                                                                                                                                      |

Table I-1. Tactical radio specifications (continued)

| Nomenclature/<br>Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Frequency Bands |      |                      |      |                        |      | Frequency<br>Ranges | CryptoMod<br>Compliant | Select<br>Features,<br>Waveforms,<br>and Modes of<br>Operation                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|----------------------|------|------------------------|------|---------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | HF<br>2–30 MHz  |      | VHF<br>30–300<br>MHz |      | UHF<br>300–3000<br>MHz |      |                     |                        |                                                                                                                                                                                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | LOS             | BLOS | LOS                  | BLOS | LOS                    | BLOS |                     |                        |                                                                                                                                                                                                  |
| AN/PRC-117G<br><br>Wideband<br>Manpack and<br>Vehicle<br>Mountable<br>Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |      | X                    | X    | X                      | X    | 30.00–2.00<br>MHz   | No                     | Voice and data<br>radio capable<br>of operating<br>SINCGARS,<br>Havequick I/II<br>(SATURN<br>capable),<br>VHF/UHF AM<br>and FM,<br>ANW2, and<br>TACSAT, MIL-<br>STD-188-<br>181C.<br>(Mode: FM). |
| <b>Legend:</b><br>AM                    amplitude modulation<br>ATC                  air traffic control<br>BLOS                beyond line of sight<br>CryptoMod        cryptographic modernization<br>DAMA              demand-assigned multiple access<br>ECCM                electromagnetic counter-countermeasures<br>FM                    frequency modulation<br>GHz                  gigahertz<br>HF                    high frequency<br>LOS                  line of sight<br>MHz                  megahertz<br>MIL-STD            military standard<br>MUOS                mobile user objective system<br>SATURN             Second Generation Anti-jam Tactical Ultrahigh Frequency for North Atlantic Treaty Organization<br><br>SINCGARS        single-channel ground and airborne radio system<br>SSB                   single-sideband<br>TACSAT             tactical satellite<br>UHF                   ultrahigh frequency<br>VHF                   very high frequency |                 |      |                      |      |                        |      |                     |                        |                                                                                                                                                                                                  |

Table I-1. Tactical radio specifications (continued)

| Nomenclature/<br>Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Frequency Bands |      |                      |      |                        |      | Frequency<br>Ranges | CryptoMod<br>Compliant | Select<br>Features,<br>Waveforms,<br>and Modes of<br>Operation                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|----------------------|------|------------------------|------|---------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | HF<br>2–30 MHz  |      | VHF<br>30–300<br>MHz |      | UHF<br>300–3000<br>MHz |      |                     |                        |                                                                                                                                                                                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | LOS             | BLOS | LOS                  | BLOS | LOS                    | BLOS |                     |                        |                                                                                                                                                                                 |
| AN/PRC-119<br><br>SINGARS<br>Dismounted<br>Manpack                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 |      | X                    |      |                        |      | 30.00–<br>87.95 MHz | No                     | Voice and data<br>radio capable<br>of operating<br>ASIP,<br>SINGARS,<br>VHF, FM, LOS<br>in SC and FH.<br>(Mode: FM)                                                             |
| AN/VRC-87<br>AN/VRC-88<br>AN/VRC-89<br>AN/VRC-90<br>AN/VRC-91<br>AN/VRC-92<br><br>Legacy<br>SINGARS<br>ASIP Vehicle<br>Mounted Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 |      | X                    |      |                        |      | 30.00–<br>87.96 MHz | No                     | Vehicular<br>short-range<br>and long-range<br>radios with<br>select<br>dismountable<br>or retransmit<br>capability.<br>Supports voice<br>and data in SC<br>or FH.<br>(Mode: FM) |
| <b>Legend:</b><br>AM                    amplitude modulation<br>ATC                  air traffic control<br>BLOS                beyond line of sight<br>CryptoMod        cryptographic modernization<br>DAMA              demand-assigned multiple access<br>ECCM                electromagnetic counter-countermeasures<br>FM                    frequency modulation<br>GHz                  gigahertz<br>HF                    high frequency<br>LOS                  line of sight<br>MHz                  megahertz<br>MIL-STD            military standard<br>MUOS                mobile user objective system<br>SATURN            Second Generation Anti-jam Tactical Ultrahigh Frequency for North Atlantic Treaty Organization<br><br>SINGARS           single-channel ground and airborne radio system<br>SSB                   single-sideband<br>TACSAT             tactical satellite<br>UHF                   ultrahigh frequency<br>VHF                   very high frequency |                 |      |                      |      |                        |      |                     |                        |                                                                                                                                                                                 |

Table I-1. Tactical radio specifications (continued)

| Nomenclature/<br>Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Frequency Bands |      |                      |      |                        |      | Frequency<br>Ranges                       | CryptoMod<br>Compliant | Select<br>Features,<br>Waveforms,<br>and Modes of<br>Operation                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|----------------------|------|------------------------|------|-------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | HF<br>2–30 MHz  |      | VHF<br>30–300<br>MHz |      | UHF<br>300–3000<br>MHz |      |                                           |                        |                                                                                                                                                                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | LOS             | BLOS | LOS                  | BLOS | LOS                    | BLOS |                                           |                        |                                                                                                                                                                              |
| AN/PRC-148<br><br>Enhanced<br>MBITR<br>Handheld                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |      | X                    | X    | X                      | X    | 30.00–<br>512.00<br>MHz                   | No                     | Voice and data<br>radio capable<br>of operating<br>SINGARS,<br>TACSAT/<br>DAMA/IW, and<br>Havequick I/II.<br>(FM Mode)                                                       |
| AN/PRC-148C<br>AN/PRC-148D<br><br>2-Channel<br>HMS Leaders<br>Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 |      | X                    | X    | X                      | X    | 30.00–<br>512.00<br>MHz<br>1.1–2.5<br>GHz | Yes                    | VHF and UHF<br>voice and<br>networking<br>data 2-channel<br>radio capable<br>of operating<br>SINGARS,<br>TACSAT IW,<br>TSM, WREN<br>narrowband,<br>and MUOS.<br>(AM/FM Mode) |
| <b>Legend:</b><br>AM                    amplitude modulation<br>ATC                  air traffic control<br>BLOS                beyond line of sight<br>CryptoMod        cryptographic modernization<br>DAMA              demand-assigned multiple access<br>ECCM                electromagnetic counter-countermeasures<br>FM                    frequency modulation<br>GHz                  gigahertz<br>HF                    high frequency<br>LOS                  line of sight<br>MHz                  megahertz<br>MIL-STD            military standard<br>MUOS                mobile user objective system<br>SATURN            Second Generation Anti-jam Tactical Ultrahigh Frequency for North Atlantic Treaty Organization<br><br>SINGARS           single-channel ground and airborne radio system<br>SSB                   single-sideband<br>TACSAT            tactical satellite<br>UHF                   ultrahigh frequency<br>VHF                   very high frequency |                 |      |                      |      |                        |      |                                           |                        |                                                                                                                                                                              |

Table I-1. Tactical radio specifications (continued)

| Nomenclature/<br>Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Frequency Bands |      |                      |      |                        |      | Frequency<br>Ranges                     | CryptoMod<br>Compliant | Select<br>Features,<br>Waveforms,<br>and Modes of<br>Operation                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|----------------------|------|------------------------|------|-----------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | HF<br>2–30 MHz  |      | VHF<br>30–300<br>MHz |      | UHF<br>300–3000<br>MHz |      |                                         |                        |                                                                                                                                                                      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | LOS             | BLOS | LOS                  | BLOS | LOS                    | BLOS |                                         |                        |                                                                                                                                                                      |
| AN/PRC-150<br><br>Handheld and<br>Vehicle<br>Mountable<br>Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | X               | X    | X                    |      |                        |      | 1.6–28.80<br>MHz<br>30.00–<br>60.00 MHz | No                     | Voice and data<br>radio capable<br>of operating<br>HF 2G/3G ALE<br>and SC VHF<br>(up to 60.00<br>MHz), 19.2<br>Kbps to 75 bps<br>data.<br>(AM/FM Mode)               |
| AN/PRC-152<br><br>Handheld and<br>Vehicle<br>Mountable<br>Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |      | X                    | X    | X                      | X    | 30.00–<br>512.00<br>MHz                 | No                     | Voice and data<br>radio capable<br>of operating<br>SINGARS,<br>VHF/UHF LOS<br>in both AM/FM,<br>Havequick I/II,<br>and TACSAT;<br>MIL-STD-188-<br>181C.<br>(FM Mode) |
| <b>Legend:</b><br>AM                    amplitude modulation<br>ATC                  air traffic control<br>BLOS                beyond line of sight<br>CryptoMod        cryptographic modernization<br>DAMA              demand-assigned multiple access<br>ECCM                electromagnetic counter-countermeasures<br>FM                   frequency modulation<br>GHz                  gigahertz<br>HF                    high frequency<br>LOS                  line of sight<br>MHz                  megahertz<br>MIL-STD            military standard<br>MUOS                mobile user objective system<br>SATURN             Second Generation Anti-jam Tactical Ultrahigh Frequency for North Atlantic Treaty Organization<br><br>SINGARS           single-channel ground and airborne radio system<br>SSB                   single-sideband<br>TACSAT              tactical satellite<br>UHF                   ultrahigh frequency<br>VHF                   very high frequency |                 |      |                      |      |                        |      |                                         |                        |                                                                                                                                                                      |

Table I-1. Tactical radio specifications (continued)

| Nomenclature/<br>Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Frequency Bands |      |                      |      |                        |      | Frequency<br>Ranges       | CryptoMod<br>Compliant | Select<br>Features,<br>Waveforms,<br>and Modes of<br>Operation                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|----------------------|------|------------------------|------|---------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | HF<br>2–30 MHz  |      | VHF<br>30–300<br>MHz |      | UHF<br>300–3000<br>MHz |      |                           |                        |                                                                                                                                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LOS             | BLOS | LOS                  | BLOS | LOS                    | BLOS |                           |                        |                                                                                                                                        |
| AN/PRC-154<br><br>HMS Rifleman<br>Handheld<br>Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |      |                      |      | X                      | X    | 300.00<br>MHz–2.00<br>GHz | No                     | Voice and<br>networking<br>data radio<br>capable of<br>operating in<br>the UHF and<br>Commercial L-<br>Band.<br>(FM Mode)              |
| AN/PRC-155<br><br>HMS 2-<br>Channel<br>Manpack Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                 |      | X                    | X    | X                      | X    | 30.00<br>MHz–3.00<br>GHz  | Yes                    | Voice and<br>networking<br>data 2-channel<br>radio capable<br>of operating<br>SINCGARS,<br>TACSAT IW<br>SRW, and<br>MUOS.<br>(FM Mode) |
| <b>Legend:</b><br>AM                    amplitude modulation<br>ATC                   air traffic control<br>BLOS                beyond line of sight<br>CryptoMod        cryptographic modernization<br>DAMA              demand-assigned multiple access<br>ECCM              electromagnetic counter-countermeasures<br>FM                   frequency modulation<br>GHz                gigahertz<br>HF                   high frequency<br>LOS                line of sight<br>MHz                megahertz<br>MIL-STD          military standard<br>MUOS              mobile user objective system<br>SATURN            Second Generation Anti-jam Tactical Ultrahigh Frequency for North Atlantic Treaty Organization<br><br>SINCGARS        single-channel ground and airborne radio system<br>SSB                single-sideband<br>TACSAT            tactical satellite<br>UHF                ultrahigh frequency<br>VHF                very high frequency |                 |      |                      |      |                        |      |                           |                        |                                                                                                                                        |

Table I-1. Tactical radio specifications (continued)

| Nomenclature/<br>Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Frequency Bands |      |                      |      |                        |      | Frequency<br>Ranges                                                                        | CryptoMod<br>Compliant | Select<br>Features,<br>Waveforms,<br>and Modes of<br>Operation                                                                                          |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | HF<br>2–30 MHz  |      | VHF<br>30–300<br>MHz |      | UHF<br>300–3000<br>MHz |      |                                                                                            |                        |                                                                                                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | LOS             | BLOS | LOS                  | BLOS | LOS                    | BLOS |                                                                                            |                        |                                                                                                                                                         |
| AN/PRC-158<br>HMS 2-<br>Channel<br>Manpack Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                 |      | X                    | X    | X                      | X    | 30–225<br>MHz<br>225–520<br>MHz<br>762–874<br>MHz<br>1.3–2.5<br>GHz<br>selected L-<br>band | Yes                    | 2-channel<br>voice and<br>networking<br>data radio<br>capable of<br>operating<br>SINGARS,<br>TACSAT/IW<br>TSM, MUOS;<br>SATURN<br>capable.<br>(FM Mode) |
| <b>Legend:</b><br>AM                      amplitude modulation<br>ATC                     air traffic control<br>BLOS                    beyond line of sight<br>CryptoMod            cryptographic modernization<br>DAMA                  demand-assigned multiple access<br>ECCM                  electromagnetic counter-countermeasures<br>FM                      frequency modulation<br>GHz                     gigahertz<br>HF                      high frequency<br>LOS                     line of sight<br>MHz                    megahertz<br>MIL-STD               military standard<br>MUOS                  mobile user objective system<br>SATURN                Second Generation Anti-jam Tactical Ultrahigh Frequency for North Atlantic Treaty Organization<br><br>SINGARS               single-channel ground and airborne radio system<br>SSB                      single-sideband<br>TACSAT                tactical satellite<br>UHF                      ultrahigh frequency<br>VHF                      very high frequency |                 |      |                      |      |                        |      |                                                                                            |                        |                                                                                                                                                         |

Table I-1. Tactical radio specifications (continued)

| Nomenclature/<br>Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Frequency Bands |      |                      |      |                        |      | Frequency<br>Ranges | CryptoMod<br>Compliant | Select<br>Features,<br>Waveforms,<br>and Modes of<br>Operation                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|----------------------|------|------------------------|------|---------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | HF<br>2–30 MHz  |      | VHF<br>30–300<br>MHz |      | UHF<br>300–3000<br>MHz |      |                     |                        |                                                                                                                                                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | LOS             | BLOS | LOS                  | BLOS | LOS                    | BLOS |                     |                        |                                                                                                                                                              |
| AN/PRC-160<br><br>Manpack and<br>Vehicle<br>Mountable<br>Advanced HF<br>Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | X               | X    | X                    |      |                        |      | 1.50–59.99<br>MHz   | Yes                    | Voice and data<br>capable<br>(2G/3G/4G<br>ALE<br>compliant),<br>VHF up to<br>60.00 MHz,<br>wideband HF<br>data capable<br>up to 128<br>Kbps.<br>(AM/FM Mode) |
| <b>Legend:</b><br>AM                      amplitude modulation<br>ATC                     air traffic control<br>BLOS                    beyond line of sight<br>CryptoMod            cryptographic modernization<br>DAMA                  demand-assigned multiple access<br>ECCM                   electromagnetic counter-countermeasures<br>FM                      frequency modulation<br>GHz                     gigahertz<br>HF                       high frequency<br>LOS                      line of sight<br>MHz                     megahertz<br>MIL-STD               military standard<br>MUOS                   mobile user objective system<br>SATURN                Second Generation Anti-jam Tactical Ultrahigh Frequency for North Atlantic Treaty Organization<br><br>SINCGARS             single-channel ground and airborne radio system<br>SSB                      single-sideband<br>TACSAT                tactical satellite<br>UHF                      ultrahigh frequency<br>VHF                      very high frequency |                 |      |                      |      |                        |      |                     |                        |                                                                                                                                                              |

Table I-1. Tactical radio specifications (continued)

| Nomenclature/<br>Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Frequency Bands |      |                      |      |                        |      | Frequency<br>Ranges      | CryptoMod<br>Compliant | Select<br>Features,<br>Waveforms,<br>and Modes of<br>Operation                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|----------------------|------|------------------------|------|--------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | HF<br>2–30 MHz  |      | VHF<br>30–300<br>MHz |      | UHF<br>300–3000<br>MHz |      |                          |                        |                                                                                                                                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | LOS             | BLOS | LOS                  | BLOS | LOS                    | BLOS |                          |                        |                                                                                                                                                          |
| AN/PRC-163<br><br>HMS 2-<br>Channel<br>Handheld<br>Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |      | X                    | X    | X                      | X    | 30.00<br>MHz–2.50<br>GHz | Yes                    | 2-channel<br>voice and<br>networking<br>data radio<br>capable of<br>operating<br>SINCGARS,<br>TACSAT/IW<br>TSM, MUOS;<br>SATURN<br>capable.<br>(FM Mode) |
| <b>Legend:</b><br>AM                    amplitude modulation<br>ATC                  air traffic control<br>BLOS                beyond line of sight<br>CryptoMod        cryptographic modernization<br>DAMA              demand-assigned multiple access<br>ECCM               electromagnetic counter-countermeasures<br>FM                   frequency modulation<br>GHz                  gigahertz<br>HF                   high frequency<br>LOS                  line of sight<br>MHz                  megahertz<br>MIL-STD           military standard<br>MUOS               mobile user objective system<br>SATURN            Second Generation Anti-jam Tactical Ultrahigh Frequency for North Atlantic Treaty Organization<br><br>SINCGARS        single-channel ground and airborne radio system<br>SSB                   single-sideband<br>TACSAT            tactical satellite<br>UHF                   ultrahigh frequency<br>VHF                   very high frequency |                 |      |                      |      |                        |      |                          |                        |                                                                                                                                                          |

Table I-1. Tactical radio specifications (continued)

| Nomenclature/<br>Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Frequency Bands |      |                      |      |                        |      | Frequency<br>Ranges     | CryptoMod<br>Compliant | Select<br>Features,<br>Waveforms,<br>and Modes of<br>Operation                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|----------------------|------|------------------------|------|-------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | HF<br>2–30 MHz  |      | VHF<br>30–300<br>MHz |      | UHF<br>300–3000<br>MHz |      |                         |                        |                                                                                                                                                                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | LOS             | BLOS | LOS                  | BLOS | LOS                    | BLOS |                         |                        |                                                                                                                                                                        |
| AN/PSC-5<br><br>Spitfire<br>Manpack<br>VHF/UHF<br>SATCOM<br>Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                 |      | X                    | X    | X                      | X    | 30.00–<br>420.00<br>MHz | No                     | Voice and Data<br>SATCOM-<br>DAMA and<br>capable of<br>operating VHF<br>and UHF LOS<br>in AM and FM<br>modes;<br>supports ATC<br>and maritime<br>nets.<br>(AM/FM Mode) |
| <b>Legend:</b><br>AM                    amplitude modulation<br>ATC                  air traffic control<br>BLOS                beyond line of sight<br>CryptoMod        cryptographic modernization<br>DAMA              demand-assigned multiple access<br>ECCM                electromagnetic counter-countermeasures<br>FM                    frequency modulation<br>GHz                  gigahertz<br>HF                    high frequency<br>LOS                  line of sight<br>MHz                 megahertz<br>MIL-STD            military standard<br>MUOS                mobile user objective system<br>SATURN             Second Generation Anti-jam Tactical Ultrahigh Frequency for North Atlantic Treaty Organization<br>SINCGARS        single-channel ground and airborne radio system<br>SSB                   single-sideband<br>TACSAT            tactical satellite<br>UHF                   ultrahigh frequency<br>VHF                   very high frequency |                 |      |                      |      |                        |      |                         |                        |                                                                                                                                                                        |

Table I-1. Tactical radio specifications (continued)

| Nomenclature/<br>Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Frequency Bands |      |                      |      |                        |      | Frequency<br>Ranges     | CryptoMod<br>Compliant | Select<br>Features,<br>Waveforms,<br>and Modes of<br>Operation                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|----------------------|------|------------------------|------|-------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | HF<br>2–30 MHz  |      | VHF<br>30–300<br>MHz |      | UHF<br>300–3000<br>MHz |      |                         |                        |                                                                                                                                                                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | LOS             | BLOS | LOS                  | BLOS | LOS                    | BLOS |                         |                        |                                                                                                                                                                        |
| AN/PSC-5C<br><br>Shadowfire<br>Multiband<br>Manpack and<br>Vehicle<br>Mountable<br>Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                 |      | X                    | X    | X                      | X    | 30.00–<br>420.00<br>MHz | No                     | Voice and Data<br>SATCOM-<br>DAMA and<br>capable of<br>operating VHF<br>and UHF LOS<br>in AM and FM<br>modes;<br>supports ATC<br>and maritime<br>nets.<br>(AM/FM Mode) |
| <b>Legend:</b><br>AM                    amplitude modulation<br>ATC                  air traffic control<br>BLOS                beyond line of sight<br>CryptoMod        cryptographic modernization<br>DAMA              demand-assigned multiple access<br>ECCM              electromagnetic counter-countermeasures<br>FM                  frequency modulation<br>GHz                gigahertz<br>HF                  high frequency<br>LOS                line of sight<br>MHz                megahertz<br>MIL-STD          military standard<br>MUOS              mobile user objective system<br>SATURN            Second Generation Anti-jam Tactical Ultrahigh Frequency for North Atlantic Treaty Organization<br>SINCGARS        single-channel ground and airborne radio system<br>SSB                single-sideband<br>TACSAT            tactical satellite<br>UHF                ultrahigh frequency<br>VHF                very high frequency |                 |      |                      |      |                        |      |                         |                        |                                                                                                                                                                        |

Table I-1. Tactical radio specifications (continued)

| Nomenclature/<br>Configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Frequency Bands |      |                      |      |                        |      | Frequency<br>Ranges     | CryptoMod<br>Compliant | Select<br>Features,<br>Waveforms,<br>and Modes of<br>Operation                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|----------------------|------|------------------------|------|-------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | HF<br>2–30 MHz  |      | VHF<br>30–300<br>MHz |      | UHF<br>300–3000<br>MHz |      |                         |                        |                                                                                                                                                                        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | LOS             | BLOS | LOS                  | BLOS | LOS                    | BLOS |                         |                        |                                                                                                                                                                        |
| AN/PSC-5D<br><br>Improved<br>Multiband<br>Manpack and<br>Vehicle<br>Mountable<br>Radio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |      | X                    | X    | X                      | X    | 30.00–<br>512.00<br>MHz | No                     | Voice and data<br>SATCOM-<br>DAMA and<br>capable of<br>operating VHF<br>and UHF LOS<br>in AM and FM<br>modes;<br>supports ATC<br>and maritime<br>nets.<br>(AM/FM Mode) |
| <b>Legend:</b><br>AM                    amplitude modulation<br>ATC                  air traffic control<br>BLOS                beyond line of sight<br>CryptoMod        cryptographic modernization<br>DAMA              demand-assigned multiple access<br>ECCM                electromagnetic counter-countermeasures<br>FM                   frequency modulation<br>GHz                  gigahertz<br>HF                    high frequency<br>LOS                  line of sight<br>MHz                  megahertz<br>MIL-STD            military standard<br>MUOS                mobile user objective system<br>SATURN            Second Generation Anti-jam Tactical Ultrahigh Frequency for North Atlantic Treaty Organization<br>SINCGARS        single-channel ground and airborne radio system<br>SSB                   single-sideband<br>TACSAT            tactical satellite<br>UHF                   ultrahigh frequency<br>VHF                   very high frequency |                 |      |                      |      |                        |      |                         |                        |                                                                                                                                                                        |

## SOFTWARE-DEFINED RADIOS

I-29. Software-defined radios function as a family of advanced multiband, multimode, net-centric-capable radios. The software-defined radio's operating system enables the interaction between the radio hardware components and waveform application software. With legacy radios, the hardware determined the function and operation of the radio. For software-defined radios, the hardware and software are less rigidly coupled. This provides greater interoperability with waveform applications and network management tools while minimizing interoperability issues associated with upgrades to the radio's operating system. Current software-defined radios include program of record systems and commercial acquisition solutions.

## RADIO WAVEFORMS

I-30. Software-defined radio waveforms serve as software applications that provide the capabilities needed to operate over-the-air communications, including interoperability with most legacy military radios. Waveforms include network configurations, frequency bands, electromagnetic counter-countermeasures and operating modes, channel presets, voice and data processing, and signal processing. Waveforms designs support all domains including air, land, and maritime platforms, command posts, Soldier portable and carried systems, and software-defined radios that directly support special applications and legacy functions such as SINCGARS. Examples of current advanced waveforms include—

- TSM.
- WREN-TSM.
- WREN-Narrowband.
- MUOS.

### Tactical Scalable Mobile Ad Hoc Network

I-31. TSM is a commercial development that supports networked voice, data and position location information capabilities. TSM provides a resilient, self-healing, wideband mesh configuration to support talk groups over a single-channel. TSM variants include an enhanced mode with greater throughput, flexible bandwidth, network adaptability, and waveform portability. TSMx, is a vendor assigned designator, which is specifically designed to support interface with National Security Agency-certified type-1 security architectures up to and including secret level classification.

### Warrior Robust Enhanced Network Tactical Scalable Mobile Ad Hoc Network

I-32. The WREN-TSM is an evolution of TSM waveform used for transmissions supporting secret and below classifications. WREN-TSM provides the same functions of TSM but is security hardened with National Security Agency type 1 cryptography for government use. Porting WREN-TSMs advanced commercial waveform into the Army's tactical radios gives Soldiers the option to run both sensitive but unclassified and secret communications on different channels, or one at a time, while maintaining complete data and voice isolation among the different security classification levels.

### Warrior Robust Enhanced Network Narrowband

I-33. WREN-Narrowband is an Army program of record waveform on multiple vendor platforms, and answers the battlefield needs of today and tomorrow. WREN-Narrowband has wide frequency coverage as well as massive scalability. It provides mitigation against intentional and unintentional electromagnetic interference. The waveform includes secure Internet protocol network capabilities, extended network range, and interoperability between different radio platforms. WREN-Narrowband provides a highly scalable and robust waveform with simplified network management for operations in congested or contested environments.

### Mobile User Objective System

I-34. MUOS is an advanced narrowband satellite communications replacement for the existing UHF TACSAT constellation. The MUOS waveform operates on handheld, manpack or mounted ground terminals to provide beyond line of sight or over-the-horizon voice and data communications at the tactical edge. MUOS provides a ten-fold increased capacity and more robust support by adapting a commercial 3G wideband code division multiple access cellular phone architecture, using geosynchronous satellites in place of cell towers. The MUOS waveform encapsulates all network configurations, addressing, and DODIN access protocols and is currently ported on select handheld, manpack, small form fit program for manpack radios and updated aviation radios.

## SECTION II – RADIO FORM FACTORS

I-35. Form factors are tactical radio assets and capabilities that adhere to desired performance parameters to include specified standards for hardware and software components. Implementing industry and Army

standards for system size, weight and power, and cooling supports the Army's concepts on interoperability and ensures commonality across multiple platforms and operational assets. When prototypes and fielded systems meet defined technical requirements, it reduces the need for platform-specific integration, enables fielding of evolutionary capabilities, and significantly improves reliability, reusability, and interchangeability of software modules, devices, and system components.

## RADIO CONFIGURATIONS

I-36. The Army's communications assets are as diverse as the requirements, operational need, and environmental conditions for the mission being supported. Tactical radio configurations play a vital role in Army operations and combat effectiveness allowing troops to coordinate movement and exchange critical information to effectively carry out mission objectives. Typical Army radios consist of a transmitter and receiver and an antenna that radiates the electromagnetic energy for signal transmission and communication exchange. Tactical radio configurations are also equipped with user controls that allow frequency, power, and channel adjustment to augment capabilities and may be mounted on various platforms allowing operations in moderate or harsh weather conditions and obscure or unconventional terrain.

I-37. Tactical radios have evolved into highly sophisticated communications devices making them indispensable assets during all phases from pre-planning to execution of operations. With a variety of tactical radio configurations, commanders and signal Soldiers have access to a myriad of functionality and features to suit their needs. The Army inventory includes radio configurations offering voice and data transmission with advanced encryption, anti-jamming, and enhancement features such as frequency hopping for added transmission security. Configurations may include single or dual channel sidebands or both. Radio systems may also operate in varying frequency bands or ranges including HF, VHF, or UHF enabling Soldiers to stay connected with each other and their commanders at all times.

I-38. Having such a diverse suite of radios allows Soldiers and units flexibility for long- or short-range communication while multiple host platforms—including ground (portable), airborne, maritime, and transportable—enable synchronization of military operations between dismounted troops and aircraft, land vehicles, warships or submarines. Radio configuration types include—

- Portable.
  - Handheld and small form fit.
  - Manpack.
  - Ground.
- Vehicular.
- Airborne.
- Transportable.
- Narrowband (single-channel) tactical satellite.

## PORTABLE CONFIGURATIONS

I-39. Portable radio communications configurations are typically single, lightweight radio systems designed for the dismounted short-range communications. They consist of handheld and man-portable radios used by individual users for point-to-point communities and information sharing. These configurations typically include a battery box for rechargeable or non-rechargeable batteries, short and medium range antennas, and other related accessories, including handsets, headphones, and carrying equipment, for operating in a dismounted mode. Some portable radio communications configurations may support vehicle configurations using specialized mounting adapters and supplementary equipment. Portable radio communications configurations include systems such as—

- PRC-117F (manpack multiband).
- PRC-117G (manpack multiband).
- PRC-119 (manpack, legacy single-channel SINCGARS).
- PRC-148 (handheld, original MBITR single-channel multiband).
- PRC 150 (manpack, single-channel high frequency).
- PRC-152 (multiband, handheld).
- PRC-154 (handheld Rifleman; operates Soldier Radio Waveform).
- PRC-158 (two-channel, multiband manpack).

- PRC 160 (improved manpack, single-channel high frequency).
- PRC-162 (two-channel, multiband).
- PRC-163 (two-channel, handheld Leader Radio).

I-40. Handheld, manpack, and small form fit radios provide tactical radio communications for on-the-move, at-the-halt, and stationary line of sight and beyond line of sight communications. These handheld devices offer reduced footprint and include additional waveforms supporting simultaneous voice, data, and video communications.

## HANDHELD AND SMALL FORM FIT RADIOS

I-41. Handheld and small form fit radios include systems that are easily carried or worn by Soldiers with personal tactical equipment or Nett Warrior ensembles normally used in dismounted operations. While some legacy handheld radios are single-channel systems, others provide two simultaneous operating channels supporting voice and data and enable line of sight and beyond line of sight communications. These radios are compatible with most legacy radios and provide joint interoperability. Two-channel handheld radios such as the Leader Radio enables net-centric operations and provide multiband and multimode functionality to deliver reliable and secure tactical communications down to the squad level. The Leader Radio can support secure voice and data using various waveforms including SINCGARS, VHF and UHF line of sight, and narrowband (single-channel) TACSAT. The Leader Radio also operates TSM and WREN advanced networking waveforms that provide connectivity to the Nett Warrior end-user device. Examples include the AN/PRC-148C/D/F and AN/PRC-163 radios.

## MANPACK RADIOS

I-42. Manpack radios provide single-channel and two-channel secure voice and data via multiple waveforms and consist of both legacy systems and software-defined radios such as the AN/PRC-158 and the AN/PRC-162. These radios enable communication exchange in austere environments between units and higher headquarters without having to rely on a fixed infrastructure.

### AN/PRC-104

I-43. The AN/PRC-104 is a legacy HF radio that operates between 3 and 30 MHz for long-range communication. It operates in both upper sideband and lower sideband modes for voice communications; supports continuous wave for Morse code; and enables frequency shift keying for transmission of teletype signals or other data. The AN/PRC-104 can be configured for vehicular or fixed-station use with proper accessories. The basic manpack configuration is comprised of—

- AM-6874/AN/PRC-104 (20-watt RF amplifier/coupler).
- CY-7541 (battery case and battery pack).
- RT-1209/URC (power receiver transmitter).
- AN-PRC 217A (whip antenna).
- Handset H-189/GR and telegraph key (KY-872/PRC-104).

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**Note.** The legacy AN/PRC-104 is replaced by AN/PRC-150 and-160 with ITN modernization upgrades.

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### AN/PRC-117F and AN/PRC-117G, Falcon III Family of Multiband Radios

I-44. Both the AN/PRC-117F and AN/PRC-117G radios are single-channel software-defined manpack systems supporting voice and data for mounted and dismounted operations. They are designed to provide reliable tactical communications in a small, lightweight package that can maximize user mobility. The AN/PRC-117F systems support multiple waveforms including—

- SINCGARS and VHF line of sight.
- Single-channel UHF (30 MHz–512) narrowband (single-channel) TACSAT.
  - DAMA.
  - Integrated waveform.
- Havequick II (supports ground-to-air voice communications).

I-45. The AN/PRC-117G supports new capabilities offering significant technological upgrades to the AN/PRC-117F. These capabilities provide advantages including—

- SINCGARS and VHF line of sight.
- Single-channel UHF line of sight and SATCOM.
  - DAMA.
  - Integrated waveform.
- MUOS-ready hardware.
- Harris Advanced Networking Waveform (GEN I and II).
- L-band network (up to 2 GHz).

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**Note.** The AN/PSC-5 C/D, AN/PRC-117F, and AN/PRC-117G radios do not meet cryptographic modernization requirements as mandated by DOD and the National Security Agency and will be replaced by the AN/PRC-158 or AN/PRC-162 handheld, manpack and small form fit radios.

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I-46. The AN/PRC-117G radio is capable of simultaneously transmitting Voice over Internet Protocol and digital data such as file transfers, chat, streaming video, and position location information on a single channel. The AN/PRC-117G also supports SINCGARS networks and SATURN .which replaces Havequick II for air-ground voice communications.

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**Note.** The AN/PRC-117F and AN/PRC-117G radios were limited purchases of commercial off-the-shelf) multiband multi-mission radios requisitioned specifically to fill vacancies for narrowband (single-channel) TACSAT radios, normally requiring AN/PSC-5 UHF SATCOM authorizations.

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### **AN/PRC-119 and RT-1523 Series**

I-47. The AN/PRC-119 is included in the SINCGARS family of radios that uses the RT-1523. The AN/PRC-119 is designed to ensure commonality among various ground (vehicular), airborne, and manpack configurations. The individual components of SINCGARS radios are interchangeable from one configuration to the next, a design feature that reduces the burden on the logistics system for repair parts. It has evolved from a hardware-driven module to the ASIP radio configuration that includes a software base to complement smaller and lighter design requirements and the ability to reprogram the SINCGARS to further improve the radio overtime.

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**Note.** The RT-1523 radio series includes a diverse set of systems including the RT-1523/A/B and E/F and the AN/PRC-119 radios. It will continue to support the SINCGARS waveform and its associate future upgrades. The combat net radio will replace the AN/PRC-119 (RT-1523) series legacy radios through attrition. The RT-1523 C/D radio models are no longer fielded to Army units.

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### **AN/PRC-148, Leader Radio Series**

I-48. The AN/PRC-148 MBITR and handheld, manpack small form fit Leader Radio series are handheld, Soldier-worn two channel radios that provide interoperable communications to Soldiers needing an on-the-move or at-the-halt capability, whether mounted or dismounted.

I-49. The legacy AN/PRC-148 MBITR is used for special operations forces and company size networks depending on command guidance and mission requirements. When used as a handheld radio, the AN/PRC-148 MBITR supports the secure requirements for a platoon, squad, or team. The AN/PRC-148 MBITR enables small-unit leaders' adequate control of subordinate elements activities and can perform ground-to-air, ship-to-shore, DAMA TACSAT, civil, military, and multinational communications. The AN/PRC-148 MBITR consists of a portable, battery-operated transceiver capable of operating in non-secure analog or secure digital voice and data. The AN/PRC-148 MBITR is software upgradeable to add the following capabilities—

- SINCGARS (including Retransmission).

- Narrowband (single channel) TACSAT.
  - DAMA.
  - Integrated waveform.
- Advanced narrowband digital voice terminal/KYV-5.
- Havequick I/II.

I-50. Planners and operators can configure the AN/PRC-148 MBITR manually using the radio control panel or through a software-based application. The software-based application has an interface that allows uploading and downloading assigned frequency lists, waveform types, and radio power settings. Once configured, the AN/PRC-148 MBITR can distribute configuration information or clone to another AN/PRC-148 MBITR. This cloning feature allows the G-6 or S-6 spectrum manager to disseminate technical information down the tactical echelons to each radio. The AN/PRC-148 MBITR system is suited for urban operations where communication signals are challenged by terrain or man-made structures. For these conditions, the AN/PRC-148 MBITR system provides a retransmission capability called back-to-back. The retransmission configuration requires two AN/PRC-148 MBITRs and a small cable kit with electronic filters. The radios repeat the transmission with no loss of signal quality.

I-51. The AN/PRC-148D/F series radios are versions of the handheld, manpack, and small form fit two-channel Leaders Radio and are enhanced versions of the legacy AN/PRC-148 MBITR radio. Its functionality is comparable to the existing MBITR radio operating SINCGARS and TACSAT integrated waveform on one channel, but includes a second channel capable of operating TSM and WREN narrowband networking waveforms.

I-52. Handheld, manpack and small form fit Leader Radios provide voice communications including support for data services such as text, control graphics, and imagery through interoperability with tactical end-user devices or Nett Warrior systems. Leader Radio systems include the full rate production two-channel AN/PRC-148D/F series, AN/PRC-163(v)2, and AN/PRC-163(v)3 systems. These systems operate SINCGARS and networking waveforms such as the TSM, WREN, single-channel TACSAT, and SATURN. Some configurations support mounting the leader radio to extend its communication range which provides additional amplification on the narrowband SINCGARS channel and TSM wideband channel.

## AN/PRC-150

I-53. The AN/PRC-150 HF radio provides tactical forces, homeland defense forces, and emergency operations elements with secure, terrain-independent line of sight and beyond line of sight voice and data communications. The AN/PRC-150 radio does not rely on retransmission or SATCOM. The radio provides long distance, wide area, gap free, fixed or on the move, ground, ground to air communications via plain text, secure analog voice with robust data, and digital voice modes and advanced serial tone electromagnetic counter-countermeasures modem Internet Protocol networking.

I-54. The AN/PRC-150 supports red and black key management and ALE link protection. The AN/PRC-150 vehicular radio systems provide units with beyond line of sight communications without having to rely on satellite availability on a saturated tactical communications network. The systems' manpack and vehicular configurations have reliable communications while on the move and allow for rapid transmission of data and imagery to display the common operational picture. The legacy AN/PRC-150 is replaced by AN/PRC-160 with ITN modernization upgrades.

I-55. When employed, the AN/PRC-150 radio supports single-channel radio operations, and frequency hopping radio operations. The AN/PRC-150 has the following characteristics and capabilities—

- Operates from 1.6–29.9999 MHz using skywave modulation with selectable low, medium, and high output power. It also operates from 20.0000–59.9999 MHz frequency modulation with a maximum output of 10.0 watts.
- Uses embedded type 1 multinational cryptographic algorithms for secure voice and data communications between ground and aircraft.
- Interfaces with SINCGARS cryptographic ignition key embedded in the removable keypad.
- Uses advanced electromagnetic counter-countermeasures serial-tone frequency hopping to improve communications reliability in jamming environments.
- Supports frequency hopping in HF narrowband, wideband, and list.

- Employs programmable system presets for one-button operation.
- Provides internal tuning unit that matches a wide variety of whip, dipole, and long-wire antenna automatically.
- Uses internal, high-speed serial-tone modem that provides data operation up to 9,600 bits per second.
- Uses embedded ALE digital voice 600 simplifies HF operation by automatically selecting an accepted channel.
- Supports North Atlantic Treaty Organization STANAG 4538 automatic radio control system link setup and data link protocols in third-generation ALE radio mode.
- Supports point-to-point protocol or ethernet networking capabilities.
- Supports North Atlantic Treaty Organization STANAG 4538-compliant wireless Internet Protocol data transfer.
- Supports frequency hopping in HF narrowband, wideband, and list.

### **AN/PRC-152A**

I-56. The AN/PRC-152A radio is a wideband networking handheld radio that provides simultaneous voice, video, and high-speed data in a portable form factor. The AN/PRC-152A radio delivers ad hoc, self-healing, and adaptive networking capabilities to dismounted Soldiers on the move. The voice and data capabilities of the AN/PRC-152A are National Security Agency-certified to encrypt networks classified up to top secret. The AN/PRC-152 includes an embedded GPS receiver to display local position and to provide automatic position reporting for situational awareness. The vehicular version of the AN/PRC-152 is the AN/VRC-110.

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**Note.** The AN/PRC-152A does not meet cryptographic modernization requirements as mandated by the DOD and the National Security Agency and is being replaced by the AN/PRC-148C/D or AN/PRC-163 Leader Radios.

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### **AN/PRC-158, GEN II Handheld, Manpack and Small Form Fit Manpack**

I-57. The upgraded AN/PRC-158 is a two-channel software-defined radio that hosts several waveforms and provides secure Internet Protocol-based voice and data communication. The radio set configurations provide networking transport to extend communication from brigade and battalion to company and platoon radios. The upgraded AN/PRC-158 replaces the legacy AN/PRC-158, Falcon IV as well as the handheld, manpack, and small form fit, GEN I, AN/PRC-155.

I-58. The legacy AN/PRC-158 radio is a multichannel manpack radio equipped with dual channel connectivity to provide forward-deployed Soldiers with simultaneous send and receive capabilities in a compact and lightweight form factor. The AN/PRC-158 routing and crossbanding technologies support communications redundancy, voice, and data communications. The AN/PRC-158 is National Security Agency-certified radio that is secure up to top secret security classification. The AN/PRC-158 covers the multiband frequency range, with up to 10 watts of transmit power for narrowband waveforms and up to 20 watts for SATCOM and wideband waveforms. The AN/PRC-158 supports the following waveforms:

- SINCGARS.
- VHF and UHF line of sight.
- SATURN.
- MUOS.
- UHF SATCOM.
  - Integrated waveform.
  - DAMA.
- TSM.
- WREN.

### **AN/PRC-160(V)**

I-59. The AN/PRC-160(V) expands secure HF technology by adding advanced encryption capabilities. This radio system provides enhanced wideband data performance and interoperability with legacy radios,

AN/PRC-104, AN/PRC-150, and AN/PRC-160. The AN/PRC-160(V) is a single-channel, half-duplex, tactical, HF transceiver that provides secure voice and data transmission. The AN/PRC-160(V) radio uses a single battery, which reduces the overall weight and provides 20-watt HF and 10-watts VHF transmission with continuous coverage from 1.5–59.9999 MHz. The AN/PRC-160(V) transmits data at frequencies from 3 kHz–24 kHz, at data rates exceeding 120 kilobits per second. The AN/PRC-160(V) uses software programmable encryption for both legacy and modern voice and data. The AN/PRC-160(V) replaces legacy AN /PRC 160 manpack radio which provides accurate position location and enhanced situational awareness using embedded selective availability anti-spoofing module and GPS receivers and secure voice and data confidentiality protection up to top secret classification level.

### **AN/PRC-162, Gen II Handheld, Manpack, and Small Form Fit Manpack**

I-60. The AN/PRC-162 is a two channel manpack radio that provides line of sight and beyond line of sight, high-bandwidth waveforms for on-the-move voice and data communications for mounted or dismounted Soldiers. The AN/PRC-162 can be mounted in a vehicle configuration that provides the capability for real-time sharing of data, images, voice, and video between assets on the battlefield. It supports retransmission service and can crossband voice or data between two separate networks. The AN/PRC-162 provides high-speed mobile ad hoc and networked communications, point-to-point data and voice networking, next-generation satellite, and MUOS communications. The AN/PRC-162 offers interoperability with legacy waveforms and has an open-architecture design to facilitate upgrades. The AN/PRC is interoperable with the following:

- SINCGARS.
- VHF and UHF line of sight.
- SATURN.
- MUOS.
- UHF SATCOM.
  - Integrated waveform.
  - DAMA.
- TSM.
- WREN.

I-61. The legacy AN/PRC-162 radio provides high-speed mobile ad hoc networked communications, point-to-point data and voice next-generation SATCOM, and MUOS and can be configured for vehicular and airborne platforms. The AN/PRC-162 radio provides the capability for real-time sharing of data, images, voice, and video between assets on the battlefield; offers interoperability with legacy waveforms and has an open-architecture design to facilitate upgrades. Handheld, manpack, vehicular, and airborne versions are available to support army tactical radio operations.

### **AN/PRC-163(V)2, Leader Radio**

I-62. The AN/PRC-163(V)2 provides crossbanding, interoperability, and redundancy in a small form factor. The AN/PRC-163(V)2 provides Soldiers with status information at a glance, which allows them to access basic functions without removing the radio from its holster. The radio provides two discrete radio modules, each capable of multi-band, multi-mode operation. The AN/PRC-163(V)2 replaces legacy AN/PRC-163 handheld radio which allows Soldiers to receive mission critical information by viewing a liquid crystal display screen. The intelligence, surveillance, and reconnaissance video module provides full-motion video capabilities for advanced situational awareness. The AN/PRC-163 is interoperable with—

- TSM waveform.
- SATURN.
- SINCGARS system improvement program and ASIP.
- Integrated waveform.
- SATURN.

**AN/PRC-166**

I-63. The AN/PRC-166 provides real-time, continuous position location information. The AN/PRC-166 is designed to track personnel and assets and provide situational awareness to prevent fratricide. It easily connects to tablets or laptops and is mountable onto robotic or unmanned aerial vehicles.

**AN/PRC-168(V)1**

I-64. The AN/PRC-168(V)1 includes an expanded frequency range, enhanced network throughput to stream high-definition videos. It is interoperable with compatible devices that offer TSM waveform functionality for tactical operations.

**AN/PRC-170**

I-65. The AN/PRC-170 is a small form factor TSM radio system. It includes enhancements which provide Soldiers with interoperable voice, streaming video, and data communications. The radio supports long wavelength band, satellite band frequencies (1250–2600 MHz) and UHF (225–450 MHz) communications.

**AN/PRC-171**

I-66. The AN/PRC-171 is a small form factor, wideband mobile ad hoc network radio that operates in the UHF and long wavelength band frequency range (225–2600 MHz) with a power range of up to 3.2 watts. The AN/PRC-171 radio is compatible with the AN/PRC-163(V)2 handheld Leader Radio batteries and accessories.

**AN/PRC-2064**

I-67. The AN/PRC-2064 is an interoperability system to interface otherwise incompatible systems compatible. A tactical voice bridge enables interoperability between different radio networks operating over any frequency range or waveform.

**GENERAL GROUND CONFIGURATIONS**

I-68. The ground radio communications configurations are single or multiple radio system configurations that provide high power, long-range communications for permanent or semi-permanent fixed sites, operating bases, or installations to include airfield or forward operations base. Ground radio communications configurations are not typically designed for transit or transportation and rely on permanent installation hardware, commercial power, and larger antennas. Ground radio communications configurations include the AN/GRC-100 series single-channel VHF radio used for ground to air communications.

**VEHICULAR CONFIGURATIONS**

I-69. The vehicular radio communications configurations are characterized by their rugged and enduring performance capabilities. Available for single-or dual-radio mounting, the vehicular-mounted tactical radios provide interoperability and high performance to units in a variety of operational environments, including remote environments characterized by harsh conditions. Configurations are customizable for a variety of missions requirements and include mobile assets, both ground and airborne. The Army inventory of vehicular-mounted assets includes the AN/PRC-117 (L3 Harris family), AN/PRC-148 C/D/F (manpack and mounted Leader Radios), AN/PRC-150, AN/PRC-158, AN/PRC-162, AN/PRC-163, AN/VRC-100, and the RT-1523 series.

**AN/VRC-100**

I-70. The AN/VRC-100 is an improved legacy HF, AM ground and vehicular radio that provides multifunctional, full digital signal processing. It is used for a variety of ground or mobile applications. Its primary role in the Army is for supporting tactical air traffic control and operations centers needing beyond line of sight ground to air communications with aircraft equipped with HF systems such as the AN/ARC-220 radio. The AN/VRC-100 radio is a fully integrated plug-and-play multi-mode voice or data communications system configured in a portable case that allows substantial distance communications beyond line of sight by

providing users the ability to maintain contact during short, mid, and long-range operations. As an advanced data communications system, the AN/VRC-100 provides reliable digital connectivity.

I-71. The AN/VRC-100 radio uses the RT, power amplifier coupler, and control display unit line replaceable units of the AN/ARC-220 system without modification, within an aluminum-structured, bracketed case. The AN/VRC-100 has a portable, metal case, with a removable top, that provides easy access for removal of line replaceable units. All controls and the radio input and output are located on the front panel. The AN/VRC-100 increases the situational awareness of aviation assets. The AN/VRC-100 has capabilities that are identical to the AN/ARC-220 HF radio, which makes the AN/VRC-100 radio an ideal radio to support AN/ARC-220 HF radio equipped airborne platforms. Key features of the AN/VRC-100 are—

- Full digital signal processing with embedded ALE, electromagnetic protection, and data modem.
- Spare card slot in the RT provides for future growth.
- Operates on 28 volts direct current (and is compatible with 24 volts direct current vehicular power) or from 115 or 220 volts alternating current 50 or 60 Hertz (Hz) power source.
- A personal computer or laptop connectivity.
- E-mail messaging using the local RS-232 interface.
- Capability to tune a variety of antennas.

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**Note.** The AN/VRC-100 requires the KY-100M to provide encryption capabilities. The KY-100M is cryptographic modernized according to DOD and National Security Agency requirements.

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I-72. Table I-2 lists the three configurations of the AN/VRC-100. Refer to the operator technical manual for more information on the AN/VRC-100(V) 1/2/3.

**Table I-2. AN/VRC-100 configurations**

| <b>Configuration</b> | <b>Description</b>                                                                                 |
|----------------------|----------------------------------------------------------------------------------------------------|
| AN/VRC-100(V) 1      | Consists of three-line replaceable units housed in a metal casing with a power supply and speaker. |
| AN/VRC-100(V) 2      | Consists of the AN/VRC-100(V) 1 mounted in a wheeled vehicle.                                      |
| AN/VRC-100(V) 3      | Consists of the AN/VRC-100(V) 1 with the AS-3791/G broadband antenna and is used at theater level. |

### ***RT-1523 SINCGARS***

I-73. Either the RT-1523 A/B/C/D or the RT-1523E (ground version receiver transmitter) comprise the core component of all ground-based radio sets. The RT-1523 series has internal COMSEC circuits, so they do not require additional cryptographic hardware for secure communications. The ground versions have a whisper mode for noise restriction during patrols or while in defensive positions. The radio operator whispers into the handset while being heard at the receiver in a normal voice. There are six ground unit versions of the SINCGARS (RT-1523/A/B and E/F). Table I-3 on page 174 compares the components of several versions of SINCGARS radios.

Table I-3. Comparison of SINCGARS versions and components

|           | <i>Short Range<br/>(consists of<br/>1 radio)</i> | <i>Long Range<br/>(consists of<br/>1 radio)</i> | <i>Power<br/>Amplifier</i> | <i>Dismount<br/>Manpack</i> | <i>Vehicular<br/>Amplifier<br/>Adapter<br/>(AM-7239C/E)</i> |
|-----------|--------------------------------------------------|-------------------------------------------------|----------------------------|-----------------------------|-------------------------------------------------------------|
| AN/VRC-87 | X                                                |                                                 |                            |                             | X                                                           |
| AN/VRC-88 | X                                                |                                                 |                            | X                           | X                                                           |
| AN/VRC-89 | X                                                | X                                               | X                          |                             | X                                                           |
| AN/VRC-90 |                                                  | X                                               | X                          |                             | X                                                           |
| AN/VRC-91 | X                                                | X                                               | X                          | X                           | X                                                           |
| AN/VRC-92 |                                                  | X (2)                                           | X (2)                      |                             | X                                                           |

## AIRBORNE CONFIGURATIONS

I-74. Airborne radio configurations play a vital role in providing communications between ground and airborne elements and communications between individual aircraft and formations. These radios are critical enablers essential to close air support, search and rescue, resupply, medical and casualty evacuation mission objectives. Airborne radio configurations are used for aviation platforms and provide line of sight and beyond line of sight communications supporting all aviation operations. line of sight communications systems supporting air-to-air and air-to-ground aviation operations. These single system configurations include adapters, installation hardware, and integration components to enable crew operations while on the aircraft, during flight, or while stationary on the ground. The airborne radio configurations supports traditional cockpit controls or human machine interface settings. The airborne radio configurations include such systems as—

- ARC 201, 201D (SC legacy SINCGARS radio).
- ARC 210 (SC multiband radio).
- ARC 220 (SC HF radio).
- ARC 231 (SC multiband, software-defined radio).

## AN/ARC-201

I-75. The AN/ARC-201 SINCGARS radio is a tactical airborne radio subsystem that provides secure, anti-jam voice and data communications. Ground and airborne versions are interoperable even though they each have distinct physical characteristics. The significant change in the airborne mode is the faceplate attached to the various configurations and the add-on modules that change each version's capabilities. Airborne versions RT-1476, RT-1477A/B/C, and RT-1478 require the KY-58 security equipment for ciphertext operations.

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**Note.** A new cryptographic compliant air ground network radio will replace the AN/ARC-201 radio. The air ground network will continue to support the SINCGARS waveform and provide MUOS and TSM to airborne rotary wing platforms.

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## RT-1476

I-76. The RT-1476, AN/ARC-201, is the base radio in all three versions; each version operates in the single-channel and frequency hopping modes. The RT-1476, AN/ARC-201 typically mounts in the cockpit of an aircraft.

## RT-1477

I-77. The RT-1477 mounts in an isolated equipment compartment on the aircraft and consists of radio and a radio set control, C-11466. The radio set control is co-located with the pilot and can remotely control from the cockpit. Dedicated cables transmit control and status signals back and forth between the RT-1477 and radio set control. The RT-1477 is capable of retransmission operation.

**RT-1478**

I-78. The RT-1478 mounts in the remote equipment compartment of the aircraft and operates in conjunction with the aircraft's system control display unit. The optional data rate adapter enables the radio to process 1,200 and 2,400 Hz frequency. The operation of the data rate adapter is automatic, and no operator interface is necessary.

**AN/ARC-210**

I-79. The AN/ARC-210 offered in several models, which when coupled with ancillary equipment, provides the aviation community with exceptional long-range capability. The RT-1556B provides line of sight VHF and UHF capability, SINCGARS, MUOS, and SATURN electromagnetic counter-countermeasures waveforms. The RT-1794I, RT-1824I, RT-1851I, and RT-1851AI are network capable and include embedded cryptographic algorithms, 5 kHz and 25 kHz and DAMA SATCOM.

I-80. The AN/ARC-210 radio provides air-to-air and air-to-ground, two-way voice communications via UHF and VHF. The embedded SATCOM functions that operate in the UHF radio band data and voice communications. The AN/ARC-210 is integrated on Army fixed wing platforms only. It provides the following key features—

- 30–400 MHz frequency range provides VHF and UHF in all radios; 121.5 and 243.0 MHz guard channels, and four-channel scan.
- 30–512 MHz frequency range providing VHF and UHF in the RT-1851AI; 121.5 and 243.0 MHz guard channels, four-channel scan.
- Tuning capability: 5 kHz with remote control, 2.5 kHz via 1553 bus.
- Optional power amplifiers, mounts, and low noise amplifier and diplexer.

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**Note.** A new cryptographic compliant air ground network radio will replace the AN/ARC-210 radio. The air ground network will continue to support the SINCGARS waveform and provide MUOS and TSM to airborne rotary wing platforms.

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**AN/ARC-220**

I-81. The AN/ARC-220 radio system is a microprocessor-based HF communications system intended for airborne applications. The ground version of the AN/ARC-220 is the AN/VRC-100. The AN/ARC-220 radio system uses advanced digital signal processor technology to provide two-way communication.

I-82. The AN/ARC-220 consists of three interchangeable units; an RT (RT-1749/URC or RT-1749A/URC), power amplifier coupler (AM-7531/URC), and control display unit (C-12436/URC). The AN/ARC-220 has embedded ALE, serial tone data modem, and anti-jamming functions. The AN/ARC-220 provides the electrical interface with other AN/ARC-220 and associated aircraft systems. The AN/ARC-220 can program up to twenty-five free text data messages in real time and retrieve data messages to be stored for later viewing. The AN/ARC-220 radio system provides the following capabilities—

- Frequency range: 2–29.9999 MHz in 100 Hz steps.
- Twenty user programmable simplex or half-duplex channels.
- Twelve programmable electromagnetic countermeasures hop sets.
- Certified for ALE.
- An integrated data modem that enables communication in noisy environments.
- Built-in integration with external GPS units allows position data report submission with the push of a button.
- Embedded ALE, electromagnetic countermeasures, and data.
- Rapid and efficient tuning to a variety of antennas.

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**Note.** The AN/ARC-220 requires the KY-100M to provide encryption capabilities. The KY-100M is cryptographic modernized according to DOD and National Security Agency requirements.

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## AN/ARC-231

I-83. The AN/ARC-231 is a multiband multi-mission airborne VHF and UHF radio capable line of sight and SATCOM radio system that has multiband multi-mission, secure anti-jam voice and data capabilities. The RT-1808 is the primary RT for the AN/ARC-231 that capitalizes on the AN/PSC-5 Spitfire family's expandable architecture and permits users to upgrade as new requirements drive new capabilities. The AN/ARC-231, used in the Army Airborne Command and Control System, provides communications capabilities to corps, division maneuver brigades, commander's airborne tactical command posts, or individual attack and lift helicopters. AN/ARC-231 radios will implement the SATURN waveform to replace Havequick I and II capabilities. The AN/ARC-231 has the following characteristics and capabilities—

- SINGARS communications modes.
- DAMA and integrated waveform tactical SATCOM modes.
- Frequency ranges of—
  - 30–87.975 MHz VHF and SINGARS.
  - 108–173.995 MHz VHF AM and FM modes.
  - 225–399.995 MHz UHF AM and UHF SATCOM band.
  - 403–511.995 MHz UHF modulation, public service band.
- Embedded COMSEC and transmission security keys with transmitting and receive over-the-air rekeying.
- 148 channel presets.
- Embedded data control waveform analog to digital converter and tactical Internet Protocol.
- Access to 5 kHz and 25 kHz UHF SATCOM channels.
- 8.33 kHz air traffic control channelization coverage to 512 Hz.
- Minimal size and weight are suitable for installation in rotary wing and fixed-wing platforms.

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**Note.** The AN/ARC-231 (RT-1808) radio will be replaced by cryptographic compliant AN/ARC-231A (RT-1987) radio.

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## TRANSPORTABLE (GROUND) CONFIGURATIONS

I-84. The transportable (ground) radio communications configurations include single or multiple radio systems designed to support communications for command posts operations or fixed locations and are not capable of operation while in transit. Configurations are transportable modular systems housed in self-contained transit cases and include lightweight adapters or docking stations, installation hardware, and power supplies designed for movement, when necessary, to support various tactical requirements. Transportable radio communications configurations include the AN/TRC 214 series.

## NARROWBAND (SINGLE-CHANNEL) TACTICAL SATELLITE

I-85. Narrowband (single-channel) TACSAT radios are small, lightweight, manpack, multiband, multimode VHF and UHF radios. Narrowband (single-channel) TACSAT radios also provide secure voice communications and situational awareness for special operations forces, airborne, air assault, infantry divisions, and infantry brigades.

I-86. The Army uses narrowband (single-channel) TACSAT radio and other satellite capabilities to provide long haul, worldwide communications coverage to support critical command and control communications to ground and mobile operating forces.

I-87. Narrowband (single-channel) TACSAT radios provide range extension for voice and data. The beyond line of sight range extension capability is used in the Army's SATCOM on-the-move OE-563 functionality in moving vehicular platforms, versus stationary. Narrowband terminals are preferred for initial communications in contingency situations because of mobility and size advantages. Their disadvantages include difficulty in obtaining access to the UHF range of the EMS and the lack of anti-jam capability for threat mitigation.

I-88. Narrowband (single-channel) TACSAT radios offer interoperability with integrated waveform terminals and DAMA when accessing non-processed narrowband SATCOM and offer greater satellite access and automated channel sharing, though at reduced data rates. This limits normal access to 2400 bits per

second with advanced narrowband digital voice terminal provided voice and data; however, with voice encoder and mixed excitation linear prediction upgrades, voice quality and clarity are vastly improved. Integrated waveform provides access improvements of up to four times the rate seen in DAMA on a 25 kHz channel. Capabilities associated with integrated waveform include—

- Network communications.
- Preplanned support to operations.
- Ad hoc communications.
- Point-to-point calls

### **Shadowfire, AN/PSC-5C**

I-89. The Shadowfire is a field upgrade of the AN/PSC-5 Spitfire terminal. The upgrade provides all the capabilities of the AN/PSC-5 and additional capabilities for SATURN and SINCGARS anti-jam. The ability to receive and transmit over-the-air rekeying or over-the-air transfer, extended 30–420 MHz frequency range, access to 5 kHz and 25 kHz UHF SATCOM channels, and embedded data control waveform controller.

I-90. Additional features include range extension and mixed excitation linear prediction voice coding, 142 preset channels, advanced key loading, DS-101 fill capability and embedded tactical Internet Protocol and the cryptographic algorithm. The Shadowfire operates in the VHF and UHF spectrum and supports line of sight with frequency agile modes, SATCOM, DAMA, and maritime operation. Voice and data operations are available in each of these modes. The Shadowfire also provides enhanced mixed excitation linear prediction, vocoder and improved linear predictive coding, anti-jam communications, and over-the-air rekeying and over-the-air transfer capabilities.

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**Note.** DS-101 is an interface standard for electronic transfer devices or fill units developed and endorsed by the US National Security Agency.

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### **AN/PSC-5D**

I-91. The multiband multi-mission radio, AN/PSC-5D, offers a higher frequency range than the Spitfire and Shadowfire. The AN/PSC-5D provides lightweight, secure, network-capable, multiband multi-mission, anti-jam, voice, imagery, and data communications capability in a single package. The AN/PSC-5D operates in the VHF and UHF spectrum band and supports line of sight with frequency agile modes, SATCOM, DAMA, and maritime operation. Voice and data operations are available in each of these modes. Table I-4 on page 178 outlines a line of sight interoperability comparison of the AN/PSC-5 family of radios and the AN/PRC-117F. For more information on narrowband (single-channel) TACSAT and DAMA, refer to the equipment operator manual.

Table I-4. AN/PSC-5/C/D, AN/PRC-117F and AN/ARC-231 line of sight interoperability

| <i>Radio Item</i>                                                                                                                                                                                                                                                                                                                                               | <i>AN/PSC-5C<br/>Shadowfire</i>                    | <i>AN/PSC-5D<br/>and<br/>AN/ARC-231</i>            | <i>AN/PRC-117F</i>                                                                                                       | <i>AN/PRC-117G</i>                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Frequency Range MHz</b>                                                                                                                                                                                                                                                                                                                                      | 30–420 MHz                                         | 30–512 MHz                                         | 30–512 MHz                                                                                                               | 30MHz–2.00GHz                                                                                                            |
| <b>Voice 12 kilobits per second</b>                                                                                                                                                                                                                                                                                                                             | FASCINATOR                                         | FASCINATOR                                         | FASCINATOR                                                                                                               | FASCINATOR                                                                                                               |
| <b>Voice 16 kilobits per second</b>                                                                                                                                                                                                                                                                                                                             | VINSON                                             | VINSON                                             | VINSON                                                                                                                   | VINSON                                                                                                                   |
| <b>Data 16 kilobits per second</b>                                                                                                                                                                                                                                                                                                                              | VINSON, 3 or 4 KG-84C                              | VINSON, 3 or 4 KG-84C                              | VINSON, 3 or 4 KG-84C                                                                                                    | VINSON, 3 or 4 KG-84C                                                                                                    |
| <b>Data over 16 kilobits per second</b>                                                                                                                                                                                                                                                                                                                         | 1-4 KG-84C (3 KG-84C)-up to 48 kilobits per second | 1-4 KG-84C (3 KG-84C)-up to 48 kilobits per second | KG-84C - up to 56 kbps (25kHz channel)<br>ANW2 - 50 kbps to 5 Mbps                                                       | KG-84C - up to 56 kbps (25kHz channel)<br>ANW2 - 50 kbps to 5 Mbps                                                       |
| <b>CTCSS</b>                                                                                                                                                                                                                                                                                                                                                    | Yes                                                | Yes                                                | Selectable tone, and digital (CDCSS, CTCSS)                                                                              | Selectable tone, and digital (CDCSS, CTCSS)                                                                              |
| <b>SINCGARS Frequency Hopping</b>                                                                                                                                                                                                                                                                                                                               | No                                                 | Yes                                                | Yes                                                                                                                      | Yes                                                                                                                      |
| <b>Guard frequency</b>                                                                                                                                                                                                                                                                                                                                          | Yes                                                | Yes                                                | Yes                                                                                                                      | Yes                                                                                                                      |
| <b>Channel Spacing</b>                                                                                                                                                                                                                                                                                                                                          | 5 and 25 kHz                                       | 5, 6.25, 8.33, 12.5, 25 kHz                        | Narrowband: 8.33, 12.5, 25 kHz<br>SATCOM: 5, 25 kHz<br>Wideband: 500 kHz, 1.2, 2.5, 5 MHz<br>FM deviation: 5, 6.5, 8 kHz | Narrowband: 8.33, 12.5, 25 kHz<br>SATCOM: 5, 25 kHz<br>Wideband: 500 kHz, 1.2, 2.5, 5 MHz<br>FM deviation: 5, 6.5, 8 kHz |
| <b>Legend:</b><br>ANW2 Adaptive Networking Wideband Waveform<br>CDCSS continuous digital-coded squelch system<br>CTCSS continuous tone-codes squelch system<br>FM frequency modulation<br>Hz hertz<br>kb/s kilobits per second<br>kHz kilohertz<br>MHz megahertz<br>SATCOM satellite communications<br>SINCGARS single-channel ground and airborne radio system |                                                    |                                                    |                                                                                                                          |                                                                                                                          |

I-92. Table I-5 on page 179 outlines a DAMA line of sight 5 kHz and 25 kHz interoperability comparison of the AN/PSC-5 family of radios and the AN/PRC-117F. For more information on narrowband (single-channel) TACSAT, refer to the operator technical manual.

Table I-5. AN/PSC-5/C/D, AN/ARC-231 and AN/PRC-117F

| <i>Terminal Mode</i>               | <i>AN/PSC-5</i>       | <i>AN/PSC-5C</i>                                  | <i>AN/PSC-5D<br/>and<br/>AN/ARC-231</i>            | <i>AN/PRC-117</i>                                                                                                                           |
|------------------------------------|-----------------------|---------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Voice 16 kilobits per second       | VINSON                | VINSON                                            | VINSON                                             | VINSON                                                                                                                                      |
| Data 16 kilobits per second        | VINSON, 3 or 4 KG-84C | VINSON, 3 or 4 KG-84C                             | VINSON, 3 or 4 KG-84C                              | VINSON, 3 or 4 KG-84C                                                                                                                       |
| Data (over 16 kilobits per second) | No                    | 14 KG-84C (3 KG-84C)-up to 48 kilobits per second | 1-4 KG-84C (3 KG-84C)-up to 56 kilobits per second | 1-4 KG-84C (3 KG-84C)-up to 56 kilobits per second<br><br>*Must use 181B for interoperability (high performance waveform between 117F only) |

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**Note.** The AN/PSC-5 C/D, AN/PRC-117F, and AN/PRC-117G radios do not meet cryptographic modernization requirements as mandated by the DOD and the National Security Agency. They are being replaced by the AN/PRC-158 or AN/PRC-162 handheld, manpack and small form fit radios, which are cryptographic compliant.

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# Glossary

The glossary lists acronyms and terms with Army, multi-service, or joint definitions, and other selected terms. Terms for which ATP 6-02.53 is the proponent manual (the authority) are marked with an asterisk (\*). The proponent manual for other terms listed in parentheses after the definition.

## SECTION I – ACRONYMS AND ABBREVIATIONS

|                |                                                   |
|----------------|---------------------------------------------------|
| <b>ADP</b>     | Army doctrine publication                         |
| <b>ALE</b>     | automatic link establishment                      |
| <b>AM</b>      | amplitude modulation                              |
| <b>AR</b>      | Army regulation                                   |
| <b>ASIP</b>    | Advanced system improvement program               |
| <b>ATP</b>     | Army techniques publication                       |
| <b>CEMA</b>    | cyberspace electromagnetic activities             |
| <b>CEWO</b>    | cyberspace electromagnetic warfare officer        |
| <b>CJCSI</b>   | Chairman of the Joint Chiefs of Staff instruction |
| <b>CJCSM</b>   | Chairman of the Joint Chiefs of Staff manual      |
| <b>COMSEC</b>  | communications security                           |
| <b>DA</b>      | Department of the Army                            |
| <b>DAMA</b>    | demand-assigned multiple access                   |
| <b>dBm</b>     | decibels above one milliwatt                      |
| <b>DD</b>      | Department of Defense                             |
| <b>DODI</b>    | Department of Defense instruction                 |
| <b>DODIN</b>   | Department of Defense information network         |
| <b>DODIN-A</b> | Department of Defense information network-A       |
| <b>EMCON</b>   | emission control                                  |
| <b>EMS</b>     | electromagnetic spectrum                          |
| <b>EPM</b>     | electronic protective measures                    |
| <b>FORSCOM</b> | United States Army Forces Command                 |
| <b>FM</b>      | field manual                                      |
| <b>G-1</b>     | assistant chief of staff, personnel               |
| <b>G-2</b>     | assistant chief of staff, intelligence            |
| <b>G-3</b>     | assistant chief of staff, operations              |
| <b>G-4</b>     | assistant chief of staff, logistics               |
| <b>G-6</b>     | assistant chief of staff, signal                  |
| <b>GEN</b>     | generation                                        |
| <b>GHz</b>     | gigahertz                                         |
| <b>GPS</b>     | Global positioning system                         |

|                    |                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GTA</b>         | graphic training aid                                                                                                                      |
| <b>HF</b>          | high frequency                                                                                                                            |
| <b>ITN</b>         | integrated tactical network                                                                                                               |
| <b>JCR</b>         | Joint capabilities release                                                                                                                |
| <b>JP</b>          | joint publication                                                                                                                         |
| <b>JSIR</b>        | joint spectrum interference resolution                                                                                                    |
| <b>kb/s</b>        | kilobits per second                                                                                                                       |
| <b>kHz</b>         | kilohertz                                                                                                                                 |
| <b>MBITR</b>       | multiband inter/intra team radio                                                                                                          |
| <b>MCTP</b>        | Marine corps tactical publication                                                                                                         |
| <b>METT-TC (I)</b> | mission, enemy, terrain and weather, troops and support available, time available, civil considerations, and informational considerations |
| <b>MHz</b>         | megahertz                                                                                                                                 |
| <b>MIL-STD</b>     | military standard                                                                                                                         |
| <b>MUOS</b>        | mobile user objective system                                                                                                              |
| <b>NATO</b>        | North Atlantic Treaty Organization                                                                                                        |
| <b>Net ID</b>      | network identification                                                                                                                    |
| <b>PACE</b>        | primary, alternate, contingency, and emergency                                                                                            |
| <b>PNT</b>         | positioning, navigation, and timing                                                                                                       |
| <b>QEAM</b>        | quick erect antenna mast                                                                                                                  |
| <b>RF</b>          | radio frequency                                                                                                                           |
| <b>RT</b>          | receiver-transmitter                                                                                                                      |
| <b>S-1</b>         | battalion or brigade personnel staff officer                                                                                              |
| <b>S-2</b>         | battalion or brigade intelligence staff officer                                                                                           |
| <b>S-3</b>         | battalion or brigade operations staff officer                                                                                             |
| <b>S-4</b>         | battalion or brigade logistics staff officer                                                                                              |
| <b>S-6</b>         | battalion or brigade signal staff officer                                                                                                 |
| <b>SATCOM</b>      | satellite communications                                                                                                                  |
| <b>SATURN</b>      | second generation anti-jam tactical ultrahigh frequency radio for North Atlantic Treaty Organization                                      |
| <b>SINGARS</b>     | single-channel ground and airborne radio system                                                                                           |
| <b>SMCT</b>        | Soldier's manual of common tasks                                                                                                          |
| <b>SOP</b>         | standard operating procedure                                                                                                              |
| <b>STANAG</b>      | standardization agreement                                                                                                                 |
| <b>STP</b>         | Soldier training publication                                                                                                              |
| <b>TACSAT</b>      | tactical satellite                                                                                                                        |
| <b>TC</b>          | training circular                                                                                                                         |
| <b>TSM</b>         | tactical scalable mobile ad hoc network                                                                                                   |
| <b>UHF</b>         | ultrahigh frequency                                                                                                                       |
| <b>VHF</b>         | very high frequency                                                                                                                       |
| <b>WREN</b>        | warrior robust enhanced network                                                                                                           |

## SECTION II – TERMS

**adversary**

A party acknowledged as potentially hostile to a friendly party and against which the use of force may be envisaged. (JP 3-0)

**communications security**

Protective measures to deny unauthorized access to information of value derived from use of communications systems. (JP 6-0)

**Department of Defense information network**

The information capabilities and associated processes owned or leased by the Department of Defense for collecting, processing, storing, disseminating, and managing information, whether interconnected or stand-alone. (JP 6-0)

**Department of Defense information network-Army**

An Army-operated enclave of the Department of Defense information network that encompasses all Army information capabilities that collect, process, store, display, disseminate, and protect information worldwide. (ATP 6-02.71)

**electromagnetic interference**

Any electromagnetic disturbance, induced intentionally or unintentionally, that interrupts, obstructs, or otherwise degrades or limits the effective performance of electromagnetic spectrum-dependent systems and electrical equipment. (JP 3-85)

**electromagnetic operational environment**

A composite of the actual and potential electromagnetic energy radiation, conditions, circumstances, and influences that affect the employment of capabilities and the decisions of the commander. (JP 3-85)

**electromagnetic order of battle**

A subset of the overall order of battle that consists of the identification, strength, command structure, disposition, and operating parameters of the electromagnetic spectrum-dependent systems. (JP 3-85)

**electromagnetic jamming**

The deliberate radiation, reradiation, or reflection of electromagnetic energy for the purpose of preventing or reducing an enemy's effective use of the electromagnetic spectrum, with the intent of degrading or neutralizing the enemy's combat capability. (JP 3-85)

**electromagnetic pulse**

A strong burst of electromagnetic radiation caused by a nuclear explosion, energy weapon, or by natural phenomenon, that may couple with electrical or electronic systems to produce damaging current and voltage surges. (JP 3-85)

**emission control**

The selective and controlled use of electromagnetic, acoustic, or other emitters to optimize command and control capabilities while minimizing, for operations security: a. detection by enemy sensors, b. mutual interference among friendly systems, and/or c. enemy interference with the ability to execute a military deception plan. (JP 3-85)

**enemy**

A party identified as hostile against which the use of force is authorized. (ADP 3-0)

**\*net control station**

A communications station designated to control traffic and enforce circuit discipline within a given net.

**operational environment**

The aggregate of the conditions, circumstances, and influences that affect the employment of capabilities and bear on the decisions of the commander. (JP 3-0)

**threat**

Any combination of actors, entities, or forces that have the capability and intent to harm United States forces, United States national interests, or the homeland. (ADP 3-0)

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**31 July 2025**

By Order of the Secretary of the Army:

**RANDY A. GEORGE**

*General, United States Army  
Chief of Staff*

Official:

A handwritten signature in black ink, appearing to read 'M. Sannito', with a horizontal line extending from the end of the signature.

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